

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Course - Evaluating & Improving AI Models

Transition-Year readiness | Weeks 1-6 (Core + Lab)

STUDENT LESSON CONTENT

Week 1 - Core | AI Conceptual Understanding - How workflow designs affect outcomes

I can...

- I can explain how changing the steps in a workflow changes the outcome it gives.
- I can trace an example from input to output and show where the workflow design made the difference.
- I can convince my partner, using the example, why one workflow design gave a better outcome.

Words to know

- workflow - a planned sequence of steps that turns an input into an outcome
- design - the plan of a workflow: which steps it has and the order they run in
- check step - a step added to test or verify the work before the outcome is given
- outcome - the result a workflow produces at the end
- analyse - to break something down and reason about why it turned out the way it did

Before the new unit starts, we are making one big idea rock-solid: the way a workflow is designed decides how

good its outcome can be. Today you become the detective who traces an outcome back to the design that produced it.

The design behind the outcome

Every workflow follows a design - a plan of which steps run and in what order. That design is the hidden

ingredient. If a workflow is designed with the right steps in the right places, it usually produces a reliable

outcome. If a step is missing, out of order, or in the wrong place, the outcome has gaps - even when your input

is perfect. So when an outcome looks wrong, a good detective does not just blame the input.

They ask: how was

this workflow designed, and which step let the problem through?

Same input, different design

In a workflow the shape is always input, then process, then outcome. The design works inside the process,

deciding which steps the input passes through. That is why two workflows can take the exact same input and

give very different outcomes - their designs were different. Your job is to trace the outcome back through the

steps to the design choice that shaped it, and to explain the link out loud. Remember the charter: the examples

are made by your teacher, use no personal data, and a confident-looking outcome still has to be checked

before you trust it.

The same input runs through two designs - the added check step changes the outcome.

Why it matters

Why it matters - across the UAE, teams design AI workflows for things like sorting hospital paperwork, checking

solar-panel images for faults, and translating between Arabic and English; a workflow that skips a check step

can let a costly error through, so understanding design helps you question and improve the AI around you.

Let's try it - hands-on

1. Read the workflow card: the input, the two designs, and each outcome.
2. Map each design from input, through its steps, to the outcome.
3. Write how the design explains the outcome's quality - which step it added, moved or dropped.
4. Compare the two designs and note the single change that made the outcomes differ.
5. Explain one example to your partner, pointing to the design as the cause.

Think about it

Think about it - if two workflows take the same input but give different outcomes, what does that tell you about

how each one was designed?

My reflection

My reflection: One way a workflow's design affected an outcome I looked at today was ...

Week 1 - Lab | AI Conceptual Understanding - How workflow designs affect outcomes (Lab)

I can...

- I can investigate workflows and show how a change in design changed the outcome.
- I can map how an input flows through the steps to the outcome.
- I can justify to a partner which design gave the better outcome and back it with evidence.

Words to know

- fair test - a test where you change one thing only, so you know what caused the result
- controlled - kept the same on purpose, so it cannot be the reason for a difference
- map - to lay out each step of a workflow so you can see how it connects
- evidence - what you actually observed, used to back up your conclusion
- conclusion - the idea you decide on after weighing the evidence

In the Core period you traced outcomes back to their workflow design. Today you run the fair test yourself - keep

the input the same, change only one design choice, and prove which design makes the better outcome.

Change one thing, prove the cause

A good investigator uses a fair test. Here, the input and the underlying model are kept exactly the same for both

workflows - the only thing that changes is one design choice, such as adding a check step. So if the two

outcomes come out different, there is only one thing left to blame: the design. That is powerful. It turns 'I think

this design is better' into 'I can prove this design is better', because you controlled everything else. Mapping the steps makes the proof visible to anyone who reads it.

Map it, then justify it

On your mapping sheet you lay out the shared input, the two designs, their steps and the two outcomes. When it

is all in front of you, you can point to exactly where the design made the difference and say which design won.

Then you justify it to a partner using the map as evidence - not a feeling. Remember the charter: the examples

are made by your teacher, use no personal data, and even a fluent, confident outcome has to be checked

before you trust the design behind it.

A fair test holds everything constant except the one thing you are testing - here, the workflow design.

Why it matters

Why it matters - engineers building UAE tools, from smart-traffic systems in Abu Dhabi to Arabic-English

translators, test different workflow designs before they choose one; running a fair test on the design is exactly

how they decide which workflow to trust and improve.

Let's try it - hands-on

1. Read the first workflow pair and predict which design gives the better outcome.
2. Map the shared input, the two designs, the steps and the two outcomes.
3. Write how the design choice explains the difference in the outcomes.
4. Rotate to a new pack and map it the same way.
5. Write a conclusion naming which design won, and justify it to your partner.

Today's tool - hands-on

Tool: Teachable Machine / Scratch (workflow designs affect outcomes)

1. Run design A on a small test-set and score it.
2. Run design B on the same test-set and score it.
3. Compare the two scores and describe how the design changed the outcome.
4. Name the single design choice that mattered most.
5. Write the rule: a better design gives a better outcome.

You will make: A two-design comparison explaining how design affects outcomes.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Two workflow designs, same task, different measured outcomes.

Think about it

Think about it - if you wanted to prove that one design was more reliable, what one thing would you keep the same and what one thing would you change?

My reflection

My reflection: The workflow design that produced the better outcome today was ..., and my evidence was ...

Week 2 - Core | AI Solution Design & Development - Refines workflows by improving structure, sequencing and tool selection

I can...

- I can reorganise a messy workflow into clear stages that are easy to follow.
- I can put the steps in the right order and pick the right tool for each step.
- I can run my refined workflow and explain why it now gives a more reliable output.

Words to know

- workflow - an ordered set of steps that turns an input into a useful output
- structure - how the steps are grouped into clear stages
- sequencing - the order the steps run in
- tool selection - choosing a tool that can actually do each step
- verify - to check an output against the task and a trusted source before you trust it

Today you are a workflow engineer. You are handed a solution that already runs but is clumsy, and your job is to

refine it - regroup the steps, fix the order, and pick the right tool for each job - so the output you can actually

trust.

Three levers refine any workflow

When a workflow gives a poor result, you do not scrap it - you refine it with three levers.

Structure is how you

group the steps into stages, so a reader can follow and check the solution. Sequencing is the order the steps run

in - checking a fact before you publish, not after. Tool selection is picking a tool that can really do each step,

because a summariser can shorten text but it cannot tell you whether a figure is true. Pull the right lever and the

same steps produce a far more reliable output.

Order and tools decide whether errors get caught

Imagine a notice that looks perfect but lists the wrong temperature, because the verify step ran last - after

publishing. Moving that one step earlier means the error is caught before anyone reads it. That is the power of

sequencing. And if the wrong tool was doing the checking, no amount of reordering would help - you also need

the right tool for the job. Keep the class rules while you work: the routine is run by your teacher, there is no

personal data anywhere, and every output is verified before you rely on it.

The same steps, refined with three levers: structure, sequencing and tool selection.

Why it matters

Why it matters - across the UAE, teams building AI services spend most of their time refining the workflow

behind a tool, not the tool itself: reordering steps, adding a checking stage, swapping a tool that does not fit. The person who can improve structure, sequence and tool choice is the one who makes the whole solution reliable.

Let's try it - hands-on

1. Read the draft workflow and agree what a good final notice looks like.
2. Regroup the steps into clear stages - the structure lever.
3. Reorder the steps so verifying comes before publishing - the sequencing lever.
4. Check each tool and swap any that cannot do its job - the tool-selection lever.
5. Run the notes through your refined workflow, verify one figure, and note the change that helped most.

Think about it

Think about it - if two teams used the same steps and the same tools but in a different order, could one produce

a reliable notice and the other a misleading one? What single change would matter most?

My reflection

My reflection: The change I made that improved my workflow the most, and which lever it was, was ...

Week 2 - Lab | AI Solution Design & Development - Refines workflows by improving structure, sequencing and tool selection (Lab)

I can...

- I can design a multi-step workflow that produces a version for each audience.
- I can compare two workflow designs and improve the weaker one.
- I can justify my structure, order and tool choices and say how they make the output reliable.

Words to know

- shared stage - the steps both audiences need, done once before the workflow branches
- branch - where the workflow splits to make a different version for each audience
- reading-level tool - a tool that makes text easier or harder to read
- fit for audience - right for the people who will read it, in language and level
- justify - to give clear, evidence-backed reasons for a design choice

In the Core period you refined a workflow that made one notice. Today the job is harder: the same alert has to

reach students in plain English and parents in Arabic, both correct. You will judge two designs and improve the

better one - same three levers, bigger challenge.

Do the shared work once, then branch

When two audiences need the same facts, do not build two separate workflows. Clean, organise and verify the

facts once, in a shared stage, and then let the workflow branch - one path adjusts the reading level for students,

the other translates for parents. This is a structure choice: the shared stage means the facts are checked once

and cannot disagree between versions. Branching after the check, not before, is what keeps both outputs

saying the same true thing. More tools are not better - each tool has to earn its place by doing a job that is actually needed.

Order and tools decide whether both versions are right

Picture translating the alert first and verifying afterwards. A wrong figure gets carried into the Arabic version

before anyone catches it, so both outputs are wrong. Verify in the shared stage, before the branch, and the error

is caught once for everyone. Then each branch needs the right tool: a translator can make the Arabic version but

cannot simplify English, and a reading-level tool cannot translate. Keep the class rules while you design: the

routine is teacher-run, there is no personal data, and every output is verified against the source before you trust

it.

A shared verify stage, then a branch: one tool per audience keeps both versions correct.

Why it matters

Why it matters - public services in the UAE routinely publish the same alert in Arabic and English, for different

readers, and it has to be correct in every version. The engineer who structures the shared checking, orders the

steps and picks the right tool for each audience is the one who keeps every version trustworthy.

Let's try it - hands-on

1. Read the brief and both required outputs - a student version and an Arabic parent version.
2. Judge designs A and B against structure, sequencing and tool selection, logging each weakness.
3. Choose the stronger design and mark the shared stage and the branch point.
4. Improve it - verify in the shared stage, then match a fitting tool to each branch.
5. Run it, verify one fact in each output, and justify a key design choice.

Today's tool - hands-on

Tool: Scratch (refine a workflow's structure and sequencing)

1. Run the starter project and note what is clumsy or fails.
2. Reorder or regroup the steps to improve the structure.
3. Add one check step and choose the better-suited block/tool where relevant.
4. Re-run and confirm the output improved.
5. Explain which refinement helped most and why.

You will make: A refined Scratch workflow and an explanation of the improvement.

Stay safe: No personal data; neutral inputs only.

A clumsy workflow refined into a clearer, better-sequenced one.

Think about it

Think about it - if a new third audience were added later, where in your design would you add the branch, and

which steps would still be shared? What does that tell you about good structure?

My reflection

My reflection: One design choice I made to serve both audiences reliably, and why it worked, was ...

Week 3 - Core | Critical Evaluation & Informed Interaction - Iterating a Workflow and Explaining the Impact of Changes

I can...

- I can change one step of a workflow on purpose and say exactly what I changed and why.
- I can judge two versions of a workflow against the same criteria using evidence, not opinion.
- I can explain how a change made the output better or worse and what I would try next.

Words to know

- iterate - to improve something by repeating a loop of run, check, change, and run again
- criterion - a clear rule you agree in advance for judging whether a result is good
- evidence - what you can actually point to, like a count of correct cards, not just a feeling
- impact - the difference a change makes to the result, for better or worse
- verify-before-trust - checking an output against the task and a trusted source before believing it

Great AI solutions are almost never right first time. Today you learn the loop that makes them better: change one

thing, check it against clear rules, and explain what your change actually did.

One change at a time

When you want to improve a workflow, it is tempting to change lots of things at once. The problem is that if the

result changes, you will not know which change did it. So you change exactly one thing, run it again, and

compare. This is how you can honestly explain the impact of a change: you can point to the one thing that was

different and the difference it made. Scientists work this way too, and so do the people who build and improve

real AI models.

Judge with criteria, not vibes

A new version can look smart and still be wrong. That is why you agree your criteria before you look at the

results, so your judgement stays fair. Then you compare each version against those criteria using evidence you

can count or point to. If an output looks confident but fails the criteria, you do not trust it - you check it against

the task and a source you trust. Good evaluators are calm and honest: they follow the evidence, even when it

means their favourite version was not the best one.

The iterate loop: run, judge against criteria, change one thing, run again, compare.

Why it matters

In the UAE, teams building AI for hospitals, transport and schools improve their models the same way you will

today - one careful change at a time, judged against clear criteria - because a small improvement can matter to

thousands of people.

Let's try it - hands-on

1. Agree one clear criterion with your partner for a good sort of the cards.
2. Run version 1 of your sorting rule and record how it did against the criterion.
3. Choose one change to try, and write it down before you run it.
4. Run version 2 with only that one change and record the new result.
5. Write your impact sentence: the change was ____, the impact was ____, and my evidence was ____.

Think about it

If two versions look equally neat but one meets the criterion and one does not, which should you trust, and

why?

My reflection

My reflection: the change I am most proud of this lesson was ____ because ____.

Week 3 - Lab | Critical Evaluation & Informed Interaction - Lab: Iterating and Evaluating a Model's Outputs Against Criteria

I can...

- I can score a set of curated outputs against criteria and back up each score with evidence.
- I can complete two iteration cycles, changing one thing each time, and keep a clear log.
- I can explain how my changes added up to a better workflow and justify my final choice.

Words to know

- evaluate - to judge outputs against agreed criteria using evidence
- curated set - a group of outputs chosen for you to compare and score
- cycle - one full pass of the loop: score, change one thing, re-score
- cumulative impact - the combined difference several changes make together
- justify - to defend your judgement with reasons and evidence

Today you go deeper: you become the evaluator. You will score a set of AI outputs against fair rules, improve a

workflow over two careful cycles, and justify why your final version wins.

Score the set, not the single output

One output does not tell you much - it might have been lucky or unlucky. So you score a whole set against the

same criteria and look for a pattern. The pattern is what points you to your next change. When you score, you

always attach evidence, so anyone reading your grid can see why output B sits above output C.

This is exactly

how real teams check whether an AI model is any good: not by admiring one nice answer, but by testing many

and following the evidence.

Justify the whole journey

By the end you will have three versions of your workflow and a log of what you changed.

Justifying your final

version does not mean pointing only at your last clever tweak. It means telling the whole story: version one

scored this, your first change did that, your second change did this, and together they moved the result to here.

When you can relate all that evidence to your criteria in one clear judgement, you are doing real critical

evaluation - calm, honest, and backed up.

Why it matters

AI teams across the UAE, from smart-city services to healthcare, keep versioned logs and justify every

improvement with evidence - the same habit you are practising - because they must be able to defend why a

newer model is safer or better before it reaches real people.

Let's try it - hands-on

1. Score every output in the set against the agreed criteria and write your evidence beside each score.

2. Use the pattern in your scores to choose your first change, and log it before you run it.

3. Run cycle 1, re-score against the same criteria, and record what happened.

4. Choose an evidence-led second change, log it, and run cycle 2.

5. Write a justification that explains the cumulative impact of both changes on your result.

Today's tool - hands-on

Tool: Teachable Machine retrain + packs (improve through iteration)

1. Test the current model and record the score.

2. Make one change and retrain (an iteration).

3. Re-test and record the new score.

4. Repeat once more, so you have three scores.

5. Explain the impact of each change with evidence.

You will make: A three-iteration score record explaining the impact of each change.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Scores rising across three iterations as changes are made.

Think about it

If your second change helped but your first change hurt, is your final version still an improvement on version

one? How would you know?

My reflection

My reflection: the strongest piece of evidence in my justification was ____ because ____.

Week 4 - Core | Ethics & Governance - Spot the risk, propose responsible-use fixes

I can...

- I can read an AI scenario and name the real risks and ethical or governance concerns in it.
- I can propose sensible responsible-use measures that reduce those risks.
- I can give a reason for each measure that explains why it helps.

Words to know

- risk — a way an AI system could cause harm or an unfair result
- responsible use — putting checks around AI so its mistakes are caught and limited
- bias — when a model works worse for some people because of gaps in its data
- accountable — the person or group answerable for what the AI does
- transparent — being open about how the model works and where its limits are

Every AI model can be wrong, and sometimes wrong in unfair ways. Today you learn to spot those risks and

propose fixes that actually help, with a reason.

From worry to a real fix

Saying 'be careful with AI' does not change anything. A responsible-use measure is a concrete step that acts on

one risk: a human checks every flag, the model uses no personal data, or people can challenge a decision. The

skill is to match the fix to the risk and explain the link. Start from the risk, then choose a measure, then say

because. That because is the part that shows you really understand the problem, not just the vocabulary.

Who is responsible?

Governance is about rules and roles: who may use the model, on what data, and who answers if it goes wrong. A

clever model with nobody accountable is still unsafe. When you evaluate a model, ask who decides, who

checks, and how someone affected can push back. Good answers name a person or role, not just the machine,

because a machine cannot take responsibility for a decision on its own.

The risk - measure - because chain that turns a worry into a real fix.

Why it matters

Across the UAE, AI is being trialled in schools, clinics and city services, so knowing how to reduce its risks

responsibly is a real citizenship skill, not just a class exercise.

Let's try it - hands-on

1. Read your scenario card and underline the decision the AI influences and who it affects.
2. Write two risks you can see, in your own words.
3. For each risk, propose one responsible-use measure that would reduce it.
4. Add a because to each measure that links it to the risk.
5. Swap with another pair and improve your weakest measure using their feedback.

Think about it

Think about it: if a model is right most of the time, is that good enough when it is wrong about you?

My reflection

My reflection: one risk I can now spot and fix responsibly is ...

Week 4 - Lab | Ethics & Governance - Build a responsible-use plan for a harder case

I can...

- I can break a complex AI case into its separate risks and say who is affected by each.
- I can build a responsible-use plan with measures aimed at the most serious risks.
- I can justify my plan, including the trade-offs I accepted and why.

Words to know

- stakeholder — anyone affected by or involved in how the AI is used
- impact — how much harm a risk would cause if it happened
- likelihood — how probable it is that the risk actually happens
- trade-off — what you give up when you choose one fix over another
- appeal — a way for someone to challenge or overturn an AI decision

Hard AI cases have many risks at once and no perfect fix. Today you learn to rank the risks and build a plan that

protects people while being honest about its costs.

Rank before you fix

When a case has many risks, fixing them in the order you noticed them wastes effort. Instead, judge each risk by

two things: how bad it would be (impact) and how often it would happen (likelihood). The risks that are both

serious and likely go to the top. Aim your strongest measures there. This is how real teams improve a model

responsibly - they cannot fix everything at once, so they protect against the worst harms first and say why those were chosen.

Every fix has a cost

There is no free fix. More human review makes decisions fairer but slower. More monitoring catches harm but

reduces privacy. Blocking more posts is safer but silences more people wrongly. A strong plan does not hide

these trade-offs - it names them and argues why its balance is acceptable for this case. Being honest about the

cost is part of being responsible, because it lets others check whether they agree with the choice you made.

Why it matters

Cities and schools in the UAE increasingly use AI to moderate content, sort applications and prioritise services,

so being able to weigh risks and defend a plan is exactly the judgement these systems need from citizens.

Let's try it - hands-on

1. Read the case and list the stakeholders affected by the model.
2. Write one main risk for each stakeholder.
3. Rank the risks by impact and likelihood; circle your top two.
4. Propose one measure for each top risk and name its trade-off.
5. Write a one-line justification for your whole plan and defend it to another pair.

Today's tool - hands-on

Tool: Card decks (bias-source cards) + packs (reduce risks / responsible use)

1. Read a bias-source card and the paired outputs.
2. Identify the risk or unfairness and its likely cause in the data or design.
3. Say who could be affected.
4. Propose one concrete action to reduce the risk and improve responsible use.
5. Predict how the action would change the output.

You will make: A risk-reduction record: risk, cause, who is affected, and a proposed action.

Stay safe: Unplugged; outputs pre-curated.

A risk traced to its cause, then a concrete action to reduce it.

Think about it

Think about it: is a fix that protects most people fair if it silences a few of them unfairly?

My reflection

My reflection: the trade-off in my plan I am least sure about is ... because ...

Week 5 - Integrated Task (Design & Build) | Integrated Prerequisite Mastery Task -

Design, Build, Evaluate & Justify an AI Solution

I can...

- I can explain how changing the order of our steps or the tool we pick changes the reply our workflow produces.
- I can design and build a clear multi-step workflow that answers an enquiry and checks the reply before it is sent.
- I can record our workflow map, our tool choices and one test run so someone else can see it works.

Words to know

- workflow - the ordered set of steps our solution follows to turn an enquiry into a reply
- step - one job in the workflow, such as understand, find the facts, draft or verify
- tool - what we choose to do a step with, from a reply deck to a trusted FAQ
- simulated AI - our stand-in model: a curated deck of pre-written reply cards, no account and no live tool
- verify - to check a reply against a trusted source before we trust or send it

This week you and your partner become AI solution builders. You will design and build a small reply helper: a

workflow that takes an enquiry in, moves it through a few clear steps, and gives back a checked reply - just like a

real AI-assisted solution.

Design decides the outcome

Our reply helper is a simulated AI: the drafting step just picks from a curated deck of pre-written reply cards. The

clever part is the design - which steps you use, the order you put them in, and the tool you give each step. Draft

before you understand the enquiry and you get a confident reply to the wrong question. Verify after you have

already sent and the check is useless. Move verify before sending and a wrong reply is caught.

Same enquiry,

different design, different outcome - that is the heart of building an AI solution well.

Build it as a workflow

A strong solution has a clear shape: an input, a sequence of steps, an output. Your input is an enquiry card. Your steps might be understand, find the facts, draft with the reply deck, and verify against the FAQ. Your output is a checked reply. When you draw this map and label the tool each step uses, anyone can see how your solution works and test it. Build small, run a few enquiry cards, and always verify each reply against the trusted FAQ before you trust it - a confident reply can still be wrong. Your reply helper is a workflow: an enquiry goes in, ordered steps process it, and a checked reply comes out.

Why it matters

Why it matters - across the UAE, smart-city services and school and government help lines use AI-assisted

workflows to answer enquiries quickly; the same input-steps-output thinking you use this week sits behind

those real solutions, and so does the habit of checking a reply before sending it.

Let's try it - hands-on

1. Watch the demo and name the input, the ordered steps and the output.
2. Reorder one step in the starter workflow and see how it changes a reply.
3. Draw your input-steps-output map and choose the tool for each step.
4. Build your workflow by sequencing your steps and giving each a clear job.
5. Run three enquiry cards and verify each reply against the FAQ.

Today's tool - hands-on

Tool: Teachable Machine + Scratch (integrated: design, build, first measure)

1. In pairs, plan the small solution with the plan cards, including performance criteria.
2. Train a simple non-personal model and build the solution in Scratch.
3. Run it on a short test-set and record a first score.
4. Note the single biggest limitation to fix in Week 6.
5. Save your build and note what each strand contributed.

You will make: A built solution with a first measured score and a target limitation.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Non-personal model only.

Integrated build with a first measured score and a target to improve.

Think about it

Think about it - if you could add only one more step to your workflow, which step would most improve the replies, and where in the order would it go?

My reflection

My reflection: One design choice that changed the reply our workflow produced was ...

Week 6 - Integrated Task (Evaluate, Justify & Showcase) | Integrated Prerequisite

Mastery Task - Design, Build, Evaluate & Justify an AI Solution

I can...

- I can test our workflow against a criterion, make one improvement, and prove with evidence that it helped.
- I can name one risk in our workflow and suggest a fair, safe fix, and say why it helps.
- I can reflect on all four skills and show our solution is ready for the Grade 8 unit.

Words to know

- criterion - the clear target we judge our workflow against, such as correct, verified replies
- iteration - improving our workflow one change at a time and testing after each change
- evidence - the before-and-after scores that prove a change really helped
- bias - when a workflow is unfair to a whole type of enquiry because of the tools or replies it was given
- readiness gate - the check that shows our solution and our skills are ready for the Grade 8 unit

This week you prove your solution works. You will test your workflow, improve it, guard it against a risk, and

show it is ready - the same evaluate-and-improve habit real AI builders use every day.

Improve it, then prove it

Making a change is easy; proving it helped is the real skill. First you set a criterion - here, how many test

enquiries get a correct, verified reply. You score your workflow, change one thing - reorder a step, swap a tool,

or add the verify gate - and score it again. If the score goes up on the same test set, you have evidence the

change improved the workflow. If it does not, you learned something too, and you try a different change. This

one-change-at-a-time habit is called iteration, and it is exactly how AI builders turn a first attempt into a solution

they can trust.

Guard against the risk

Even an accurate workflow can be unfair or unsafe. If your reply deck missed a whole enquiry type, those

enquiries get a poor reply every time - that is a biased output. And a confident reply sent without checking can

be wrong, or an enquiry might contain personal data that should never be entered. So you name the risk and

propose a responsible-use fix with a reason: add a verify-before-trust gate, a 'not sure - pass to a human' route,

and a no-personal-data rule. Naming risks and fixing them is part of building AI responsibly.

Score, change one thing, score again - if the number goes up, you have evidence it improved.

Why it matters

Why it matters - the UAE expects AI to be used responsibly, from smart-city services to sustainability projects;

learning to evaluate, improve and guard your own workflow is exactly what a responsible AI builder does.

Let's try it - hands-on

1. Score your workflow on the six-enquiry test set and circle the replies that fail.
2. Change one design element, re-run, and check if your score went up.
3. Write the evidence that your change improved the workflow.
4. Name one risk in your workflow and a fair, safe fix, with a reason.
5. Reflect on all four skills and show your solution is ready.

Today's tool - hands-on

Tool: Pre-generated packs + improvement-move cards (integrated: improve, re-measure, justify)

1. Re-run the Week-5 solution and confirm the first score.
2. Apply one improvement-move card that targets the weakest case.
3. Re-measure and compare with the first score.
4. Draw a risk-scenario card and check your solution against it; confirm no personal data.
5. Justify your improvement with evidence and complete the readiness reflection.

You will make: A before-and-after score, a justified improvement, a risk check, and a readiness reflection.

Stay safe: Risk check confirms no personal data; outputs curated.

Improve, re-measure, check the risk, justify with evidence.

Think about it

Think about it - if two different changes both raised your score by the same amount, how would you decide

which is the better improvement to keep?

My reflection

My reflection: The skill I found hardest was ..., and the way I worked through it was ...

Cross-curricular links

Science link: the fair test in Week 1 (Run the Fair Test) - change ONE thing and keep everything else the same, so

you know what caused the difference. Language link: the two-audience workflow in Week 2 (One Alert, Two

Audiences) - one branch in plain English for students, one branch translated into Arabic for parents, both

checked for reading level. Maths link: the score comparisons in Week 3 (Score, Iterate, Justify) and Week 6 -

work out your score before and after each change and compare them to prove an improvement.

Key terms - consistent definitions (same across all Grade 8 packs)

Model - the trained part that turns an input into an output; change its training examples and train again and you

get a NEW, retrained model.

Process - one step (or set of steps) that acts on an input. Workflow / pipeline - the ordered sequence of steps

that turns an input into an output (same meaning).

Output - the direct result of one run (a label, number or sentence). Outcome - the real-world effect of using that

output.

Training data / test data - training data teach the model; test data are separate new examples used to check it.

One prediction runs forward; development loops through testing and retraining.

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Course - Evaluating & Improving AI Models

Transition-Year readiness | Weeks 1-6 (Core + Lab)

STUDENT WORKSHEETS

Week 1 - Core - Worksheet | AI Conceptual Understanding - How workflow designs affect outcomes

Name: _____ Class: _____ Date: _____

Follow the arrow from the outcome back through the steps to the design choice that shaped it.

Trace it back

For each example, write which design choice shaped the outcome and what effect it had.

Example The design choice (step added, moved or dropped)

Effect on the outcome

Example A

Example B

Example C

My explanation

Pick one example. In two or three sentences, explain to a reader how its workflow design shaped the outcome.

Input or design?

An outcome came out weak but the input was clear. Explain why the workflow design is the more likely cause.

Exit ticket

Exit ticket - Explain, in one clear sentence, how the workflow design affected the outcome of one example you studied today, and give your reason.

Challenge (stretch)

Challenge - Apply the skill to a more complex case and justify the approach: take a workflow with three or four steps and one weak outcome, decide where a check or evaluation step should be added, and justify why that placement would improve the outcome. This stretches toward the Proficient outcome of explaining how a workflow design can be adjusted to improve a model's performance.

Week 1 - Lab - Worksheet | AI Conceptual Understanding - How workflow designs affect outcomes (Lab)

Name: _____ Class: _____ Date: _____

Same input, same model, one changed design - map it to prove where the difference comes from.

Fair-test map

For each pair, record the one design choice that changed and its effect on the outcome.

Workflow pair Design choice (what changed) Effect on the outcome

Pair 1

Pair 2

Pair 3

My conclusion

Use your maps as evidence. Which workflow design produced the better outcome, and why?

Was it fair?

Explain how you know the difference came from the design choice and not from the input.

Tool log - Teachable Machine / Scratch (workflow designs affect outcomes)

Record as you work. Goal: A two-design comparison explaining how design affects outcomes.

Step What I did What I saw / result

1 Run design A on a small test-set and score it.

2 Run design B on the same test-set and score it.

3 Compare the two scores and describe how the design changed

4 Name the single design choice that mattered most.

5 Write the rule: a better design gives a better outcome.

Design A vs B

Test case A output B output Which better?

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Exit ticket

Exit ticket - Name which design gave the better outcome in one pair, justify it with your mapped evidence, and say why the input cannot be the cause.

Challenge (stretch)

Challenge - Apply the skill to a more complex case and justify the approach: design your own fair test for a workflow with several steps, choose one design choice to change (for example, moving the check step earlier), predict how the outcome should improve, and describe how you would map and compare the before and after to prove it. This stretches toward the Proficient outcome of evaluating how design choices shape a model's performance.

Week 2 - Core - Worksheet | AI Solution Design & Development - Refines workflows by improving structure, sequencing and tool selection

Name: _____ Class: _____ Date: _____

Before and after: reordering and one tool swap turn a clumsy workflow into a reliable one.

Spot the problems

Find three things wrong with the draft workflow. For each, name the lever: structure, sequencing or tool.

Problem in the draft workflow Lever (structure / sequencing / tool)

My refined workflow

Write your steps in the new order. For each, name the tool and say why it belongs there.

Order Tool chosen Why this tool here

1

2

3

4

Verify it

Pick one figure in your output. How did you check it was correct against the reference?

Exit ticket

Exit ticket - Name one change you made under each lever - structure, sequencing and tool selection - and explain how each

one made your output more reliable.

Challenge (stretch)

Challenge - Apply the skill to a more complex case and justify the approach: add a step that translates the notice into

Arabic, decide exactly where it belongs in your sequence and which tool does it, then explain why that placement keeps the output correct. This stretches toward the Proficient outcome of designing and justifying multi-step AI workflows for real audiences.

Week 2 - Lab - Worksheet | AI Solution Design & Development - Refines workflows by improving structure, sequencing and tool selection (Lab)

Name: _____ Class: _____ Date: _____

Design A checks twice and can disagree; design B verifies once, before the branch.

Judge the designs

Compare designs A and B. Log three weaknesses and tick the lever each belongs to.

Weakness (which design) Lever (structure / sequencing / tool)

My improved design

Describe your improved workflow: name the shared stage, the branch point, and the tool on each branch.

Justify it

Pick one choice - structure, order or tool - and explain how it makes an output reliable and fit for its audience.

Tool log - Scratch (refine a workflow's structure and sequencing)

Record as you work. Goal: A refined Scratch workflow and an explanation of the improvement.

Step What I did What I saw / result

1 Run the starter project and note what is clumsy or fails.

2 Reorder or regroup the steps to improve the structure.

3 Add one check step and choose the better-suited block/tool

4 Re-run and confirm the output improved.

5 Explain which refinement helped most and why.

Refinement log

What was clumsy Refinement made Result

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Exit ticket

Exit ticket - Name one improvement you made, the lever it used, and explain how it makes both audience outputs reliable and fit for purpose.

Challenge (stretch)

Challenge - Apply the skill to a more complex case and justify the approach: add a third audience that needs a spoken-style

summary, decide where the workflow branches for it and which tool it uses, and explain why your shared stage still only

checks the facts once. This stretches toward the Proficient outcome of designing and justifying multi-branch AI workflows

that stay reliable across audiences.

Week 3 - Core - Worksheet | Critical Evaluation & Informed Interaction - Iterating a Workflow and Explaining the Impact of Changes

Name: _____ Class: _____ Date: _____

Version log

Record each version of your workflow, the single change you made, and the result against your agreed criterion.

Version Single change made Result vs criterion (evidence)

1

2

Explain the impact

In full sentences, explain how your one change affected the result. Link the change to the impact and name your evidence.

Exit ticket

Name the single change you made and explain its impact on the result, citing one piece of evidence.

Challenge (stretch)

Challenge (stretch): plan a second iteration. Predict the impact of your next change, then say how you would test whether your prediction was correct - a step toward the Proficient outcome of designing and justifying a sequence of improvements.

Week 3 - Lab - Worksheet | Critical Evaluation & Informed Interaction - Lab:

Iterating and Evaluating a Model's Outputs Against Criteria

Name: _____ Class: _____ Date: _____

Score the whole set against fixed criteria, then let the pattern guide your next single change.

Evaluation grid

Score each output against the two agreed criteria and write the evidence you used for the score.

Output Score vs criterion 1 Score vs criterion 2 Evidence

A

B

C

Two-cycle version log

Record the single change and the result for each cycle, then write your final justification.

Cycle Single change Result vs criteria

1

2

Justify your final workflow

In full sentences, justify why your final version is the strongest, relating evidence from both cycles.

Tool log - Teachable Machine retrain + packs (improve through iteration)

Record as you work. Goal: A three-iteration score record explaining the impact of each change.

Step What I did What I saw / result

- 1 Test the current model and record the score.
- 2 Make one change and retrain (an iteration).
- 3 Re-test and record the new score.
- 4 Repeat once more, so you have three scores.
- 5 Explain the impact of each change with evidence.

Iteration scores

Iteration Change made Score / 10

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Exit ticket

Justify why your final workflow is stronger than version one, citing evidence from both iteration cycles.

Challenge (stretch)

Challenge (stretch): design a third criterion that would make the evaluation fairer, explain what it would reveal, and predict how it might change your ranking - moving toward the Proficient outcome of designing and defending an evaluation, not just applying one.

Week 4 - Core - Worksheet | Ethics & Governance - Spot the risk, propose responsible-use fixes

Name: _____ Class: _____ Date: _____

Risk - Measure - Because

Fill the table for your scenario. Each measure must reduce the risk next to it and have a reason.

Risk in the scenario Responsible-use measure Because (reason it reduces the risk)

Who is accountable?

Name the person or role who should be answerable if this AI model gets a decision wrong, and why.

Exit ticket

Exit ticket: write one complete risk - measure - because chain for any AI model you choose.

Challenge (stretch)

Challenge (stretch): take a scenario with two conflicting fixes (for example, more monitoring vs more privacy) and argue which measure you would choose and why - this pushes toward the Proficient outcome of weighing trade-offs when improving responsible use.

Week 4 - Lab - Worksheet | Ethics & Governance - Build a responsible-use plan for a harder case

Name: _____ Class: _____ Date: _____

Rank each risk by impact and likelihood, then aim your strongest measures at the top-right.

Stakeholder - Risk - Rank

List each stakeholder, their main risk, and a rank from impact and likelihood (High/Med/Low).

Stakeholder Main risk Impact x likelihood rank

Responsible-use plan

For your top two risks, write a measure, its trade-off, and a because. Then justify the whole plan.

Tool log - Card decks (bias-source cards) + packs (reduce risks / responsible use)

Record as you work. Goal: A risk-reduction record: risk, cause, who is affected, and a proposed action.

Step What I did What I saw / result

1 Read a bias-source card and the paired outputs.

2 Identify the risk or unfairness and its likely cause in th

3 Say who could be affected.

4 Propose one concrete action to reduce the risk and improve

5 Predict how the action would change the output.

Risk-reduction plan

Risk / unfairness Likely cause Who is affected Action to reduce it

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Exit ticket

Exit ticket: name the weakest trade-off in your plan and explain why you still accept it.

Challenge (stretch)

Challenge (stretch): redesign one measure so it protects people just as well but at a lower cost, and argue why your version

is better - moving toward the Proficient outcome of optimising responsible-use measures, not just proposing them.

Week 5 - Integrated Task (Design & Build) - Worksheet | Integrated Prerequisite

Mastery Task - Design, Build, Evaluate & Justify an AI Solution

Name: _____ Class: _____ Date: _____

Our workflow map

List your steps in order and the tool each one uses. Start with the enquiry and end with the checked reply.

Step (in order) Tool for this step

Our design choice

Write the input, the order of your steps and the output, then one design choice you made and why.

Test log

Run three enquiry cards and record the reply and whether it passed the verify check.

Enquiry card Reply produced Passed verify? (yes / no)

Tool log - Teachable Machine + Scratch (integrated: design, build, first measure)

Record as you work. Goal: A built solution with a first measured score and a target limitation.

Step What I did What I saw / result

1 In pairs, plan the small solution with the plan cards, inc

2 Train a simple non-personal model and build the solution i

3 Run it on a short test-set and record a first score.

4 Note the single biggest limitation to fix in Week 6.

5 Save your build and note what each strand contributed.

First measure

Test case Output Correct?

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Exit ticket

Exit ticket - Name one design choice you made (a step you added or moved, or a tool you chose) and explain how it changed

one of your workflow's replies.

Challenge (stretch)

Challenge - Add an enquiry your workflow has never seen (for example a question the FAQ does not answer) and predict

how your workflow will handle it. Explain which step should catch it and what it should do instead of guessing. This

stretches toward the Proficient idea of designing workflows that handle new, unexpected inputs.

Week 6 - Integrated Task (Evaluate, Justify & Showcase) - Worksheet | Integrated

Prerequisite Mastery Task - Design, Build, Evaluate & Justify an AI Solution

Name: _____ Class: _____ Date: _____

Score and improve

Score your workflow, make one change, and re-score. Show the evidence it improved.

Stage Score (out of 6) Change I made

Before

After

Risk and fix

Name one risk in your workflow and a responsible-use fix, and give your reason.

Four-skill reflection

Answer all three: the hardest skill and how you worked through it; one improvement you would make; one risk you would guard against.

Tool log - Pre-generated packs + improvement-move cards (integrated: improve, re-measure, justify)

Record as you work. Goal: A before-and-after score, a justified improvement, a risk check, and a readiness reflection.

Step What I did What I saw / result

1 Re-run the Week-5 solution and confirm the first score.

2 Apply one improvement-move card that targets the weakest c

3 Re-measure and compare with the first score.

4 Draw a risk-scenario card and check your solution against

5 Justify your improvement with evidence and complete the re

Improve & justify

First score Improvement move New score Evidence it helped

Safety check (tick): I used no personal data (no faces, voices or names). □

Exit ticket

Exit ticket - In one or two sentences, say why your solution is ready for the Grade 8 unit, naming one improvement you made and one risk you guarded against.

Challenge (stretch)

Challenge - Evaluate your workflow against a second criterion - speed, usability or fairness - and describe the trade-off:

what got better, what got worse, and why you would still choose it. This stretches toward the Proficient idea of evaluating solutions against more than one performance criterion.

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Course - Evaluating & Improving AI Models

Transition-Year readiness | Weeks 1-6 (Core + Lab)

TEACHER PACK

1. About this Teacher Pack

This pack contains everything needed to teach the Grade 8 bridging program: a start-of-year diagnostic and six weeks that rebuild each of the four strands to the prior-grade Advanced mastery before the Grade 8 unit begins.

Weeks 1-4 rebuild one strand each; Weeks 5-6 integrate them in a mastery task.

How to use this pack

- Teach each week as Core (period 1) then Lab (period 2) on the same outcome.
- Keep all AI use teacher-mediated and account-free; enter no personal data.
- Build the evaluate-verify-risk habit into every AI activity; provide the unplugged fallback where tools are unavailable.
- Use the per-lesson formative check to confirm the outcome before moving on.
- Pair this Teacher Pack with the Student Lesson Content and Student Worksheets (same order).

Safeguarding & responsible use

Grade 8 works with AI systems and workflows. Keep all AI use teacher-mediated and account-free for students;

prefer a board demonstration, curated or pre-generated output sets, or a card-based 'simulated AI' routine. Never

enter personal data (names, faces, voices) into a prompt or a model. Build the evaluate-verify-risk habit into every AI

activity: check each output against the task and a trusted source, because a confident output can still be wrong.

Every AI activity has a non-AI (unplugged) fallback so no lesson depends on tool access.

Materials & tools (UAE-accessible)

Tool / material What it is & why Access route Unplugged fallback

Workflow / model demo

set + cards

A board demo tied to the unit theme

(Evaluating & Improving AI Models) plus printed example cards.

School-printed; teacher-run; no accounts.

Sort and map examples on paper.

School-approved child-safe assistant

An approved generative tool, teacher-operated only, for curated demos on non-personal data.

School-approved; teacher device only.

Use pre-generated output sets.

Pre-generated output
sets

Curated strong/weak outputs from the same
task for evaluation and comparison.

Prepared by the teacher in
advance.

Same sets, printed.

Simulated-AI cards Card decks that mimic an AI step so
students practise the workflow unplugged.

School-printed. This is itself the
unplugged route.

Program at a glance

Week Strand / session Target (prior-grade
Adv.)

Focus

Wk 1 AI Conceptual Understanding AIF·CU·G7·A Explains how different workflow designs
affect

Wk 2 AI Solution Design & Development AIF·SD·G7·A Refines workflows by improving
structure,
sequ

Wk 3 Critical Evaluation & Informed
Interaction

AIF·CE·G7·A Improves workflows through iteration and
expla

Wk 4 Ethics & Governance AIF·GE·G7·A Proposes ways to reduce risks and improve
resp

Wk 5-6 Integrated Prerequisite Mastery Task AIF·CU·G7·A
AIF·SD·G7·A
AIF·CE·G7·A
AIF·GE·G7·A

Design, build, evaluate & justify an AI solution

Start-of-Year Diagnostic - prior-grade Advanced readiness (all four strands)

Administered at the very start of the year, outside timetabled teaching hours (about 30 minutes,
in small groups or stations),
before bridging teaching begins. Each strand is recorded as mastery / partial / not-yet; no grade
is reported to students.

Strand-specific routing (when strand scores differ)

The diagnostic reports a score per strand, so route per strand - not by one overall number. A
student ready on some strands

but weak on others joins only the Bridging weeks for their weak strands: W1 Conceptual
Understanding, W2 Solution

Design, W3 Critical Evaluation, W4 Ethics & Governance.

Per-strand rule of thumb: 40% or above - may skip that strand's Bridging week; 15-39% - join
that strand's focus-group work;

below 15% - needs the Primer for that strand first. Re-check any borderline strand (about 38-42%) after Week 3 and adjust.

Strand Prior-grade Advanced skill probed Baseline probe Record

AI Conceptual

Understanding

Explains how different workflow designs affect outcomes.

Can the student demonstrate:

Explains how different workflow designs affect outcomes.

mastery /

partial / not-yet

AI Solution Design &

Development

Refines workflows by improving structure, sequencing, and tool selection.

Can the student demonstrate:

Refines workflows by improving structure, sequencing, and tool selection.

mastery /

partial / not-yet

Critical Evaluation &

Informed Interaction

Improves workflows through iteration and explains the impact of changes.

Can the student demonstrate:

Improves workflows through iteration and explains the impact of changes.

mastery /

partial / not-yet

Ethics & Governance Proposes ways to reduce risks and improve responsible use of AI systems.

Can the student demonstrate:

Proposes ways to reduce risks and improve responsible use of AI systems.

mastery /

partial / not-yet

Week 1 - Core | AI Conceptual Understanding - How workflow designs affect outcomes | Interactive lesson: 'The Design Behind the Outcome'

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G7·A | 1 x 45 min

Objectives & success criteria

- Explain how a change in a workflow's design - its steps or their order - affects the outcome it produces.
- Analyse a worked input-process-output example to reason about how the workflow design shaped the result.
- Justify to a partner, using evidence from the example, why one workflow design produced a better outcome than another.

Student-facing success criteria:

- I can explain how changing the steps in a workflow changes the outcome it gives.
- I can trace an example from input to output and show where the workflow design made the difference.
- I can convince my partner, using the example, why one workflow design gave a better outcome.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: This week rebuilds a skill from earlier grades before the new unit begins. In Grade 5 you described a simple sequence of steps that reaches an outcome; in Grade 6 you explained how changing the order of the steps changes the result; in Grade 7 you explained how different workflow designs affect outcomes. Today you make that Grade 7 idea secure and apply it to this year's theme - evaluating and improving AI models.

Activate: Show two curated workflows for the same task on the board - one that jumps straight from input to output, and one that adds a check step before the output - and ask which produces the more reliable outcome and why.

Teach: Name the chain: the workflow design decides which steps run and in what order, the steps run the process, and the process shapes the outcome. Model the reasoning on the two board workflows - point to the step one design added and show how that step changes the outcome.

Apply: In pairs, students look at one more curated pair of workflows and say, in one sentence, how the difference in design explains the difference in the outcomes.

Check: Cold-call two pairs to explain the link from design to outcome. Listen for the full chain - the design sets the steps, the steps shape the outcome - not just 'one is better'. If a pair stops at the outcome, re-model tracing it back to the design choice before moving on.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A workflow is a designed sequence of steps that turns an input into an outcome. Its design - which steps are included, their

order, and where a check or evaluation step sits - sets a ceiling on how good, reliable or fair that outcome can be. Two workflows given the identical input and the same underlying model can still produce different outcomes because their designs differ: adding a verification step, reordering stages, or splitting one step into two all reshape the result. The reasoning move here is analysis - tracing an outcome back through the process to the design choice that shaped it. This underpins the unit's work of evaluating and improving AI models against clear criteria. Keep all examples pre-generated and teacher-operated, account-free, with no personal data; any model discussed is trained only on non-personal data.

Cognitive demand for this lesson: Understand/Analyse | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two curated workflows for one

task - a plain input-to-output design and a design that adds a check step before the output. Ask students which outcome they would rely on.

Study both workflows and jot one difference they notice in how the steps are arranged.

- Both workflows do the same task - so why does one give a more reliable outcome?

Teach the design-to-outcome chain /

Model the chain workflow design -> steps -> process -> outcome. Think aloud: name the step one design added

Copy the chain into their notes and label where the design choice acts on the

- At which step does the design choice do its work - and where does its effect show up in the

9 min and trace how that step reaches the outcome.

workflow. outcome?

Investigate

worked

examples / 14

min

Hand out curated workflow cards (the input, the two designs, and their outcomes). Circulate and prompt students to reason from design to outcome, not from opinion. Keep any live tool teacher-operated only.

In pairs, map each example from input to output and write how the workflow design explains the outcome's quality.

- Which single design choice would you change to improve this outcome, and why?

Explain to a

partner / 11 min

Set up a structured pair-explain: each partner justifies one example aloud using evidence from the card, then swaps. Prompt for the full relational chain from design to outcome.

Explain one example to a partner, pointing to the design choice as the cause of the outcome; listen and challenge in return.

- Convince your partner: how do you know the design - not the input - caused this outcome?

Check and close

/ 5 min

Run the exit check; collect the mapped workflow cards and the exit tickets as evidence. Flag verify-before-trust: a confident outcome can still come from a weak design.

Complete the exit ticket: one sentence linking a design choice to an outcome, with a reason.

- Explain how the workflow design affected this outcome - in one clear sentence.

Anticipated misconceptions

Misconception How to address it

If the outcome is poor, someone just gave it a bad input. Show a clear, identical input feeding two workflows with different

designs and different outcomes; the input was fine, so the cause must lie in the design. Trace it back together.

More steps in a workflow always mean a better outcome. Compare a long workflow with a redundant step against a shorter

one with a well-placed check; agree that the right steps in the right place matter, not the number of steps. Name the check step as the one that earned its place.

The workflow design and the input are the same thing. Separate the two on the board: the input is what you feed in today;

the design is the fixed plan of steps the workflow always follows.

Have pairs relabel any example that confused them.

Hands-on activity guide

Overview: Pairs investigate curated worked examples of two workflow designs for the same task, mapping each from input

to output and reasoning about how the design shaped the outcome, then justify one example aloud to a partner using evidence from the card.

Grouping / timing: Pairs for the investigation and the explain, whole class for the debrief.

Tool: Pre-generated workflow cards on paper plus an optional teacher-operated board demo (a 'simulated AI' routine).

Account-free, no logins, no personal data - students never enter names or operate a live model themselves.

Facilitation steps:

1. Read the workflow card: the input, the two designs, and the outcome each produced.
2. Map each design from input, through its steps, to the outcome.
3. Write how the design explains the quality of the outcome - which step it added, moved or dropped.
4. Compare the two designs and note the single change that made the outcomes differ.
5. Explain one example to your partner, pointing to the design choice as the cause.
6. Check each other's reasoning: could the input alone explain it, or does it have to be the design?

Unplugged fallback: The cards are already paper-based, so no device is needed. If the board demo is unavailable, read

one worked example aloud - the input, the two designs and their outcomes - and have pairs reason from the printed cards only.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Explain how the workflow design affected The second design added a check step that Look for a clear link from a design

the outcome in one worked example. caught the error before the output, so its outcome was reliable while the plain design passed the error straight through - the design choice, not the input, made the difference. choice to the outcome; the reason must name the step that was added, moved or dropped, not just the result.

Given two workflows with different designs on the same task, predict which produces the stronger outcome and justify it.

The design that evaluates its draft before the output will give the stronger outcome, because the extra check catches problems the plain design cannot.

One mark for the correct prediction, one for a justification that references the specific design difference between the two workflows.

Explain why the same input can give a poor outcome.

The input was fine, but the workflow was designed with no check step, so an error ran straight through the process to the outcome.

Accept answers that locate the cause in the workflow design, not in the input alone.

- Targets: the Grade 7 skill being rebuilt - explaining how different workflow designs affect outcomes - applied to evaluating AI models and reasoned through by analysis.
- Check: an exit ticket plus circulating listen-ins during the pair-explain, questioned against the workflow design -> steps -> process -> outcome chain.
- Success criteria: the student links a specific design choice to an outcome and justifies it with evidence from the example.
- Evidence: collected mapped workflow cards and the exit tickets.
- Feedback / adjustment: if students stop at 'one outcome is better', re-model tracing an outcome back to the design choice and have them add the missing design-to-outcome link before the next lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Linking design to outcome Explains how one design choice

affected the outcome, with support.

Names a difference in the outcomes but not the design cause.

Cannot connect the workflow design to the outcome.

Analysing the workflow Traces an example from input to output and locates where the design acts, with guidance.

Maps the example but misplaces where the design has its effect.

Cannot map the example through the workflow.

Justifying to a partner Justifies one example aloud using evidence from the card.

States an opinion without evidence from the example.

Does not justify a choice.

Representation: pair each workflow card with the visible workflow design -> steps -> process -> outcome diagram; pre-teach

'workflow', 'design' and 'check step' with simple icons.

Action & expression: let students show the design-to-outcome link by annotating the card, speaking it aloud to a scribe, or writing it - accept any mode for the exit ticket.

People of Determination: provide a reduced set of cards with picture support, sentence starters ('This outcome is more reliable because the design added ...'), extra processing time, and a peer partner for the explain step.

Week 1 - Lab | AI Conceptual Understanding - How workflow designs affect outcomes (Lab) | Interactive lesson: 'Run the Fair Test'

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G7·A | 1 x 45 min

Objectives & success criteria

- Investigate a set of worked workflows to test how changing one design choice changes the outcome.
- Map how an input flows through a workflow's steps to produce its outcome.
- Justify a conclusion to a partner about which workflow design produced the better outcome, using evidence.

Student-facing success criteria:

- I can investigate workflows and show how a change in design changed the outcome.
- I can map how an input flows through the steps to the outcome.
- I can justify to a partner which design gave the better outcome and back it with evidence.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you rebuilt the Grade 7 skill - explaining how different workflow designs affect outcomes - by

tracing worked examples. That skill grew from Grade 5 (describing a sequence of steps to an outcome) and Grade 6

(explaining how the order of the steps changes the result). Today you apply it more deeply and hands-on.

Activate: Show one input feeding two curated workflows whose only difference is one design choice - one adds an evaluation step

before the output, the other does not - then reveal their two different outcomes and ask what could possibly explain the gap.

Teach: Model an investigate-and-map routine: for one worked pair, map the shared input, the two workflow designs, the steps and

the two outcomes onto a single sheet, then state which design won and why.

Apply: In pairs, students map one worked pair onto the sheet and write a one-sentence claim about which design produced the better outcome.

Check: Check that each pair has pinned the cause on the design choice, not the input; redirect any pair that credits the input, since the input was held identical.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

This Lab does not introduce a new concept - it deepens the same idea by controlled comparison. When the input and the

underlying model are held identical and only one design choice changes, any difference in the outcome must come from

that design choice. That is a fair test, and it is the clearest way to see how workflow design sets the quality and reliability of

a result. Students investigate curated workflow pairs, map how the input flows through the steps of the input-process-

outcome workflow, and justify an evidence-based conclusion to a partner. The habit - hold things constant, change one

thing, explain the result - is the foundation of the unit's evaluation and improvement work. All examples are pre-generated

and teacher-operated, account-free, with no personal data; any model discussed is trained only on non-personal data.

Cognitive demand for this lesson: Understand/Analyse | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set up

/ 5 min

Recap the workflow design -> steps ->

process -> outcome chain and the safe-use charter. Hand out the mapping sheets and the curated workflow-pair packs.

Read the first pair and predict which design will give the better outcome.

- The input is the same for both - so what is the only thing that can explain a different outcome?

Model

investigate-and-map / 8 min

Think aloud through one pair: map the shared input, the two designs, the steps and the two outcomes, then state which design won and the evidence for it.

Watch, then copy the mapping headings onto their own sheet.

- Because we changed only one design choice, what does the different outcome prove?

Investigation

station 1 / 14 min

Circulate as pairs work the first two workflow pairs; prompt for evidence

Map each pair and write how the design choice explains

- What is your evidence that the design - not chance - caused this from the design, not opinion. Keep all tools teacher-operated only.

the difference in the outcomes.

difference?

Rotate and

conclude / 13

min

Rotate pairs to a new pack; prompt them to compare across packs and build a justified conclusion.

Map the new pair, then write

a conclusion naming which design produced the better outcome and why.

- Across every pair you mapped, which kind of design choice mattered most?

Justify and close

/ 5 min

Collect the mapping sheets; take two conclusions and test them against the class evidence; flag verify-before-trust.

Complete the exit ticket: one conclusion about workflow design and outcome, with its evidence.

- Justify which design gave the better outcome, and name your evidence.

Anticipated misconceptions

Misconception How to address it

The better outcome just got lucky. Point to the fair test: the input and the model were identical, so

the result is not luck - it traces to the design choice. Have pairs name the controlled variable on their sheet.

You can tell the design is good just by reading the outcome.

Show a fluent, confident outcome that came from a design with no check step; agree the outcome alone is not proof, so we map the steps to see how it was produced. Reinforce verify-before-trust.

If two outcomes differ, the model must have changed. Clarify that the underlying model was the same - only one design

choice differed - so the design is the cause; re-map one pair to make this visible.

Hands-on activity guide

Overview: Pairs run a controlled comparison across curated workflow-pair packs, mapping how identical inputs but

different designs lead to different outcomes, then justify an evidence-based conclusion about which design produced the better result.

Grouping / timing: Pairs at rotating stations, whole class for the debrief.

Tool: Pre-generated workflow-pair packs on paper or on a teacher-operated board (a 'simulated AI' comparison routine).

Account-free, no logins, no personal data.

Facilitation steps:

1. Read the first workflow pair and predict which design gives the better outcome.
2. Map the shared input, the two designs, the steps and the two outcomes onto your sheet.
3. Write how the design choice explains the difference in the outcomes.
4. Rotate to a new pack and map it the same way.
5. Write a conclusion naming which design produced the better outcome, with your evidence.
6. Check the fair test: was the input really identical, so the design is the true cause?

Unplugged fallback: The packs are already paper-based, so no device is required. If the board demo is unavailable, use

the printed workflow pairs only and map the steps by hand.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine / Scratch (workflow designs affect outcomes)

Teacher setup: Prepare two designs for the same task: design A (fewer/lower-quality examples or a shorter workflow) and

design B (more balanced examples or an extra check).

What students do:

1. Run design A on a small test-set and score it.
2. Run design B on the same test-set and score it.
3. Compare the two scores and describe how the design changed the outcome.
4. Name the single design choice that mattered most.
5. Write the rule: a better design gives a better outcome.

Students produce: A two-design comparison explaining how design affects outcomes.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Unplugged fallback: Compare two rule-designs with the human-classifier and tally each.

Two workflow designs, same task, different measured outcomes.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Map one workflow pair and state which design gave the better outcome.

Both used the same input, but the design with an evaluation step produced the reliable outcome while the plain design let an error through; the design with the evaluation step won.

One mark for a correct map, one for naming the design choice as the cause with evidence.

Explain why a different outcome proves something about the design here.

The input and the model were kept the same, so the only thing left to explain the different

outcome is the one design choice that changed.

Look for reference to the controlled fair test - identical input and model, so the design is the cause.

Write a conclusion about which kind of design choice mattered most, using your maps as evidence.

Adding a check or evaluation step before the output mattered most, because in every pair it caught problems the plain design let through.

Accept a conclusion that cites the mapped evidence and names a specific design choice, not just 'the better one'.

- Targets: the rebuilt Grade 7 skill - explaining how different workflow designs affect outcomes - applied through a controlled comparison and justified with evidence.

- Check: circulating evidence-prompts during the stations plus an exit ticket stating one conclusion about workflow design and outcome.

- Success criteria: the student maps the workflow, pins the outcome difference on the design choice, and justifies a conclusion with mapped evidence.

- Evidence: collected mapping sheets and the exit tickets.

- Feedback / adjustment: if pairs credit the input or chance, re-show the fair test - identical input, one changed design choice - and have them relocate the cause before the next lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Investigating the fair test Recognises that only one design Notices the outcomes differ Cannot identify what

choice changed and uses that to explain the outcome, with support.

but not that the design was the only change.

caused the difference.

Mapping the workflow Maps input, the two designs, the steps and the outcomes onto the sheet, with guidance.

Maps some stages but misplaces where the design

acts.

Cannot complete a
usable map.

Justifying a conclusion States a conclusion about the
design backed by mapped
evidence.

States a conclusion but
without evidence from the
maps.

No conclusion offered.

Representation: side-by-side workflow cards, a pre-formatted mapping sheet with the stages
labelled, and the workflow design ->

steps -> process -> outcome diagram on display.

Action & expression: pairs may record their map in writing, by annotating the cards, or by
explaining it to a scribing teacher.

People of Determination: fewer workflow pairs, a partly completed mapping sheet, sentence
starters ('The better outcome came
from ... because the design ...'), and extra time to rotate.

Week 2 - Core | AI Solution Design & Development - Refines workflows by
improving structure, sequencing and tool selection | Interactive lesson: 'Three
Levers to Refine a Workflow'

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G7·A | 1 x 45 min
Objectives & success criteria

- Refine a given AI-assisted workflow by improving its structure - grouping the steps into
sensible stages - so the solution is
easier to follow and check.
- Improve the sequencing and tool selection of the workflow, putting the steps in the right order
and choosing the right tool
for each job.
- Run the refined workflow on a teacher-mediated or simulated routine and explain how each
change made the output
more reliable.

Student-facing success criteria:

- I can reorganise a messy workflow into clear stages that are easy to follow.
- I can put the steps in the right order and pick the right tool for each step.
- I can run my refined workflow and explain why it now gives a more reliable output.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you refined prompts to improve the quality of an output. In Grade 6 you
built structured AI-assisted
solutions using guided inputs and a small set of data. In Grade 7 you learned to refine whole
workflows by improving their
structure, the order of the steps, and the tools chosen for each step. This week rebuilds that
Grade 7 skill before the new
unit so it is secure.

Activate: Put a clumsy draft workflow on the board - a set of steps that makes a school notice but in a poor order, with one wrong tool doing the job. Ask the class what looks off before anyone touches it.

Teach: Name the three levers of refinement: structure (how steps are grouped into stages), sequencing (the order they run in), and tool selection (the right tool for each step). Walk the draft workflow once and show where each lever could improve it.

Apply: In pairs, students circle one thing they would change in the draft workflow and label it structure, sequencing or tool selection.

Check: Cold-call two pairs to read their change and the lever it belongs to. If a pair only says the output is bad without naming a stage or a step, redirect them to point at the exact step and choose the lever before moving on.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A workflow is an ordered set of steps that turns an input into a useful output. Refining a workflow means improving three

things: its structure - how steps are grouped into stages; its sequencing - the order they run in; and its tool selection - the

right tool for each step. A neat-looking output can still be wrong if a checking step runs too late, or the wrong tool does a job

- a summariser cannot verify a fact. Improving one part at a time and testing the effect is an Apply/Create task: students

restructure and re-sequence a workflow and justify each choice rather than follow fixed steps.

Keep every routine teacher-

operated and account-free, use only non-personal data, and verify each output against a trusted source. Cognitive demand

for this lesson: Apply/Create | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show a real task - turning raw sports-

day notes into a clear heat-safety notice

- alongside a draft workflow that works

but is clumsy: steps in a poor order and

one wrong tool. Ask what looks off.

Spot two things they would

change and jot why, without

fixing anything yet.

- Which step is in the wrong place,

and which tool is doing a job it

should not?

Model the three Name structure, sequencing and tool

selection. Think aloud as you regroup

Copy the three levers and

label, on the draft, where

- Why must the verify step come

levers / 10 min the steps into stages, reorder them so

verifying comes before publishing, and

swap the wrong tool for the right one.

each one applies. before publish and not after?

Refine in pairs /

10 min

Hand out the draft workflow as tool

cards and step slots. Guide pairs to

rebuild it - group into stages, fix the

order, choose the right tool per step -

and to justify each change.

In pairs, rearrange the cards

into a refined workflow and

note the reason for each

change.

- What does grouping the steps

into stages let a reader check that

a flat list does not?

Build and run /

14 min

Run the simulated routine (a teacher-

operated card-and-slot 'workflow deck',

or a board demo of a school-approved,

teacher-run tool). Circulate and prompt

verify-before-trust.

Run the raw notes through

their refined workflow,

record the output, and check

it against the heat-index

reference.

- Your notice reads well - what is

your evidence the figures in it are

actually correct?

Check and close

/ 5 min

Collect the refined workflows and

outputs. Run the exit check and flag

that the Lab will refine a harder, two-

audience workflow.

Complete the exit ticket:

name one change you made
under each lever and how it
improved the output.

- Explain how better structure,
order or tool choice made your
output more reliable.

Anticipated misconceptions

Misconception How to address it

If the output looks tidy, the workflow must be fine. Show a tidy notice that carries a wrong figure because the verify

step ran too late. Establish that a neat output still has to be
checked against a trusted source - appearance is not
correctness.

Any tool can do any step if you word it well. Have a summariser card try to verify a fact, and
show it only

shortens - it cannot check truth. Tool selection means matching
each step to a tool that can actually do that job.

The order of the steps does not really matter. Run the same steps in two orders - verify before
publish, then

publish before verify - and compare the outputs. Show that
sequencing changes whether an error is caught in time.

Hands-on activity guide

Overview: Pairs take a clumsy draft workflow for a heat-safety notice and refine it - regrouping
the steps into stages, fixing

the order, and choosing the right tool for each step - then run it on a simulated routine and verify
the output.

Grouping / timing: Pairs for the refine and build, whole class for the model and the debrief.

Tool: A teacher-operated 'workflow deck': printed TOOL cards (Cleaner, Organiser, Verifier,
Summariser, Translator), each

showing what it takes in and what it gives out, plus step slots pairs arrange on the desk.

Optionally a board demo of a

school-approved, teacher-run tool. Account-free, no logins, no personal data - the sports-day
notes are generic, with no

names or faces.

Facilitation steps:

1. Read the task and the draft workflow, and agree what a good final notice looks like.
2. Regroup the steps into clear stages - this is the structure lever.
3. Reorder the steps so each runs at the right time, with verifying before publishing - the sequencing lever.
4. Check each step's tool and swap any that cannot do its job - the tool-selection lever.
5. Run the raw notes through your refined workflow and record the output.
6. Verify one figure in the output against the heat-index reference, then note which change helped most.

Unplugged fallback: The workflow deck already runs on paper cards and step slots, so no device is needed. If the board demo is unavailable, pairs move the cards by hand and read each tool's stated output - the refine-the-workflow skill is unchanged.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Point to one step in the draft workflow and say which lever would improve it.

The publish step comes before the verify step; moving verify earlier fixes it - this is the sequencing lever.

Award the mark only if the named lever matches the actual problem, not just 'it is wrong'; the step must be correctly identified.

Swap one tool in the workflow and justify the new choice.

I replaced the Summariser on the fact-check step with the Verifier, because a summariser only shortens text and cannot check whether a figure is true.

One mark for a swap that fixes a genuine mismatch, with a reason tied to what each tool can and cannot do.

Explain how regrouping the steps into stages made the workflow more reliable.

Grouping into Prepare, Check and Publish stages meant every fact was cleaned and verified before anything was published, so an error could not slip through to the reader.

Look for a clear link between the improved structure and a more reliable output; reject answers that only say it 'looks neater'.

- Targets: refining a workflow by improving its structure, sequencing and tool selection, and explaining how each change made the output more reliable.

- Check: circulating refine-and-verify prompts during the build, plus an exit ticket naming one change under each of the three levers.

- Success criteria: the student regroups the steps, fixes the order, chooses a fitting tool for each step, and explains why the

refined workflow gives a more reliable output.

- Evidence: collected refined workflows, the recorded output with its verified figure, and the exit tickets.

- Feedback / adjustment: if a pair leaves a flat step list or a mismatched tool, re-model the three levers and have them fix one stage and rerun before the Lab.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Improving structure Regroups the steps into sensible stages that are easy to follow, with support.

Reorders a few steps but leaves a flat list with no clear stages.

Cannot group the steps into stages.

Sequencing and tool selection

Puts the steps in a workable order and chooses a fitting tool for each, with guidance.

Fixes the order or the tools but not both, or leaves one clear mismatch.

Cannot order the steps or match tools to jobs.

Explaining the refinement Explains how one change under each lever made the output more reliable.

Describes the changes but not why they improved reliability.

No link between the changes and a better output.

Representation: display a workflow with the three levers labelled - structure, sequencing, tool selection - provide the tool cards in large print with icons, and pre-teach the words stage, sequence and verify with simple examples.

Action & expression: pairs may refine the workflow by moving cards, annotating a printed draft, or explaining the order aloud to the teacher who scribes; accept any mode for the exit ticket.

People of Determination: provide a part-refined workflow with the stages already labelled, fewer tool cards, sentence starters ('I

moved this step before ... because ...'), and extra processing time.

Week 2 - Lab | AI Solution Design & Development - Refines workflows by improving structure, sequencing and tool selection (Lab) | Interactive lesson:

'One Alert, Two Audiences'

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G7·A | 1 x 45 min

Objectives & success criteria

- Design a more complex AI-assisted workflow that serves two audiences, choosing and ordering several tools deliberately.
- Compare two candidate workflow designs, identify the weaker structure, sequencing or tool choices, and improve one design.
- Justify the structure, sequencing and tool selection, explaining how each choice makes the output reliable and fit for its audience.

Student-facing success criteria:

- I can design a multi-step workflow that produces a version for each audience.
- I can compare two workflow designs and improve the weaker one.
- I can justify my structure, order and tool choices and say how they make the output reliable.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you refined prompts, in Grade 6 you built structured AI-assisted solutions using guided inputs and

data, and in Grade 7 you learned to refine workflows by improving structure, sequencing and tool selection. In this week's

Core period you refined a single-notice workflow. Today you apply the same skill to a harder case - a workflow that must serve two different audiences.

Activate: Show a raw weather alert and a requirement that it must reach students in plain English and parents in Arabic, both fact-checked. Ask why a single straight line of steps will not be enough this time.

Teach: Model designing for two outputs: a shared preparation stage that cleans, organises and verifies once, then the sequence

branches - one path adjusts the reading level, the other translates. Show how a shared stage avoids checking the same facts twice.

Apply: In pairs, students sketch where their workflow should branch and name the tool each branch needs.

Check: Check each pair can point to the shared stage and the branch point, and name the tool on each branch. Redirect any pair

who plans to verify the facts separately on each branch back to a single shared checking stage.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

As a task grows, the workflow grows too, and the three levers matter more. Structure now includes what work is shared and where the workflow branches - for two audiences, you clean, organise and verify the facts once in a shared stage, then split. Sequencing sets the order within and after the branch; tool selection matches each branch to a fitting tool - a reading-level adjuster for students, a translator for parents. A good design is judged against the task: is each output correct and fit for its audience? This is Apply/Create at DOK 3 - students build and justify a more complex structured solution, not meet a new idea. Keep the routine teacher-operated and account-free, use only non-personal data, and verify every output against a trusted source. Cognitive demand for this lesson: Apply/Create | DOK 3 | SOLO: Relational. Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set up

/ 5 min

Recap the three levers and the safe-use rules. Hand out the brief - one weather alert, two required outputs - and two candidate workflow designs, A and B, to compare.

Read the brief and both designs, and predict which one will produce reliable outputs.

- Which design checks the facts once, and which risks checking them twice or not at all?

Model the shared stage and branch / 8 min

Think aloud through a good design: a shared Prepare-and-Verify stage, then a branch - reading-level tool for students, Copy the shared-stage-then-branch shape and mark the tool on each branch.

- Why verify in the shared stage and not separately on each branch?

translator for parents. Show why the verify step sits before the branch.

Compare station

/ 14 min

Circulate as pairs analyse designs A and B against the three levers, marking each weak structure, sequence or tool choice with evidence from the brief.

Judge both designs, logging at least three weaknesses and the lever each belongs to.

- What is your evidence this is a tool problem and not just a wrong order?

Improve and justify / 13 min

Support pairs to rebuild the stronger design into an improved workflow and run it on the routine; prompt them to verify each audience output against the source.

Produce one improved design, run it, and record whether each audience output is correct.

- Justify one change - how does it make an output more reliable or more fit for its audience?

Debrief and close / 5 min

Collect the improved designs. Take two justifications and test the reasoning against the class evidence; reinforce verify-before-trust.

Complete the exit ticket: one improvement and the lever it used, with its justification.

- Explain why your improved design serves both audiences reliably.

Anticipated misconceptions

Misconception How to address it

For two audiences I need two completely separate workflows.

Show that cleaning, organising and verifying the facts is the same work for both, so it belongs in one shared stage before the branch; two full workflows just check the same facts twice and can

disagree.

The best design is the one with the most tools. Compare a design with an extra tool that adds nothing to one with

fewer, well-chosen tools; show that each tool must earn its place by doing a needed job, or it only adds risk and delay.

If I translate first, I can verify the translated version afterwards.

Run that order and show a wrong fact gets translated before it is caught, so both outputs carry it; verifying in the shared stage, before the branch, catches the error once for everyone.

Hands-on activity guide

Overview: Pairs compare two candidate workflow designs for a two-audience weather advisory, judge each against

structure, sequencing and tool selection, then rebuild the stronger design into an improved workflow, run it, and justify their choices.

Grouping / timing: Pairs at the compare station, whole class for the model and the debrief.

Tool: The same teacher-operated 'workflow deck' as the Core period, now with extra TOOL cards (Reading-level adjuster,

Splitter) and two printed candidate designs to judge. Optionally a teacher-run board demo of a school-approved tool.

Account-free, no logins, no personal data - the weather brief is generic, with no names or faces.

Facilitation steps:

1. Read the brief and the two required outputs - a plain-English student version and an Arabic parent version.

2. Judge designs A and B against the three levers, logging each weakness and the lever it belongs to.

3. Choose the stronger design and mark the shared stage and the branch point.

4. Improve it: place the verify step in the shared stage, and match a fitting tool to each branch.

5. Run your improved workflow and record both audience outputs.

6. Verify one fact in each output against the source, then write one sentence justifying a key design choice.

Unplugged fallback: The designs and tool cards are already on paper, so no device is needed. If any board demo is

unavailable, pairs move the cards by hand to build and trace their improved design - the compare-and-improve skill is unchanged.

Lab tool integration (practical, hands-on)

Tool: Scratch (refine a workflow's structure and sequencing)

Teacher setup: Provide a Scratch project that works but is poorly structured (steps in an awkward order, no check).

What students do:

1. Run the starter project and note what is clumsy or fails.

2. Reorder or regroup the steps to improve the structure.

3. Add one check step and choose the better-suited block/tool where relevant.

4. Re-run and confirm the output improved.

5. Explain which refinement helped most and why.

Students produce: A refined Scratch workflow and an explanation of the improvement.

Data-safety: No personal data; neutral inputs only.

Unplugged fallback: Re-sequence block-cards on the desk to improve the structure.

A clumsy workflow refined into a clearer, better-sequenced one.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Judge one weakness in design A or B and name the lever it belongs to.

Design A verifies the facts on each branch instead of once before the split - a structure weakness, because the two checks can disagree and it repeats work.

One mark for a real weakness correctly matched to a lever, with a reason from the brief - not just 'design A is worse'.

Improve the sequence so an error is caught for both audiences, and explain the change.

I moved the Verify step into the shared stage before the branch, so a wrong figure is caught once before it is either simplified or translated, and neither output can carry it.

Award the mark only if the change genuinely stops the error reaching both branches, with a clear reason tied to the order.

Justify one tool choice on a branch, using what the tool can do.

The parent branch needs the Translator because it produces the Arabic version; the Reading-level adjuster could not, as it only changes how hard the English is to read.

Look for a justification that matches the tool to the branch's job and rules out a tool that cannot do it; reject 'it just fits'.

- Targets: designing and improving a more complex, two-audience workflow by refining structure, sequencing and tool selection, and justifying each choice.

- Check: circulating compare-and-improve prompts during the station, plus an exit ticket naming one improvement, its lever and its justification.
- Success criteria: the student judges the designs, places verification in a shared stage before the branch, matches a fitting tool to each branch, and justifies why each output is reliable and fit for its audience.
- Evidence: collected comparison logs, the improved design with both recorded outputs, and the exit tickets.
- Feedback / adjustment: if a pair duplicates the checking on each branch or picks a tool that cannot do its job, re-model the shared-stage-then-branch shape and have them rebuild one branch before the next unit lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Judging the designs Finds real weaknesses in a design and names the lever each belongs to, with support.

Notices a design is weaker but mislocates or cannot name the lever.

Cannot tell which design is weaker or why.

Improving the workflow Rebuilds a stronger design with a shared verify stage and a fitting tool on each branch, with guidance.

Improves the order or the tools but not both, or checks facts twice.

Cannot produce an improved, workable design.

Justifying the design Explains how a structure, sequence or tool choice makes an output reliable and fit for its audience.

Describes a change but not why it improves reliability or fit.

No justification linking a choice to the output.

Representation: display the shared-stage-then-branch shape as a labelled diagram, provide the tool cards and both designs in large print with icons, and pre-teach the words shared stage, branch and fit for audience with examples.

Action & expression: pairs may improve the design by moving cards, annotating the printed designs, or explaining the branch aloud

to the teacher who scribes; accept any mode for the exit ticket.

People of Determination: provide the two designs with the shared stage already marked, fewer tool cards, sentence starters ('This

design checks the facts ... so ...'), and extra time to run each branch.

Week 3 - Core | Critical Evaluation & Informed Interaction - Iterating a Workflow

and Explaining the Impact of Changes | Interactive lesson: 'Change, Judge, Improve'

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G7·A | 1 x 45 min

Objectives & success criteria

- Improve a given AI-supported workflow through at least one deliberate iteration and record what was changed.
- Judge each version of the workflow against agreed success criteria using concrete evidence.
- Explain the impact of each change on the quality of the output, using cause-and-effect reasoning.

Student-facing success criteria:

- I can change one step of a workflow on purpose and say exactly what I changed and why.
- I can judge two versions of a workflow against the same criteria using evidence, not opinion.
- I can explain how a change made the output better or worse and what I would try next.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 6 you followed a set workflow step by step to reach a result, and in Grade 7 you learned to improve a

workflow through iteration and explain the impact of the changes you made. This week revises that Grade 7 skill and applies

it to evaluating and improving AI models, the theme of this unit.

Activate: Show two versions of a short classroom workflow for sorting a set of pre-generated animal photos into 'mammal' and 'not

mammal' (all images are teacher-supplied, no personal data). Version A has a vague instruction; Version B adds one clear rule.

Ask: which one gave the cleaner result, and how do you know?

Teach: Model the loop out loud: run the workflow, judge the result against a criterion, change one thing, run it again, then compare.

Name each part as you go so students hear iterate, evidence and impact used correctly.

Apply: In pairs, students look at the two versions on the board and write one sentence: 'Version B changed ____, and the impact was ____.'

Check: Cold-call three pairs. The rung is secure if they can name the single change and link it to a specific, observed impact rather than saying 'it was just better'.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Iteration is the disciplined loop at the heart of improving any AI-supported solution: run, evaluate against criteria, change

one variable, run again, and compare. Because AI outputs are probabilistic, a single run is weak evidence; the value comes from controlled comparison across versions. Changing only one thing at a time lets you attribute any difference in quality to that change, which is what 'explaining the impact' really means - a cause-and-effect claim backed by evidence. Good evaluation needs criteria agreed before you look at the output, so judgement stays honest rather than being bent to fit a favoured version. This is also where verify-before-trust lives: a confident-looking output can still fail the criteria, so every version is checked against the task and a trusted reference, never accepted on appearance alone.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Teacher shows two versions of a curated output set from the same simple workflow and asks the class to spot which is stronger.

Students vote and give one reason each in pairs.

- Which version meets the criterion, and what evidence tells you so?

Direct teach / 10 min

Teacher names and models the iterate loop: run, judge against criteria, change one thing, re-run, compare; writes the loop on the board.

Students copy the loop and label the 'change one thing' step.

- Why change only one thing at a time?

Guided practice / 12 min

Teacher runs a teacher-operated demo (or reveals a pre-generated version)

Students record the change made and the observed

- Did the change help, hurt, or do nothing against our criterion?

after the class agrees a single change; leads comparison against criteria.

impact in a two-column
note.

Independent

practice / 12 min

Teacher circulates as pairs iterate a
paper 'simulated AI' sorting routine and
log their versions.

Students complete one full
loop and write an impact
sentence for their change.

- Can you point to the evidence for
your impact claim?

Check and close

/ 5 min

Teacher collects exit tickets and
highlights two strong impact
explanations.

Students write: the change I
made was ____; the impact on
the result was ____; my
evidence was ____.

- Does your explanation link
change to impact with evidence?

Anticipated misconceptions

Misconception How to address it

If I change lots of things at once, I will improve it faster. Show that changing several things
means you cannot tell which
one caused the difference; insist on one change per loop so
impact can be explained honestly.

The new version looks more confident, so it must be
better.

Reinforce verify-before-trust: judge against the agreed criteria and
a trusted reference, not by how polished the output appears.

Improving a workflow means getting a perfect answer in
one go.

Reframe improvement as a loop: each iteration is judged against
criteria and may reveal a new change to try; progress, not
perfection, is the goal.

Hands-on activity guide

Overview: Pairs run a short 'sort the cards' workflow, judge it against one agreed criterion, make
a single deliberate

change, run it again, and explain the impact of that change with evidence.

Grouping / timing: Pairs for the iterate-and-log build; whole class for the criteria-setting and the
debrief.

Tool: A paper 'simulated AI' card-sorting routine operated by the students, with an optional teacher-operated board demo of a school-approved, child-safe classifier on non-personal images; account-free for students and no personal data used at any point.

Facilitation steps:

1. As a class, agree one success criterion for the sort, for example 'every card ends in the correct pile'.
2. Run version 1 of the sorting rule on the teacher-supplied cards and record the result against the criterion.
3. Choose exactly one change to the rule to try to improve the result, and write it down before running it.
4. Run version 2 with only that single change and record the new result against the same criterion.
5. Compare version 1 and version 2 using your evidence, not your opinion, and decide whether the change helped, hurt, or did nothing.
6. Write one impact sentence: the change was ____, the impact was ____, and the evidence was ____.

Unplugged fallback: The card-sorting routine is already fully unplugged, so if no approved tool is available the lesson runs

unchanged using only the teacher-supplied cards and the class criterion.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Given two versions of a workflow and one criterion, state which version is stronger and cite the evidence.

Version B is stronger because it sorted 9 of 10 cards correctly against our criterion, while Version A sorted 6 of 10.

Award the mark only if a specific criterion and countable evidence are used, not a general 'it looks better'.

Describe the single change you made to your workflow and predict its impact before running it.

I added the rule 'four legs and fur means mammal'; I predict fewer wrong placements in the mammal pile.

Look for exactly one change named and a plausible, testable impact prediction.

Explain the impact of your change after

running the new version, linking change to result with evidence.

The rule fixed three cards that were wrong before, so the mammal pile went from 6 to 9 correct; the change had a clear positive impact.

Full marks require an explicit cause-and-effect link supported by the recorded evidence.

- Targets: improving a workflow through one deliberate iteration and explaining the impact of the change with evidence.
- Check: pairs submit a two-column version log plus a one-sentence impact explanation on an exit ticket.
- Success criteria: the student names a single change, judges both versions against the agreed criterion, and links the change to its impact with evidence.
- Evidence: the version log, the impact sentence, and the teacher's cold-call notes from the debrief.
- Feedback / adjustment: if impact claims lack evidence, model one strong example and have pairs redo their sentence citing a countable result before the next lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Makes a deliberate iteration Changes exactly one step on purpose and records it

Changes a step but records it vaguely or changes more than one

Changes nothing deliberate or cannot say what changed

Judges against criteria with evidence

Uses the agreed criterion and countable evidence to compare versions

Refers to the criterion but leans on opinion for the judgement

Judges by appearance with no criterion or evidence

Explains the impact Links the change to its impact with clear cause and effect

States the impact but the link to the change is unclear
States only that it was better or worse with no reasoning

Representation: provide the iterate loop as a labelled diagram and a spoken walkthrough, and give the criterion as a printed sentence strip so it stays visible during the task.

Action & expression: allow the impact explanation to be spoken, written, or shown as a simple before-and-after tally rather than in full sentences.

People of Determination: offer a pre-printed version-log template with sentence stems, larger cards with high-contrast labels, and a partner to scribe so the focus stays on the reasoning.

Week 3 - Lab | Critical Evaluation & Informed Interaction - Lab: Iterating and Evaluating a Model's Outputs Against Criteria | Interactive lesson: 'Score, Iterate, Justify'

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G7·A | 1 x 45 min
Objectives & success criteria

- Evaluate a curated set of model outputs against agreed criteria and justify each judgement with evidence.
- Run a full iteration loop on a workflow, making and logging one deliberate change per cycle across two cycles.
- Explain the cumulative impact of the changes and justify why the improved workflow is stronger.

Student-facing success criteria:

- I can score a set of curated outputs against criteria and back up each score with evidence.
- I can complete two iteration cycles, changing one thing each time, and keep a clear log.
- I can explain how my changes added up to a better workflow and justify my final choice.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: This Lab builds directly on the Core lesson and on the Grade 7 rung 'improves workflows through iteration and

explains the impact of changes', which itself follows the Grade 6 rung of following a fixed workflow. Here you apply that skill

more deeply by evaluating a set of curated outputs and iterating across two cycles in the context of improving AI models.

Activate: Show a curated set of five pre-generated outputs from a simple classifier (all on teacher-supplied, non-personal data)

and ask pairs to rank them fastest-to-slowest against one visible criterion.

Teach: Model scoring one output against a criterion aloud, showing how to attach evidence to a judgement, then show how the scores guide which single change to try next.

Apply: In pairs, students score one further output from the set and write the evidence they used for that score.

Check: Ask two pairs to defend a score. The rung is secure if the score is tied to the criterion and a specific piece of evidence, not to preference.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Evaluating a model means scoring its outputs against criteria that are fixed before you look, so the judgement is honest and

repeatable. Because outputs vary, a single example is thin evidence; you assess a set and look for a pattern, then let that

pattern point to the one change most likely to raise quality. Iterating across several cycles turns scattered tweaks into a

defensible sequence: each cycle changes one variable, is re-scored against the same criteria, and is logged, so the

cumulative impact can be traced and justified. This is critical evaluation at DOK 3 and SOLO relational level - relating

multiple pieces of evidence to a criterion to reach and defend an overall judgement.

Verify-before-trust remains central: a

persuasive output that fails the criteria is not accepted, and every judgement is checked against the task and a trusted

reference rather than the output's surface appeal.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap / 5 min Teacher recaps the iterate loop from

Core and shows the curated output set to be evaluated.

Students restate the loop

and read the agreed criteria.

- What does one full cycle of the loop include?

Model the

scoring / 8 min

Teacher models scoring two outputs against the criteria and attaching evidence to each score.

Students annotate the modelled scores with the evidence used.

- What evidence justifies this score rather than the one above it?

Lab build cycle 1

/ 12 min

Teacher supports pairs as they score

the set, choose one change, and re-run a version.

Students score the set, log one change, and record the new result against criteria.

- Which single change does your evidence point to?

Lab build cycle 2

/ 13 min

Teacher pushes pairs to justify a second, evidence-led change and

Students complete a second cycle and compare versions

- Is version three better than version one, and how do you know? compare across all versions. one, two and three.

Justify and close

/ 7 min

Teacher runs a defence round where pairs justify their final workflow; collects logs.

Students write a justification of their final workflow citing the cumulative impact of their changes.

- Can you justify your final choice using evidence from every cycle?

Anticipated misconceptions

Misconception How to address it

A model that gives a confident, detailed output is a better model.

Show a confident output that fails a criterion; reinforce that scores come from criteria and evidence, and that verify-before-trust overrides surface confidence.

I should pick my next change based on a hunch about what feels right.

Redirect to evidence: the pattern in the scores should point to the change; a hunch is a hypothesis that still has to be tested against criteria.

Two iterations means the second one cancels out the first. Clarify that changes accumulate; the log lets you trace each change so you can justify the combined, cumulative impact across cycles.

Hands-on activity guide

Overview: Pairs evaluate a curated output set against criteria, then run two iteration cycles on a paper workflow, logging

one change each cycle, and finish by justifying why their final version is the strongest.

Grouping / timing: Pairs for scoring and iterating; a short whole-class defence round for the justification.

Tool: A curated set of pre-generated outputs plus a paper 'simulated AI' workflow the students operate; any live classifier is

a school-approved, child-safe, teacher-operated board demo on non-personal data only, and students use no accounts.

Facilitation steps:

1. Read the agreed criteria and score each output in the curated set, writing the evidence beside every score.
2. Use the pattern in your scores to choose one change to your workflow, and log it before running.
3. Run cycle 1 with that single change and re-score the result against the same criteria.
4. Use the new scores to choose a second, evidence-led change, and log it before running.
5. Run cycle 2 with only that change, then compare all versions against the criteria.
6. Write a justification of your final workflow that explains the cumulative impact of both changes with evidence.

Unplugged fallback: The scoring set and paper workflow are fully unplugged; without any tool the Lab runs exactly as

written using the printed output cards and the class criteria.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine retrain + packs (improve through iteration)

Teacher setup: Have a trained model and a pack showing outputs across two or three iterations.

What students do:

1. Test the current model and record the score.
2. Make one change and retrain (an iteration).
3. Re-test and record the new score.
4. Repeat once more, so you have three scores.
5. Explain the impact of each change with evidence.

Students produce: A three-iteration score record explaining the impact of each change.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Unplugged fallback: Compare the pack's iteration outputs and tally each by hand.

Scores rising across three iterations as changes are made.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Score three curated outputs against the criteria and justify the middle score with evidence.

I scored output B in the middle: it met the

accuracy criterion but missed the completeness one, so it sits below A and above C.

Award marks only where each score is tied to the named criteria and cites specific evidence.

Log two iteration cycles, one change per cycle, and state each change clearly.

Cycle 1: added a tie-break rule; cycle 2: tightened the mammal definition; each on its own line with the result recorded.

Require exactly one change per cycle and a recorded result for each; deduct if changes are bundled.

Justify why your final workflow is stronger than version one, using cumulative evidence.

Version three fixed four cards across two changes, moving the correct count from 5 to 9 against our criterion, so it is clearly stronger.

Full marks need an overall judgement that relates evidence from both cycles, not just the last change.

- Targets: evaluating a curated output set against criteria and justifying an iterated workflow by its cumulative impact.
- Check: pairs submit a scored evaluation grid, a two-cycle version log, and a written justification of the final workflow.
- Success criteria: the student ties every score to criteria and evidence, changes one thing per cycle, and justifies the final version by relating evidence across both cycles.
- Evidence: the evaluation grid, the version log, the justification paragraph, and notes from the defence round.
- Feedback / adjustment: where justifications rely on the last change alone, prompt the student to cite version one evidence too, and re-run the defence for those pairs next session.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Evaluates against criteria
with evidence

Scores the set against agreed
criteria and cites evidence for each

score

Scores against criteria but
evidence is thin or missing
on some items

Scores by preference
with no criteria or
evidence

Iterates across two cycles Completes two cycles, one logged
change each, with results recorded

Completes cycles but
bundles changes or logs
them incompletely

Does not complete a
second cycle or cannot
show the changes

Justifies the improved
workflow

Justifies the final version by relating
evidence across both cycles

Justifies mainly from the last
change without the full
picture

Asserts it is better with
no justification

Representation: provide the evaluation grid with criteria pre-printed and a worked scoring
example, plus the iterate loop visible as
a labelled diagram throughout.

Action & expression: let pairs present their justification as a spoken defence, a written
paragraph, or an annotated comparison
table.

People of Determination: supply a large-print scoring grid with sentence stems for justifications,
a scribe partner where needed,
and extra time for the two-cycle build so the reasoning is not rushed.

Week 4 - Core | Ethics & Governance - Spot the risk, propose responsible-use
fixes | Interactive lesson: 'Spot the Risk, Fix It'

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G7·A | 1 x 45 min

Objectives & success criteria

- Analyse a fresh AI-model scenario to identify its main risks and ethical or governance concerns.
- Propose at least two responsible-use measures that would reduce those risks, tied to the scenario.
- Justify each proposed measure with a clear reason linked to the risk it addresses.

Student-facing success criteria:

- I can read an AI scenario and name the real risks and ethical or governance concerns in it.

- I can propose sensible responsible-use measures that reduce those risks.
- I can give a reason for each measure that explains why it helps.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you sorted AI uses into helpful and harmful, in Grade 6 you named a single risk in an AI use and one way to reduce it, and in Grade 7 you began proposing ways to reduce risks and improve responsible use. This week revises that Grade 7 Advanced rung to full mastery before the Grade 8 unit on evaluating and improving models.

Activate: Show one short scenario card on the board: a school trials an AI model that predicts which students might fall behind,

using their quiz scores. Ask: what could go wrong here, and who is responsible if it does?

Teach: Model a two-step move on the board: first name the risk (for example, the model could be wrong for some students and quietly label them), then name a responsible-use measure that reduces it (a teacher checks every flag before acting), and say the reason out loud.

Apply: In pairs, students take the same card and add one more risk and one more measure with a reason, writing each as risk - measure - because on a sticky note.

Check: Cold-call three pairs for a risk-measure-because chain; the rung is secure when a pair can state a measure and the reason it reduces the named risk, not just a general 'be careful'.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Responsible use of AI means putting checks around a model so that its mistakes and harms are caught and limited, not just

hoped away. Every model can be wrong, and it can be wrong unevenly - performing worse for some groups because of gaps

or bias in its training data. Governance is the set of rules and roles that decide who may use a model, on what data, and who

is accountable for outcomes. Key measures include keeping a human in the loop for consequential decisions, limiting the

model to non-personal data, being transparent about how it works and its limits, offering a way to challenge a decision, and

testing for bias regularly. At this stage students learn to reason from a specific risk to a specific measure and to justify the

link, rather than listing generic cautions.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Teacher shows the 'predict who falls behind' scenario card and thinks aloud

about one thing that could go wrong.

Students call out risks;
teacher lists them without
judging yet.

- What decision does this model
influence, and who feels it?

Frame the move

/ 8 min

Teacher models the risk - measure -
because chain on the board and names
the governance question: who is
accountable.

Students copy the sentence
frame and label a worked
example.

- Why is naming a reason different
from just naming a fix?

Guided analysis /

12 min

Teacher hands each pair a different
scenario card and circulates, pushing
for reasons.

In pairs, students find two
risks and one responsible-
use measure per risk, each
with a because.

- Which of your risks is most
serious, and how do you know?

Share and Teacher runs a gallery swap and Students critique another - Does the measure
match the risk,
pressure-test /

12 min

challenges weak measures ('would that
actually reduce the risk?').

pair's chains and suggest
one stronger measure.

or is it just good advice?

Check and close

/ 7 min

Teacher collects sticky notes and
names the success criteria for the exit
ticket.

Students write one risk-
measure-because chain on

the exit ticket independently.

- Can you justify one measure without help?

Anticipated misconceptions

Misconception How to address it

If the AI is very confident, its answer must be right, so there is not much risk.

Show a confident but wrong pre-generated output; explain confidence is not correctness, which is exactly why we verify and keep a human check.

Responsible use just means telling people to be careful with AI.

Contrast a vague 'be careful' with a concrete measure (human review of every flag) and ask which one actually changes what happens; measures must act on the specific risk.

Once we remove names, the model has no ethics or governance problems left.

Point out a model can still be biased, unaccountable, or unchallengeable even with no names; removing personal data is one measure, not the whole solution.

Hands-on activity guide

Overview: Pairs work a bank of short, paper AI-model scenarios, building risk - measure - because chains, then compare and strengthen each other's proposals in a gallery swap.

Grouping / timing: Pairs for the analysis and build, whole class for the gallery swap and debrief.

Tool: A 'simulated AI' paper routine: pre-printed scenario cards with an example model output already on them, so no live

tool and no accounts are needed. Any optional demo of a real model output is a teacher-operated, school-approved board demo only, on non-personal data.

Facilitation steps:

1. Read your scenario card and underline the decision the AI model influences and who is affected.
2. Name two risks or ethical/governance concerns in the scenario; write each in your own words.
3. For each risk, propose one responsible-use measure that would reduce it.
4. Add a because to each measure explaining how it acts on that exact risk.
5. Swap cards with another pair and tick any measure that truly matches its risk; flag any that do not.
6. Rewrite your weakest measure into a stronger one using the feedback you received.

Unplugged fallback: The whole task is already non-AI: it runs on printed scenario cards and student reasoning, so it works

with no device or tool access at all.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Given a new one-line scenario (an AI app suggests which library books a pupil should read), name one risk in it.

The model might narrow a pupil's choices or push biased suggestions based on limited data.

Accept any genuine risk tied to the scenario; reject vague answers like 'AI is bad'.

Propose one responsible-use measure for that scenario.

Let a librarian review the suggestions and let pupils browse freely beyond the list.

Measure must plausibly reduce a named risk, not be generic advice.

Justify your measure in one sentence starting 'because'.

Because a human check catches unfair or narrow suggestions before they limit the pupil.

Award the mark only if the reason links the measure to the specific risk.

- Targets: proposing and justifying responsible-use measures that reduce risks in a fresh AI-model scenario (objectives 2 and 3).
- Check: exit-ticket risk-measure-because chain plus cold-call justification during the gallery swap.
- Success criteria: student names a real scenario risk, proposes a matching measure, and gives a reason that links the two.
- Evidence: collected sticky notes and exit tickets showing at least one complete, justified chain per student.
- Feedback / adjustment: if reasons are missing or generic, re-model one chain and have students rewrite using the risk - measure - because frame.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Identifies risk Names a real risk tied to the scenario

Names a risk but keeps it vague or generic

No genuine risk identified

Proposes measure Proposes a measure that reduces

the named risk

Proposes a measure only

loosely linked to the risk

Proposes only 'be

careful' advice

Justifies with reason Gives a reason linking the measure to the risk

Gives a reason but the link is unclear

No reason given

Representation: provide the scenario as text and as a simple icon strip, and give a word bank (risk, bias, accountable, transparent, human check) with kid-friendly meanings.

Action & expression: students may respond in writing, as a labelled sticky-note chain, or by explaining their chain aloud to the teacher.

People of Determination: offer sentence frames (This is risky because... / A fair fix is... because...), enlarged print cards, extra thinking time, and a partner scribe where needed.

Week 4 - Lab | Ethics & Governance - Build a responsible-use plan for a harder case | Interactive lesson: 'Rank Risks, Build a Plan'

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G7·A | 1 x 45 min

Objectives & success criteria

- Analyse a more complex, multi-stakeholder AI-model case for layered risks and governance concerns.
- Design a short responsible-use plan proposing measures that reduce the most serious risks.
- Justify the plan by explaining trade-offs and why the chosen measures are the strongest available.

Student-facing success criteria:

- I can break a complex AI case into its separate risks and say who is affected by each.
- I can build a responsible-use plan with measures aimed at the most serious risks.
- I can justify my plan, including the trade-offs I accepted and why.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Building on Grade 5 (helpful vs harmful), Grade 6 (one risk, one fix) and Grade 7 (proposing ways to reduce risks

and improve responsible use), and directly on this week's Core lesson where you built single risk - measure - because

chains. The Lab extends that to a case with several risks at once.

Activate: Re-show a Core scenario card, then reveal a harder version with more stakeholders: an AI model that flags social posts

as safe or unsafe across Arabic and English on a school's shared tablets.

Teach: Model how to map a complex case: list the stakeholders, list a risk for each, then rank risks by how serious and how likely

they are before proposing measures - so effort goes to the biggest risks first.

Apply: In pairs, students list three stakeholders and one risk each for the flagging model, then mark the risk they think is most serious.

Check: Ask two pairs to justify their most-serious pick; the rung is secure when they reason from impact and likelihood, not just from which risk they noticed first.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Real deployments rarely have a single risk. A content-flagging model, for example, can be biased across languages, can

wrongly block harmless posts (false positives) or miss harmful ones (false negatives), can be opaque so users cannot see

why they were flagged, and can lack any route to appeal. Improving responsible use means prioritising: estimating each risk

by its impact and likelihood, then targeting the worst with concrete measures such as human review of borderline cases,

testing performance separately for each language, publishing what the model does and does not do, and giving an appeal

channel. Measures also trade off against each other - more monitoring can mean less privacy, more human review can

mean slower decisions. A strong plan names these trade-offs and justifies the balance it strikes rather than pretending a

perfect, cost-free fix exists.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and raise

the stakes / 6

min

Teacher links back to the Core chain

and introduces the harder multi-

stakeholder flagging case.

Students recall the risk -

measure - because move

and read the new case.

- What makes this case harder

than yesterday's card?

Map the case / 9

min

Teacher models a stakeholder-and-risk

map and a simple impact x likelihood

ranking on the board.

In pairs, students list

stakeholders and one risk each, then rank the risks.

- Which risk deserves the most attention, and why that one?

Build the plan /

14 min

Teacher circulates, insisting each measure names the risk it targets and the trade-off it accepts.

Pairs draft a three-measure responsible-use plan aimed at their top-ranked risks.

- What does this measure cost, and is it worth it here?

Justify and challenge / 10

Teacher runs a short defence round where another pair probes the trade-

Pairs present one measure and defend the trade-off;

- Could a different measure protect people better for less cost?

min offs. challengers propose an alternative.

Consolidate and close / 6 min

Teacher names the success criteria and sets the individual exit reflection.

Students write which trade-off in their plan they are least sure about and why.

- Can you name the weakest point in your own plan?

Anticipated misconceptions

Misconception How to address it

A good responsible-use plan removes every risk completely.

Explain that risks are reduced and managed, not erased; a strong plan targets the worst risks and is honest about what remains.

The best plan is simply the one with the most rules and monitoring.

Show that more monitoring can harm privacy and trust; the best plan balances protection against its costs and justifies the balance.

Whichever risk we noticed first is the most important one to fix.

Re-teach impact x likelihood ranking so priority comes from how serious and how likely a risk is, not from the order it was spotted.

Hands-on activity guide

Overview: Pairs take one complex, multi-stakeholder AI-model case, map its stakeholders and risks, rank the risks, then

build and defend a short responsible-use plan that names its trade-offs.

Grouping / timing: Pairs for the mapping and plan, then a whole-class or pair-vs-pair defence round for the debrief.

Tool: A 'simulated AI' case dossier on paper: the case, three sample model outputs (two reasonable, one wrong) and a

blank plan template, so no accounts or live tools are needed. Any real model output shown is a teacher-operated, school-

approved board demo on non-personal data only, and every sample is verified against the task before students trust it.

Facilitation steps:

1. Read the case dossier and list the stakeholders and who is affected by the model.
2. Write one main risk for each stakeholder in your own words.
3. Rank your risks by impact and likelihood and circle your top two.
4. Propose one responsible-use measure for each top risk, naming the trade-off it accepts.
5. Write a because for each measure and a one-line justification for the whole plan.
6. Defend one measure to another pair and note one improvement they suggest.

Unplugged fallback: Already unplugged: the case dossier, ranking and plan are all done on paper with student reasoning,

so the Lab runs fully without any device or tool access.

Lab tool integration (practical, hands-on)

Tool: Card decks (bias-source cards) + packs (reduce risks / responsible use)

Teacher setup: Print bias-source cards and include a pair of outputs where one is risky or unfair due to a known cause.

What students do:

1. Read a bias-source card and the paired outputs.
2. Identify the risk or unfairness and its likely cause in the data or design.
3. Say who could be affected.
4. Propose one concrete action to reduce the risk and improve responsible use.
5. Predict how the action would change the output.

Students produce: A risk-reduction record: risk, cause, who is affected, and a proposed action.

Data-safety: Unplugged; outputs pre-curated.

Unplugged fallback: This is the unplugged route.

A risk traced to its cause, then a concrete action to reduce it.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

From the flagging case, list two stakeholders and one risk each.

Students who post (risk: unfair blocking of harmless posts) and teachers who review flags (risk: over-trusting the model).

Accept any two valid stakeholders with a real, distinct risk each.

Rank those risks and name the more serious one with a reason.

Unfair blocking is more serious because it silences pupils often and is hard to appeal.

Reason must use impact and/or likelihood, not just personal preference.

Propose one measure for the top risk and state its trade-off.

Add a human appeal step; trade-off is slower decisions, but it prevents unfair silencing.

Award full marks only if both the measure and an honest trade-off are given.

- Targets: designing and justifying a responsible-use plan for a complex case, including trade-offs (objectives 2 and 3).
- Check: pair plan template plus a defence round where trade-offs are probed by peers and teacher.
- Success criteria: student ranks risks by impact and likelihood, targets the worst with matched measures, and justifies the trade-offs.
- Evidence: completed plan templates and the exit reflection naming each pair's least-certain trade-off.
- Feedback / adjustment: if plans ignore trade-offs, re-model one measure with its cost and have pairs annotate their own plan's trade-offs.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Maps the case Identifies several stakeholders and a distinct risk for each

Identifies risks but misses stakeholders or overlaps them

Lists only one vague risk

Prioritises risks Ranks risks by impact and likelihood and targets the worst

Ranks risks but reasoning is unclear

No ranking; fixes the first

risk noticed

Justifies trade-offs Names the trade-off in each measure and justifies the balance

Mentions a trade-off but

does not justify it

No trade-offs considered

Representation: supply the case as text plus a stakeholder-risk map template and an impact x likelihood grid, with the key terms defined in a side glossary.

Action & expression: pairs may submit the plan as a filled template, a labelled grid, or a spoken defence recorded by the teacher as notes.

People of Determination: pre-printed stakeholder cards to sort rather than generate, sentence frames for trade-offs (This measure helps... but it costs... and that is worth it because...), extra time, and a scribe option.

Week 5 - Integrated Task (Design & Build) | Integrated Prerequisite Mastery Task - Design, Build, Evaluate & Justify an AI Solution | Interactive lesson: 'Design, Build, Test'

Strand: Integration (all four prior-grade strands) | Framework ref(s): AIF·CU·G7·A AIF·SD·G7·A AIF·CE·G7·A AIF·GE·G7·A | 1 x 45 min

Objectives & success criteria

- Analyse how different workflow designs - the order of steps and the tool chosen for each step - shape the outcome a simulated AI reply-helper produces.
- Design and build, in pairs, a structured multi-step workflow that turns a fictional club enquiry into a correct, verified reply.
- Document the workflow map, the tool chosen for each step and one test run as evidence the solution works.

Student-facing success criteria:

- I can explain how changing the order of our steps or the tool we pick changes the reply our workflow produces.
- I can design and build a clear multi-step workflow that answers an enquiry and checks the reply before it is sent.
- I can record our workflow map, our tool choices and one test run so someone else can see it works.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Weeks 1-4 you rebuilt four Grade 7 Advanced skills: explaining how different workflow designs affect

outcomes (CU), refining a workflow by improving its structure, sequencing and tool selection (SD), improving a workflow

through iteration and explaining the impact of the changes (CE), and proposing ways to reduce risks and improve

responsible use of AI systems (GE). This two-week task brings all four together in one solution.

Activate: Show a board demo of a simulated reply-helper: feed one fictional enquiry card in, run it through two different step orders, and reveal that one order sends an unverified reply while the other verifies against the FAQ first. Ask which design is better and why.

Teach: Model the whole workflow shape once - an enquiry card in, then ordered steps (understand the enquiry, find the facts, draft with the simulated AI reply deck, verify against the FAQ), then the reply out. Show live how moving the verify step changes the outcome.

Apply: In pairs, students reorder one step in a starter workflow and predict how it changes one reply.

Check: Ask two pairs to name the input, the steps in order, and the output, and to state one way the design changed a reply; re-model if a pair confuses the enquiry with a workflow step.

Critical & analytical thinking focus

Analyse, Evaluate and Create - sustained higher-order thinking across all four strands (Analyse/Evaluate/Create | DOK 3-4 |

SOLO: Relational to Extended abstract)

The technical concept (teacher background)

A workflow, or pipeline, is an ordered set of steps that turns an input into an output; in an AI-assisted solution one step calls

a model while the others prepare the input and check the result. Here the model is simulated - a curated deck of pre-

generated reply cards, teacher-prepared and checked - so students see how structure, sequencing and tool selection drive

quality without any account or live tool. The same enquiry can yield very different replies depending on the design: verifying

against a trusted FAQ before sending catches a confident but wrong reply, whereas drafting last or skipping the check lets

errors through. Building the workflow - input, ordered steps, output - rehearses structured solution design. All work is

account-free and uses only fictional, non-personal enquiry cards, never real names or personal data. Cognitive demand for

this lesson: Analyse/Create | DOK 2-3 | SOLO: Multistructural to Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Launch the task /

6 min

Introduce the integrated mastery task and the paired brief. Run the two-order board demo of the simulated reply-helper and reveal how the design changed the reply.

Watch, then name the input,
the ordered steps and the
output on a mini whiteboard.

- Which design gave the trustworthy reply - and what made the difference?

Analyse the
design / 12 min

Hand each pair a starter workflow and enquiry cards. Model reordering one step, such as moving the verify step, and how it changes a reply; prompt pairs to test a change (CU step).

In pairs, reorder one step and record how it changes one reply and why.

- Which change to your step order or tools would most improve the reply, and how do you know?

Design the
workflow / 10
min

Model drawing the input-steps-output map and choosing a tool for each step. Circulate to check the maps (SD step).

Draw their own workflow map and label the tool each step uses.

- What does each step do, in what order, and which tool did you choose for it?

Build and test /
12 min

Support pairs to finish their workflow and run three enquiry cards through it, logging each reply and whether it was verified. Keep any live tool teacher-operated.

Build the workflow, run three enquiry cards, and log the reply and verify-check for each.

- Did every reply pass the verify check - and if not, which step let it

through?

Check and close

/ 5 min

Collect the build sheets. Run the exit check and preview Week 6 (evaluate, justify, risks, reflection and showcase).

Complete the exit ticket:

name one design choice and how it shaped a reply.

- Name one thing about your step order or tool choice that changed the reply your workflow produced.

Anticipated misconceptions

Misconception How to address it

The AI does all the work - the order of our steps doesn't really matter.

Run the same enquiry through two step orders and show one sends an unverified, wrong reply; make clear the design is what makes the output trustworthy, not the AI step alone.

A step is only worth adding if it uses the AI. Show that the non-AI steps - understanding the enquiry and

verifying against the FAQ - are what catch errors; drop the verify step and watch a confident wrong reply go out.

More steps always means a better workflow. Add several steps that repeat the same job and show the

workflow gets slower without getting better; agree that the right steps in the right order matter, not just the number.

Hands-on activity guide

Overview: Across the week's two periods, pairs analyse how a workflow's design shapes the reply a simulated AI helper

produces, then design and build their own multi-step reply workflow and test it against sample enquiry cards.

Grouping / timing: Pairs for the analyse-design-build; whole class for the launch demo and the close.

Tool: A 'simulated AI' reply-helper - a curated deck of pre-generated reply cards plus fictional enquiry cards and a trusted

FAQ sheet, run by hand or on a teacher-operated board. Account-free, no logins, and only fictional, non-personal enquiries

- never real names or personal data.

Facilitation steps:

1. Watch the launch demo and name the input, the ordered steps and the output of the simulated reply-helper.

2. Analyse the starter workflow: reorder one step and record how it changes one reply (CU step).

3. Draw your input-steps-output map and choose the tool each step will use (SD step).
 4. Build your workflow by sequencing your steps and giving each a clear job.
 5. Run three enquiry cards through your workflow and log the reply and the verify-check for each.
 6. Check each reply against the trusted FAQ and fix any step that let a wrong reply through.
- Unplugged fallback: The whole build already runs on paper cards, a reply deck and a hand-run FAQ, so no device is needed. If the board demo is unavailable, route the enquiry cards by hand through the step order and log the results on the build sheet.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine + Scratch (integrated: design, build, first measure)

Teacher setup: Provide the full kit: Teachable Machine open, Scratch starter, plan cards.

What students do:

1. In pairs, plan the small solution with the plan cards, including performance criteria.
2. Train a simple non-personal model and build the solution in Scratch.
3. Run it on a short test-set and record a first score.
4. Note the single biggest limitation to fix in Week 6.
5. Save your build and note what each strand contributed.

Students produce: A built solution with a first measured score and a target limitation.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Non-personal model only.

Unplugged fallback: Build with card-decks and block-cards; score with the human-classifier.

Integrated build with a first measured score and a target to improve.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Reorder one step in your workflow and explain how it changes one reply.

We moved the verify step to before sending, so the 'meeting time' reply is now checked against the FAQ and the wrong time is caught instead of sent.

One mark for a correct linked change (step moved, then changed reply) with a reason that references the design.

Draw or describe the input, the ordered steps and the output of your reply workflow.

Input: an enquiry card. Steps: understand the enquiry, find the facts, draft with the reply

deck, verify against the FAQ. Output: a checked reply.

One mark each for a correct input, an ordered set of steps, and an output that fit a workflow shape.

Run one enquiry card and say whether the reply passed the verify check, using the FAQ.

The 'trip cost' card was drafted, then checked against the FAQ - the amount matched, so the reply passed.

Accept a logged result checked against the trusted FAQ; it must reference verify-before-trust, not a guess.

- Targets: the CU-analyse and SD-build steps - analysing how workflow design shapes outcomes, and designing and building a working input-steps-output reply workflow.
- Check: circulating design-and-build prompts during the pair work plus an exit ticket naming one design choice and its effect.
- Success criteria: the pair sequences a clear multi-step workflow, assigns a tool to each step, and links one design choice to a changed reply.
- Evidence: collected build sheets - the workflow map, the tool choices, and the three logged test runs with their verify-checks.
- Feedback / adjustment: if a pair cannot link design to outcome, re-run one enquiry with the verify step moved and have them log the difference before Week 6.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Analysing workflow design Links one design choice - a step order or a tool - to a changed reply, with support.

Notices a change but cannot say why it happened.

Cannot connect the design to the reply.

Building the workflow Maps and builds a working input-steps-output workflow with a tool assigned to each step.

Builds part of the workflow, or the sequence is unclear.

No usable workflow, or the steps are mixed up.

Testing and recording Runs three enquiry cards and logs each reply and verify-check, with guidance.

Runs a test but does not check the reply against the FAQ.

Does not produce a usable test record.

Representation: display the input-steps-output map, a worked example workflow, and the trusted FAQ; pre-teach 'workflow', 'step', 'tool' and 'verify' with simple icons.

Action & expression: pairs may build with physical cards, on a printed workflow strip, or on the board, and may explain a design choice by speaking, drawing or writing.

People of Determination: provide a partly sequenced workflow, a reduced card set, sentence starters ('When we moved the ... step, the reply ...'), and extra time to run tests, with a peer scribe where helpful.

Week 6 - Integrated Task (Evaluate, Justify & Showcase) | Integrated Prerequisite

Mastery Task - Design, Build, Evaluate & Justify an AI Solution | Interactive lesson:

'Evaluate and Prove It'

Strand: Integration (all four prior-grade strands) | Framework ref(s): AIF·CU·G7·A AIF·SD·G7·A AIF·CE·G7·A AIF·GE·G7·A | 1 x 45 min

Objectives & success criteria

- Evaluate the pair's reply workflow against a clear performance criterion, then improve it through iteration and justify the change with evidence.
- Identify one risk in the workflow and propose a responsible-use mitigation, giving reasons.
- Complete a structured reflection across the four skills and present the solution at the readiness showcase.

Student-facing success criteria:

- I can test our workflow against a criterion, make one improvement, and prove with evidence that it helped.
- I can name one risk in our workflow and suggest a fair, safe fix, and say why it helps.
- I can reflect on all four skills and show our solution is ready for the Grade 8 unit.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Week 5 you designed and built your simulated reply workflow and logged three test runs. You also rebuilt the

Grade 7 Advanced skills for improving a workflow through iteration and explaining the impact of the changes (CE) and for proposing ways to reduce risks and improve responsible use of AI systems (GE). This week you use them to evaluate, improve and justify your solution.

Activate: Show a workflow that gets four of six test enquiries right and ask: does it meet our criterion, and how would you improve

it?

Teach: Model the evaluate-improve-justify loop - score the workflow against a performance criterion, change one thing (reorder a step, swap a tool, or add the verify gate), re-score, and state the evidence that the change helped.

Apply: In pairs, students score their Week 5 workflow on a six-enquiry test set and mark which replies fail.

Check: Ask two pairs for their score and one failed enquiry; check they can name a design change that would fix it, not just say 'try again'.

Critical & analytical thinking focus

Analyse, Evaluate and Create - sustained higher-order thinking across all four strands

(Analyse/Evaluate/Create | DOK 3-4 |

SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Evaluation turns a build into an improvement cycle: you set a performance criterion (here, the proportion of test enquiries

that get a correct, verified reply), measure the workflow against it, change one part of the design, and measure again.

Because the workflow is transparent and the model is a simulated card deck, the cause of any failure is visible - usually a

missing step, a poor order, or the wrong tool for a step - so students justify each change with evidence rather than

guesswork and explain its impact. Risk work sits alongside: a reply sent without verification can be confidently wrong, a

narrow reply deck can be biased against a whole enquiry type, and enquiries must carry no personal data, so responsible-

use measures include a verify-before-trust gate, a human-in-the-loop route, and a

no-personal-data rule. All work stays

account-free. Cognitive demand for this lesson: Evaluate | DOK 3 | SOLO: Relational.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set

the criterion / 6

min

Recap the Week 5 builds. Agree the performance criterion and hand out the six-enquiry test set and the risk-and-reflection sheet.

Score their workflow on the six enquiries and circle any that fail.

- Does your workflow meet the criterion yet - and which enquiries

let it down?

Evaluate and

improve / 12 min

Model changing one design element -
reorder a step, swap a tool, or add the
verify gate - to fix a failed enquiry, then
re-scoring. Circulate to prompt
Change one design element,
re-run the failed enquiries,
and log the new score.

- What is your evidence that your
change improved the workflow, not
just altered it?
evidence-based changes (CE step).

Find the risk / 10

min

Model spotting a biased or unsafe
output - an unverified reply, or a reply
deck that misses a whole enquiry type -
and proposing a mitigation with a
reason. Prompt pairs to inspect their
own workflow (GE step).
Identify one risk in their
workflow and write a
responsible-use fix with a
reason.

- Where could your workflow send
something wrong or unfair, and
how would you guard against it?

Reflect and gate

/ 12 min

Guide the four-skill reflection and run
the readiness gate check against the
criteria. Set up the showcase.

Complete the structured
reflection and self-check
their solution against the
readiness gate.

- Across the four skills, which was
hardest, and how did you work
through it?

Showcase and

close / 5 min

Run a short pair-to-pair showcase;

collect all evidence sheets as the readiness record.

Present the solution: input-steps-output, one improvement with evidence, and one risk and its fix.

- In one line, why is your solution ready for the Grade 8 unit?

Anticipated misconceptions

Misconception How to address it

A change that alters the reply means the workflow improved.

Compare scores before and after against the criterion; only a change that raises the score on the same test set counts as an improvement. Keep the evidence.

If the workflow is usually right, we don't need to worry about fairness or safety.

Show one biased case - a whole enquiry type the reply deck never answers well - and one unsafe case - a confident wrong reply sent without verifying; agree these need mitigations even at high accuracy.

Reflection just means saying it went well. Model a real reflection naming the hardest skill, how they worked

through it, one improvement and one risk; require evidence, not praise.

Hands-on activity guide

Overview: Across the week's two periods, pairs evaluate their Week 5 workflow against a performance criterion, make one evidence-based improvement through iteration, identify a risk and propose a responsible-use fix, then reflect across the four skills and showcase the solution at the readiness gate.

Grouping / timing: Pairs for the evaluate-improve-risk-reflect work; pair-to-pair for the showcase; whole class for the close.

Tool: The same 'simulated AI' reply workflow from Week 5 - a curated reply deck, fictional enquiry cards and a trusted FAQ

run by hand or on a teacher-operated board - plus a fixed six-enquiry test set. Account-free, no logins, and only fictional, non-personal enquiries.

Facilitation steps:

1. Score your workflow on the six-enquiry test set and circle any replies that fail.
2. Change one design element - reorder a step, swap a tool, or add the verify gate - then re-run and log the new score (CE step).
3. Write the evidence that your change improved the workflow, checked against the FAQ.

4. Identify one risk - a biased or unverified reply, or personal data - and propose a responsible-use fix with a reason (GE step).
 5. Complete the four-skill reflection: the hardest skill and how you worked through it, one improvement, one risk to guard against.
 6. Showcase your solution to another pair and self-check it against the readiness gate.
- Unplugged fallback: The evaluation, risk work and reflection all run on paper with the card sets, so no device is needed. If the board is unavailable, score and re-run the enquiry cards by hand and present from the evidence sheets.

Lab tool integration (practical, hands-on)

Tool: Pre-generated packs + improvement-move cards (integrated: improve, re-measure, justify)

Teacher setup: Have each pair's Week-5 build, an evaluation pack, improvement-move cards and a risk-scenario card.

What students do:

1. Re-run the Week-5 solution and confirm the first score.
2. Apply one improvement-move card that targets the weakest case.
3. Re-measure and compare with the first score.
4. Draw a risk-scenario card and check your solution against it; confirm no personal data.
5. Justify your improvement with evidence and complete the readiness reflection.

Students produce: A before-and-after score, a justified improvement, a risk check, and a readiness reflection.

Data-safety: Risk check confirms no personal data; outputs curated.

Unplugged fallback: Improve and re-score the paper version by hand.

Improve, re-measure, check the risk, justify with evidence.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Score your workflow, make one change, and show with evidence that it improved.

We scored 4 out of 6; the 'meeting time' reply went out unverified and wrong. We added a verify-against-FAQ step before sending and re-scored 6 out of 6, so the change improved accuracy against the test set.

One mark for a before-and-after score plus a change justified by the test evidence, not opinion.

Identify one risk in your workflow and propose a responsible-use fix with a reason.

Our reply deck had no cards for cost questions, so those enquiries always got a vague reply - a biased output. We added balanced reply cards and a 'not sure - pass to a teacher' route, because a confident reply

can still be wrong.

Accept a real risk (bias, safety or personal data) with a mitigation and a reason tied to fairness, safety or verify-before-trust.

Reflect across the four skills: name the hardest and how you worked through it.

Evaluating was hardest because our first change did not raise the score; we compared scores and found the real weak step - drafting before understanding - and reordered it. One improvement: add a verify gate. One risk: sending replies without checking.

Look for the hardest skill named with a concrete strategy, plus one improvement and one risk.

- Targets: the CE-evaluate and GE-risk steps plus the four-skill reflection and readiness gate - evaluating against a performance criterion and improving through justified iteration, identifying a risk with a reasoned mitigation, and confirming readiness for the Grade 8 unit.
- Check: circulating evidence-and-reason prompts during the pair work, the four-skill reflection, and a pair-to-pair showcase scored against the readiness gate.
- Success criteria: the pair raises its performance score with a justified change, names one risk with a reasoned responsible-use fix, and completes an honest four-skill reflection.
- Evidence: collected test-score logs (before and after), the risk-and-mitigation note, the structured reflection, and the showcase readiness record.
- Feedback / adjustment: if a pair changes the design without improving the score, re-model comparing scores against the test set and have them redo one evidence-based change before the readiness sign-off.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Evaluating and justifying Scores against the criterion, improves the workflow through one change, and cites evidence it helped, with support.

Makes a change but the evidence is missing or it does not raise the score.

No evaluation, or changes are guesses

with no evidence.

Risk and mitigation Identifies one real risk and proposes a responsible-use fix with a reason.

Names a risk but the fix or the reason is unclear.

Cannot identify a risk in the solution.

Reflection and readiness Completes the four-skill reflection and shows the solution meets the readiness gate.

Reflects partly, or the readiness check is incomplete.

No usable reflection or readiness evidence.

Representation: display the performance criterion, the six-enquiry test set, a risk-matrix prompt (bias, safety and personal data), and reflection sentence stems.

Action & expression: pairs may present the showcase by speaking, by walking through their sheets, or with a simple poster; reflections may be written or spoken to a scribe.

People of Determination: provide a partly completed test log, a reduced test set, a risk card bank to choose from, reflection stems ('The hardest skill was ... We worked through it by ...'), and extra time for the showcase.

Readiness Gate

Level What it looks like Action

Mastery Meets the prior-grade Advanced bar independently within the integrated task.

Proceed to framework content (Week 7).

Partial Meets the bar with support or on some steps of the task. Targeted Prerequisite Primer on that strand in the relevant Week 7+ lesson.

Not yet Does not yet meet the bar on that strand. Monitored small-group Primer support into Term 1; re-checked in context.

Standards & references

This pack is designed against the following international standards for curriculum and assessment quality.

Reference Used for

ADEK (2026). K-12 AI Fluency Framework (Draft V4). Source of all learning outcomes, strands, tiers and codes.

AERA, APA & NCME (2014). Standards for Educational and Psychological Testing.

Validity, reliability, and fairness of assessment.

Assessment Reform Group (2002). Assessment for Learning: 10

Principles.

Formative assessment that supports learning.

OECD (2005). Formative Assessment. Evidence base for embedding formative assessment.

Anderson & Krathwohl (2001). Revised Bloom's Taxonomy. Cognitive-demand classification of objectives.

Webb, N. (1997/2002). Depth of Knowledge. Depth of cognitive processing and grading of demand.

Biggs & Collis (1982). SOLO Taxonomy. Structure of observed learning; anchors success criteria.

CAST (2018). Universal Design for Learning Guidelines v2.2. Access, representation, and multiple means of engagement.

Reviewer-response addenda (external technical review, Term 1)

Key terms - consistent definitions (use these across all Grade 8 packs)

Model - the trained part of an AI system that has learned patterns from examples and turns an input into a

prediction/output. Changing its training examples and training again produces a NEW, retrained (updated)

model - not the same model.

Machine learning (ML) - building a model by learning patterns from a dataset instead of writing fixed rules by hand.

Dataset - the whole collection of examples, split into training data and test data.

Training data / test data - training data are the examples the model learns from; test data are separate, new

examples held back to measure performance. Never test on training examples.

Training / inference / evaluation - training teaches the model from data; inference is one forward run that

produces an output from an input; evaluation measures output quality against criteria using the test data.

Process - a single step, or set of steps, that acts on an input (e.g. the model step, a translate step, a check step).

Workflow / pipeline - the planned, ordered sequence of steps (processes) that turns an input into an output. In

these materials 'pipeline' and 'workflow' mean the same thing.

Output - the direct result a single run produces (a label, number, sentence or image).

Outcome - the real-world effect of using the output (e.g. a parent receives a correct heat-safety alert). An output

can be good while the outcome is still poor if the workflow around it is weak.

Forward vs loop - one prediction runs forward (input -> model -> output); development runs in loops of testing,

evaluation and retraining, so AI building is not linear.

Prerequisite knowledge (maths & computing) - required before this unit

Students are assumed to be able to: (1) sequence steps and follow/trace a workflow (computing); (2) calculate

simple percentages and read them as accuracy, e.g. correct / total x 100 (maths); (3) compare two results and

describe the difference (maths); (4) run a controlled/fair test by changing ONE factor while keeping all other test

conditions the same (science/computing). Use the Week 1 Prerequisite Primer to activate or reteach any gap;

the Bridging course rebuilds these before the main unit.

Label each activity by what students actually improve (Design & Creation)

MODEL TRAINING = changing the training examples and training again -> a new/retrained model (e.g. Teachable

Machine train/retrain labs). WORKFLOW DESIGN = adding or reordering steps; the model is unchanged (e.g.

Scratch multi-step solutions). PROMPT / INPUT REFINEMENT = changing the input given to the model; the

model is unchanged (e.g. change-one-factor labs). INTERFACE DEVELOPMENT = changing how a user interacts

with the solution. State the label in each lab header so students know whether they are changing the model or

the surrounding workflow.

Fitting the Teachable Machine + Scratch labs into 45 minutes (Accessibility)

Combined train-build-test-reflect labs are too long for one period. Use ONE of these: (a) provide a PRE-TRAINED

Teachable Machine model and a STARTER Scratch file so students go straight to testing and improving; (b) split

setup and evaluation across two prepared stations (Station A trains/sets up, Station B evaluates); or (c) divide

the activity across two periods. The ESSENTIAL task is: run new test examples -> measure accuracy -> make

one improvement -> re-measure. Extra classes, extra test cases and interface polish are optional/extension.

Reduce over-scoped test sets (e.g. 6 test cases, 2 per class, instead of 10) when time is short.

Educational tools, not production platforms (Industry Relevance)

Teachable Machine and Scratch simulate key stages of professional AI development (classify, train, test,

prototype) but are not full production platforms. Real industry systems use more advanced tools for large-scale

training, deployment, monitoring and security. Present them as an honest introduction to the concepts.

Technical sources of bias (name the mechanisms in L5)

Go beyond 'one-sided data'. Name at least: (1) training-data/sampling bias; (2) selection bias; (3)

label/measurement bias; (4) design/threshold bias. Each can produce unfair or unreliable outputs. Full

procedure in the Fairness and Bias Technical Note.

Fairness evaluation step - add to the Signature Solution rubric (L7 planning sheet)

Require every student to test whether their workflow performs consistently for at least TWO input types or user groups. They record each group's accuracy or error count under the SAME test conditions, note any performance disparity, and propose a mitigation if a disparity is found. 'Cannot determine' is a valid result when the evidence is too limited. Full method and rubric wording in the Fairness and Bias Technical Note.

Explainability note (use in L2/L3)

Structured, traceable workflows are more EXPLAINABLE than black-box models: each output can be traced to the step that produced it, which makes the system auditable and trustworthy. This traceability is a design advantage. See the Explainability Note.

Model record card (Future Readiness)

From L2 onward, have students complete the one-page Model Record Card (data source, purpose, version/changes, known limitations, test results, and checks needed after the system is put into use). This builds the habit of recording provenance and monitoring performance after deployment.

Cross-curricular labels for lesson headers (Interdisciplinary Connections)

Add these visible labels to lesson headers so subject coordinators can plan across subjects:

Science link - heat safety (heat-safety notices, sports-day weather alerts); Language link - Arabic-English translation and reading-level (two-audience reply workflow); Maths link - accuracy percentages and before-and-after comparison;

Science link - fair test (change one factor). Spam filtering and litter-sorting classifiers connect to computing/science classification.

Cross-curricular links - applied to Bridging activities (Interdisciplinary Connections)

Science link - fair test: Week 1 Lab (Run the Fair Test) - change ONE thing and keep everything else the same, so you know what caused the difference in outcome. Language link - Arabic-English translation and reading-level:

Week 2 Lab (One Alert, Two Audiences) - one branch stays in plain English for students, one branch is translated into Arabic for parents, both checked for reading level. Maths link - accuracy/score comparison and before-and-after evidence: Week 3 Lab (Score, Iterate, Justify) and Week 6 (Evaluate and Prove It) - score the workflow, change one thing, re-score, and compare before-and-after to prove an improvement.

Printable Card Decks

The card decks referenced by the labs and unplugged fallbacks. Print, cut and laminate. No devices, no personal data.

How to use these decks

Each deck is the default unplugged fallback for its activity. Print one set per group. Cards use objects, neutral text and scenarios only - never names, faces or voices.

Deck 1 - Workflow-step cards

Students lay these in order to build or trace a workflow (input -> process -> output, plus checks). Also used for the 'human classifier' unplugged model.

- INPUT - the item that goes in (text, image or reading)
- TRANSLATE - convert the input to another language
- CLASSIFY / MODEL - label the input using a learned rule
- CHECK STEP - verify the output against the task or a source
- SUMMARISE - shorten the output for the reader
- OUTPUT - the direct result of the run
- RETRAIN - change the training examples and train again (NEW model)
- FALLBACK - what to do when the check step fails

Deck 2 - Output-verdict cards

Students hold up a verdict card when judging an output against a criterion.

- MEETS - fully meets the criterion
- PARTLY - meets it in part; note the evidence
- FAILS - does not meet the criterion
- CANNOT DETERMINE - not enough evidence to decide
- VERIFY FIRST - looks right but must be checked against a source
- ACCURATE / NOT ACCURATE
- RELIABLE / NOT RELIABLE
- FAIR / BIASED

Deck 3 - Bias / fairness scenario cards

Each card is a short scenario. Students name the bias mechanism (sampling, selection, label/measurement, design/threshold), say who is affected, and propose a check or fix.

- A litter classifier trained only on daylight photos fails at dusk. (sampling/selection)
- A reply deck has no card for one common enquiry type, so those people get no answer. (design)
 - Examples were labelled quickly and some 'recycling' items were tagged 'litter'. (label/measurement)
- A translator works well for one dialect but poorly for another. (sampling)
- The 'yes' threshold is set so high that borderline correct cases are rejected. (design/threshold)
 - A heat-safety alert is only tested for the school, not for parents at home. (selection)
 - A spam filter was trained mostly on English mail and misses Arabic spam. (sampling)
 - Test photos were reused from the training set, so the score looks too good. (evaluation error)
- Borderline cards (two mechanisms at once - argue how they interact):
- A heat-alert model is trained only on daytime data AND its 'alert' threshold is set very

high - both push the same way. (sampling + design/threshold)

- A translator is trained mostly on one dialect AND its examples were labelled in a rush - few examples for one group plus noisy labels. (sampling + label/measurement)
- A litter classifier reuses training photos in its test set AND those photos are all daylight - an inflated score for a case that also under-samples dusk. (evaluation error + sampling)

Fairness routine for Deck 3:

Split the scenario's test cases into two groups, count correct in each under the same conditions, compare the accuracy percentages, record any disparity (or 'cannot determine'), and name one mitigation.

Deck 4 - Improvement-move cards (support deck)

Used in the build-test-improve loop. Each names WHAT is being improved so students label the change.

- INPUT REFINEMENT - clearer prompt or cleaner input (model unchanged)
- WORKFLOW DESIGN - add or reorder a step (model unchanged)
- MODEL TRAINING - add/clean examples and retrain (NEW model)
- INTERFACE - change how the user interacts (model unchanged)
- ADD A CHECK - insert a verify step before trusting the output

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Explainability - Note

Black-box vs explanation: why a model made a prediction, tracing behaviour, and the limits of explanation.

Black box vs explanation

A black-box model shows only its input and its output - you cannot see why it decided what it did. An explanation answers the question 'WHY did the system produce this output?'

Explainable AI (XAI) is the field that builds systems whose decisions can be explained.

Why traceable workflows are more explainable

A structured workflow is built from separate steps (input -> process/model -> check -> output). Because each step is separate, you can TRACE any output back to the exact step that produced it and say why. That makes the system auditable (someone can inspect it) and more trustworthy. A single black-box model has no such steps to inspect, so it is harder to explain and to trust.

Tracing behaviour - how

1. Take one output you want to understand.
2. Follow it backwards through the workflow: which step produced it, what went into that step, and what rule or model it used.
3. Name the step and the choice that shaped the result ('the classify step labelled it litter because the confidence was above the threshold').

The limits of explanation

- Even a full trace tells you WHICH step decided, not always the deep reason a learned model weighted one pattern over another.
- A model can give a correct output for the wrong reason, so an explanation must be checked against a trusted source, not just believed.
- More detail is not always clearer - a good explanation is one a human can actually

understand and act on.

- Some systems are only partly explainable; being honest about what you cannot explain is itself part of responsible design.

Why it matters

Explainability is a design advantage: a system whose outputs can be traced to steps can be audited, corrected and trusted. This is one reason the structured, workflow-based approach in this unit is safer for beginners than an opaque black box.

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Fairness and Bias - Technical Note

A concrete fairness-evaluation procedure and the technical sources of bias (Emerging tier).

Teacher-mediated; no personal data.

Why this note

Grade 8 students already treat biased outputs as a risk. This note gives them the precise technical vocabulary and a concrete, repeatable procedure so 'is it fair?' becomes something they can measure, not just feel.

Part A - A concrete fairness-evaluation procedure

Use this whenever you evaluate a model or workflow. The teacher operates any live tool; students direct, observe and record.

1. Choose a characteristic that splits your test set into groups - for example short cards vs long cards, morning photos vs evening photos, or English inputs vs Arabic inputs. Pick something relevant to who or what the system serves.
2. Split the SEPARATE test set (new examples only) into at least two groups by that characteristic. Keep the test conditions identical for every group - same task, same instructions, same run.
3. Run each group through the system and count how many outputs are CORRECT in each group.
4. Work out each group's accuracy as a percentage: $\text{correct} / \text{total} \times 100$. Record correct and wrong counts too, not just the percentage.
5. Compare the groups. If one group's accuracy is clearly lower, you have found a possible bias (a performance disparity). Record the size of the gap.
6. If the groups are too small or the scores are too close to be sure, record 'cannot determine' and collect more examples before deciding.
7. Write one mitigation for any disparity found (e.g. add more examples of the weaker group, relabel unclear cases, or adjust the design/threshold), then re-test.

Worked example

A litter-sorting classifier is tested on 20 new photos: 10 taken in daylight and 10 at dusk. Daylight: 9/10 correct = 90%. Dusk: 5/10 correct = 50%. The 40-percentage-point gap is a performance disparity - the model is biased toward daylight conditions. Mitigation: add balanced dusk examples and retrain (a new, retrained model), then re-test both groups.

Part B - Technical sources of bias (name the mechanism)

Bias is not only 'one-sided data'. Students should be able to name where it comes from:

- Training-data / sampling bias - the training examples do not represent everyone the system will serve, so a whole group is under-served.

- Selection bias - the way the data was chosen skews it (e.g. only daytime photos, only one dialect).
- Label / measurement bias - the human labels are wrong or unfair, or the thing is measured in a way that disadvantages some cases.
- Design / threshold bias - a design decision or a cut-off value (e.g. the confidence threshold for 'yes') favours some cases over others.

Any of these can produce unfair or unreliable outputs even when the code is correct. Fixing fairness usually means fixing the data, the labels or the design choice - not just the code.

Part C - Fairness evaluation step to add to the Signature Solution rubric
(L7 planning sheet)

Add this required step to the L7 planning sheet and the assessment rubric:

Fairness evaluation (required):

- Test whether the workflow performs consistently for at least TWO input types or user groups.
- Record each group's accuracy or error count under the SAME test conditions.
- Document any performance disparity (state the gap; 'cannot determine' is acceptable when evidence is too limited).
- Propose one mitigation if a disparity is found, and re-test if time allows.

Cross-curricular links: Maths - percentages and comparison; Science - fair test (change one factor, hold the rest constant).

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder
Main Course, Term 1 - Evaluating & Improving AI Models
Emerging tier | 10 weeks (Core + Lab)
Staying safe with AI in this course

- We never enter personal data - no names, faces or voices. We use objects, words, drawings or neutral text only.
- De-identification means removing anything that could identify a person before data is ever used; we go further and use no personal data at all.
- The teacher runs any live AI tool. You direct it, but you do not log in or make accounts.
- We always check the output. AI can be confidently wrong, so we verify every result against the task and a trusted source.
- If something looks wrong, unexpected or unfair, tell your teacher - that is exactly what a responsible AI user does.

STUDENT LESSON CONTENT

Week 1 - Core | Unit Launch - Evaluating & Improving AI Models

I can...

- I can ask at least two good questions about how AI turns an input into an output and how we judge that output.
- I can sort examples that relate to evaluating and improving AI models, and say why.

- I can help write our unit success criteria and our safe, responsible-use rules for AI tools.

Words to know

- output - what the AI system gives back after it processes your input
- criteria - the clear measures we agree on to judge whether an output is good, such as accuracy or usefulness
- evaluate - to judge something fairly against the criteria, not just by how it looks
- improve - to change the input or the process on purpose so more outputs meet the criteria
- responsible use - using AI safely and fairly, with no personal data and always checking the result

Today we launch a new unit. You already know an AI turns an input into an output - now we ask the harder

question: is that output any good, and how could we make the model better?

From 'it works' to 'how well does it work'

An AI system still follows the shape you know: input, process, output. This unit adds a new job on top. Instead of

just asking 'did it give me something back', you ask 'is what it gave me good enough for the task'. To answer that

fairly you need criteria - clear measures like accuracy, usefulness and fairness that you agree on before you

judge. Judging against criteria is called evaluating. It turns a gut feeling into a decision you can explain and

defend to someone else.

Judge it, then improve it

Once you can evaluate an output, you can improve the system behind it - but be precise about WHICH part you

improve. You can improve the INPUT (a clearer prompt or cleaner example data - the trained model itself does

not change); you can improve the WORKFLOW (add or reorder steps, such as a check step - the model still does

not change); or you can improve the MODEL itself by changing its training examples and training again, which

produces a new, RETRAINED (updated) model. So a clearer input improves the output, a better-designed

workflow improves the outcome, and only retraining changes the model itself. Improving does not mean nagging

the AI until it gives a nicer answer. It means changing something on purpose - a clearer input, a better example,

a tuned step in the process - and then re-testing against the same criteria to see if the output really got better.

Keep the charter in mind: no real names, faces or voices go into any tool, an adult runs it for us, and a confident-

looking output still has to be checked before you trust it.

The workflow still runs input to output - now we add a step that judges the output against clear criteria.

Why it matters

Why it matters - across the UAE, teams evaluate and improve AI every day: tuning an Arabic-English translator

for government service desks, checking a smart-parking predictor in Abu Dhabi, or testing a recycling sorter

until it meets a clear accuracy target; learning to judge and improve outputs helps you understand and question

the tools around you.

Let's try it - hands-on

1. Watch the class stimulus and note one thing you noticed and one thing you wonder.
2. Turn your wondering into two clear questions about judging or improving AI outputs.
3. Sort each example card as an output, a way to judge, or a way to improve, with your partner.
4. Write one reason for each card you sorted.
5. Add one word to the vocabulary wall and suggest one safe-use rule.

Think about it

Think about it - if two outputs both look neat, how would you decide fairly which one is actually better for the task?

My reflection

My reflection: One question I still have about evaluating or improving AI models is ...

Week 1 - Lab | Unit Launch - Evaluating & Improving AI Models (Lab)

I can...

- I can compare two AI outputs for the same task and write down how they differ.
- I can judge each output against a simple criterion like accuracy or usefulness.
- I can use what I observed to say what made one output better.

Words to know

- compare - look closely at two outputs to find how they are alike and different
- criterion - one clear measure you judge an output against, such as accuracy or usefulness
- difference - something that is not the same between the two outputs
- evidence - what you actually observed, used to back up your judgement
- conclusion - the idea you decide on after weighing the evidence against the criterion

In the Core period you learned that AI outputs can be judged against clear criteria. Today you become an

investigator - comparing real outputs and using evidence to say which one is better.

Same task, different outputs

Two AI outputs can be made for the same task and still come out different. Maybe one input was clearer, or the

process handled it differently. On its own, 'I like this one better' is not enough. So you pick a criterion - say

accuracy or usefulness - and judge each output against it. When you compare them side by side and trace

where the difference began, you can say which output wins and prove it. That is detective work: you are finding

evidence for a judgement, not guessing.

Log it, judge it, then trust it

A good investigator writes things down. In your log you note each difference, the stage it came from - input,

process or output - and how it affects the criterion. Then, before you trust a conclusion, you check it against the

task: would it still be true for a new example? Remember the charter - the outputs are made by the teacher, with

no personal data, and a confident-looking output still has to be checked against a trusted source before you rely

on it.

Same task, small change, different output - your job is to judge which better meets the criterion.

Why it matters

Why it matters - engineers in the UAE constantly compare AI outputs against criteria before they trust them,

whether tuning a smart-traffic system in Abu Dhabi or checking an Arabic-English translation for accuracy;

learning to judge and log differences is the first step of that job.

Let's try it - hands-on

1. Read the first output pair and the named criterion, and predict which output wins.
2. Judge each output against the criterion.
3. Log at least two differences and mark the stage each came from.
4. Rotate to a new pack with a new criterion and log again.
5. Write one conclusion about what made an output better against its criterion.

Today's tool - hands-on

Tool: Card decks (evaluation-criteria cards)

1. Deal the deck and read each criterion and its 'how to check' line aloud.
2. Rank the six criteria for a chosen everyday AI tool (for example, a translation app) from most to least important.
3. Agree, as a pair, what 'good' would look like for your top two criteria.
4. Write one measurable check for each of your top two (for example, 'count how many of 10 answers are correct').
5. Share your ranking with another pair and defend the order.

You will make: A ranked criteria layout and two measurable checks the class will reuse all term.

Stay safe: Fully unplugged; no devices or data.

Evaluation-criteria cards ranked, each with a way to measure it.

Think about it

Think about it - if you changed only the input and kept everything else the same, could you make the weaker

output meet the criterion? How would you test it fairly?

My reflection

My reflection: One difference I logged today, the criterion it changed, and where it came from, was ...

Week 2 - Core | AI Conceptual Understanding - Understanding the System

I can...

- I can name the stages an AI system uses to turn an input into an output.
- I can trace a worked example and explain to a partner how the input became the output.
- I can use words like input, model and output correctly, and say how they help me evaluate a solution.

Words to know

- input - the data that goes into the system, such as text, an image or a sensor reading
- model - the trained process in the middle that turns an input into an output using patterns learned from data
- output - what comes out: a label, a prediction or a new piece of content
- pipeline - the stages joined in order, from input through the model to output
- evaluate - to judge how good an output is by checking it against the task and a trusted source

Every AI tool you use turns an input into an output using a model in the middle. Today you will open up that

shape so you can explain - and later judge - how any tool works.

The pipeline: input, model, output

Look inside almost any AI tool and you find the same pipeline. First an input goes in - text, an image or a sensor

reading. Then a model changes it - this is the part you cannot see, and it learned its behaviour from many past

examples. Last, an output comes out - a label, a prediction or something new. During one prediction the arrows

only point one way, so the order of the stages matters for that single run. This does NOT mean building AI is

linear: developing a model is a loop. Teams test the outputs, evaluate them against criteria, and feed what they

learn back to improve the data, the workflow or the model - then retrain and run again. One prediction runs

forward; development runs in repeated testing, feedback and retraining cycles. Once you can name the three

stages, you can explain how a tool works and later decide whether it works well.

The model is not magic

The model stage can feel like magic, but it is not. Inside, it has learned patterns from many past examples, and it

uses those patterns to turn your input into a likely output. It does not think or understand - it predicts. That is

why an unusual or unclear input can lead to a strange output, and why the same model behaves differently as

its input changes. Knowing this is the first step to evaluating a system: you check the output instead of trusting

it, and you look for the stage that shaped the result.

How a model learns: training, inference and evaluation

Machine learning (ML) means building a model by learning patterns from a dataset instead of writing fixed rules

by hand. The dataset is split into TRAINING DATA (the examples the model learns from) and TEST DATA

(separate, new examples kept back to check it - never test on a training example). Three different actions

matter: TRAINING is teaching the model from the training data; INFERENCE is one forward run that turns a new

input into an output; EVALUATION is measuring the outputs against your criteria using the test data. If you

change the training examples and train again, the result is a new, retrained model - not the same model. You will

train your first model in today's Lab.

One pipeline, three stages: an input flows through a trained model to an output.

Why it matters

AI services across the UAE - from Arabic-English translation apps to smart-parking sensors and traffic-flow tools

- all follow the input-model-output pipeline. Knowing the shape lets you understand, question and improve the tools around you.

Let's try it - hands-on

1. Look at the worked example your teacher shows on the board.
2. Write down the input - the exact data that went in.
3. Describe what the model does to it and write it next to Process.
4. Write down the output - the exact result that came out.
5. Explain to your partner how the input became the output, in order.

Think about it

If you wanted a different output, which stage would you change - the input, the model, or nothing at all? Why?

My reflection

My reflection: the stage I understand best is ... because ...

Week 2 - Lab | AI Conceptual Understanding - Understanding the System (Lab)

I can...

- I can build a pipeline with an input, a model and an output, and trace one item through it.
- I can change a factor, compare the two outputs, and explain why the output changed.
- I can say which factor most affected the result and how I would check the output against a trusted source.

Words to know

- trace - to follow one item of data through input, model and output
- factor - something you can change that affects the result, such as the input or the example data
- compare - to run different factors through the same pipeline and see how the output changes
- verify - to check an output against a trusted answer before trusting it
- black box - a model whose inside we cannot see, only its inputs and outputs

You already know the pipeline. Today you become the tester - build a small system, then change one factor at a

time and work out why the output moves.

Why traceable workflows are easier to explain (explainability)

A structured workflow is more EXPLAINABLE than a single black-box model. Because each step is separate, you

can trace any output back to the exact step that produced it and say WHY that step gave that result. A black box

only shows its input and output, so you cannot see why it decided as it did. Being able to trace outputs to steps

is what makes a system auditable and trustworthy - a real design advantage. Real AI research calls this

explainable AI (XAI).

Trace one item

Tracing means picking one item and following it all the way through the pipeline. You watch it go in as the input,

see what the model turns it into, and read what comes out as the output. Writing down each stage for the same

item keeps you honest - it stops you mixing up two different items. A clear trace is how testers prove they know

what a system really does, not just what they hope it does. It is also the record you compare against when you start changing factors.

Change one factor, compare the output

Here is the tester's method: keep everything the same except one factor, then run it again and compare. Change

only the input, or swap the example data the model learned from, and watch the output. If the output moves,

that factor affects performance. Change one thing at a time so you know what caused the change. Some factors

move the output a lot, others barely at all - that is exactly what you are trying to find out. Before you trust any

output, check it against something you know is correct.

Why it matters

Testers who change factors on purpose find problems before real users do. This is how teams in the UAE check

an AI service performs fairly - across different inputs and data - before it is switched on for the public.

Let's try it - hands-on

1. Set up your three stages: input, model and output.
2. Send one item through and write what happens at each stage.
3. Change one factor - the input, or swap the example set - and run it again.
4. Compare the two outputs - what changed, and why?
5. Check one output against a trusted answer before you trust it.

Today's tool - hands-on

Tool: Google Teachable Machine (train, test, measure)

1. Name the three classes and add about 15 example images each (objects or hands only).
2. Ask your teacher to press Train (the teacher operates the tool; you direct and observe). Say clearly: the examples are the TRAINING DATA, training builds the MODEL, and if the examples are changed and trained again the result is a NEW, retrained model - not the same one.
3. Run the held-back test examples one by one and record correct / wrong for each.
4. Count the score (for example 7 out of 10) - this is your first accuracy measure.
5. Discuss where it failed and why, and predict one change that would raise the score.

You will make: A trained model and a first accuracy measure with a note on where it failed.
Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Train, then test on new examples, then measure the score.

Think about it

You changed only one factor - so which factor moved the output the most, and how do you know?

My reflection

My reflection: changing a factor taught me that ...

Week 3 - Core | AI Solution Design & Development - Designing & Building

I can...

- I can define a specific task and design a simple AI-powered workflow to solve it.
- I can build my workflow step by step and explain why each step is there and what it produces.
- I can use words like input, process, output, prompt and workflow correctly to describe my solution.

Words to know

- workflow — an ordered chain of steps that turns an input into a useful output
- input — the information you feed into the first step
- process step — the step that changes the input, here a simulated AI step
- output — what a step hands on, and what the whole workflow produces
- check — the step where you verify the output against the task and a trusted source before you trust it

Today you stop treating AI as a magic box and start designing a real solution - a clear chain of steps that solves

one specific task and that you can actually check.

A solution is a chain, not a single answer

An AI-powered solution is not one clever reply - it is a workflow. You start with a task, choose an input, run a

process step that transforms it, get an output, and then check that output. Because the steps are laid out in

order, you can run an example through and see exactly what each box does. If something goes wrong, you can

point to the box that caused it. That is what makes a solution something you can trust and improve, instead of a guess you just hope is right.

Design safely, then build

Good design happens before building. You decide the one specific task, then the input - and you only ever use

non-personal sample data, never real names, faces or voices. The AI step in class is simulated or teacher-run

and needs no account, so you can focus on the shape of your solution. The most important box is the check: a

confident-looking output can still be wrong for your task, so you always compare it against the task and a

trusted source before you rely on it.

A five-part AI solution workflow, with the check step highlighted.

Why it matters

Across the UAE, teams building tools like smart-city waste sorting or Arabic-English translation services

succeed because they design tested, traceable workflows - the same skill you are practising today, at

classroom scale.

Let's try it - hands-on

1. Write your one specific task in a single clear sentence.
2. Draw five boxes: task, input, process step, output, check.
3. Choose an input using only non-personal sample data.
4. Run one sample input through the simulated AI step and write the output it returns.
5. Check the output against the task and a trusted source, then say which step you would change if it did not fit.

Think about it

If your workflow gave a confident output that was wrong, which single box would you look at first, and why?

My reflection

My reflection: the step in my workflow I am most confident about is ... because ...

Week 3 - Lab | AI Solution Design & Development - Designing & Building (Lab)

I can...

- I can build my whole workflow and run more than one sample input through it.
- I can spot which step causes a wrong output and change that step to improve the result.
- I can describe my testing and my fix using the correct vocabulary.

Words to know

- test — running several inputs through your workflow to see where it works and where it fails
- sample input — a piece of non-personal data you feed in to try the workflow
- iteration — changing one step and re-testing to improve the solution
- failing step — the single box responsible for a wrong output
- reliable — a solution you can trust because it passes a range of inputs, not just one

In the Core period you designed a workflow. Today you build it fully and stress-test it - because a solution you can trust is one you have tried to break and then improved.

One pass is not proof

A workflow that gets one input right might still be unreliable. Real testing means running several sample inputs, including a tricky or ambiguous one, and watching for where the output fails the task. Each failure is useful:

because your workflow is an ordered chain, a wrong output can be traced back to the exact box that caused it.

So instead of hoping your solution works, you find out where it breaks - and that is the first step to making it

something people can actually depend on.

Iterate: change one step, then re-test

Improving a solution does not mean starting again. You find the box causing the most failures - maybe the input

you chose, the process step, or the check - and you change just that one step. Then you run the same sample

inputs again to see whether the output really improved. Keeping everything else the same is what lets you tell if

your fix actually helped. The AI step stays simulated and account-free, and every input is non-personal, so all

your effort goes into building, testing and improving the workflow itself.

Why it matters

Engineers on UAE projects - from Arabic speech tools to smart-logistics routing - ship reliable products by

testing against many cases and fixing the exact step that fails, which is precisely the build-test-improve loop

you run today.

Let's try it - hands-on

1. Rebuild your full five-part workflow so every box is connected.
2. Choose three or more non-personal sample inputs, including one tricky one.
3. Run each input through the simulated AI step and log every output as fit or not fit.
4. For each misfit, write the single box responsible.
5. Change that box, re-run the same inputs, and note whether the outputs improved.

Today's tool - hands-on

Tool: Scratch + Teachable Machine (develop a solution for a task)

1. Agree the task and what a good result looks like (link to your L1 criteria).
2. Use or train a model for the task on non-personal objects.
3. In Scratch, build the solution so an input runs through the model and gives a labelled output.
4. Run it on five real inputs and record the outputs.
5. Note one limitation you already notice, ready to fix next lesson.

You will make: A working AI-powered solution for a specific task and a first limitations note.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Only non-personal objects.

An input runs through a model inside a Scratch solution to give a labelled output.

Think about it

Why does re-running the exact same inputs after a change tell you more than trying brand-new inputs?

My reflection

My reflection: the step I changed today was ... and I knew it helped because ...

Week 4 - Core | Critical Evaluation & Informed Interaction - Judging & Improving Outputs

I can...

- I can name the limitations in an AI output instead of assuming it is correct.
- I can judge an output against agreed criteria and back my decision with evidence.
- I can use words like criteria, evidence, accuracy and reliability correctly when I explain a judgement.

Words to know

- limitation - a weakness in an output you can name and evidence, such as a wrong fact or an answer that only works

sometimes

- criteria - the agreed things you check an output against, like accuracy, relevance and reliability
- evidence - the exact word, fact or step you point to that proves your judgement
- accuracy - whether the output is actually correct when you check it against a trusted source
- reliability - whether an output or solution holds up across different inputs, not just the easy ones

Today you become an evaluator. Instead of trusting an AI output because it sounds sure, you will name its

limitations, judge it against agreed criteria, and prove every decision with evidence.

Judge with criteria, not gut feeling

When two outputs look different, it is tempting to pick the one you like. That is not evaluation.

First you agree

criteria - the things everyone will check for, such as accuracy, relevance and reliability. Then you judge each

output against every criterion and give a verdict. Because everyone uses the same criteria, your judgements are

fair and someone else can check them. Liking an output is an opinion; naming a limitation against a criterion,

with evidence, is evaluation - and it is the first step to improving the solution.

Confident is not the same as reliable

AI outputs often sound smart and sure. But a model works by predicting likely patterns, not by knowing the

truth, so a neat, confident answer can still be wrong. That is why you verify before you trust: check the key fact

against a source you already trust. A limitation is more than one slip - it is a weakness you can see repeat. So

test the solution on more than one input: if it works on easy cases but fails on unusual ones, reliability is a

limitation you can name and, later, improve.

A limitation is a pattern, not a one-off - test the solution across several inputs.

Why it matters

Across the UAE, people use AI to draft messages, translate and answer questions in seconds - but a wrong

figure in a school report or a solution that only works on easy cases can cause real problems.

Naming an

output's limitations before you use it keeps you in charge of the result.

Let's try it - hands-on

1. Agree the three criteria with your class: accuracy, relevance and reliability.
2. Read the task the outputs were meant to do, so you know what a good result looks like.
3. Judge each output against every criterion and write meets, partly or fails.
4. For every verdict, write one piece of evidence - the exact word, fact or step.
5. Verify one output against a trusted source, then decide whether each limitation is a one-off slip or a repeating weakness.

Think about it

Think about it: could an output meet the accuracy criterion on one input but fail the reliability criterion across

several? What would that tell you about the solution?

My reflection

My reflection: the hardest limitation to prove was ... because ...

Week 4 - Lab | Critical Evaluation & Informed Interaction - Judging & Improving Outputs (Lab)

I can...

- I can apply agreed criteria to a larger, messier set of outputs and name each limitation.
- I can rank outputs from best to worst and justify each placement with evidence.
- I can weigh accuracy, reliability and bias and suggest one way to improve a weak output.

Words to know

- rank - to put outputs in order from best to worst against your criteria
- reliability - how well an output or solution holds up across different inputs, not just easy ones
- bias - when a solution treats some cases or groups unfairly, even if some facts are correct
- evidence - the exact fact, output or step you point to that justifies a placement
- improvement - a specific change that targets a named limitation in the solution

Welcome to the evaluation clinic. Today you do not just judge one output - you compare a whole set, rank them

best to worst, and prove every placement with evidence about its limitations.

From verdicts to a ranking

Judging one output is a start; a clinic compares many. To rank them, line up your verdicts and ask which output

has the less serious limitation against the criteria. Often two are close - one is accurate but unreliable, another

fuller but slightly biased. That is when you use a deciding lens: for this task, does reliability or bias matter more?

Because every placement rests on evidence, your ranking is not just an opinion - someone else could follow

your evidence and reach the same order.

Accuracy, reliability and bias can pull apart

The best output is not always the longest or the most detailed. A short answer that is fair and holds up across

inputs can beat a long one that hides a wrong or unfair result. So you weigh three lenses:

accuracy (is it

correct?), reliability (does it hold up across inputs?) and bias (is it fair to all cases?). When an output ranks low,

do not just reject it - name the limitation and trace it back to the solution. Adding harder examples or fixing an

unfair rule can lift the whole result.

Why it matters

In UAE workplaces and schools, people often get several AI drafts and must choose one fast.

The skill of ranking

outputs by their limitations and improving the weakest is exactly what keeps the final choice accurate, reliable

and fair.

Let's try it - hands-on

1. Lay out the wider set of outputs and re-read the task they were meant to do.
2. Judge each output against accuracy, relevance and reliability, adding bias for the trickier ones.
3. Rank them from best to worst on your ranking strip.
4. Write the evidence beside each placement, using reliability or bias to decide close calls.
5. Take the weakest output, name its limitation, and propose one improvement to the solution.

Today's tool - hands-on

Tool: Pre-generated output packs (find and measure limitations)

1. Agree the evaluation criteria (reuse the L1 cards).
2. Read each output and mark it correct / unclear / incorrect, naming the limitation.
3. Score the pack: count how many meet the criteria.
4. Sort the limitations into types (wrong fact, unclear, unfair, incomplete).
5. Choose the most common limitation and say what change would reduce it.

You will make: A scored pack, a limitations-by-type tally, and one targeted improvement idea.

Stay safe: All outputs are teacher-curated; students touch no live tool.

Outputs marked against criteria, limitations sorted into types.

Think about it

Think about it: could an output be the most accurate yet still deserve a low rank because it is biased or unreliable? What evidence would prove that?

My reflection

My reflection: my ranking changed when I noticed ... , which taught me ...

Week 5 - Core | Ethics & Governance - Risks, Ethics & Responsible Use

I can...

- I can name the ethical concerns in an AI use case, such as bias, privacy or misuse.
- I can read a scenario, name an ethical concern, and suggest a responsible-use measure with a clear reason.
- I can use words like bias, privacy, transparency and governance correctly, and explain how a concern changes how

I would evaluate an AI solution.

Words to know

- ethical concern — a way an AI use could harm people, such as being unfair or invading privacy
- bias — when a result is unfair because the data it learned from was one-sided
- privacy — keeping personal data such as names, faces and voices out of a system
- transparency — being open that a decision was made by AI and how it can be questioned
- governance — the rules and roles that keep AI use responsible in a school or workplace

An AI solution can hit its accuracy target and still be unfair, invasive or misused. Today you learn to spot the

ethical concerns in an AI use case before anyone releases it to real people.

Accurate is not the same as ethical

When you evaluate an AI solution you check whether it does its job well. But a solution can score highly and still

cause harm. It might be biased against one group because its data was one-sided. It might collect faces or

names it does not need. It might hide the fact that a computer, not a person, made a decision.

Ethics adds new

criteria to your evaluation: a responsible solution is not only accurate, it is also fair, private, transparent and used as intended.

Name the concern, then design the fix

You cannot fix a problem you cannot name, so the first skill is naming the concern precisely: is this bias,

privacy, transparency, or misuse and over-reliance? Once you have named it, you can design a responsible-use

measure that actually reduces it, and give a reason. Keeping personal data out of a system protects privacy.

Testing on a wide range of people reduces bias. Telling users a decision was made by AI supports transparency.

Governance is simply the shared rules that make these habits normal for everyone.

A responsible solution must pass ethical criteria as well as accuracy.

Why it matters

Schools, hospitals and companies across the UAE are building AI into real decisions, so being the person who

can name an ethical concern and suggest a responsible fix makes you genuinely useful and trusted.

Privacy mechanisms

Keeping personal data out is the first rule. Real systems also use these mechanisms, matching UAE's data

protection law (PDPL) and ADEK policy: data minimisation - only collect what the task actually needs. De-

identification - remove or hide anything that could identify a person before it is stored or used.

Data retention -

delete data when it is no longer needed (in class: clear your project at the end of the lesson).

Access control -

only the people who need the data can reach it (in class: only your teacher owns the account or key). Encryption

- scrambling stored or sent data so only someone with the key can read it.

Let's try it - hands-on

1. Read your scenario card and underline the AI use.
2. Decide the ethical concern: bias, privacy, transparency, or misuse and over-reliance.
3. Write down who could be affected and how.
4. Decide which criterion this concern adds when you evaluate the solution.
5. Finish this stem: A responsible measure is... because...

Think about it

An AI solution passes its accuracy test but quietly stores every user's face. Should it be released? Why or why not?

My reflection

My reflection: the ethical concern I will always check for is... because...

Week 5 - Lab | Ethics & Governance - Risks, Ethics & Responsible Use (Lab)

I can...

- I can audit a real scenario and sort its ethical concerns into bias, privacy and misuse.
- I can rate each concern for how likely it is and how much harm it would do, then propose a mitigation with a reason.
- I can argue which mitigation to build in first, using words like bias, privacy, mitigation and governance correctly.

Words to know

- audit — a careful, organised check that sorts ethical concerns and rates how serious each one is
- likelihood — how often a concern is expected to happen
- impact — how much harm a concern would cause when it happens
- mitigation — a specific change to a solution that lowers a concern's likelihood or impact
- residual concern — the concern that is left over even after a mitigation is added, and must still be watched

Spotting an ethical concern is step one. Today you become an auditor: you sort the concerns into bias, privacy and misuse, rate how likely and harmful each is, and decide which fix a real AI solution should build in first.

Sort, then rate, then fix

A long list of ethical concerns is hard to act on until it is organised. Auditors first sort concerns into types - bias,

privacy, and misuse or over-reliance - so none is missed. Then they rate each one two ways: likelihood, how

often it is expected, and impact, how much harm it does. Plotting them on a simple grid makes the priority

obvious. A concern that is both likely and harmful sits in the top corner and gets fixed first. This is how real

teams spend their limited time where it protects people most.

A mitigation changes the solution

A mitigation is not a warning label, it is a real change to how the solution works. It lowers either the likelihood of

a concern or its impact. Good examples are testing on diverse data to reduce bias, removing a field so no

personal data is collected, or restricting a tool to its intended task. Each mitigation protects an evaluation

criterion - fairness, privacy or intended use. Even after a strong mitigation some residual concern usually

remains, so governance means naming who owns it and keeps watching.

Why it matters

When a UAE school, clinic or company builds an AI solution into a real decision, someone has to decide which

ethical concerns to fix first. That auditor role is exactly the skill you are practising today.

Let's try it - hands-on

1. List every ethical concern you can find in your scenario's AI use.
2. Sort each concern into bias, privacy, or misuse and over-reliance.
3. Rate each: likelihood low, medium or high, and impact low, medium or high, then plot it on the matrix.
4. Write a specific mitigation for each and name the evaluation criterion it protects.
5. Choose the one mitigation to build in first and finish: ...because...

Today's tool - hands-on

Tool: Card decks (ethics-scenario cards) + pre-generated packs

1. Draw a scenario card and read it in your group.
2. Name the ethical concern and who could be harmed.
3. Check the two flagged outputs and label the concern each shows.
4. Decide whether the concern is about fairness, privacy, or honesty.
5. Propose one responsible-use action that would address it.

You will make: An ethics record: the concern, who is affected, its type, and an action.

Stay safe: Unplugged; outputs pre-curated.

An ethical concern traced to who it affects, its type, and a responsible action.

Think about it

Two concerns have the same impact, but one is far more likely. Which should the team mitigate first, and how

would you explain your choice to the person building the solution?

My reflection

My reflection: the mitigation I would build in first is... because...

Week 6 - Core | Bringing the Strands Together

I can...

- I can complete a guided task that uses all four strands to test and improve a model.
- I can explain how the parts of my workflow connect to make a result and check it against a target.
- I can name one responsible choice I made while doing the task.

Words to know

- test set — a group of items whose correct answers are already known, used to score a model
- accuracy — how many predictions the model got right out of the total
- performance criterion — the target a model must reach, such as eight correct out of ten
- error analysis — looking at which kinds of item a model gets wrong so you can aim an improvement
- integration — joining several skills into one working task

Today you bring together everything from this term - understanding, building, evaluating and using AI responsibly

- to test one classifier against a clear target, then make one change to improve it.

Four skills, one model

Until now you have practised four skills in separate lessons: understanding how a classifier works, building the

steps, evaluating an output, and using AI responsibly. Today they click together on a single model. You set a

target, run a test set whose answers you already know, score how many the model got right, and compare that

score with the target. Nothing here is brand new - you are using all four skills at once, which is exactly how real

AI solutions are judged and improved.

A score you can trust, then improve

A classifier gives every data a label, and it can sound very sure. Sure is not the same as correct. That is why the

test set matters: because you already know the true label of each card, you can mark every prediction right or

wrong and count the accuracy. If it misses the target, you look at which kind of data failed most and add one

rule aimed at that, then re-test to see if the score rises. You keep the task safe by using only fruit photos - never

a person's face - so evaluating and improving stay responsible from the start.

One model, four strands: set a target, run the test set, score it, then improve and re-test.

Why it matters

Across the UAE, date growers and sorting lines are trialling classifiers to grade fruit quickly - and they only help if

their accuracy is tested against a clear target and their data is handled responsibly, just like in your task.

Let's try it - hands-on

1. Write your target at the top: at least eight correct out of ten.
2. Read the classifier's prediction for each of ten date cards that show only fruit, no people.
3. Flip each card to its true label, mark right or wrong, and count your accuracy.
4. Compare your score with the target and note whether the model meets it.
5. Name the kind of date it missed most, add one rule to fix it, re-run those cards, and write one responsible choice you made.

Think about it

If a classifier scores nine out of ten on ripe dates but only four out of ten on dark ripe dates, is its overall

accuracy telling the whole story? What would you change first?

My reflection

My reflection: the part of my workflow I trust most, and the part I would improve first, is ... because ...

Week 6 - Lab | Bringing the Strands Together (Lab)

I can...

- I can run my whole evaluate-and-improve workflow from start to finish and write down every score I make.
- I can use my results to say how reliably the model met the target and where it went wrong.
- I can add a new rule or risk to my model and re-test it responsibly.

Words to know

- end-to-end — running every stage of the evaluation in order on a full test set
- results log — a table where you record each prediction, its true label, and whether it was right
- reliability — how well the model met the target, counted out of the total
- constraint — a new rule or limit you add, such as a second criterion or a risk to watch for
- iteration — improving a model by testing it, analysing errors, changing one thing, and re-testing

Today you run your whole evaluate-and-improve workflow properly from start to finish, keep a full record of

every score, then add one new rule and re-test to see how your model copes.

From one card to a full test

In the Core lesson you scored the model on a few cards. Today you run it end-to-end on twelve or more, and you

write down every score in a log. That log is powerful: it turns 'it seemed to work' into real evidence. You can total

how many predictions were right, compare that with the criterion, and spot patterns - maybe dark ripe dates trip

it up most. That total is your model's reliability, and it tells you exactly how far it is from the target and where to aim your improvement.

Improve, add a constraint, re-test

A real model rarely stays the same - you improve it as you learn. Today you make one aimed change, then add

one constraint: either a second criterion, like also labelling each date bruised or not, or a risk rule, like flagging

any date that might be mouldy. Then you re-score only the cards the new rule touches and check your accuracy

before against accuracy after. Notice you do not start over - you adapt the model you already have. All the while

you keep the safe rules: only fruit photos go in, and every prediction is verified against its true label before you

trust it.

Why it matters

Date growers and sorting lines in the UAE keep improving their grading systems by logging errors and testing

each change against a clear target - exactly the run, evaluate and improve loop you are practising today.

Let's try it - hands-on

1. Choose twelve to fifteen date cards that show only fruit, no people.
2. Run every card through the classifier and record its prediction in your results log.
3. Flip each card to its true label, mark right or wrong, and total your accuracy before any change.
4. Compare with the criterion and name the kind of date the model got wrong most.
5. Add one improvement and one constraint, re-score the cards they affect, and record your accuracy after.

Today's tool - hands-on

Tool: Teachable Machine + Scratch + packs (build, evaluate, improve loop)

1. Run the solution and record its outputs on five inputs.
2. Evaluate the outputs against the criteria and score them.
3. Make one improvement (add cleaner examples, or add a check in Scratch).
4. Re-run the same five inputs and re-score.
5. Record one responsible-use choice you made while improving it.

You will make: A before-and-after score showing an improvement, plus a responsible-use note.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Reuse only the non-personal model.

One loop: build, evaluate, improve, re-test - with the score going up.

Think about it

If your model gets 11 out of 15 right but your target is 12, is one small rule enough to close the gap? How would you prove your change worked?

My reflection

My reflection: after running it end-to-end, the change that improved my model most was ... because ...

Week 7 - Core | Signature Solution - Planning

I can...

- I can write a clear goal and set out my solution as ordered input-process-output steps with a check step at the end.
- I can name the inputs, the teacher-operated tool and the non-personal data my solution needs.
- I can state one performance criterion, name one risk, and show the check step that measures the output and catches the risk.

Words to know

- goal — the one task your solution is built to complete
- structure — the ordered steps of your workflow: input, process, output
- performance criterion — a measurable bar for a good output, such as at least 8 of 10 correct
- check step — a step that measures the output against your criterion, not just that the tool ran
- risk — something that could go wrong, such as a confident but wrong answer

Today you become the architect of your signature solution: before you build anything, you plan how it works and

set the exact bar it must clear to count as good.

Architect first, set the bar second

A signature solution is a small workflow that finishes one task using input, process and output steps. Planning it

means designing the whole chain before you build, and deciding how good the output has to be.

You write a

clear goal, order your steps, and name what each step needs - the inputs, the teacher-operated tool, and non-

personal data only. Then you set a performance criterion: a measurable bar, like 8 correct labels out of 10. A

criterion you can measure is what lets you improve your solution on purpose later, instead of just hoping it works.

The check step does the measuring

The most important part of your plan is the check step, and this year it does more than ask did it work. It

measures your output against your performance criterion - counting, comparing, or scoring - because an AI

output can look confident and still be wrong. Never trust it on sight. Think about one risk in your plan, then place

a check step that would catch it by measuring the output against your bar and a trusted source.

If you want to go

further, add a fallback: what your solution does when the output misses the bar.

A signature-solution plan: goal and performance criterion on top, an input-process-output chain, with a check step measuring the output.

Why it matters

Across the UAE, engineers judge smart systems - from traffic sensors in Abu Dhabi to

farm-monitoring tools -

against clear numbers before anyone relies on them. Setting a measurable bar is what lets a team improve an AI

solution instead of guessing.

Let's try it - hands-on

1. Write your goal in one sentence: what task will your solution complete?
2. Draw three boxes - Input, Process, Output - and write the ordered steps inside them.
3. List the inputs, the teacher-operated tool and the non-personal data each step needs.
4. Set one measurable performance criterion: how good must the output be?
5. Add a check step that measures the output against your criterion, and name one risk it catches.

Think about it

If your check step showed the output missed your criterion, what should your solution do next instead of

trusting it?

My reflection

My reflection: the performance criterion I set is ... and the check step I still need to strengthen is ...

Week 7 - Lab | Signature Solution - Planning (Lab)

I can...

- I can walk my whole workflow through once, step by step, and confirm the output reaches the check step.
- I can test my plan with a hard input and use my check step to measure whether the output meets my criterion.
- I can add a fallback path so my solution does something safer when the output misses the bar.

Words to know

- prototype — a first working version you walk through by hand to see if it holds together
- walk-through — tracing one input through every step, in order, to the output and check step
- pressure-test — feeding a hard or tricky input on purpose to see if your check catches an output that misses the

criterion

- fallback path — what your solution does when the output misses the criterion, instead of trusting it

- test input — the example, non-personal input you use to try your workflow

Your plan meets reality today: you will run it as a prototype, try to break it on purpose, and make it safer when

the output misses your bar.

A plan is only a hypothesis

Your Core plan looks neat on paper, but paper hides gaps. A prototype is the first time you actually run it: you

take one real, non-personal input and walk it through every step in order until it reaches the output and the

check step. Say each step out loud. At the check step, measure the output against your performance criterion.

Where the input gets stuck, or a step does not do what you expected, you have found a gap - and finding it now,

by hand, is exactly the point. Prototyping turns a guess into something you have seen work.

Try to break your own plan

Anyone can pass an easy test. To really trust your solution, feed it a tricky input designed to fool it - a blurry

photo, a strange example - then measure the output at your check step. Did the output still meet your criterion,

or did the check catch a result below the bar? If a poor output slipped through, tighten the check. Then build a

fallback path: decide exactly what your solution does the moment the output misses the criterion, such as

rerouting to a safer step or flagging it for a human. That is how real engineers make solutions safe.

Why it matters

Engineers testing self-driving features or medical tools in the UAE deliberately try to break their systems, and

measure the results against strict numbers, before people rely on them. Pressure-testing and fallbacks are how

a smart solution earns trust.

Let's try it - hands-on

1. Choose one normal, non-personal test input and set up your Core plan.
2. Walk that input through every step aloud until it reaches the check step.
3. Measure the output against your performance criterion at the check step.
4. Now send a tricky input through and measure it too; if it slips below the bar, tighten your check.
5. Add a fallback path: write what your solution does the moment the output misses the criterion.

Today's tool - hands-on

Tool: Plan cards + Scratch storyboard (plan an evaluate-improve project)

1. Lay out the plan cards and fill each for your signature solution.
2. Write your performance criteria as numbers you can measure.
3. Decide exactly how you will test and score the solution.
4. Name one improvement you expect to make and how you will check it worked.
5. Sketch the solution as Scratch blocks on the storyboard.

You will make: A measurable plan with a stated evaluate-and-improve method.

Stay safe: Planning only; no live tool or data.

Plan cards including performance criteria and an evaluate-then-improve method.

Think about it

Which single input would be most likely to push your output below its criterion, and does your check step catch it?

My reflection

My reflection: the gap my prototype revealed was ... and the fallback I added to handle it is ...

Week 8 - Core | Signature Solution - Building

I can...

- I can build my solution step by step by following our plan and structure.
- I can test my solution and judge its outputs against our success criteria.
- I can make one clear improvement and explain what evidence led to it.

Words to know

- signature solution — the small AI-powered workflow you designed to do one clear task
- criteria — the fixed rules that decide whether an output is good enough
- evaluate — to judge an output against the criteria, using evidence not feelings
- iterate — to run, check, change, and run again to make the solution better
- evidence — the specific result from testing that shows something passed or failed

Today you turn your plan into a real, running solution - then you test it honestly and make it better with evidence, not guesswork.

Building is testing, not just making

Building your solution does not mean finishing it in one go. You assemble the steps from your plan, run them on

test inputs, and read what comes out. Then you compare that output to your criteria - one by one - and mark

each as pass or not-yet. A solution that runs is only halfway there; a solution you have tested against clear

criteria is one you can actually trust. Any AI step is run by your teacher at the board, and you never enter

personal data such as names, faces, or voices.

A confident output can still be wrong

AI outputs often sound sure of themselves. Fluent and detailed is not the same as correct. That is why you verify

before you trust: check each output against your criteria and against a source you already trust.

When

something fails, that failure is useful - it points to exactly what to improve. A good improvement is one you can

trace back to evidence: this criterion failed, so I changed this, and on the re-run it passed.

Random changes just

move the problem around.

The build loop: run the solution, evaluate each output against the criteria, then improve and run again.

Why it matters

Across the UAE, teams building smart tools - from Arabic-language helpers to traffic and energy systems - only

ship them after testing outputs against strict criteria. Learning to evaluate and improve, not just build, is exactly how that real work is done.

Let's try it - hands-on

1. Copy your workflow steps and your criteria list onto your build sheet.
2. Make three test inputs using only non-personal data - no names, faces, or voices.
3. Run the solution on each input (ask your teacher to operate the AI step) and record the output.
4. Score every output against every criterion: pass or not-yet, plus one word of evidence.
5. Change one thing to fix the criterion that failed most, re-run, and note the before-and-after.

Think about it

How could an output pass one of your criteria while completely failing another - and which one matters more for your task?

My reflection

My reflection: the strongest evidence that my improvement worked was ...

Week 8 - Lab | Signature Solution - Building (Lab)

I can...

- I can finish a working solution that runs end to end on test inputs.
- I can keep an evaluation log that scores each output against every criterion.
- I can make more than one evidence-based improvement and show the log that justifies each.

Words to know

- evaluation log — your written record of each input, its output, and how it scored on every criterion
- end to end — the solution runs through every step of the workflow without a gap
- edge case — an unusual or tricky input that can catch a solution out
- fail gracefully — respond safely and clearly to a bad input instead of giving a confident wrong answer
- iterate — run, log, change, and run again to keep improving

This Lab is your build-and-test workshop: finish your solution so it runs end to end, keep an honest log, and

improve it more than once using what the log tells you.

The log is what turns testing into improvement

Anyone can change their solution; the log is what makes a change worth trusting. For each test input you write

the output, a pass or not-yet against every criterion, and one word of evidence. When you look down the log and

see the same criterion failing again and again, you know exactly where to work next. That is how real teams

prioritise - by evidence, not by which problem is loudest. Keep any AI step teacher-run, use no personal data,

and verify each output against the criteria and a trusted source before you believe it.

Good solutions fail safely

A strong solution is not one that never meets a hard input - it is one that handles a hard input well. Feed yours something tricky on purpose: a blank field, an unknown value, something outside what you planned for. A weak solution gives a confident wrong answer; a strong one says clearly that it cannot match the input and asks you to check. When you find a case like that, fix it and write the fix in your log so the solution fails gracefully next time. Documenting the fix is part of the engineering, not an extra.

Why it matters

Whether it is a delivery app, a health screening tool, or a smart-city sensor in the UAE, the difference between a demo and a product is a tested solution with a log and safe handling of odd inputs. That is the standard you are practising here.

Let's try it - hands-on

1. Finish the steps still unbuilt so your solution runs end to end on every test input.
2. Set up an evaluation log: one row per input, one column per criterion.
3. Run at least five non-personal test inputs and fill in every row with scores and evidence.
4. Improve the criterion that fails most, re-run, and log the before-and-after.
5. Feed in one edge-case input, make your solution fail gracefully, and document the fix.

Today's tool - hands-on

Tool: Scratch + improvement-move cards (build and iterate)

1. Build the solution following your plan.
2. Run it and score the outputs against your performance criteria.
3. Draw an improvement-move card (change the data, change a step, add a check, change the tool) and apply it.
4. Re-run and re-score to see if the move helped.
5. Log the move, the before score and the after score.

You will make: A working solution and a log showing at least one evidence-based improvement.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Only the non-personal model is used.

Pick an improvement move, apply it, and check the score changed.

Think about it

When your log shows two different criteria failing on the same input, how do you decide which to fix first?

My reflection

My reflection: the log row that most changed how I improved my solution was ...

Week 9 - Core | Signature Solution - Refining & Finishing

I can...

- I can finish my AI solution so it meets our agreed performance criteria.

- I can evaluate my solution for accuracy, usefulness and risk against those criteria.
- I can explain one change I made that improved it, and show the evidence it worked.

Words to know

- performance criteria - the agreed, measurable list that says when your solution is good enough
- refine - to improve something by changing one part and re-measuring it
- accuracy - whether the output is correct when you check it against a trusted source
- risk - the chance the output could mislead, exclude or expose someone
- verify-before-trust - checking an output before you believe it, even when it sounds confident

Today you finish your AI solution and prove it is good enough to share. Finishing is not adding more - it is making

one smart change and measuring that it worked.

Change one thing, then measure

When you improve a solution, change only one variable at a time - the prompt, the input data, or a single

process step. Then run it again and compare the new output with the old one against your performance criteria.

If you change three things at once, you will not know which one helped. Keep both the before and after so you

can prove what improved. Small, tested changes are how real AI solutions get finished and trusted.

Accurate, useful and safe

A finished solution is judged three ways. Accuracy: is it correct when you check it against a trusted source? A

confident output can still be wrong. Usefulness: does it actually complete the task for its user?

Risk: could a

wrong output mislead, exclude or expose someone? Before you share, do a quick risk check - name who could

be affected and how you would catch a mistake. That is what makes a solution trustworthy, not how polished it

looks.

The improvement loop: change one variable, re-run, then measure against your criteria.

Why it matters

Across the UAE, teams finishing an AI tool - from a hospital scheduler to a customer chatbot - measure the

same accuracy and risk criteria before launch; learning to finish and evaluate well is a real workplace skill.

Let's try it - hands-on

1. Read your performance criteria and tick the ones you already meet.
2. Pick ONE variable to change - the prompt, the input data, or one step.
3. Ask your teacher to re-run the AI step (or run your paper version) and keep the before and after.
4. Measure the new output against a trusted source - do not trust it just because it sounds sure.
5. Do the risk check and write one sentence about the change that improved your solution.

Think about it

Think about it: your output sounds confident and reads well, but you have not checked it - should you trust it yet, and why not?

My reflection

My reflection: the one change that most improved my solution was ... and I know it helped because ...

Week 9 - Lab | Signature Solution - Refining & Finishing (Lab)

I can...

- I can run one final iteration and choose the version I will show.
- I can rehearse my demo and explain each step of my solution and its checks clearly.
- I can justify that my solution is trustworthy and low-risk using the checks I ran.

Words to know

- iteration - one round of changing and re-testing your solution
- version-lock - choosing the output you will show and not changing it again
- demo - a short, clear showing of your solution from input to output
- justify - to back up a claim with specific evidence, not just say it
- trustworthy - deserving belief because it has been checked, not because it sounds confident

Today you lock your final version and rehearse showing it well. A great demo does not just show the answer - it

shows the performance checks that make it trustworthy.

Lock it, do not add to it

Right before a showcase, the temptation is to add one more feature. Do not. A last-minute change is untested,

and untested means risky - it can quietly move a performance criterion you already passed.

Instead, run one

final iteration, confirm it still meets every criterion, then version-lock: pick the output you can reproduce and

agree not to touch it again. A locked, reproducible version is one you can show with confidence, because you

know exactly how it behaves and what checks it passed.

Show the checks, not just the answer

A strong demo narrates your workflow: here is the task, the input, the process step, and the output. Then it

shows evidence - the performance checks you ran. To justify that your solution is trustworthy and low-risk, point

to two specific checks: you verified the output against a trusted source, and your risk check named who could

be affected and how you would catch a mistake. 'It works' is not a justification; named checks are.

The demo spine: five lines that show your workflow and its checks.

Why it matters

When a UAE team demos a new AI service to leaders, they are judged on the checks they can show, not on how

polished it looks; rehearsing an evidence-based demo builds exactly that skill.

Let's try it - hands-on

1. Run one final iteration and confirm it still meets all your performance criteria.
2. Version-lock the output you can reproduce and agree not to change it.
3. Write your five demo sentences - task, input, process, output, one check.
4. Rehearse twice with your partner, swapping speaker and checker.
5. Finish with your justification: why it is trustworthy and low-risk, citing two checks.

Today's tool - hands-on

Tool: Teachable Machine retrain + packs (measure, improve, re-measure)

1. Run the solution across the 10-input test-set and count correct outputs.
2. Identify the two weakest cases and the limitation behind them.
3. Improve (retrain with cleaner examples, or add a check) targeting those cases.
4. Re-run the full test-set and count correct outputs again.
5. Record the before and after accuracy and confirm no personal data is used.

You will make: A before-and-after accuracy measure with a completed risk check.

Stay safe: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Risk check is part of the task.

Accuracy measured before and after an improvement, side by side.

Think about it

Think about it: two teams show the same output, but only one names the checks they ran - which solution would you trust, and why?

My reflection

My reflection: I can show my solution is trustworthy because ... and low-risk because ...

Week 10 - Core | Showcase, Assessment & Reflection

I can...

- I can show my solution to the class and explain two decisions I made, saying why each one improved it.
- I can answer a short task that samples all four strands - concept, design, evaluation and ethics.
- I can name one thing I learned this term and write one clear goal for next term.

Words to know

- showcase — presenting your finished solution and explaining the choices behind it
- design choice — a decision you made when building or improving the solution
- evaluation — judging your solution against clear, measurable performance criteria
- performance criterion — a measurable standard, such as accuracy on a sample, used to test the solution
- reflection — an honest look back at what you learned and what to improve next

Today you show what you built this term and prove it works: you present your AI solution, justify your best

choices with evidence, and set a goal for what comes next.

Show the evidence, not just the result

A strong showcase is more than a solution that looks finished. It is you explaining two real choices you made and proving each one helped. For every choice, name the performance criterion it was meant to improve, say what you measured, and compare the result before and after the change. Then be honest: name one limitation that still remains. A builder who admits what is not yet perfect is trusted more, not less, because it shows you evaluated your own work instead of just hoping it was good.

Reflect, then aim

Reflection is a skill, not a formality. Look back honestly: what is one thing you genuinely learned, and one thing that was hard? Then turn that into a goal for next term. A weak goal is get better at AI. A strong goal is specific and measurable - improve my solution's accuracy on the hardest category, or learn to test for fairness across groups. Remember the rules that kept your work safe all term: the tool stayed teacher-operated, no personal data ever went in, and every output was checked before you trusted it.

A showcase panel: each design choice links to a criterion and a before-and-after measurement, with one limitation named.

Why it matters

Across the UAE, teams building AI for hospitals, transport and government must present and defend their systems with evidence before anyone relies on them. Showing your reasoning and admitting limits is exactly how real AI work earns trust.

Let's try it - hands-on

1. Name your solution's goal and the performance criteria you evaluated it against.
2. Present two design or evaluation choices, each with a before-and-after measurement.
3. Admit one limitation that still remains and one risk you would watch.
4. Answer the four-strand task: concept, design, evaluation and ethics, one each.
5. Complete your reflection: one thing learned, one thing hard, and one specific goal for next term.

Think about it

Which of your two choices are you most confident about, and could you convince a doubter using only the evidence you measured?

My reflection

My reflection: the strongest choice I made this term was ... and my specific goal for next term is ...

Week 10 - Lab | Showcase, Assessment & Reflection (Lab)

I can...

- I can review a peer's solution against clear criteria and give one strength and one specific, kind next step.
- I can take the feedback I received and make my justification of my choices clearer and better evidenced.
- I can turn my reflections into a single measurable Term-2 goal and name the first step I will take.

Words to know

- peer review — judging a classmate's solution helpfully against the same criteria everyone uses
- carousel — a rotation where you review several solutions in turn at different stations
- criterion-anchored feedback — a comment tied to a named performance criterion, not to personal taste
- consolidate — bringing many scattered notes together into one clear point
- Term-2 goal — one specific, measurable target you set yourself for next term

Today you become a reviewer: you judge other solutions fairly against shared criteria, use the feedback on your

own work to make it stronger, and set one clear goal for next term.

Fair feedback beats nice feedback

Good peer review is kind and honest at the same time. Instead of just saying nice work, you judge each solution

against the same shared criteria everyone is using, so your feedback is fair and comparable.

For every solution

you visit, write one genuine strength and one specific next step, and tie each to a named criterion. And stay a

careful evaluator: if a sheet shows a confident result, check what it was measured against before you trust it. A

ninety-five per cent score means little if you do not know the sample behind it.

From many notes to one goal

By the end of the carousel you will have collected several comments on your own work. The skill now is to

consolidate: read them all, find the pattern, and turn it into a single Term-2 goal that is specific and measurable.

Weak: get better at AI. Strong: improve my solution's accuracy on the hardest category to a set target. Then

name the very first step you will take - something small and concrete you could do next week. A goal with a first

step is a goal you will actually start.

Why it matters

In UAE research labs and start-ups, engineers review each other's AI systems against shared standards before

launch, then set clear next targets. Fair critique and focused goals are how good teams keep improving.

Let's try it - hands-on

1. Take a review frame and the shared criteria to your first station.
2. Judge each of three solutions one criterion at a time.

3. Write one strength and one specific next step for each, tied to a criterion.
4. Read all the feedback on your own sheet and find the pattern.
5. Write one measurable Term-2 goal and the first step you will take toward it.

Today's tool - hands-on

Tool: Showcase the improved solution + peer-review cards

1. Demonstrate your solution and show your before-and-after score.
2. Explain two evaluation or improvement choices you made.
3. Visit two other solutions and complete a peer-review card for each, noting their improvement.
4. Read the feedback you received.
5. Write one Term-2 goal based on the feedback.

You will make: A demonstrated improved solution, peer-review cards, and a Term-2 goal.

Stay safe: Showcase uses only the non-personal solution; no data shared.

A showcase carousel with peer-review cards noting each solution's improvement.

Think about it

Looking at the feedback you received, what single change would improve your solution the most next term?

My reflection

My reflection: the most useful feedback I received was ... and my measurable Term-2 goal with its first step is ...

How a model learns: training, inference and evaluation

Machine learning (ML) means building a model by learning patterns from a dataset instead of writing fixed rules

by hand. The dataset is split into TRAINING DATA (the examples the model learns from) and TEST DATA

(separate, new examples kept back to check it). Three different actions matter: TRAINING is teaching the model

from the training data; INFERENCE is one forward run that turns a new input into an output; EVALUATION is

measuring the outputs against your criteria using the test data. If you change the training examples and train

again, the result is a new, retrained model - not the same model.

Checking fairness: compare two groups (Maths link)

Fairness is not just a feeling - you can measure it. Split your test set by one characteristic (for example short

cards vs long cards, or morning photos vs evening photos). Run the SAME test on each group under the SAME

conditions. Count how many the model got right in each group and work out the accuracy as a percentage

(correct / total x 100). Compare the two percentages. If one group scores clearly lower, you have found a

possible bias; if the counts are too small or too close to be sure, it is fair to record 'cannot determine' and gather

more examples. Always report the disparity and suggest one fix.

Cross-curricular links

Maths link: work out accuracy as a percentage and compare before-and-after scores. Science link: use a fair

test - change ONE factor and keep all other test conditions the same, so you know the change caused any

difference. Language link: the two-audience reply workflow (English for students, Arabic for parents) is a

translation and reading-level task.

Glossary - key words for this term

- pipeline - an AI workflow seen as connected stages: input -> process (model) -> output.
- factor - something you can change (data, input, or design) that affects how well a system performs.
- criteria - the agreed measures you judge an output against (e.g. accuracy, bias, reliability).
- accuracy - how many results are correct out of the total, as a fraction or percentage.
- reliability - the workflow keeps working across more than one input.
- bias - a repeating, systematic error that makes a system work worse for one group than another.
- fairness - treating comparable cases or groups consistently; measured by comparing accuracy across groups.
- mitigation - a specific change that reduces a risk (bias, privacy or misuse).
- explainability - being able to trace an output back to the step that produced it.
- de-identification - removing details that could identify a person from data.
- data minimisation - collecting only the data a task actually needs, and no more.
- encryption - scrambling stored or sent data so only someone with the key can read it.
- human in the loop - the person who decides, checks and can overrule the AI's output.
- responsible use - using AI safely and fairly: no personal data, always check the result.

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder

Main Course, Term 1 - Evaluating & Improving AI Models

Emerging tier | 10 weeks (Core + Lab)

STUDENT WORKSHEETS

Week 1 - Core - Worksheet | Unit Launch - Evaluating & Improving AI Models

Name: _____ Class: _____ Date: _____

Sort each example card into the correct column and give a reason.

My two questions

Write at least two questions about how AI outputs are made, judged or improved. Aim at least one at how we decide an output is good.

Sort it: output, judge or improve

Write each example in the right column and add one reason.

Example Output / Way to judge / Way to improve

Reason

A finished Arabic-English translation

Checking the translation for accuracy against the task

Giving the model a clearer input, then re-testing

Our safe-use charter

Write one responsible-use rule for the class and why it keeps us safe.

Exit ticket

Exit ticket - Name one example, say whether it is an output, a way to judge, or a way to improve a model, and explain why.

Then write one question you still have.

Challenge (stretch)

Challenge - Pick a real UAE situation (for example a smart-parking app or an Arabic-English translator) and predict what a

strong AI solution would need to get right. Name two performance criteria you would judge it against and one change you

might make to improve it. This stretches toward the Proficient idea of connecting the unit to a real context and predicting

what a strong solution needs.

Week 1 - Lab - Worksheet | Unit Launch - Evaluating & Improving AI Models (Lab)

Name: _____ Class: _____ Date: _____

Judge each output against the criterion, then log the difference and where it began.

Comparison log

For each output pair, write one difference, name the criterion it affects, and mark the stage where it began.

Output pair (criterion) Difference I observed Stage (input / process / output)

Pair 1

Pair 2

Pair 3

My conclusion

Use your log as evidence. What one change most improved an output against its criterion?

Verify it

How could you check your conclusion is right for a new example on the same task?

Tool log - Card decks (evaluation-criteria cards)

Record as you work. Goal: A ranked criteria layout and two measurable checks the class will reuse all term.

Step What I did What I saw / result

1 Deal the deck and read each criterion and its 'how to chec

2 Rank the six criteria for a chosen everyday AI tool (for e

3 Agree, as a pair, what 'good' would look like for your top

4 Write one measurable check for each of your top two (for e

5 Share your ranking with another pair and defend the order.

Our evaluation criteria

Criterion Why it matters here How we will measure it

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use reflection

• Could the inputs this system usually receives disadvantage anyone? What would you check?

• Where did a person decide or check in today's workflow (not the AI)?

Exit ticket

Exit ticket - Name one difference you logged, the criterion it affected, and the workflow stage it came from, and say whether

it made the output better or worse against that criterion.

Challenge (stretch)

Challenge - Design your own fair test: choose one thing to change in the input, name the criterion you will judge against,

predict how the output should change, and describe how you would compare the before and after to prove it. Then explain what a strong UAE solution built on this idea would need to get right. This stretches toward the Proficient idea of predicting what a strong solution needs and evaluating outputs against criteria such as accuracy and usefulness.

Week 2 - Core - Worksheet | AI Conceptual Understanding - Understanding the System

Name: _____ Class: _____ Date: _____

Trace the pipeline

Fill in the three stages for the worked example your teacher showed.

Stage What happens in the worked example

Input

Process (model)

Output

Explain the reasoning

Write one or two sentences explaining how the input became the output. Use because or so.

Spot the stage

Read each line and write which stage it describes: input, process or output.

Example Which stage?

A voice recording sent to a speech app

The model matches the sounds to words it learned

The typed sentence appears on screen

Exit ticket

Name the stages of an AI pipeline in order, give one example of an input, and say what the model in the middle does.

Challenge (stretch)

Challenge (stretch) - Explain how three different factors - the input given, the data the model learned from, and the way the

system was designed - could each change how well the output performs. Use the word pipeline.

Week 2 - Lab - Worksheet | AI Conceptual Understanding - Understanding the System (Lab)

Name: _____ Class: _____ Date: _____

Same model, one changed INPUT - watch the output move. You change the input only, so the model is NOT retrained and stays the same model. (If instead you changed the training examples and trained again, you would get a new, retrained/updated model.)

Build and trace

Set up your pipeline and follow one item through it. Record each stage.

Stage Your pipeline

Input

Model

Output

Change one factor, compare the output

Keep everything the same except one factor. Change it, run again, and compare.

First run Second run

Factor changed

Input

Output

What changed and why

Which factor mattered most?

Write which factor changed the output the most and how you would check the result.

Tool log - Google Teachable Machine (train, test, measure)

Record as you work. Goal: A trained model and a first accuracy measure with a note on where it failed.

Step What I did What I saw / result

1 Direct your teacher to name three classes and add BALANCED example sets: the same number of examples per class (about 15 each), varied conditions, and NO duplicate examples. Keep separate, brand-new examples aside for testing.

2 Ask your teacher to press Train (the teacher operates the

tool; you direct and observe). Say clearly: the examples are the TRAINING DATA, training builds the MODEL, and if the examples are changed and trained again the result is a NEW, retrained model - not the same one.

3 Direct your teacher to run the SEPARATE held-back test examples one by one (never test on a training example); you record each output and whether it was correct.

4 Count the score (for example 7 out of 10 = 70%) - this is your first accuracy measure. Record correct and wrong counts.

5 Discuss where it failed and why, and predict one change that would improve accuracy (retrain with better/cleaner examples, or refine the input).

Test-and-measure log

Test example Model's output Correct? (Y/N)

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use reflection

- Could the inputs this system usually receives disadvantage anyone? What would you check?

- Where did a person decide or check in today's workflow (not the AI)?

Exit ticket

You changed one INPUT and the output changed, but the model was NOT retrained, so it stayed the same model. Name the input you changed and explain in one sentence why the output moved. (Changing the training examples and training again would instead give a new, retrained model.)

Challenge (stretch)

Challenge (stretch) - Explain how the data a model learned from, the input it is given, and the way the system was designed can each affect how well it performs. Which would you change first to improve a weak output, and why? Use the word pipeline.

Building

Name: _____ Class: _____ Date: _____

Design your workflow

Fill each box for one specific task. Use only non-personal sample data in the input box.

Task (one sentence)

Input (sample data)

AI process step

Output

Check (against what?)

Run and trace

Run one sample input through your workflow, then explain the reasoning behind the output you got.

Verify-before-trust

Write where in your workflow you check the output, and one reason a confident output could still be wrong.

Exit ticket

In one sentence, name the specific task your workflow solves and explain the reasoning behind one step of its output.

Challenge (stretch)

Challenge (stretch): add a second step or tool to your workflow so it combines multiple steps (for example a translate step before the classify step). Draw the new order and explain why the order matters for the result.

Week 3 - Lab - Worksheet | AI Solution Design & Development - Designing & Building (Lab)

Name: _____ Class: _____ Date: _____

The build-test-improve loop: test several inputs, find the failing step, fix it, re-test.

Test log

Run three or more non-personal inputs through your workflow. For each, record the output, whether it fits, and the box responsible for any misfit.

Sample input Output Fit? (Y/N) Responsible step

Your fix

Write which single step you changed and why, then what happened when you re-ran the same inputs.

Reliability check

Explain in one or two sentences why testing several inputs makes your solution more trustworthy than testing one.

Tool log - Scratch + Teachable Machine (develop a solution for a task)

Record as you work. Goal: A working AI-powered solution for a specific task and a first limitations note.

Step What I did What I saw / result

- 1 Agree the task and what a good result looks like (link to
- 2 Use or train a model for the task on non-personal objects.
- 3 In Scratch, build the solution so an input runs through th
- 4 Run it on five real inputs and record the outputs.
- 5 Note one limitation you already notice, ready to fix next

Solution run log

Input Output the solution gave Looks right?

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use reflection

- Could the inputs this system usually receives disadvantage anyone? What would you check?

- Where did a person decide or check in today's workflow (not the AI)?

Exit ticket

Name the step you changed and explain the reasoning behind how you know your fix improved the result.

Challenge (stretch)

Challenge (stretch): design a workflow that combines multiple steps or tools - for example translate, then classify, then summarise. Test the whole chain and explain how a failure in an early step changes the final output.

Week 4 - Core - Worksheet | Critical Evaluation & Informed Interaction - Judging & Improving Outputs

Name: _____ Class: _____ Date: _____

A limitations grid: judge each output against every criterion, then record the evidence for each limitation.

Limitations grid

For each output, write a verdict (meets / partly / fails) for every criterion, then write the evidence for any limitation.

Output Accuracy Relevance Reliability Evidence

A

B

C

Slip or limitation?

Pick one output with a problem. Tick whether it is a one-off slip or a repeating limitation, and explain in one line.

Verify one fact

Choose one fact from an output. Write the trusted source you checked it against and whether it held up.

Exit ticket

Exit ticket: choose one output, name its clearest limitation, state the criterion it breaks, and write the single piece of evidence that decided it.

Challenge (stretch)

Challenge (stretch): add bias as a fourth criterion and re-judge the outputs using accuracy, bias and reliability together.

Name one output whose limitation is bias rather than accuracy - where it treats some cases or groups unfairly - and explain your evidence.

Week 4 - Lab - Worksheet | Critical Evaluation & Informed Interaction - Judging & Improving Outputs (Lab)

Name: _____ Class: _____ Date: _____

A ranking ladder: order the outputs top to bottom and weigh accuracy, reliability and bias to decide close calls.

Ranking strip

Write each output in order from best (1) to worst, then give the evidence and the limitation that decided each place.

Rank Output Evidence and limitation for this place

- 1
- 2
- 3
- 4

Decide the close call

Pick two outputs that were close. Explain which ranks higher for the task and whether reliability or bias decided it.

Improve the weakest

For your lowest-ranked output, tick which limitation it fails on - accuracy, reliability or bias - and write one specific improvement to the solution.

Tool log - Pre-generated output packs (find and measure limitations)

Record as you work. Goal: A scored pack, a limitations-by-type tally, and one targeted improvement idea.

Step What I did What I saw / result

- 1 Agree the evaluation criteria (reuse the L1 cards).
- 2 Read each output and mark it correct / unclear / incorrect
- 3 Score the pack: count how many meet the criteria.

4 Sort the limitations into types (wrong fact, unclear, unfair)
5 Choose the most common limitation and say what change would

Limitation tally

Output # Limitation named Type

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Responsible-use reflection

• Could the inputs this system usually receives disadvantage anyone? What would you check?

• Where did a person decide or check in today's workflow (not the AI)?

Exit ticket

Exit ticket: name your best and worst output and give one piece of evidence for each placement, including the limitation of the worst.

Challenge (stretch)

Challenge (stretch): write a short evaluation of your set using accuracy, bias and reliability together. Argue why your top output would still win even if bias were weighted most heavily, and name the one improvement that would most raise your bottom output's reliability.

Week 5 - Core - Worksheet | Ethics & Governance - Risks, Ethics & Responsible Use

Name: _____ Class: _____ Date: _____

Four ethical concerns to check in every AI use case.

Ethical concern analysis

For each scenario, name the ethical concern, say who is affected, and note which evaluation criterion it adds to the solution.

Scenario Ethical concern Who is affected Criterion it adds

1

2

Propose a responsible-use measure

Choose one scenario above. Write a responsible-use measure and justify it using the word because.

Personal-data check

Tick or write whether the AI use in scenario 1 collects any personal data it does not need, and how you would remove it.

Exit ticket

Name one ethical concern you found today and the one responsible-use measure you would never skip, and explain why in one sentence.

Challenge (stretch)

Challenge (stretch): choose one scenario and carry out a structured evaluation of it across bias, privacy and misuse - name how each shows up, decide which concern is the most serious, and justify which one you would fix first.

Week 5 - Lab - Worksheet | Ethics & Governance - Risks, Ethics & Responsible Use (Lab)

Name: _____ Class: _____ Date: _____

Sort concerns by type, then plot each by how likely and harmful it is; fix the top-right first.
Structured concern audit

List each concern, sort it into bias, privacy or misuse, then rate its likelihood and impact as low, medium or high.

Concern Type (bias/privacy/misuse) Likelihood Impact

Mitigation plan

For your two highest-priority concerns, write a specific mitigation, name the evaluation criterion it protects, and justify it with because.

Priority and owner

State which mitigation you would build in first and who in the solution's governance would own and monitor it.

Tool log - Card decks (ethics-scenario cards) + pre-generated packs

Record as you work. Goal: An ethics record: the concern, who is affected, its type, and an action.

Step What I did What I saw / result

1 Draw a scenario card and read it in your group.

2 Name the ethical concern and who could be harmed.

3 Check the two flagged outputs and label the concern each
s

4 Decide whether the concern is about fairness, privacy, or

5 Propose one responsible-use action that would address it.

Ethics record

Scenario / output Concern + who is affected Type (fair/privacy/honest) Action

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Exit ticket

Name the one mitigation you would build into this solution first, the ethical concern it reduces, and whether it lowers
likelihood, impact or both.

Challenge (stretch)

Challenge (stretch): conduct a full structured evaluation showing how bias, privacy and misuse each appear in your
scenario, then write one governance rule naming who owns and monitors the residual concern after your top mitigation is in
place.

Week 6 - Core - Worksheet | Bringing the Strands Together

Name: _____ Class: _____ Date: _____

Meets the target, keep it; misses the target, find the failing kind and fix one rule.

My target and workflow map

Write your performance criterion, then fill in what goes in each stage of your evaluate-and-improve task.

Stage What goes here

Target (criterion)

Test set (input)

Classifier (process)

Score vs target (output)

My score sheet

Record each card's prediction, its true label, and whether the prediction was right, then total your accuracy.

Card Prediction True label Right? (Y/N)

Total correct out of 10

My improvement and responsible-use choice

Name the kind of item that failed most, the one rule you added, your re-score, and one responsible choice you made.

Exit ticket

In one sentence, explain how your workflow made a result and checked it against the target, give your score before and after your change, and name one responsible choice you made.

Challenge (stretch)

Challenge (stretch): add a constraint to your model - a second criterion, such as also labelling each date bruised or not, or a

risk rule, such as flagging any item that might be mouldy - then adapt your rules and re-score against both targets. Explain

how adding the constraint changed your workflow and your accuracy. This is a step toward the Proficient target of adapting an approach when a new constraint is added.

Week 6 - Lab - Worksheet | Bringing the Strands Together (Lab)

Name: _____ Class: _____ Date: _____

Run every card, log each score, total accuracy before and after, then add a rule and re-test.

Full results log

Record every card you run: its prediction, its true label, and whether the prediction was right.

Card Prediction True label Right? (Y/N)

Accuracy before and after

Write your accuracy before your change and after your change, compare each with the criterion, and name the weakest kind of data.

My added constraint

Name the constraint you added and describe how you re-evaluated the affected cards responsibly.

Tool log - Teachable Machine + Scratch + packs (build, evaluate, improve loop)

Record as you work. Goal: A before-and-after score showing an improvement, plus a responsible-use note.

Step What I did What I saw / result

1 Run the solution and record its outputs on five inputs.

2 Evaluate the outputs against the criteria and score them.

3 Make one improvement (add cleaner examples, or add a check

4 Re-run the same five inputs and re-score.

5 Record one responsible-use choice you made while improving

Before / after

Input Output before Output after Better?

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Exit ticket

State your model's accuracy before and after your change, whether each meets the criterion, name one weak spot, and

describe the constraint you added and how you kept the task responsible.

Challenge (stretch)

Challenge (stretch): add a second constraint on top of the first - for example, combine a bruised-or-not criterion with a flag-mould risk rule - then predict, and test, which change affects accuracy more. Explain how you would adapt the model further. This pushes toward the Proficient target of adapting an approach as constraints grow.

Week 7 - Core - Worksheet | Signature Solution - Planning

Name: _____ Class: _____ Date: _____

My signature solution: goal and criterion

Write your goal in one sentence, then write one measurable performance criterion for a good output.

My workflow structure

Complete the table: write the ordered steps and what each step needs. Use non-personal data only.

Step What happens What it needs (input / tool / data)

Input

Process

Output

Check step

Risk and check

Name one risk in your plan, then explain how your check step measures the output and reduces the risk.

Exit ticket

In one sentence, link your goal, your performance criterion and your check step: what is your solution for, and how will you measure whether its output is good?

Challenge (stretch)

Challenge (stretch): design a fallback path - write what your solution should do when the check step shows the output misses your criterion, so it reroutes to a safer step instead of trusting a wrong output.

Week 7 - Lab - Worksheet | Signature Solution - Planning (Lab)

Name: _____ Class: _____ Date: _____

Pressure-testing a workflow: a tricky input flows to the check step, which measures the output against the criterion and either passes it or triggers a fallback path.

Prototype log

Walk one normal input through your workflow. Tick pass or fail for each step and note how the output measured against your criterion at the check step.

Step What happened Pass / Fail

Input

Process

Output

Check step

Pressure-test

Describe the tricky input you used and whether the output met your criterion at the check step. If a poor output slipped through, write how you tightened the check.

Fallback path

Write exactly what your solution does when the output misses your criterion.

Tool log - Plan cards + Scratch storyboard (plan an evaluate-improve project)

Record as you work. Goal: A measurable plan with a stated evaluate-and-improve method.

Step What I did What I saw / result

1 Lay out the plan cards and fill each for your signature so

2 Write your performance criteria as numbers you can measure

3 Decide exactly how you will test and score the solution.

4 Name one improvement you expect to make and how you will c

5 Sketch the solution as Scratch blocks on the storyboard.

Plan summary

Card Our decision

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Creator's reflection

- What did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

In one sentence, describe one gap your prototype revealed and the fallback path you added to handle it.

Challenge (stretch)

Challenge (stretch): design two fallback paths for different failures - one for an output below the criterion and one for a missing or unusable input - and state which check step triggers each.

Week 8 - Core - Worksheet | Signature Solution - Building

Name: _____ Class: _____ Date: _____

How to score one output: pass or not-yet against each criterion, with evidence.

Evaluation log

Run your solution on each test input and score the output against every criterion. Write one word of evidence for each score.

Test input (non-personal)

Output summary Criterion 1 Criterion 2 Criterion 3 Evidence

My one improvement

Name the criterion that failed, the change you made, and what the re-run showed.

Safe-use check

Write the one rule about data you followed while building, and why it matters.

Exit ticket

Name one improvement you made to your solution and the exact evaluation evidence that led to it.

Challenge (stretch)

Challenge (stretch): feed your solution a deliberately tricky or edge-case input. Show how it handles a wrong or unexpected output gracefully, then document the fix so the solution fails safely next time.

Week 8 - Lab - Worksheet | Signature Solution - Building (Lab)

Name: _____ Class: _____ Date: _____

A worked evaluation-log row: input, output, a score for every criterion, and the evidence.

Evaluation log (full)

Complete one row per test input. Score every criterion pass or not-yet and add one word of evidence for each.

Test input (non-personal)

Output summary Criterion 1 Criterion 2 Criterion 3 Evidence

Two improvements, from evidence

For each improvement, name the log row that justifies it, the change, and the re-run result.

Edge-case fix

Describe the tricky input, how your solution now fails gracefully, and where you documented the fix.

Tool log - Scratch + improvement-move cards (build and iterate)

Record as you work. Goal: A working solution and a log showing at least one evidence-based improvement.

Step What I did What I saw / result

1 Build the solution following your plan.

2 Run it and score the outputs against your performance crit

3 Draw an improvement-move card (change the data, change a s

4 Re-run and re-score to see if the move helped.

5 Log the move, the before score and the after score.

Improvement log

Improvement move Score before Score after Kept it?

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Creator's reflection

- What did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Point to the two log rows that justified your two improvements and state, for each, the change and the re-run result.

Challenge (stretch)

Challenge (stretch): design a second edge-case that could still slip past your fix, predict how the solution would respond,

and add a guard so it also fails gracefully - logging your reasoning as evidence for a

Proficient-level, robust solution.

Week 9 - Core - Worksheet | Signature Solution - Refining & Finishing

Name: _____ Class: _____ Date: _____

Three checks before you share: accuracy, usefulness, risk.

Performance-criteria check

Write your performance criteria, tick if met, and give one piece of measurable evidence for each.

Performance criterion Met? (Y/N) Evidence

Refine and re-measure

Record the one change you made and what happened when you re-ran it.

The one change I made Before output (short) After output (short)

Risk check

Answer both questions in full sentences: who could be affected if the output were wrong, and how would you catch it?

Exit ticket

Exit ticket: name one change that improved your solution and the check that proves it helped.

Challenge (stretch)

Challenge: write a short justification arguing that your solution is trustworthy and low-risk, citing at least two specific checks you ran - accuracy against a source, and your risk check.

Week 9 - Lab - Worksheet | Signature Solution - Refining & Finishing (Lab)

Name: _____ Class: _____ Date: _____

My five-sentence demo

Write one sentence for each part of your demo.

Part My sentence

Task

Input

Process

Output

One check I ran

Version-lock

Record the version you will show and why you chose it.

Rehearsal feedback

After each rehearsal, note one glow and one grow from your partner.

Rehearsal Glow (worked) Grow (fix)

1

2

Tool log - Teachable Machine retrain + packs (measure, improve, re-measure)

Record as you work. Goal: A before-and-after accuracy measure with a completed risk check.

Step What I did What I saw / result

1 Run the solution across the 10-input test-set and count co

2 Identify the two weakest cases and the limitation behind t

3 Improve (retrain with cleaner examples, or add a check) ta

4 Re-run the full test-set and count correct outputs again.

5 Record the before and after accuracy and confirm no

person

Accuracy: before vs after

Correct / 10 Notes

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Creator's reflection

- What did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

Exit ticket: in one sentence, justify why your solution is trustworthy and low-risk, naming the two checks you ran.

Challenge (stretch)

Challenge: prepare a 20-second answer to a tough question - 'How do you know your AI output is not misleading anyone?' -

citing your accuracy check and your risk check by name, and stating which performance criterion each one measures.

Week 10 - Core - Worksheet | Showcase, Assessment & Reflection

Name: _____ Class: _____ Date: _____

My showcase: two justified choices

Fill the table. For each choice, name the criterion it improved and give the result before and after.

Design or evaluation choice Criterion it improved Before After

Four-strand task

Answer one item per strand: explain a concept, describe a design improvement, evaluate one limitation, and name one ethics or data safeguard.

Reflection and next-term goal

Write one thing you learned, one thing that was hard, and one specific, measurable goal for next term.

Exit ticket

In one sentence, give the design choice you are proudest of and the before-and-after evidence that proves it worked.

Challenge (stretch)

Challenge (stretch): explain how your solution would need to change for a harder task or a higher-stakes context such as a medical or safety setting - name the performance criterion that would matter most and one new safeguard you would add.

Week 10 - Lab - Worksheet | Showcase, Assessment & Reflection (Lab)

Name: _____ Class: _____ Date: _____

A peer-review carousel: a reviewer rotates past three solutions, judging each against shared criteria, then consolidates feedback into one

Term-2 goal.

Carousel review frame

For each solution you visit, judge it against a named criterion and write one strength and one specific next step.

Solution Criterion used Strength One specific next step

1
2

3

Feedback I received

Write the pattern across the feedback on your own sheet, and how you will sharpen the justification of one choice.

My Term-2 goal

Write one specific, measurable goal for next term and the first step you will take.

Tool log - Showcase the improved solution + peer-review cards

Record as you work. Goal: A demonstrated improved solution, peer-review cards, and a Term-2 goal.

Step What I did What I saw / result

1 Demonstrate your solution and show your before-and-after s

2 Explain two evaluation or improvement choices you made.

3 Visit two other solutions and complete a peer-review card

4 Read the feedback you received.

5 Write one Term-2 goal based on the feedback.

Peer review received

From group Two stars One wish

Safety check (tick): I used no personal data (no faces, voices or names). ☐

Creator's reflection

- What did you build or design today, and how is it different from what an AI would produce on its own?

Exit ticket

In one sentence, give your Term-2 goal and the very first step you will take toward it.

Challenge (stretch)

Challenge (stretch): pick one solution you reviewed and explain how it would have to change for a higher-stakes context,

then set a Term-2 goal that names the new criterion and the new safeguard that context would demand.

Teacher notes and dataset rules for the training lab (Week 2 / L2)

Fit to 45 minutes: run this as TWO prepared stations or across the setup and evaluation halves of the period.

Station A (setup): the teacher's model is already trained on balanced classes, or the class trains it together.

Station B (evaluation): students run the held-back test set and score it. To save time, reduce the test set to 6

examples (2 per class) rather than 10 if needed; the essential task is train -> test on new examples -> measure

accuracy. Extra classes or extra test cases are optional extension work.

Dataset rules (use every time you train): 1) Balanced classes - the same number of examples in each class. 2)

Varied conditions - different backgrounds, angles or wording. 3) No duplicates - never repeat the same example.

4) Separate test set - test only on brand-new examples the model has never seen. 5) Report results by class - list

correct/total and the accuracy percentage for each class, not just an overall score.

Simple accuracy and error counts (Maths link): $\text{accuracy} = \text{correct} / \text{total} \times 100$. Record correct and wrong

counts for each class, then compare the before-and-after accuracy after any change.

Fairness check (add to any evaluation): split the test set by one characteristic (e.g. short vs long items), count

correct in each group under the same conditions, and compare the accuracy percentages. If one group scores

clearly lower, record the disparity and a fix; if the counts are too small to be sure, record 'cannot determine'.

What kind of change is each activity? Label your work so you know what you improved: MODEL TRAINING =

changing the training examples and training again (new/retrained model); WORKFLOW

DESIGN = adding or

reordering steps (model unchanged); PROMPT / INPUT REFINEMENT = changing the input given (model

unchanged); INTERFACE DEVELOPMENT = changing how a user interacts with the solution (model unchanged).

Cross-curricular labels: Week 2/6 training & accuracy = Maths link (percentages, comparison) and Science link

(fair test - change one factor). Week 4-5 reply/heat-safety scenarios = Science link (heat safety) and Language

link (Arabic-English translation, reading level).

Fairness evaluation step (add to your Signature Solution plan, Week 7): test whether your workflow performs

consistently for at least TWO input types or user groups. Record each group's accuracy or error count under the same test conditions, note any performance disparity, and propose one mitigation if a disparity is found. It is acceptable to record 'cannot determine' when the evidence is too limited.

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder
Main Course, Term 1 - Evaluating & Improving AI Models
Emerging tier | 10 weeks (Core + Lab) |
TEACHER PACK

1. About this Teacher Pack

Unit map - how the ten weeks build the evaluate-and-improve cycle

The term runs one cycle - produce, judge, improve - and deepens it each week. W1 Launch: move from 'it works' to 'how well?'. W2 Understand the pipeline and how factors (data, input, design) affect performance. W3 Design and build a workflow. W4 Judge outputs against criteria (accuracy, bias, reliability). W5 Ethics - 'accurate is not ethical'. W6 Bring the four strands together on one model. W7-W9 Signature Solution: plan -> build -> refine. W10 Showcase and reflect.

Strand emphasis by week: CU leads W2; SD leads W3, W7-W9; CE leads W4, W8-W9; GE leads W5; W1, W6 and W10 integrate all four. Every Lab also carries a responsible-use reflection so ethics runs throughout, not only in W5.

Technology note - a technology-agnostic design

The tools named here (Teachable Machine, Scratch, the pre-generated output packs) are examples that make the pipeline and the evaluate-improve cycle visible; they are not the curriculum. If a tool is unavailable or blocked, substitute a school-approved equivalent (any no-account image-classifier or block editor) or run the unplugged card-deck fallback in every lesson. The learning outcome - designing, evaluating and improving a workflow - stays the same whichever tool is used.

This pack contains everything needed to teach the Grade 8 main course, Term 1: ten weeks of Emerging-tier

lessons on the unit 'Evaluating & Improving AI Models', each delivered as a Core period and a paired Lab period

on the same outcome. Weeks 7-10 run the signature solution end-to-end.

How to use this pack

- Teach each week as Core (period 1) then Lab (period 2) on the same outcome.
- Keep all AI use teacher-mediated and account-free; enter no personal data.
- Build the evaluate-verify-risk habit into every AI activity; provide the unplugged fallback where tools are unavailable.
- Use the per-lesson formative check to confirm the outcome before moving on.
- Pair this Teacher Pack with the Student Lesson Content and Student Worksheets (same order).

Safeguarding & responsible use

Grade 8 works with AI systems and workflows. Keep all AI use teacher-mediated and account-free for students;
 prefer a board demonstration, curated or pre-generated output sets, or a card-based 'simulated AI' routine. Never
 enter personal data (names, faces, voices) into a prompt or a model. Build the
 evaluate-verify-risk habit into every AI
 activity: check each output against the task and a trusted source, because a confident output
 can still be wrong.
 Every AI activity has a non-AI (unplugged) fallback so no lesson depends on tool access.
 Materials & tools (UAE-accessible)
 Tool / material What it is & why Access route Unplugged fallback
 Workflow / model demo
 set + cards
 A board demo tied to the unit theme
 (Evaluating & Improving AI Models) plus
 printed example cards.
 School-printed; teacher-
 run; no accounts.
 Sort and map examples
 on paper.
 School-approved child-
 safe assistant
 An approved generative tool, teacher-
 operated only, for curated demos on non-
 personal data.
 School-approved; teacher
 device only.
 Use pre-generated output
 sets.
 Pre-generated output
 sets
 Curated strong/weak outputs from the same
 task for evaluation and comparison.
 Prepared by the teacher in
 advance.
 Same sets, printed.
 Simulated-AI cards Card decks that mimic an AI step so
 students practise the workflow unplugged.
 School-printed. This is itself the
 unplugged route.
 Program at a glance
 # Lesson Strand Framework ref(s)
 L1 Unit Launch - Evaluating & Improving AI Models Integration AIF·CU·G8·E AIF·SD·G8·E
 AIF·CE·G8·E AIF·GE·G8·E

L2 AI Conceptual Understanding - Understanding the System

AI Conceptual Understanding

AIF·CU·G8·E

L3 AI Solution Design & Development - Designing & Building

AI Solution Design & Development

AIF·SD·G8·E

L4 Critical Evaluation & Informed Interaction - Judging & Improving Outputs

Critical Evaluation & Informed Interaction

AIF·CE·G8·E

L5 Ethics & Governance - Risks, Ethics & Responsible Use

Ethics & Governance AIF·GE·G8·E

L6 Bringing the Strands Together Integration AIF·CU·G8·E AIF·SD·G8·E

AIF·CE·G8·E AIF·GE·G8·E

L7 Signature Solution - Planning Integration AIF·CU·G8·E AIF·SD·G8·E

AIF·CE·G8·E AIF·GE·G8·E

L8 Signature Solution - Building Integration AIF·CU·G8·E AIF·SD·G8·E

AIF·CE·G8·E AIF·GE·G8·E

L9 Signature Solution - Refining & Finishing Integration AIF·CU·G8·E AIF·SD·G8·E

AIF·CE·G8·E AIF·GE·G8·E

L10 Showcase, Assessment & Reflection Integration AIF·CU·G8·E AIF·SD·G8·E

AIF·CE·G8·E AIF·GE·G8·E

Week 1 - Core | Unit Launch - Evaluating & Improving AI Models | Interactive lesson: 'From 'It Works' to 'How Well''

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Generate at least two focused questions about how AI systems process inputs, generate outputs, and can be evaluated and improved.
- Explore and classify examples related to evaluating and improving AI models, justifying each choice.
- Co-construct the unit success criteria and the class responsible-use norms for AI tools.

Student-facing success criteria:

- I can ask at least two good questions about how AI turns an input into an output and how we judge that output.
- I can sort examples that relate to evaluating and improving AI models, and say why.
- I can help write our unit success criteria and our safe, responsible-use rules for AI tools.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 7 you learned that an AI system is a workflow - an input goes in, a process runs, and an output comes

out - and that a confident output can still be wrong, so results must be checked. You have used a teacher-operated AI tool

and helped agree class rules for safe use.

Activate: Show two curated outputs made for the same simple task on the board and ask: which one does the job better, and how

could you decide fairly rather than just by feel?

Teach: Recap input - process - output, then add the new idea for this unit: an output can be judged against clear performance

criteria, and a model can be improved so its outputs meet those criteria more often.

Apply: In pairs, students look at the two board outputs and name one thing that would make an output count as 'good' for that task.

Check: Cold-call two pairs to state their 'good output' idea; listen for a criterion tied to the task (accuracy, usefulness, fairness)

rather than 'it looks nicer'. If a pair judges by appearance only, re-model with the task in view before moving on.

Critical & analytical thinking focus

Question & analyse - surface prior knowledge and frame inquiry (Understand/Analyse | DOK 1-2 | SOLO: Uni to

Multistructural)

The technical concept (teacher background)

An AI system takes an input, runs a process built on a model trained from many examples, and returns an output. This unit

adds a question on top of that workflow: is the output any good, and how do we know? To answer it we set clear

performance criteria - measures such as accuracy, usefulness and fairness that fit the task - and we judge outputs against

them rather than by first impression. Improving a model then means changing something in the input or the process so

more outputs meet the criteria. This launch lesson surfaces what students know and frames good questions; it does not

build or evaluate a model yet. Keep all tool use teacher-operated, account-free and free of personal data, and verify every

output before trusting it. Cognitive demand for this lesson: Understand/Analyse | DOK 1-2 | SOLO: Uni to Multistructural.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show a short, curated stimulus for the

unit: the same task done by an AI two

ways, one clearly better. Ask students

what went in, what came out, and

which output does the job.

Watch, then jot one thing
they noticed and one thing
they wonder on a sticky note.

- The task is the same - so why is one output better than the other?

Frame the

inquiry / 8 min

Recap input - process - output, then introduce evaluating against performance criteria and improving a model. Model turning a wondering into a clear question (What... How... Why...). Turn their 'I wonder' note into at least two questions about evaluating or improving AI outputs.

- Does your question ask about the input, the process, the output, or how we judge it?

Explore and sort Hand out a curated card set of In pairs, sort each card as an • Why is this a criterion for judging

/ 14 min examples (outputs, criteria, and improvement moves). Circulate and prompt justifications; keep any live tool teacher-operated only.

output, a way to judge an output, or a way to improve a model, writing one reason for each sort.

and not the output itself? Convince your partner.

Co-build wall

and charter / 12

min

Scribe a shared vocabulary wall (input, process, output, model, criteria, evaluate, improve) and co-write the class safe-use charter. Preview the signature activity: an AI solution evaluated and improved against clear performance criteria.

Contribute definitions in their own words and propose one responsible-use rule (no

personal data, verify before trust, teacher-operated tools).

- What is one rule that keeps us safe when we test an AI tool in class?

Check and close

/ 5 min

Run the exit check; collect the sticky-note questions and the sort cards as evidence.

Complete the exit ticket: one classified example with a reason, plus one question they still have.

- Give one example and say whether it is an output, a criterion, or an improvement - and why.

Anticipated misconceptions

Misconception How to address it

A better-looking output is always the better output. Put a tidy but wrong output next to a plain but correct one and

check both against the task; agree that 'good' means meeting the criteria, not looking polished.

Improving a model just means asking the AI to try again. Show that improvement is a deliberate change to the input or the process, then re-testing against the same criteria - not simply hoping for a nicer result.

If the AI is confident, the output must be right. Display a curated output that is confidently wrong, check it

against a trusted source, and establish verify-before-trust as a class norm.

Hands-on activity guide

Overview: The class opens the unit with a curated stimulus, then explores and sorts examples into outputs, judging criteria and improvement moves, before co-building the vocabulary wall and safe-use charter and previewing the signature activity.

Grouping / timing: Pairs for the sort, whole class for the wall, charter and debrief.

Tool: A teacher-operated board demo plus a pre-printed, curated card set (a 'simulated AI' paper routine). Account-free

and no personal data - students never log in or enter names, faces or voices.

Facilitation steps:

1. Watch the stimulus and note one thing you noticed and one thing you wonder.

2. Turn your wondering into at least two clear questions about evaluating or improving AI outputs.
3. With your partner, sort each card as an output, a way to judge an output, or a way to improve a model.
4. Write one reason for each card - why does it belong there?
5. Add one word and its meaning to the class vocabulary wall.
6. Propose one responsible-use rule for our safe-use charter.

Unplugged fallback: The whole lesson already runs unplugged on paper cards and a board demo. If the demo tool is unavailable, describe one worked pair of outputs aloud, name the task, and have pairs sort the printed cards - no device needed.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write two questions about how AI outputs are made, judged or improved.

How do we decide fairly whether an output is good enough for the task? What could we change to make the model's outputs better?

Accept any two clear questions that touch on the output, a criterion, or an improvement.

Look for question form, not perfect wording.

Classify three examples as an output, a A finished translation = output. Checking it One mark per correct sort with a

way to judge an output, or a way to improve a model, and give a reason.

for accuracy against the task = a way to judge. Giving the model clearer input then re-testing = a way to improve.

matching reason; the reason must reference producing, judging, or improving.

Suggest one responsible-use rule and say why it keeps us safe.

Never type a real name, face or voice into the tool, because that is personal data and it should stay private; the teacher runs the tool for us.

Accept any rule aligned to no-personal-data, verify-before-trust, or teacher-operated tools, with a reason.

- Targets: the three launch objectives - questioning, classifying with justification, and co-constructing the success criteria and responsible-use norms - on the outcome that AI systems process inputs and generate outputs.
- Check: exit ticket plus circulating listen-ins during the pair sort, questioned against the produce-judge-improve frame.
- Success criteria: the student generates two on-topic questions, sorts examples correctly with reasons, and contributes to the charter.
- Evidence: collected sticky-note questions, completed sort cards, and the co-built vocabulary wall and charter.
- Feedback / adjustment: if pairs judge outputs by appearance, re-model checking one output against the task and a trusted source, and re-sort two cards before the next lesson.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Questioning the unit ideas Generates two on-topic questions about how outputs are made, judged or improved, with guidance.

Generates one on-topic question, or needs prompting.

No question, or questions unrelated to the unit.

Classifying with justification Sorts examples as output, judging criterion or improvement and gives a simple reason for each, with support.

Sorts correctly but reasons are missing or vague.

Sorts incorrectly and cannot give a reason.

Responsible-use contribution

Suggests one sensible rule for the charter using the safe-use ideas modelled.

Repeats a rule when prompted.

Does not contribute a rule.

Representation: pair the spoken stimulus with the printed cards, the vocabulary wall and a visible produce-judge-improve diagram; pre-teach criteria, evaluate and improve with simple icons.

Action & expression: let students show sorting by placing cards, speaking their reason aloud, or writing - accept any mode for the exit ticket.

People of Determination: provide a reduced card set with picture support, sentence starters ('This is a way to judge because...'), extra processing time, and a peer scribe where helpful.

Week 1 - Lab | Unit Launch - Evaluating & Improving AI Models (Lab) | Interactive lesson: 'Two Outputs, One Verdict'

Human in the loop: a person decides which factor or input to change and interprets what the output change means - the AI does not choose or judge for you.

Name where the human decides in this Lab.

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Compare paired examples of AI outputs for the same task and record the differences you observe.
- Judge each output against one or two simple performance criteria, noting where the difference comes from.
- Draw one evidence-based conclusion about what makes an output better against the criteria.

Student-facing success criteria:

- I can compare two AI outputs for the same task and write down how they differ.
- I can judge each output against a simple criterion like accuracy or usefulness.
- I can use what I observed to say what made one output better.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you learned that AI outputs can be judged against clear performance criteria and that a model can be improved on purpose. You also agreed the class safe-use charter.

Activate: Show two curated outputs made for the same task (one from a clear input, one from a vague input) and ask which better meets a stated criterion, and how you would prove it.

Teach: Model a compare-and-log routine: put two outputs side by side, judge each against one named criterion, and record the difference and where it came from in a log table.

Apply: In pairs, students compare one pair of outputs and log a single difference against the criterion.

Check: Check that each pair has judged against the criterion, not appearance, and located where the difference began (usually the input or the process); redirect any pair that logged only 'it looks better'.

Critical & analytical thinking focus

Question & analyse - surface prior knowledge and frame inquiry (Understand/Analyse | DOK 1-2 | SOLO: Uni to

Multistructural)

The technical concept (teacher background)

Evaluating an AI output means comparing it against clear performance criteria rather than judging by first impression. In this Lab students investigate the same outcome more deeply: they compare curated pairs of outputs made for one task, judge each against a simple criterion such as accuracy or usefulness, and log the differences as evidence. Two outputs from nearly identical inputs can differ because one input was clearer or the process handled it differently - and naming which output better meets the criterion is the first move of evaluation. No new concept is introduced; this builds the habit behind later evaluation and improvement work: notice a difference, judge it against a criterion, and locate where it came from. All examples are pre-generated and teacher-operated, account-free, with no personal data.

Cognitive demand for this lesson:

Understand/Analyse | DOK 1-2 | SOLO: Uni to Multistructural.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set up

/ 5 min

Recap evaluating against criteria and the safe-use charter. Hand out the comparison log and the curated output-pair packs, each with one named criterion.

Read the first output pair and predict which better meets the criterion.

- Before we judge closely, which output do you expect to win on this criterion, and why?

Model compare-

and-log / 8 min

Think aloud through one pair: judge each output against the named criterion, log the difference, and mark the stage - input, process or output - where it began.

Watch, then copy the routine's headings into their own log.

- At which stage did the difference that changed the result begin?

Exploration

station 1 / 14 min

Circulate as pairs work the first two output pairs; prompt for evidence against the criterion, not opinion. Keep all tools teacher-operated only.

Judge each output against the criterion, then log at least two observed differences with the stage where each appears.

- What is your evidence, against the criterion, that this output is better - not just that you prefer it?

Rotate and log /

13 min

Rotate pairs to a new pack with a different criterion; prompt them to compare across packs and start a conclusion.

Log differences for the new pair and begin a one-sentence conclusion about what made an output meet the criterion better.

- Across the pairs you have seen, what one change most improved the output against its criterion?

Debrief and

close / 5 min

Collect the logs; take two conclusions and test them against the class evidence; flag verify-before-trust.

Complete the exit ticket: one logged difference, the criterion it affected, and the stage it came from.

- Name one difference you logged, the criterion it changed, and where in the workflow it started.

Anticipated misconceptions

Misconception How to address it

A different output means the AI is broken. Show that small input changes cause different-but-valid outputs;

the point is to judge them against the criterion, not to declare a fault.

You can judge an output without a criterion. Have pairs try to say which output is 'better' with no criterion, then

again with one named criterion; agree the criterion makes the judgement fair and provable.

Better just means longer or fancier. Compare a long vague output with a short accurate one against

an accuracy criterion; agree that 'better' means meeting the criterion, checked against a trusted source.

Hands-on activity guide

Overview: Pairs work through curated output-pair packs, each with a named criterion, judging each output against it,

logging observed differences as evidence, then drawing one conclusion about what made an output meet the criterion

better.

Grouping / timing: Pairs at rotating stations, whole class for the debrief.

Tool: Pre-generated output-pair packs on paper or on a teacher-operated board (a 'simulated AI' comparison routine).

Account-free, no logins, no personal data.

Facilitation steps:

1. Read the first output pair and the named criterion, and predict which output better meets it.
2. Judge each output against the criterion.
3. Log at least two differences you observe, and mark the stage where each begins.
4. Rotate to a new pack with a different criterion and log differences again.
5. Compare across packs and write one conclusion about what improved an output against its criterion.
6. Check your conclusion against the task - would it still hold for a new example?

Unplugged fallback: The packs are already paper-based, so no device is required. If the board demo is unavailable, use

the printed output pairs only and judge and log by hand against the named criterion.

Lab tool integration (practical, hands-on)

Tool: Card decks (evaluation-criteria cards)

Teacher setup: Print one evaluation-criteria deck per pair (6 cards: accuracy, usefulness, clarity, fairness, reliability, speed), each with a plain definition and a 'how to check it' line.

What students do:

1. Deal the deck and read each criterion and its 'how to check' line aloud.
2. Rank the six criteria for a chosen everyday AI tool (for example, a translation app) from most to least important.
3. Agree, as a pair, what 'good' would look like for your top two criteria.
4. Write one measurable check for each of your top two (for example, 'count how many of 10 answers are correct').
5. Share your ranking with another pair and defend the order.

Students produce: A ranked criteria layout and two measurable checks the class will reuse all term.

Data-safety: Fully unplugged; no devices or data.

Unplugged fallback: This is already the unplugged activity; extend by adding a blank criterion card.

Evaluation-criteria cards ranked, each with a way to measure it.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Log two differences between an output pair and mark the workflow stage for each.

Output A read more accurately because its input named the language (stage: input);

Output B's process cut off part of the text (stage: process).

One mark per difference correctly described and located at a plausible stage.

Judge one output pair against a named criterion and say which wins and why.

Against accuracy, Output A wins because it matches the source text; Output B adds a word that is not in the original.

Accept a judgement that names the criterion and cites a logged difference as evidence, not a preference.

Write one conclusion about what made an output better against its criterion, using your log as evidence.

The clearer input gave a more accurate output, because the system had exactly the information the task needed.

Accept a conclusion that cites a logged difference and the criterion; it must reference evidence, not opinion.

- Targets: comparing AI outputs, judging them against a performance criterion, and logging located differences with an evidence-based conclusion, on the same outcome that AI systems process inputs and generate outputs.

- Check: circulating evidence-prompts during the stations plus an exit ticket naming one logged difference, its criterion and its stage.

- Success criteria: the student logs at least two located differences, judges against a named criterion, and states one

conclusion backed by their log.

- Evidence: collected comparison logs and the exit tickets.
- Feedback / adjustment: if pairs judge by appearance, re-model judging one pair against a named criterion before the next lesson and have them re-log one difference.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Comparing outputs Logs two differences between an output pair, with support.

Logs one difference, or needs prompting.

Does not log a usable difference.

Judging against a criterion Names which output better meets the criterion and gives a reason, with guidance.

Judges but by appearance rather than the criterion.

Cannot link the judgement to a criterion.

Evidence-based conclusion States one conclusion that cites a logged difference and the criterion.

States a conclusion but without evidence.

No conclusion offered.

Representation: side-by-side output cards, a pre-formatted log table with stage and criterion labels, and the produce-judge-improve diagram on display.

Action & expression: pairs may record differences in writing, by annotating the cards, or by speaking to the teacher who scribes.

People of Determination: fewer output pairs, a partly completed log, sentence starters ('Against accuracy, ... wins because ...'), and extra time to rotate.

Week 2 - Core | AI Conceptual Understanding - Understanding the System |

Interactive lesson: 'Inside the Pipeline'

See the Explainability Note for teacher background on tracing a workflow and the limits of explanation (black-box vs traceable steps).

Real-World Connection (UAE): Abu Dhabi environmental monitoring runs as a pipeline - sensors (input), a processing/model step, and readings (output).

Changing a factor (a sensor, or the training data) changes the reading, just like the classroom activity.

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G8·E | 1 x 45 min

Objectives & success criteria

- Describe how an AI system processes inputs and generates outputs, naming each stage of the pipeline.
- Apply this understanding in a hands-on task by mapping how an input becomes an output, and explain the reasoning behind the result to a partner.
- Use the key vocabulary of AI Conceptual Understanding accurately, and connect the input-model-output idea to building or evaluating a solution.

Student-facing success criteria:

- I can name the stages an AI system uses to turn an input into an output.
- I can trace a worked example and explain to a partner how the input became the output.
- I can use words like input, model and output correctly, and say how they help me evaluate a solution.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 7 you met the input-process-output shape and followed simple step-by-step instructions. You know a

computer does exactly what its steps tell it to do, in order.

Activate: Show two familiar apps side by side on the board - a voice assistant and a photo search - and ask what each one takes in and gives back.

Teach: Name the stages of the pipeline out loud - input, model (the process), output - and stress that the middle stage learned its behaviour from data, it was not simply programmed with the answer.

Apply: In pairs, jot the input and the output for one app on a mini whiteboard, then say in one line what the model in the middle must be doing.

Check: Cold-call two pairs; the rung is secure when a pair can name the input and output and describe the middle stage as a trained model, not magic.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

An AI system is a pipeline. It receives an input (data such as text, an image or a sensor reading), passes it through a process

- a trained model that applies patterns learned from many examples - and returns an output (a prediction, a label or generated content). Understanding the system means naming each stage and reasoning about how one becomes the next:

the model does not grasp meaning, it maps inputs to statistically likely outputs. For Grade 8 this matters because the unit's

goal is to evaluate and improve models. You cannot judge whether an output is good, or improve it, until you can see the

stages it came from and identify where the result was shaped - by the input given, the data the model learned from, or the way the system was designed.

Training, inference and evaluation (teacher background)

Machine learning (ML) means building a model by learning patterns from a dataset instead of writing fixed rules by hand. The

dataset splits into TRAINING DATA (the examples the model learns from) and TEST DATA (separate, new examples kept

back to check it - never test on a training example). Three actions matter: TRAINING is teaching the model from the training

data; INFERENCE is one forward run that turns a new input into an output; EVALUATION is measuring the outputs against

your criteria using the test data. If you change the training examples and train again, the result is a NEW, retrained model -

not the same one. Students train their first model, and meet these terms in practice, in today's Lab.

The same pipeline inside one worked example.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two contrasting worked examples

on the board - a translation and an image classifier - and think aloud about what goes in and what comes out of each.

Watch and call out the input and the output for each example.

What is happening in the middle that you cannot see?

Direct teach / 12 min

Draw the pipeline - input, model, output - as a labelled diagram; we-do one worked example together, naming what the trained model did to the input.

Copy the pipeline diagram and help label the worked example.

Where did the model get its behaviour from?

Guided investigate / 13 min

Hand out three pre-generated worked

examples; model tracing the first, then release pairs to map input, process and output on the other two.

In pairs, trace each example and write the input, what the model did, and the output.

How did this exact input turn into that exact output?

Pair explain / 9

min

Prompt partners to explain one example using reasoning words (because, so, since); circulate and listen for a clear cause-and-effect link.

Take turns explaining to a partner how the input became the output, in order.

Which stage would you change to get a different output?

Check and close

/ 5 min

Collect the exit ticket; recap the pipeline and preview that the Lab will trace and compare how the system behaves when a factor changes.

Complete the exit ticket naming the stages and one example input.

Name the stages of the pipeline in order - go.

Anticipated misconceptions

Misconception How to address it

The model in the middle understands the question the way a person does.

Explain that the model maps inputs to likely outputs using patterns learned from data; it matches, it does not understand.

Re-voice it as 'it predicts, it does not know'.

The output is correct because a computer produced it. Show a confident but wrong pre-generated output next to a correct one and model verify-before-trust: check every output against the task and a trusted source, because a confident output can still be wrong.

The system is one solid black box with no stages worth Trace one item slowly on the board so students see three distinct

separating. stages; stress that separating them is exactly what lets you find and fix where a result went wrong.

Hands-on activity guide

Overview: Students investigate three teacher-provided worked examples, map how each input becomes an output through

a trained model, and explain the reasoning to a partner.

Grouping / timing: Pairs for the investigate and explain; whole class for the debrief.

Tool: A teacher-operated board demo with a curated, pre-generated set of example inputs and outputs - no accounts and

no live typing by students - so the class studies safe, prepared results and no personal data is ever entered.

Facilitation steps:

1. Look at the worked example your teacher shows on the board.
2. Write down the input - the exact data that went in.
3. Describe what the model does to it and write it next to Process.
4. Write down the output - the exact result that came out.
5. Explain to your partner how the input became the output, in order.
6. Check one output against the task and a trusted source - does it really answer it?

Unplugged fallback: Simulated-AI cards: a 'model' envelope holds a fixed rule matched to example cards. A student posts

an input card in and the envelope returns the matching output card - the same pipeline with no device and no accounts.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Label a new worked example (an email spam filter) with input, process and output.

Input: the incoming email text. Process: a trained model compares it with patterns from many past emails. Output: the label spam or not spam.

Award full marks when all three stages are correct and in order; naming only input and output is partial.

Explain to a partner how the input became the output for the board example.

The input went into the model, which used patterns learned from data to change it, so the output appeared; a different input would have produced a different output.

Listen for cause-and-effect reasoning linking the stages, not

just the three names.

Use input, model and output correctly in one sentence about a solution you could build or evaluate.

In a litter-sorting idea the input is a photo of an item, the model decides the material, and the output is the correct bin - I would evaluate it by checking the outputs against the real items.

Vocabulary used accurately and in order with a link to building or evaluating; gently correct any stage that is swapped.

- Targets: the Emerging outcome - describes how an AI system processes inputs and generates outputs.
- Check: partner explanation plus a labelled exit ticket, sampled by cold-call.
- Success criteria: the student names each stage in order and explains how the input became the output through the model.
- Evidence: the completed trace table and the one-sentence explanation on the worksheet.
- Feedback / adjustment: if a stage is swapped or the model is called 'magic', re-run the board example and ask the student to name what the model did before writing again.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Names the stages Names input, process (model) and output in the correct order

Names the stages but muddles the order

Names only one or two stages

Explains the sequence Explains clearly how the input became the output through the model

Describes the stages but the link is vague

Lists stages with no link between them

Uses vocabulary and connects

Uses input, model and output accurately and links them to building or evaluating a solution

Uses the words with prompting

Avoids or misuses the words

Representation: pair the pipeline diagram with a concrete worked example so the stages are available visually, verbally and through a traced item.

Action and expression: let students show understanding by labelling, speaking to a partner or arranging physical cards, not only by writing; extend confident pairs by asking which factor - the input, the data the model learned from, or the design - most changed the output.

People of Determination: provide a pre-labelled stage strip and sentence starters ('The input is ...', 'The model ...'); allow extra time and a calm partner for the explain step.

Week 2 - Lab | AI Conceptual Understanding - Understanding the System (Lab) |

Interactive lesson: 'Change One Factor'

Human in the loop: a person decides which factor or input to change and interprets what the output change means - the AI does not choose or judge for you.

Name where the human decides in this Lab.

Retrain vs reuse: where a worksheet step says 'Train', read it as 'ask the teacher to press Train'. Be explicit with students - reusing an already-trained model on a NEW input does NOT change the model; only re-training with new examples does.

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G8·E | 1 x 45 min

Objectives & success criteria

- Build and trace a simple input-model-output pipeline on a hands-on task, following one item through every stage.
- Compare how the same system behaves when a factor is changed - the input or the example data - and explain why the output changed.
- Use pipeline vocabulary to reason about which factor most affects performance and how you would check an output.

Student-facing success criteria:

- I can build a pipeline with an input, a model and an output, and trace one item through it.
- I can change a factor, compare the two outputs, and explain why the output changed.
- I can say which factor most affected the result and how I would check the output against a trusted source.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you named the stages of an AI pipeline - input, model and output - and traced one worked example.

Activate: Replay the Core pipeline diagram and ask one pair to narrate it in ten seconds.

Teach: Show that changing one factor - the input, or the examples the model learned from - can change the output while the pipeline shape stays the same, using two prepared cases on the board.

Apply: In pairs, predict the output for a third input before it is revealed.

Check: Thumbs up or down - did your prediction match? Ask one pair whose prediction missed to say why.

Critical & analytical thinking focus

Analyse - reason about how the system/workflow produces outcomes (Understand/Analyse | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Tracing a pipeline means following one item of data through each stage and naming what happens: what went in, what the

model did with it, and what came out. Comparing means changing one factor - usually the input, or the set of examples the

model learned from - while keeping the rest the same, and watching how the output responds.

This turns the model from a

black box into something students can reason about: a small change in a factor can produce a large change in output, and

different factors affect performance in different ways. It also shows where a result can go wrong - an unclear input, narrow

example data, or an output read without checking. The Lab needs no accounts and no live tools; a simulated-AI card routine

or a teacher-run prepared demo reproduces the behaviour safely, with verify-before-trust built into the routine.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set-up / 5 min

Replay the Core pipeline diagram; hand each pair a simulated-AI model card/envelope and a small set of input cards.

Recall the stages and lay out their input, model and output spaces.

What are the stages of the pipeline, in order?

Model the trace / 10 min

Think aloud tracing one input card through the model to its output; write each stage under the diagram.

Watch and copy the trace for the demonstrated item.

Which single item are we following all the way through?

Build and trace / 12 min

Release pairs to send their own item through and record each stage; circulate and check the trace stays consistent.

In pairs, send one item through your pipeline and record input, model and output.

Did the same item stay the same all the way through, or did it change?

Change a factor and compare /

12 min

Prompt pairs to change one factor - first the input, then a swapped example set - keeping the rest the same, and

Change one factor, run it again, and compare the two outputs; check one against a

You changed only one factor - so why did the output change?

compare the outputs; model verify-before-trust on one output. trusted answer.

Share and check

/ 6 min

Take two comparisons to the board; collect the exit ticket naming which factor most affected the output.

Share your comparison and complete the exit ticket.

Which factor changed your output the most - and how do you know?

Anticipated misconceptions

Misconception How to address it

If the output changed, the model must have changed too. Show that only the input was swapped while the model card stayed identical; the output changes because the model responds to its input.

Every factor changes the output by the same amount. Compare a small input change with a swapped example set;

explain that different factors - input, data, design - affect performance in different ways.

Once the pipeline runs, the output can be trusted without checking.

Model verify-before-trust: hold one output against a trusted answer; a confident output can still be wrong, so every output is checked.

Hands-on activity guide

Overview: Students build a small input-model-output pipeline, trace one item through it, then change one factor at a time and compare how the output behaves - ending with a verify step and a judgement about which factor mattered most.

Grouping / timing: Pairs for the build, trace and compare; whole class for the board share.

Tool: A simulated-AI card deck (a fixed 'model' rule in an envelope, plus a swappable set of example cards) or a teacher-run

prepared example set - fully account-free and device-free, so students test behaviour without entering any personal data.

Facilitation steps:

1. Set up your three spaces: input, model and output.
2. Send one item through and write what happens at each stage.
3. Change one factor - the input, or swap the example set - and send it through again.
4. Compare the two outputs and write what changed.
5. Decide which factor changed the output the most, and explain why.
6. Check one output against a trusted answer before you trust it.

Unplugged fallback: The card deck is already the unplugged version: no screens are required. If an approved live demo is

used instead, the same card routine is the fallback so every pair can still trace and compare.

Lab tool integration (practical, hands-on)

Tool: Google Teachable Machine (train, test, measure)

Teacher setup: Open Teachable Machine (Image Project). Prepare three NON-personal classes (objects or hand gestures)

plus a small held-back TEST set of new examples the model has not seen.

What students do:

1. Name the three classes and add about 15 example images each (objects or hands only).
2. Direct your teacher to press Train (the teacher operates the tool; you observe). Say clearly: the examples are the TRAINING DATA, training builds the MODEL, and if the examples are changed and trained again the result is a NEW, retrained model - not the same one.
3. Run the held-back test examples one by one and record correct / wrong for each.
4. Count the score (for example 7 out of 10) - this is your first accuracy measure.
5. Discuss where it failed and why, and predict one change that would raise the score.

Students produce: A trained model and a first accuracy measure with a note on where it failed.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Unplugged fallback: Run a card-deck 'human classifier' and tally hits and misses on 10 new cards.

Train, then test on new examples, then measure the score.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Trace one item of data through your built pipeline and record each stage.

Input: the word brilliant. Model: the deck matches the word to a category using its example cards. Output: the label positive.

Full marks when the same item is followed through all three stages consistently.

Change one factor, then record how the output changed and why.

When I swapped the example set for one with fewer positive words, brilliant was labelled neutral, because the data the model learned from changed.

Look for a correct comparison and a reason tied to the factor changed, not a random guess.

Say which factor most affected the output and how you would check the result.

Swapping the example data changed the output more than a small input change, so the data matters most here; I would check the output against a trusted answer before trusting it.

Accept any well-reasoned choice of factor with a sensible verify-before-trust check.

- Targets: the same Emerging outcome, applied by tracing and comparing how the system behaves when a factor changes.
- Check: pair prediction plus a change-a-factor comparison, observed during the build.
- Success criteria: the student traces one item through every stage and explains why a changed factor changed the output.
- Evidence: the build-and-trace table and the compare-the-output table on the worksheet.
- Feedback / adjustment: if the comparison is a guess, replay both runs side by side and ask the student to name the one factor that changed.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Builds and traces pipeline and traces one item through it

Builds the pipeline but the trace skips a stage

Needs support to assemble the three stages

Compares behaviour Explains why a changed factor changed the output

Notices the output changed but the reason is unclear

Does not connect the change to a factor

Reasons about performance Judges which factor most affected the output and adds a check

Names a factor but no judgement or check

Does not identify a factor

Representation: show a compare panel (same model, one changed factor, two outputs) so the cause of a changed output is visible, not only described.

Action and expression: accept a spoken trace, a card arrangement or a written table as evidence of the same reasoning; extend confident pairs to change two factors and separate their effects.

People of Determination: pre-build the model stage so the student focuses on changing one factor; provide a comparison frame ('When I changed ... the output ... because ...').

Week 3 - Core | AI Solution Design & Development - Designing & Building |

Interactive lesson: 'Design the Workflow'

Real-World Connection (UAE): UAE traffic-management systems chain several steps (detect -> predict -> decide signal timing) - a structured multi-step workflow like the one students design this week.

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G8·E | 1 x 45 min

Objectives & success criteria

- Develop an AI-powered solution for a specific, clearly defined task using a structured workflow.
- Build the solution hands-on with a teacher-mediated or simulated routine, and explain the reasoning behind each step of the result.
- Use the key vocabulary of AI Solution Design and Development accurately to describe how the parts of the solution connect.

Student-facing success criteria:

- I can define a specific task and design a simple AI-powered workflow to solve it.
- I can build my workflow step by step and explain why each step is there and what it produces.
- I can use words like input, process, output, prompt and workflow correctly to describe my solution.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 7 you learned that an AI system takes an input, runs a process, and returns an output, and that you can

steer a tool by writing a clearer prompt.

Activate: Show a one-line task on the board - 'sort these five recycling items into paper, plastic or metal' - and ask the class to

name the input, the process and the output before any building starts.

Teach: Remind students that a solution is a chain of steps, not a single magic answer - each step takes something in and hands

something on. Draw a three-box input-process-output strip and label it together.

Apply: In pairs, students take a second everyday task (for example, 'greet a visitor in the right language') and quietly label its input, process and output on a mini whiteboard.

Check: Cold-call two pairs to read back their three labels. The rung is secure when a pair can point to the process box and say what it changes the input into.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

An AI-powered solution is a structured workflow: an ordered chain of steps that turns a defined input into a useful output for

one specific task. Designing it means deciding the task, the data the system needs, the AI step that does the transforming

(such as classifying, generating or ranking), and the check that confirms the output is fit for purpose. Building it means

assembling those steps so each one feeds the next. At Emerging level the AI step is teacher-mediated or simulated with

cards, so students focus on the shape of the workflow rather than on operating a live model.

Good design is testable: you

can run an example through, inspect the output, and trace any error back to the step that caused it. This traceability is what

makes a solution improvable rather than a black box.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two versions of the same task

attempt: one where an AI step is

dropped in with no plan, one where it

sits inside a clear input-process-output

workflow. Ask which is easier to fix and why.

Compare the two, vote, and give one reason a planned workflow is easier to trust.

- Which version could you debug, and which is a guess?

Direct teach / 10

min

Define a structured AI solution as task, then input, then AI process step, then output, then check. Model designing one workflow on the board for 'sort

Copy the five-part workflow strip and annotate what each box does.

- What is the one specific task this workflow solves?

short messages as urgent or not urgent', naming the account-free simulated AI step.

Guided design /

10 min

Hand each pair a task card and walk them through drafting their own five-part workflow, stressing no names or personal data go into any step and every output will be checked.

In pairs, draft a workflow for the task card, labelling input, process, output and check.

- Where in your workflow do you verify the output before trusting it?

Build & test / 13

min

Run the simulated AI as a card deck or curated pre-generated output set. Pairs feed a sample input through their workflow and record the output the simulated step returns.

Build the workflow, run one sample input through it, and note whether the output fits the task.

- Did the output match the task, and how do you know?

Check & close /

6 min

Lead a debrief: two pairs trace an output back to the step that produced it. Collect exit tickets and confirm the vocabulary was used correctly.

Explain the reasoning behind one step of their result and hand in the exit ticket.

- Trace one output back to the exact step that made it.

Anticipated misconceptions

Misconception How to address it

AI just gives the whole answer, so I do not need steps in between.

Show that the confident-looking answer came from an unseen chain of steps; make the steps visible and have students point to where an error could enter, reinforcing that a solution is a traceable workflow.

If the AI output looks right, it is right, so there is no need to check.

Give a curated output that is fluent but wrong for the task.

Students catch it only by comparing against the task and a trusted source, proving verify-before-trust is a required step, not an optional one.

A better solution just means a better prompt to the AI. Reframe: the prompt is one step inside the workflow. Improve a

solution by fixing the right box - the input chosen, the process step, or the check - not only the wording sent to the model.

Hands-on activity guide

Overview: Pairs design a five-part workflow for a single specific task, then build and run it once using a simulated, account-free AI step, tracing the output back to its step.

Grouping / timing: Pairs for the design and build; whole class for the trace-back debrief.

Tool: A 'simulated AI' card deck or a teacher-curated pre-generated output set for the classify step - account-free, no login, no personal data - chosen so students focus on workflow shape rather than operating a live model.

Facilitation steps:

1. Read your task card and write the one specific task in a single sentence.
2. Draw the five boxes: task, input, AI process step, output, check.
3. Decide the input - use only non-personal sample data, never real names or faces.

4. Run one sample input through the simulated AI step and write down the output it returns.
5. Check the output against the task and a trusted reference; mark it fit or not fit.
6. If it is not fit, name which box you would change and why, then note it on your sheet.

Unplugged fallback: The simulated AI step is already unplugged - a card deck or paper output set - so the whole activity

runs with no device. If cards are unavailable, a partner acts as the 'model', returning a fixed labelled response so the workflow can still be built and traced.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Label the four working parts of this given workflow strip (input, process, output,

Input = the incoming message; Process = simulated AI classify step; Output = the

Award for correct placement; the

check box is the key discriminator

check) for the task 'sort messages as urgent or not urgent'.

urgent or not-urgent label; Check = compare the label against the message meaning and a trusted rule.

of Emerging mastery.

Write one sentence explaining the reasoning behind the output your workflow produced.

The classify step read my sample message, matched it to the urgent pattern, and returned the urgent label, which fits because the message asked for help today.

Look for a cause naming a specific step, not just 'the AI decided'.

Name one place in any workflow where you must verify before trusting the output, and say why.

At the check step, because a confident output can still be wrong for the task, so it must be compared to the task and a trusted source.

Full marks require both the location (check step) and the reason (confident can still be wrong).

• Targets: the Emerging outcome - develops an AI-powered solution for a specific task as a traceable input-process-output-

check workflow.

- Check: exit ticket plus a live trace-back where a pair links one output to the step that produced it.
- Success criteria: student designs a five-part workflow for one specific task and explains the reasoning behind at least one step using accurate vocabulary.
- Evidence: the completed workflow strip, the recorded sample output, and the one-sentence reasoning on the exit ticket.
- Feedback / adjustment: if a pair skips or misplaces the check step, re-run one curated wrong output so they experience verify-before-trust before the Lab period.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Designs a workflow for a specific task

Lays out all five parts for one clear task with input, process, output and check in order

Lays out most parts but the task or the check is vague

Lists steps with no clear order or missing the task

Explains reasoning behind the result

Explains one step by naming what it took in and handed on

Explains the result in general terms without naming a step

States the output only, with no reasoning

Uses key vocabulary accurately

Uses input, process, output, prompt and workflow correctly in context

Uses some terms, one or two loosely

Terms absent or used incorrectly

Representation: provide the five-box strip pre-drawn with box headings, plus a word bank of the key terms; offer the task read aloud and as an icon strip for readers who need it.

Action & expression: let students show the workflow as a drawn strip, spoken trace-back, or arranged cards - any mode that

reveals the ordered steps counts as evidence.

People of Determination: enlarge the strip and cards, allow a scribe or verbal response for the reasoning sentence, extend build

time, and pair with a peer for the run-and-trace step while keeping the student as the decision-maker.

Week 3 - Lab | AI Solution Design & Development - Designing & Building (Lab) |

Interactive lesson: 'Build, Test, Improve'

Human in the loop: a person decides which factor or input to change and interprets what the output change means - the AI does not choose or judge for you.

Name where the human decides in this Lab.

Retrain vs reuse: where a worksheet step says 'Train', read it as 'ask the teacher to press Train'. Be explicit with students - reusing an already-trained model on a NEW input does NOT change the model; only re-training with new examples does.

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G8·E | 1 x 45 min

Objectives & success criteria

- Extend the Core workflow into a fuller solution and construct it end to end using a teacher-mediated or simulated routine.
- Test the solution against several sample inputs, record where it fails, and explain the reasoning behind each result.
- Use the key vocabulary of AI Solution Design and Development accurately while iterating the build toward a more reliable workflow.

Student-facing success criteria:

- I can build my whole workflow and run more than one sample input through it.
- I can spot which step causes a wrong output and change that step to improve the result.
- I can describe my testing and my fix using the correct vocabulary.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you designed a five-part workflow - task, input, process, output, check - for one specific task

and ran a single sample input through a simulated AI step.

Activate: Ask each pair to hold up their Core workflow strip and say, in one breath, the task it solves and the one output they already tested.

Teach: Introduce testing as running many inputs, not one. Model feeding three different sample inputs through a board workflow

so students see one succeed and one fail, then trace the failure to a step.

Apply: In pairs, students pick two fresh non-personal sample inputs they have not tried yet and predict whether each will pass the check.

Check: A pair is ready to build when they can name a step where they expect a failure and give a reason for their prediction.

Critical & analytical thinking focus

Apply & create - design and build a structured solution/workflow (Apply/Create | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Testing turns a designed workflow into a reliable solution. A single passing example proves little; running a range of inputs

reveals where a workflow breaks - the boundary cases, the ambiguous inputs, the outputs that look confident but fail the

task. Each failure is diagnostic: because the workflow is an ordered chain, a wrong output can be traced to the step

responsible - the input chosen, the process step, or the check applied - and that step can be revised. This is iteration: build,

test against several inputs, locate the failing step, fix it, and re-test. At Emerging level the AI step stays simulated or teacher-

run and account-free, and all sample data is non-personal, so students spend their effort on constructing, stress-testing

and improving the workflow rather than on operating live software. A solution that has been tested and traced is one a

builder can defend and keep improving.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap & set up /

5 min

Restate the same outcome and today's

deeper goal: not to invent a new

solution but to build the Core workflow

fully and stress-test it. Redistribute the

Core strips and the simulated AI cards.

Retrieve their Core workflow

and set out three sample

inputs to test.

- What one specific task does your workflow still solve?

Model testing / 8

min

On a board workflow, run three inputs

live via the simulated step - one clear

pass, one clear fail, one ambiguous -

and trace each output back to its step, naming no personal data was used.

Watch, then mark which of

their own inputs they expect

to pass or fail.

- Which input is ambiguous, and why might it fail?

Extended build &

test / 18 min

Circulate as pairs construct the full workflow and run at least three sample inputs, logging each output as fit or not fit and which step is responsible for any misfit.

Build the full workflow, test three or more inputs, and log every result with the step that caused it.

- For each misfit, which exact box is responsible?

Iterate & improve

/ 9 min

Prompt pairs to change the one step causing most failures, then re-test the same inputs to see whether the output improved, keeping the AI step simulated and account-free.

Revise the failing step, re-run the inputs, and record whether the fix worked.

- Did your fix improve the output, and how do you know?

Trace, close & assess / 5 min

Two pairs present a before-and-after: the failing output, the step they changed, the improved output. Collect the test logs as evidence and confirm accurate vocabulary.

Explain the reasoning behind one fix and hand in the test log.

- Justify your fix by tracing it to a specific step.

Anticipated misconceptions

Misconception How to address it

One test passed, so my whole solution works. Have pairs run an ambiguous or boundary input; when it fails, they

see that reliability is shown by testing a range of inputs, not by a single lucky pass.

If an output is wrong, the AI is broken and there is nothing I can do.

Trace the wrong output back through the chain to a specific box

the student controls - input, process or check - so they see the fix is a design decision, not a dead end.

Fixing means rewriting everything from scratch. Show that iteration changes one step and re-tests. Improving the

responsible box while keeping the rest is faster and lets you tell whether the change actually helped.

Hands-on activity guide

Overview: Pairs build their Core workflow in full, stress-test it against three or more non-personal sample inputs, log each

failure to its step, then change one step and re-test to see if the solution improved.

Grouping / timing: Pairs for the build, test and iterate cycle; whole class for the before-and-after trace-back close.

Tool: The same simulated AI card deck or teacher-curated pre-generated output set as Core - account-free, no login, no personal data - now used repeatedly to return outputs for several sample inputs so the workflow can be stress-tested.

Facilitation steps:

1. Rebuild your five-part workflow so every box is complete and connected.
2. Gather three or more non-personal sample inputs, including one tricky or ambiguous one.
3. Run each input through the simulated AI step and log the output as fit or not fit.
4. For every misfit, write which single box is responsible.
5. Change the box that caused the most failures, then re-run the same inputs.
6. Record whether the fix improved the outputs and explain the reasoning behind one result.

Unplugged fallback: The build and every test run use paper cards or a printed output set, so the Lab needs no device at all.

If cards run short, a partner plays the 'model' and returns fixed labelled responses so testing and iteration still work.

Lab tool integration (practical, hands-on)

Tool: Scratch + Teachable Machine (develop a solution for a task) [WORKFLOW DESIGN]

Teacher setup: Choose one concrete task (for example, sort classroom recycling into paper / plastic / metal). Have the

Teachable Machine model and a Scratch starter ready.

What students do:

1. Agree the task and what a good result looks like (link to your L1 criteria).
2. Use or train a model for the task on non-personal objects.
3. In Scratch, build the solution so an input runs through the model and gives a labelled output.
4. Run it on five real inputs and record the outputs.
5. Note one limitation you already notice, ready to fix next lesson.

Students produce: A working AI-powered solution for a specific task and a first limitations note.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Only non-personal objects.

Unplugged fallback: Build the solution with block-cards and run the human-classifier by hand. An input runs through a model inside a Scratch solution to give a labelled output.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Show a test log of three inputs with two passes and one fail. State which single step is responsible for the fail and why.

The process step is responsible: it returned a not-urgent label for a message that asked for help today, so the classify step mismatched the task; the input and check boxes were correct.

Award only if the student names one specific box and gives a task-based reason, not 'the AI was wrong'.

After changing one step, explain how you know your fix improved the solution.

I re-ran the same three inputs and the previously failing one now passes the check while the others still pass, so the fix improved the output without breaking the rest.

Look for re-testing the same inputs as the evidence, not opinion.

Explain why testing several inputs is more trustworthy than testing one.

One input can pass by luck; several inputs, including a tricky one, reveal where the workflow breaks, so the solution is more reliable when it passes a range.

Full marks require the idea that a range exposes failure a single case can hide.

- Targets: the same Emerging outcome - develops an AI-powered solution for a specific task - assessed through end-to-end building, testing and iteration.

- Check: the test log plus a before-and-after trace-back where a pair links a fix to the step it changed.

- Success criteria: student builds the full workflow, tests three or more inputs, locates a failing step, revises it, and explains the reasoning behind the improved result.

- Evidence: the completed test log showing inputs, outputs, fit judgements, the step changed, and the re-test result.

- Feedback / adjustment: if a pair changes a step but cannot show it helped, prompt them to re-run the identical inputs so the improvement is evidenced before moving on.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Builds and tests the full workflow

Constructs all five parts and tests three or more inputs, logging each result

Builds the workflow but tests only one or two inputs

Workflow incomplete or untested

Locates and fixes a failing step

Traces a wrong output to one step, changes it, and re-tests the same inputs

Identifies a fault but does not re-test after the change

Cannot link a wrong output to a step

Explains iteration with accurate vocabulary

Uses workflow, test, output and step correctly to justify the fix

Uses some terms loosely when explaining the fix

Terms absent or used incorrectly

Representation: provide a pre-formatted test-log table with columns for input, output, fit, and responsible step, plus the key-term word bank from the Core period.

Action & expression: accept the test log as a written table, a set of arranged result cards, or a spoken before-and-after trace - any mode that shows testing and the fix counts as evidence.

People of Determination: supply the enlarged log, allow a scribe or verbal explanation of the fix, extend build-and-test time, and pair for the physical card handling while the student keeps the decision on which step to change.

Week 4 - Core | Critical Evaluation & Informed Interaction - Judging & Improving

Outputs | Interactive lesson: 'Judge with Criteria'

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G8·E | 1 x 45 min

Objectives & success criteria

- Identify limitations in a set of AI outputs, judging each against agreed criteria such as accuracy, relevance and reliability.
- Apply the criteria in a hands-on evaluation task and justify each judgement with clear evidence.
- Use the key vocabulary of critical evaluation accurately, and connect a limitation you find to improving an AI solution.

Student-facing success criteria:

- I can name the limitations in an AI output instead of assuming it is correct.
- I can judge an output against agreed criteria and back my decision with evidence.
- I can use words like criteria, evidence, accuracy and reliability correctly when I explain a judgement.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 7 you learned that an AI workflow has an input, a process and an output, and that a fluent, confident output can still be wrong. You judged simple curated outputs against criteria like accuracy and pointed to evidence.

Activate: Show two curated outputs from the same AI solution on the board, one clearly weaker.

Ask: which one is stronger, and what exactly makes you say so?

Teach: Introduce limitation - a weakness in an output you can name and evidence, not just a feeling. Model judging one output

aloud: name the criterion, point to the evidence, then state the limitation.

Apply: In pairs, students look at one short curated output and name one strength and one possible limitation.

Check: Cold-call three pairs for one limitation each. A rung is secure when students name a limitation and link it to a criterion, not just say it is bad.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Every AI output has limitations - the boundaries of what a model can reliably produce. A model generates outputs by predicting likely patterns from its training data, not by understanding truth, so it can be fluent yet inaccurate, or work well on familiar cases and fail on unusual ones. Evaluation means judging an output against explicit criteria rather than by impression. Useful criteria for an AI solution include accuracy (is it correct against a trusted source?), relevance (does it answer the task?) and reliability (does it hold up across different inputs?). A limitation is not the same as a single slip: it is a weakness you can name, evidence and often see repeated. Locating a limitation, and citing the exact output that shows it,

lets a builder improve the solution rather than trust it blindly.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two curated outputs from the

same AI solution on the board, one with

a clear flaw. Think aloud: I will not

guess which is better - I will agree what

to check first.

Vote on which output is

stronger, then note that a

vote is not evidence.

- What would we need to agree on before we can fairly judge these outputs?

Direct teaching /

10 min

Introduce three agreed criteria:

accuracy, relevance and reliability.

Model naming a limitation - name the

criterion, point to the evidence in the

output, state the verdict. Stress verify-

before-trust against a trusted source.

Copy the three criteria and

follow the modelled

judgement, marking where

the evidence sits.

- Which criterion does this limitation break, and what is the evidence for it?

Guided practice /

8 min

Hand out a curated output set (teacher

pre-generated, account-free, no

personal data). Walk one judgement

with the class using an evaluation grid.

In pairs, complete one row of

the grid together: criterion,

verdict, evidence.

- Is this a one-off slip, or a limitation you would expect to see again?

Independent

hands-on / 14

min

Circulate as pairs judge the remaining curated outputs against the criteria.

Prompt for evidence, not opinion, and for where each limitation sits.

Judge each output against every criterion and write one evidence note per judgement.

- Point to the exact word, fact or step that made you decide that.

Check & close / 7

min

Collect three verdicts on the same output and compare evidence. Confirm the success criteria and preview the Lab evaluation clinic.

Share one judgement with its evidence and self-check against the I can statements.

- Two pairs disagree on the same output - whose evidence is stronger, and why?

Anticipated misconceptions

Misconception How to address it

If the AI sounds confident and the writing is neat, the output must be correct.

Show a fluent curated output containing a clear factual error.

Contrast fluency with accuracy and rehearse verify-before-trust against a trusted source.

Finding a limitation just means saying whether I like the output.

Insist every verdict names a criterion and points to evidence.

Reject I do not like it and re-ask: which criterion, and what is your evidence?

One wrong output means the whole AI solution is broken. Separate a single slip from a limitation. Test the same solution

across a few curated inputs so students see whether the weakness repeats and where it sits.

Hands-on activity guide

Overview: Pairs act as an evaluation panel, judging a curated set of AI outputs for one task against three agreed criteria and recording evidence for every limitation they name.

Grouping / timing: Pairs for the judging; whole class for the agree-the-criteria opener and the evidence debrief.

Tool: A teacher pre-generated output set on paper or the board (account-free, no student logins). Outputs are produced in advance by the teacher on a school-approved, child-safe tool, so no personal data is entered and no student uses a live model.

Facilitation steps:

1. As a class, agree the three criteria you will check: accuracy, relevance and reliability.
2. Read the task the outputs were meant to complete, so you know what a good result looks like.
3. For each curated output, judge it against every criterion and write a verdict: meets, partly or fails.
4. For every verdict, write one piece of evidence - the exact word, fact or step that shows the limitation.
5. Verify one output against a trusted source to test whether a confident answer is actually accurate.
6. Decide whether each limitation is a one-off slip or a repeating weakness, then share your strongest judgement with the class.

Unplugged fallback: If no tool or board display is available, use the printed card set: each card is a pre-written output for the same task, and pairs sort the cards into meets, partly and fails piles with an evidence sticky note on each.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Given one curated output and the three criteria, write a verdict for each criterion with evidence.

Accuracy: fails - it states the Sheikh Zayed Mosque has 40 domes, but the trusted source says 82. Relevance: meets - it answers the task. Reliability: partly - it was correct on an easy input but wrong here.

Award the mark when every verdict names the criterion and cites specific evidence, not opinion.

Explain the difference between a one-off slip and a limitation in an AI output.

A slip happens once; a limitation is a weakness you can see repeat - for example the solution keeps failing on unusual inputs, so reliability is a limitation.

Look for the student distinguishing a single error from a repeating, nameable weakness

tied to a criterion.

Explain why a confident, fluent output can still contain a limitation.

A model predicts likely words, so it can sound sure while stating something false.

That is why we verify before we trust and check across more than one input.

Credit any answer that separates fluency or confidence from accuracy and mentions checking against a source.

- Targets: identifying limitations in AI outputs, judged against agreed criteria with evidence (AIF·CE·G8·E - Identifies limitations in outputs).

- Check: evidence-tagged evaluation grid completed in pairs, plus cold-call verdicts in the close.

- Success criteria: student names each limitation, states the criterion it breaks, and cites specific evidence.

- Evidence: the completed evaluation grids and one written verdict-with-evidence per pair collected at the door.

- Feedback / adjustment: if verdicts lack evidence, re-model point to the exact word or step next lesson before the ranking clinic begins.

Differentiation

Criterion	Emerging (expected)	Approaching	Not yet
Identifying limitations	Names a clear limitation in an output		

Names a weakness but is vague about what it is
--

Cannot yet spot a limitation without prompting
--

Using criteria	Judges against all three agreed criteria
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Uses one or two criteria	Judges by opinion, not criteria
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Evidence	Cites specific evidence for each verdict
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Gives evidence for some verdicts

Gives verdicts without evidence

Representation: provide the criteria as an icon-labelled checklist and read the task aloud.

Pre-teach accuracy, relevance and

reliability with one worked example each, and add bias as an extra card for pairs ready to stretch.

Action & expression: let students record verdicts by ticking a grid, annotating a printed output, or explaining orally to the teacher rather than writing full sentences.

People of Determination: enlarged high-contrast output cards, sentence starters (This fails ... because ...), extra time, and a partner who scribes evidence for a student who reasons aloud.

Week 4 - Lab | Critical Evaluation & Informed Interaction - Judging & Improving Outputs (Lab) | Interactive lesson: 'The Evaluation Clinic'

Human in the loop: a person decides which factor or input to change and interprets what the output change means - the AI does not choose or judge for you.

Name where the human decides in this Lab.

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G8·E | 1 x 45 min
Objectives & success criteria

- Apply the agreed criteria to a wider set of AI outputs, identifying the limitations in each.
- Rank the outputs from best to worst and justify each placement with evidence.
- Weigh accuracy, reliability and bias together to decide which output best serves a task, and propose one improvement to the weakest.

Student-facing success criteria:

- I can apply agreed criteria to a larger, messier set of outputs and name each limitation.
- I can rank outputs from best to worst and justify each placement with evidence.
- I can weigh accuracy, reliability and bias and suggest one way to improve a weak output.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you judged a small curated set of outputs against agreed criteria - accuracy, relevance and reliability - and backed each named limitation with evidence.

Activate: Show the Core criteria and one pair of outputs from last period. Ask a pair to explain which output had the more serious limitation and how they know.

Teach: Model turning separate verdicts into a ranking: when two outputs both partly meet the criteria, use the evidence to decide which limitation matters more, then add bias and reliability as deciding lenses.

Apply: In pairs, students re-order two outputs they judged last period and say what evidence moved one above the other.

Check: Listen for a ranking that rests on evidence and a named criterion, not preference. A rung is secure when the deciding reason is justified with evidence.

Critical & analytical thinking focus

Evaluate - judge outputs and solutions against criteria, with evidence (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Ranking is relative evaluation: instead of judging one output in isolation, you compare several and order them. This is harder because outputs fail on different criteria - one may be accurate but unreliable, another fluent but biased - so you must decide which limitations matter most for the task. Accuracy asks whether the content is correct; reliability asks whether it holds up across different inputs; bias asks whether the solution treats some cases or groups unfairly. These can pull apart: an output can be accurate on one input yet unreliable overall, or clear yet biased. An evaluation clinic weighs these lenses, orders the outputs, and traces each limitation back to where it could be improved in the solution. Justifying every placement with evidence keeps a ranking defensible and repeatable rather than a matter of taste.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap / 5 min Post the agreed criteria and one Core verdict. Frame today: from single verdicts to ranking a wider set of outputs.

Recall the three criteria and one limitation they named last period.

- When two outputs both partly meet the criteria, what decides which ranks higher?

Model the ranking move / 8 min

Demonstrate ordering three outputs: line up their verdicts, then use evidence, reliability and bias to decide the order.

Follow the model and note the moment bias or reliability decided a placement.

- Which limitation matters more here - a wrong fact or an unfair one - and why?

Evaluation clinic / 16 min

Issue the wider curated output set (teacher pre-generated, account-free,

no personal data). Circulate and push for evidence at every placement.

In pairs, judge the wider set, then rank all outputs from best to worst on a ranking strip.

- Point to the evidence that put this one above that one.

Improve the weakest / 10 min

Ask each pair to take their lowest-ranked output and propose one concrete improvement, naming which limitation the fix targets.

Write one improvement and say whether it targets accuracy, reliability or bias.

- Would your improvement change its rank - and can you prove it?

Clinic debrief & close / 6 min

Compare two pairs' rankings of the same set and resolve differences using evidence. Confirm the success criteria.

Defend one placement with evidence and self-check against the I can statements.

- Your rankings differ - whose evidence should the class trust, and why?

Anticipated misconceptions

Misconception How to address it

Ranking is just putting the outputs in the order I prefer. Require each placement to rest on evidence and a named criterion. Ask which piece of evidence moved one output above the next.

The most detailed or longest output is always the best. Contrast a long output that hides a wrong fact with a short accurate one. Use the task and the criteria to decide which limitation matters more.

A biased output is fine as long as the facts are correct. Show a curated output that is accurate but treats some cases

unfairly. Establish that bias is a limitation in its own right and can outweigh a small accuracy gain.

Hands-on activity guide

Overview: Pairs run an evaluation clinic: they judge a wider set of curated outputs against the agreed criteria, rank them

from best to worst with evidence, then propose one improvement to the weakest.

Grouping / timing: Pairs for the clinic and ranking; whole class for the deciding-lens debrief.

Tool: A larger teacher pre-generated output set on cards or a handout (account-free, no student logins). All outputs are

made in advance by the teacher on a school-approved, child-safe tool, with no personal data entered and no live model

used by students.

Facilitation steps:

1. Lay out the wider set of curated outputs and re-read the task they were meant to complete.
2. Judge every output against the agreed criteria - accuracy, relevance and reliability - and add bias as a lens for the trickier ones.
3. Rank the outputs from best to worst on your ranking strip.
4. Beside each placement, write the evidence that put it there; use reliability or bias to decide close calls.
5. Verify the top-ranked output against a trusted source before you confirm its place.
6. Take your lowest-ranked output, name which limitation it fails on, and propose one improvement to the solution.

Unplugged fallback: No display needed - use the printed output cards: pairs physically slide the cards into a top-to-

bottom order on the desk and clip an evidence note to each, then move one card up after writing its improvement.

Lab tool integration (practical, hands-on)

Tool: Pre-generated output packs (find and measure limitations)

Teacher setup: Prepare a pack of 8 outputs for one task, several with clear limitations (a wrong fact, an unclear answer, an

unfair assumption). Print one pack per group.

What students do:

1. Agree the evaluation criteria (reuse the L1 cards).
2. Read each output and mark it correct / unclear / incorrect, naming the limitation.
3. Score the pack: count how many meet the criteria.
4. Sort the limitations into types (wrong fact, unclear, unfair, incomplete).
5. Choose the most common limitation and say what change would reduce it.

Students produce: A scored pack, a limitations-by-type tally, and one targeted improvement idea.

Data-safety: All outputs are teacher-curated; students touch no live tool.

Unplugged fallback: This is already tool-free; add limitation-spotting cards for more examples.

Outputs marked against criteria, limitations sorted into types.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Rank three given outputs best to worst

and justify the top and bottom

placements with evidence.

Best: Output C - accurate and reliable across the sample inputs (evidence: correct 82 domes, correct on all three inputs). Worst: Output A - states a wrong figure and fails on hard inputs (evidence: 40 domes, wrong on two of three).

Award when the ranking is ordered and both named placements cite specific evidence for the limitation.

Two outputs are close. One is slightly less accurate but fair; the other is accurate but biased. Explain which should rank higher for the task and why.

The fair output should rank higher: a small accuracy gap matters less than a bias that treats some cases unfairly, which would harm real users.

Credit reasoning that weighs bias against accuracy for the specific task, not length or fluency alone.

For a low-ranked output, name which limitation it fails on and propose one improvement to the solution.

It fails on reliability - it only works on easy inputs. Fix: add examples of harder inputs when building the solution so it holds up more widely.

Look for the limitation named against a criterion and an improvement that targets that limitation.

- Targets: identifying limitations across a wider output set by ranking with evidence (AIF·CE·G8·E - Identifies limitations in outputs).
- Check: completed ranking strip with an evidence note per placement, plus a proposed improvement for the weakest output.
- Success criteria: student ranks the set, justifies each placement with specific evidence, and names one limitation with a targeted fix.
- Evidence: the ranking strips and the improvement notes collected at the close.
- Feedback / adjustment: if rankings rest on preference, re-run the deciding move with reliability and bias modelled before the

next evaluation task.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Ranking a wider set Orders the full set best to worst Orders most of the set with some gaps

Cannot yet order beyond a pair

Justifying with evidence Cites evidence for each placement Cites evidence for some placements

Ranks by preference, with no evidence

Weighing and improving Weighs accuracy, reliability and bias and proposes a targeted fix

Uses one lens or a vague fix No fix, or a fix unrelated to the limitation

Representation: provide a physical ranking strip with numbered slots and a criteria reminder card that adds bias and reliability.

Model one full ranking before pairs begin.

Action & expression: allow ranking by sliding cards, numbering a list, or explaining the order aloud; evidence may be spoken to the teacher.

People of Determination: fewer, high-contrast cards, sentence starters (I placed this above that because ...), extra time, and a scribe for the improvement note.

Week 5 - Core | Ethics & Governance - Risks, Ethics & Responsible Use |

Interactive lesson: 'Accurate Is Not Ethical'

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Identify the ethical concerns raised by an AI use case, such as bias, privacy loss, lack of transparency or misuse.
- Analyse a scenario for its ethical and governance concerns in a hands-on task, then propose a responsible-use measure and explain the reasoning behind it.
- Use the key vocabulary of Ethics and Governance accurately, and connect an ethical concern to how it would change the way you build or evaluate an AI solution.

Student-facing success criteria:

- I can name the ethical concerns in an AI use case, such as bias, privacy or misuse.
- I can read a scenario, name an ethical concern, and suggest a responsible-use measure with a clear reason.
- I can use words like bias, privacy, transparency and governance correctly, and explain how a concern changes how I would evaluate an AI solution.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier Grade 8 lessons you learned to evaluate an AI solution against clear performance criteria and to improve it through iteration, treating outputs as drafts to be checked rather than finished answers.

Activate: Show one AI solution the class has met before - for example an image sorter or a text helper - and ask: apart from whether it is accurate, what else should we judge it on before letting real people use it?

Teach: Explain that a solution can hit its performance target and still be unfair, invasive or misused. Ethical concerns are extra criteria we must evaluate a solution against, not optional add-ons.

Apply: In pairs, students list one way the shown solution could still harm someone even if its accuracy score is high.

Check: Cold-call two pairs. The prior rung is secure if they can name at least one harm that a high accuracy score would not catch.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

AI ethics is the study of how an AI system can help or harm people, and governance is the set of rules and roles that keep its use responsible. Four concerns recur. Bias: if training data over-represents some groups, outputs can be unfair or stereotyped. Privacy: systems can collect, store or expose personal data such as names, faces or voices. Transparency: users may not know a decision was made by AI or why, so they cannot challenge it. Misuse and over-reliance: a tool built for one purpose is used for another, or an output is trusted without checking. For a Grade 8 solution builder these concerns are evaluation criteria: a solution is only responsible if it performs well and treats people fairly, protects their data, is honest about being AI, and is used as intended. Naming the concern is the first step to designing a measure that reduces it.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Display three short pre-generated

scenarios about AI solutions in a UAE

school or city; each hits its accuracy

target but raises a different ethical

concern - a biased result, a privacy

leak, a hidden AI decision. Read them

aloud without labelling the problems.

Students decide with a

thumb whether they would

deploy each solution and jot

a first hunch about what feels wrong.

- This solution is accurate, so what could still make it unsafe to release?

Teach / 10 min Introduce four ethical concerns - bias, privacy, transparency, and misuse or over-reliance - and frame each as a criterion a responsible solution must pass. Model naming the concern in scenario one and stating who it could affect. Students record the four concerns and a one-line definition of responsible use in their own words.

- What makes an accurate solution still an ethical problem? affect.

Guided practice /
12 min

Hand out curated scenario cards. Work one card together: name the ethical concern, say who is affected, and propose one responsible-use measure with a reason.

In pairs, students annotate the shared card and rehearse the stem An ethical concern here is... so a responsible measure is... because...

- Does your measure actually reduce the concern you named, or just hide it?

Independent / 12
min

Circulate as pairs analyse two more scenario cards, prompting them to justify each measure and to link the concern back to how they would evaluate the solution.

Pairs complete the analysis grid for two cards, naming the concern, who is affected,

and a justified responsible-use measure for each.

- Which evaluation criterion does your concern add to the solution?

Check & close / 5

min

Collect one analysis grid per pair as the exit ticket. Restate that responsibility sits with the human designers and users, guided by governance, not with the tool.

Students write one sentence: the ethical concern they will always check for and why.

- Name one ethical concern and the one responsible-use measure you would never skip.

Anticipated misconceptions

Misconception How to address it

If a solution is accurate, it is automatically ethical and ready to use.

Show an accurate output that is still unfair or invasive. Explain that accuracy is only one criterion; a responsible solution must also be fair, protect privacy, be transparent and be used as intended. Ethics adds criteria, it does not replace them.

Bias means the AI is being unkind or unfair on purpose. Explain that bias usually comes from patterns in the training data, not intent. Use a simple example where one-sided data leads to an unfair result, and separate the idea of a data problem from the idea of blame.

Protecting privacy is just about keeping passwords secret. Reframe privacy as not collecting or exposing personal data such as names, faces and voices in the first place. Responsible design keeps personal data out of the system rather than only guarding it after the fact.

Privacy mechanisms (teacher background)

Beyond the class rule of no personal data, five mechanisms protect data whenever some information must be

kept, aligned to UAE PDPL and ADEK policy: DATA MINIMISATION - collect and keep only the data the task

actually needs. DE-IDENTIFICATION / ANONYMISATION - remove or hide anything that identifies a person

(names, faces, voices, ID numbers) before data is stored or used. DATA RETENTION - decide in advance how long data is kept and delete it when it is no longer needed (in class: clear every trained model and project at the end of the lesson). ACCESS CONTROL - only the people who need the data can reach it (in class: the teacher owns any account or key; students never log in). ENCRYPTION - scrambles stored or sent data so only someone with the key can read it, protecting data 'at rest' and 'in transit'. See the Privacy Mechanisms Note for full detail.

Hands-on activity guide

Overview: Students analyse curated scenario cards to identify the ethical and governance concerns in an AI use case, then propose a responsible-use measure with a reason for each, building toward justified evaluation of a solution.

Grouping / timing: Pairs for the scenario analysis, whole class for the shared worked card and the debrief.

Tool: Curated scenario cards plus a small set of pre-generated AI output samples shown on the board by the teacher only.

This is account-free and teacher-mediated: students never sign in and never enter any data; they reason about outputs that have already been safely produced, and no personal data appears on any card.

Facilitation steps:

1. Read your scenario card carefully and underline the AI use it describes.
2. Decide which ethical concern it raises - bias, privacy, transparency, or misuse and over-reliance - and write it down.
3. Name who could be affected and what the impact on them might be.
4. Decide which evaluation criterion this concern adds when judging the solution.
5. Propose one responsible-use measure and finish the stem ...because... to justify it.
6. Swap cards with another pair and check whether their measure really reduces the concern they named.

Unplugged fallback: No live AI is used at any point. If the pre-generated output samples cannot be displayed, run the whole activity from the printed scenario cards alone; the reasoning steps are identical.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Read a scenario where an AI marking helper scores essays but consistently rates one dialect lower. Name the ethical concern and one responsible-use measure.

Concern: bias, from one-sided training data.

Measure: test the helper on a wide range of writing and add a human-review step before

any score is final.

Award the mark for naming bias and a measure that actually reduces unfairness. A measure that only speeds up marking does not gain the mark.

An AI attendance camera stores photos of every student's face. Name the ethical concern and where the problem sits.

Concern: privacy. The system collects personal data - faces - that it does not need to store. Measure: do not store faces; use a non-personal method or delete images at once.

Full credit requires naming privacy and identifying the collection of personal data as the problem, not only its later security.

Explain how one ethical concern would change the way you build or evaluate an AI solution, using one key term correctly.

Example: transparency means our evaluation must add a criterion that users are told a decision was made by AI and can appeal it, or the solution fails on fairness.

Look for a correct key term (bias, privacy, transparency, governance, responsible use) linked to a specific build or evaluation decision.

- Targets: Identifies ethical concerns in AI use and proposes a justified responsible-use measure (AIF·GE·G8·E, Emerging).

- Check: Pair analysis grid plus a short cold-call justification using the ...because... stem.

- Success criteria: Student names a correct ethical concern, who it affects, and a responsible-use measure with a valid reason linked to how the solution is evaluated.

- Evidence: One completed scenario analysis grid per pair and the closing exit sentence.

- Feedback / adjustment: If pairs name concerns but cannot justify measures, model one more think-aloud with the stem; if justification is secure, extend to a structured evaluation across bias, privacy and misuse.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Identify ethical concern Names the correct ethical concern in a use case and who it affects

Names a concern but not
who it affects

Cannot yet name an
ethical concern in the
use case

Propose measure with
reason

Proposes a responsible-use
measure and justifies it with a clear
because

Proposes a measure but the
reason is vague or missing

Proposes no measure or
one unrelated to the
concern

Use of vocabulary Uses bias, privacy, transparency or
governance accurately and links it
to evaluating the solution

Uses a key term but with a
minor slip

Key terms are absent or
used incorrectly

Representation: provide the four ethical concerns on a reference strip with a symbol and a
one-line definition each, and offer the
scenario cards as both text and a read-aloud so meaning does not depend on reading speed.

Action & expression: allow students to record their analysis as annotations, a spoken
justification, or the written grid, and provide
the ...because... sentence stem as scaffolding.

People of Determination: pre-teach the four concern words with matching symbols, pair a fluent
reader with the student for the
card, and accept a verbal or symbol-supported justification as full evidence.

Week 5 - Lab | Ethics & Governance - Risks, Ethics & Responsible Use (Lab) |

Interactive lesson: 'The Ethics Audit'

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Carry out a structured risk-and-impact audit of a real AI use scenario, sorting its ethical
concerns into bias, privacy and
misuse.

- Rate each concern for likelihood and impact using a simple matrix, then propose a specific
mitigation and explain the
reasoning behind it.

- Use Ethics and Governance vocabulary to justify which mitigation the solution should build in
first, connecting the
concern to the way the solution is evaluated.

Student-facing success criteria:

- I can audit a real scenario and sort its ethical concerns into bias, privacy and misuse.
- I can rate each concern for how likely it is and how much harm it would do, then propose a mitigation with a reason.
- I can argue which mitigation to build in first, using words like bias, privacy, mitigation and governance correctly.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you learned to name the ethical concerns in an AI use case - bias, privacy, transparency, and

misuse or over-reliance - and to propose a responsible-use measure with a reason.

Activate: Recap the Core exit tickets on the board: show two or three responsible-use measures students wrote and ask which ethical concern each one was fixing.

Teach: Introduce the structured audit: an audit does not just spot concerns, it sorts them by type and rates each for likelihood and impact so a team knows which to fix first.

Apply: In pairs, students place one Core concern onto a two-by-two likelihood-and-impact grid and defend where they put it.

Check: Check that each pair can explain why a high-impact, high-likelihood concern is the priority to mitigate.

Critical & analytical thinking focus

Evaluate & justify - reason about risk, ethics and governance (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

A structured risk-and-impact audit turns a list of ethical concerns into an action plan. First the concerns are sorted by type -

bias, privacy, and misuse or over-reliance - so nothing is missed. Then each is rated on two axes: likelihood, how often it is

expected to occur, and impact, how much harm it would cause. Plotting concerns on a likelihood-by-impact matrix reveals

priorities: high-likelihood and high-impact concerns are mitigated first. A mitigation is a specific change to the solution that

lowers the likelihood or impact of a concern, for example testing on diverse data to reduce bias, removing a personal-data

field to protect privacy, or restricting a tool to its intended task to prevent misuse. Governance records who owns each

mitigation and who monitors the residual concern that remains. Auditing teaches Grade 8 builders that responsible use is a

design decision built into a solution and its evaluation criteria, not an afterthought.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Present one real-style scenario: a

school deploys an AI helper that

recommends which students get extra

tutoring. Show two pre-generated recommendation samples, one that quietly favours one group and one that stores personal details it does not need. Students quickly list every ethical concern they can see in the two samples.

- Which of these concerns would matter most if this helper went live unchecked?

Model the audit /

10 min

Model sorting one concern into bias, privacy or misuse, rating it on a likelihood-and-impact matrix, and writing a matched mitigation, thinking aloud about which evaluation criterion. Students copy the audit template and follow along with the modelled concern.

- Why does sorting and rating change which concern we fix first? the fix protects.

Team audit / 15 min

Give each team a scenario pack with pre-generated outputs. Circulate, pushing teams to sort every concern by type, justify each rating, and make mitigations specific to the solution.

Teams audit their scenario: sort concerns into bias, privacy and misuse, rate each on the matrix, and write a mitigation with a reason for each.

- Is your mitigation lowering the likelihood, the impact, or both?

Prioritise & justify / 10 min

Facilitate teams choosing their top-priority mitigation and preparing a 30-second justification using governance vocabulary.

Each team ranks its mitigations and drafts the sentence We would build in... first because...

- Who in the solution's governance would own your top mitigation?

Share & close / 5 min

Two teams present their top mitigation; the class critiques whether it truly reduces the rated concern. Collect the audit sheets.

Students write one sentence on how their audit would change the way the solution is built or evaluated.

- Name the one mitigation you would build in first and the concern it removes.

Anticipated misconceptions

Misconception How to address it

Every ethical concern is equally serious, so we should try to fix them all at once.

Use the matrix to show that concerns differ in likelihood and impact. Explain that a structured audit means prioritising: a high-impact, high-likelihood concern is fixed before a rare, minor one, because time and attention are limited.

A mitigation is just a warning telling users to be careful. Distinguish a real mitigation, which changes the solution to lower likelihood or impact, from a mere notice. Model turning a vague warning into a concrete step such as removing a personal-data field or adding a diverse-data test.

Once we add a mitigation, the ethical concern is gone for good.

Explain that mitigations reduce a concern, they rarely remove it entirely, and governance means someone keeps monitoring the residual concern. Some risk always remains and must still be tracked and owned.

Hands-on activity guide

Overview: Teams run a full structured risk-and-impact audit of a real-style AI use scenario: sorting concerns into bias, privacy and misuse, rating each for likelihood and impact, proposing specific mitigations, and prioritising which to build in first.

Grouping / timing: Small teams of three or four for the audit, whole class for the prioritisation share-out.

Tool: Printed scenario packs with pre-generated AI output samples plus a paper likelihood-and-impact matrix. Everything is teacher-mediated and account-free: no student logs in and no live prompt is sent, and the packs contain no real personal data; teams reason about outputs the teacher has already produced safely.

Facilitation steps:

1. Read your scenario pack and list every ethical concern you can find in the AI use.
2. Sort each concern into a type: bias, privacy, or misuse and over-reliance.
3. Rate each concern's likelihood as low, medium or high, and do the same for its impact.
4. Plot each concern on the likelihood-and-impact matrix.
5. For each concern, write a specific mitigation and name the evaluation criterion it protects.
6. Rank your mitigations, choose the one to build in first with a because reason, and note who would own it.

Unplugged fallback: The audit is already fully paper-based; no live AI is used. If the printed output samples are missing, teams audit from the written scenario description alone using the same categories, matrix and steps.

Lab tool integration (practical, hands-on)

Tool: Card decks (ethics-scenario cards) + pre-generated packs

Teacher setup: Print an ethics-scenario deck and include two outputs that raise an ethical concern (biased, privacy-risky, or misleading).

What students do:

1. Draw a scenario card and read it in your group.
2. Name the ethical concern and who could be harmed.
3. Check the two flagged outputs and label the concern each shows.
4. Decide whether the concern is about fairness, privacy, or honesty.
5. Propose one responsible-use action that would address it.

Students produce: An ethics record: the concern, who is affected, its type, and an action.

Data-safety: Unplugged; outputs pre-curated.

Unplugged fallback: This is the unplugged route; add a privacy-focused scenario card.

An ethical concern traced to who it affects, its type, and a responsible action.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Given three ethical concerns in a scenario, sort them into bias, privacy or misuse and place them on a likelihood-and-impact matrix, then state which is the priority.

Correct sorting and plotting with the high-likelihood, high-impact concern placed top-right and named as the priority to mitigate

first.

Award the mark for defensible sorting and placements and for choosing the top-right concern as priority, with a reason referring to likelihood and impact.

Turn this weak mitigation - tell users to be careful - into a real mitigation for a privacy concern where a tool stores faces it does not need.

Remove the face-storage step so no personal data is collected, or delete images immediately, lowering both likelihood and impact of the privacy harm.

Full credit for a concrete change to the solution that stops collecting the personal data. A restated warning with no design change does not gain the mark.

Explain, using one key term, how bias, privacy or misuse in this solution would change the criteria you evaluate it against.

Example: bias in the training data means evaluation must add a fairness criterion tested across groups, or accurate-looking results stay unfair.

Look for a correct term (bias, privacy, misuse, governance, mitigation) tied to a specific evaluation criterion and a design consequence.

- Targets: Identifies ethical concerns in AI use and audits them structurally to a justified, prioritised mitigation (AIF·GE·G8·E, Emerging, stretching to Proficient).
- Check: Completed likelihood-and-impact audit sheet with concerns sorted by type, plus a 30-second team justification of the top mitigation.
- Success criteria: Team sorts concerns into bias, privacy and misuse, rates each defensibly, proposes a specific mitigation, and justifies a first priority using key vocabulary.
- Evidence: One audit matrix and mitigation sheet per team, and the closing sentence on how the audit changes the build or evaluation.

- Feedback / adjustment: If teams rate concerns but write vague mitigations, model one concrete design fix; if audits are strong, push them to a full structured evaluation of bias, privacy and misuse with a residual-concern owner.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Audit and sort concerns Lists real concerns and sorts each into bias, privacy or misuse, rating for likelihood and impact

Lists concerns but sorts or rates only some

Cannot yet identify the ethical concerns in the scenario

Propose mitigation Proposes a specific mitigation linked to an evaluation criterion, with a clear reason

Proposes a mitigation but it is vague or not linked to a criterion

Proposes no workable mitigation

Prioritise & justify Chooses a top mitigation and justifies it with likelihood, impact and a key term

Chooses a top mitigation but the justification is thin

Does not prioritise or justify a choice

Representation: provide the likelihood-and-impact matrix pre-drawn with worked example placements and a bias-privacy-misuse sorting key, and offer scenario packs as text and read-aloud so the audit does not depend on reading speed.

Action & expression: teams may present the audit as a completed matrix, a labelled poster, or a spoken justification, and the We would build in... first because... stem is provided.

People of Determination: give a colour-free symbol key for low, medium and high and for the three concern types, assign a clear team role such as matrix-plotter, and accept a symbol-supported or spoken justification as full evidence.

Week 6 - Core | Bringing the Strands Together | Interactive lesson: 'One Model, Four Strands'

See the Explainability Note for teacher background on tracing a workflow and the limits of explanation (black-box vs traceable steps).

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E |
1 x 45 min

Objectives & success criteria

- Complete a guided challenge that combines all four strands to evaluate and improve one classifier against a clear performance criterion.
- Explain how the parts of the workflow connect to produce a result and check it against a performance target.
- Identify one responsible-use choice made while completing the task.

Student-facing success criteria:

- I can complete a guided task that uses all four strands to test and improve a model.
- I can explain how the parts of my workflow connect to make a result and check it against a target.
- I can name one responsible choice I made while doing the task.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Earlier this term you learned to build a simple classifier, measure its accuracy by comparing predictions with a set of correct answers, judge that accuracy against a performance criterion, and name a responsible-use rule such as keeping personal data out.

Activate: Show the four strand icons from earlier lessons - understand, build, evaluate, use responsibly - and ask which lesson this term each one came from.

Teach: Explain that today is not a new idea; it joins the four skills you already have into one short evaluate-and-improve task, like clicking four puzzle pieces together on a single classifier.

Apply: In pairs, match four earlier lesson moments to the four strands - AI Conceptual Understanding, AI Solution Design and

Development, Critical Evaluation, and Ethics and Governance - on a quick sorting card.

Check: Each pair reads back one strand and the lesson it came from; the teacher listens for correct matches before moving on.

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Evaluating a model means testing it against a clear performance criterion, not just watching it run. You need a test set -

items whose correct answers are already known - so a prediction can be marked right or wrong.

Accuracy is the count of

correct predictions out of the total, and the performance criterion is the target the model must reach, for example at least

eight correct out of ten. Where the model falls short, error analysis asks which kinds of item it gets wrong so an

improvement can be aimed, not guessed. Improving means changing one thing - a rule, a threshold, an added example -

then re-testing to see whether accuracy rises. Integration means all four strands act on one task: understand the criterion, build and run the check, evaluate honestly, and keep the data non-personal throughout.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min We-do: show two boxes of date photos

already labelled ripe or unripe and ask how we would know if a machine sorted them well; draw the empty test-set to prediction to score frame on the board.

Watch and suggest how to judge the machine; copy the frame.

- How could we prove the machine is good at this, not just fast?

Set the

performance

criterion / 8 min

I-do: label the frame with the four strands and set the criterion aloud - at least eight correct out of ten - explaining that a test set has known answers so each prediction can be marked right or wrong.

Annotate their frame with the four strand labels and write the criterion at the top.

- Why must we already know the correct answers before we can score the model?

Guided evaluate- We-do then you-do: run the simulated In pairs: predict each card, • Your model missed the criterion -

and-improve / 18

min

classifier once on the board with no personal data, then release pairs to run ten test cards, score accuracy against the criterion, find the failing kind, add one rule, and re-run the affected cards. flip it to the true label, tally correct out of ten, compare with the criterion, name the failing type, add one improvement rule, and re-

score.

which kind of item failed most, and

what one rule would help?

Debrief / 8 min Facilitate a whole-class share: collect

before and after scores and one

responsible-use choice each pair

made.

Report their score before

and after the rule and one

responsible choice, such as

using only fruit photos and

no people.

- Did your one change move you

above the criterion, and how do

you know?

Check and close

/ 5 min

Close with the exit ticket; restate that

all four strands ran on one evaluate-

and-improve task.

Complete the exit ticket

individually.

- In one sentence, how did the

parts connect to make a result and

check it against the target?

Anticipated misconceptions

Misconception How to address it

If the classifier runs and gives labels, it must be working

well.

Separate running from performing - the model always outputs a

label. Only comparing predictions with the known true labels and

the criterion tells you if it is good enough. Show a run that scores

five out of ten.

A higher accuracy always comes from a bigger change to

the model.

Show that one targeted rule, aimed by error analysis at the kind of

item that failed, can lift the score - improvement is precise, not

just more effort. Re-run only the affected cards to prove it.

Responsible use is a rule you add at the end, after the

model works.

Point to the data choice made at the input stage - only object

photos of dates, never a person's face - so responsible use

shapes the test set from the start, not just the finish.

Hands-on activity guide

Overview: Pairs evaluate a small date-ripeness classifier against a clear performance criterion, find where it fails, add one improvement, and re-test - a preview of the Term-1 signature activity of evaluating and improving a solution.

Grouping / timing: Pairs for the evaluate-and-improve; whole class for the debrief.

Tool: A simulated AI classifier run as a teacher-mediated card routine - prediction cards with the true label on the back, plus a board demo - account-free, no logins, and no personal data leaves the room. A school-approved, child-safe tool may be teacher-operated on the board instead.

Facilitation steps:

1. Set the target: write the performance criterion at the top of your sheet - at least eight correct out of ten.
2. Run the test set: for each of ten date photo cards showing only fruit, no people, read the classifier's prediction of ripe or unripe.
3. Score it: flip each card to its true label, mark the prediction right or wrong, and count how many out of ten are correct.
4. Compare with the criterion: decide whether the model meets the target of eight out of ten, and note by how much it is above or below.
5. Find and fix: name the kind of date it got wrong most, add one improvement rule, and re-run only the cards that rule affects.
6. Responsible-use note: record one responsible choice you made, such as using only fruit photos and no personal data.

Unplugged fallback: This activity is already unplugged - the classifier is simulated with cards whose true labels are on the back. If a board tool was used instead and access fails, pairs score the ten cards by hand against the true labels, which is the same test, score and improve routine.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write the performance criterion for this task and explain what a test set gives you that a normal pile of items does not.

The criterion is at least eight correct out of ten. A test set has known true labels, so every prediction can be marked right or wrong and the model can be scored.

Award full credit for a correct criterion plus the idea that known answers make scoring possible, not just 'more cards'.

Show your accuracy score and say whether the model met the criterion, then

name one change you made to improve it.
It scored six out of ten, below the criterion of eight; it missed dark ripe dates most, so I added a rule to label dark wrinkled dates as ripe and it rose to eight.

Look for a score, a clear meets or misses judgement against the criterion, and one targeted change - not a vague 'I made it better'.

Name one responsible-use choice you made and say why it mattered.

I used only photos of dates, no people, so no personal data went into the classifier.

Accept any valid choice - no personal data, verify against the true labels, teacher-operated tool - with a reason tied to safety or trust.

- Targets: objectives 1-3 - combining all four strands to evaluate and improve one classifier against a criterion, explaining how the parts connect, and naming a responsible-use choice (Analyse/Create | DOK 3 | SOLO: Relational).
- Check: observe pairs during the evaluate-and-improve and mark the exit ticket and score sheet.
- Success criteria: an accuracy score compared with the stated criterion, one targeted improvement with a re-score, and one responsible-use choice named with a reason.
- Evidence: the pair's score sheet with before and after accuracy and each student's exit ticket.
- Feedback / adjustment: if pairs change a rule without error analysis, re-model naming the failing kind first, then ask them to re-aim their rule and re-score two cards before the close.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Evaluating against the criterion

Scores the model against the true labels and states whether it meets the criterion with guidance

Scores the model but the comparison with the criterion is unclear or only when prompted

Cannot yet score the model against the true

labels without step-by-step help

Explaining how parts connect

Explains that the test set produces predictions that are scored against known answers and a target

Names the parts but the connection to the score or target is unclear

Lists parts with no connection between them

Responsible-use choice Names one responsible choice with a reason

Names a choice but gives no reason

Cannot yet name a responsible choice

Representation: provide the test-to-score frame pre-drawn, the four strand icons, and one worked example card scored right and wrong so pairs see the whole routine before starting.

Action and expression: allow the score to be shown by arranging cards into right and wrong piles, filling the tally, or explaining aloud to the teacher instead of writing.

People of Determination: offer larger high-contrast date cards, a reduced test set of six items with a criterion of five out of six, and a sentence frame for the responsible-use note ('I used ... so no ...'). Give extra processing time for the scoring step.

Week 6 - Lab | Bringing the Strands Together (Lab) | Interactive lesson: 'Run It End to End'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Run the integrated evaluate-and-improve workflow end-to-end on a larger test set, recording every prediction, score and correction made.
- Analyse the results to judge how reliably the model met the performance criterion and where it failed.
- Adapt the model by adding a constraint - a second criterion or a risk rule - and re-evaluate, keeping the responsible-use rules.

Student-facing success criteria:

- I can run my whole evaluate-and-improve workflow from start to finish and write down every score I make.
- I can use my results to say how reliably the model met the target and where it went wrong.
- I can add a new rule or risk to my model and re-test it responsibly.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you evaluated a small date-ripeness classifier against a criterion of eight out of ten, found

where it failed, added one improvement rule, and named a responsible-use choice.

Activate: Show one filled-in score sheet from the Core period and ask which kind of date the class had to fix, and by how much the score changed.

Teach: Explain that today you run the same evaluate-and-improve workflow properly end-to-end on a bigger test set, keep a full

record of every score, then add one constraint and re-evaluate to measure the change.

Apply: In pairs, predict which kinds of date the classifier is most likely to get wrong before you start.

Check: Each pair states one prediction and one measure they will record - accuracy before and after; the teacher confirms both are clear before the run begins.

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Running an evaluation end-to-end means scoring every item in a full test set, not just demonstrating one card. The value for

students is the record they keep: a results log turns 'it seemed to work' into evidence - how many predictions matched the

known labels, which were wrong, and where errors clustered. Patterns in that log, such as dark ripe dates being

misclassified most, are the heart of error analysis and reliability thinking. An improvement is judged by re-testing: accuracy

before against accuracy after shows whether the change actually helped, which is the evaluate-and-improve loop of the

signature activity. Adding a constraint - a second criterion such as bruised or not, or a risk rule such as flag possible mould -

forces students to adapt the model and re-evaluate. Throughout, governance holds: only non-personal fruit data enters the

tool, and every prediction is verified against its true label before it is trusted.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and

predict / 6 min

We-do: restate the workflow and show

the Core score sheet; ask pairs to

predict the trickiest kinds of date.

Recall the stages and write
two prediction guesses.

- Which kinds of data do you
expect the classifier to get wrong,
and why?

Set up the full
run / 7 min

I-do: model logging one card cleanly -
prediction, true label, right or wrong -
and set the test set at twelve to fifteen
cards with the criterion of at least eight
in ten correct.

Set up their results log and
select their larger test set
using fruit photos only.

- What exactly will you write in
each column of your log, and how
will you total accuracy?

End-to-end
evaluation / 16
min

You-do: pairs run all cards through the
simulated classifier, record every
prediction, mark it against the true
label, total accuracy, and compare with

Run the full test set, filling
the log for every card,
flagging wrong predictions,
and calculating accuracy

- You are halfway - how many has
the model got right so far, and
which kind is failing?
the criterion. before any change.

Improve and add
a constraint / 10
min

Facilitate: prompt one aimed
improvement, then introduce one
constraint - a second criterion (bruised
or not) or a risk rule (flag possible
mould) - and ask pairs to adapt and re-
evaluate the affected cards.

Add an aimed rule and a
constraint, adapt the model,

and re-score the cards they affect, recording accuracy after.

- Did your change lift accuracy, and how did the constraint change which cards needed a second look?

Analyse and close / 6 min

Close: pairs report accuracy before and after and one thing the constraint changed; run the exit ticket.

Complete the exit ticket, including their before and after accuracy.

- Out of all cards, how many did the model get right before your change, and how many after?

Anticipated misconceptions

Misconception How to address it

If the evaluation ran without errors, the model must be reliable.

Distinguish 'ran' from 'reliable' - the model always produces predictions; only scoring them against the true labels tells you the accuracy. Point to the wrong predictions in the log as proof.

A few wrong predictions mean the whole model is useless. Introduce reliability as a count - most predictions may be right;

the log shows exactly how many missed, so you can judge against the criterion and improve rather than reject.

The improvement worked because the model feels better after the change.

Insist on evidence - only accuracy before against accuracy after shows whether the change helped. Show a change that left the score the same and needed a different rule.

Hands-on activity guide

Overview: Pairs run the full evaluate-and-improve workflow end-to-end on twelve to fifteen cards, keep a complete results

log with accuracy before and after, then add one constraint and re-evaluate the cards it affects - recording reliability as they go.

Grouping / timing: Pairs for the run and the constraint; whole class for the reliability share.

Tool: The same simulated AI classifier as a teacher-mediated card routine - prediction cards with true labels on the back -

account-free, no logins, no personal data. A school-approved, child-safe tool may be teacher-operated on the board; the card version is the fallback.

Facilitation steps:

1. Select twelve to fifteen date photo cards, each showing only fruit - no names, faces or photos of people.
2. Run every card through the simulated classifier and record its prediction in the results log.
3. Flip each card to its true label, mark right or wrong, and total your accuracy - your score before any change.
4. Compare with the criterion of at least eight in ten, then name the kind of date the model got wrong most.
5. Add one aimed improvement and one constraint - a second criterion (bruised or not) or a risk rule (flag possible mould) - and re-score only the cards they affect, recording accuracy after.
6. Write one sentence on how your change and the constraint altered your accuracy and which cards needed a second look.

Unplugged fallback: The routine is already card-based and unplugged. If a board tool is used and access fails, pairs run the whole test set by hand against the true labels - the same end-to-end evaluation, results log and re-test.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine + Scratch + packs (build, evaluate, improve loop) [WORKFLOW DESIGN]

Teacher setup: Have the unit model, a Scratch build, and one evaluation pack per group ready.

What students do:

1. Run the solution and record its outputs on five inputs.
2. Evaluate the outputs against the criteria and score them.
3. Make one improvement (add cleaner examples, or add a check in Scratch).
4. Re-run the same five inputs and re-score.
5. Record one responsible-use choice you made while improving it.

Students produce: A before-and-after score showing an improvement, plus a responsible-use note.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Reuse only the non-personal model.

Unplugged fallback: Run the whole loop on paper: human-classifier + block-cards + printed outputs.

One loop: build, evaluate, improve, re-test - with the score going up.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Show your completed results log for at least twelve cards with the prediction,

true label, right or wrong, and accuracy totals before and after your change.

A table with twelve or more rows: each card, its prediction, the true label, a right or wrong mark, and two totals - accuracy before and accuracy after the improvement.

Full credit when every card has a scored result and both before and after totals are recorded.

State your model's reliability against the criterion and one place it was weakest.

It got 9 of 15 right before the change, below the criterion of eight in ten set as twelve of fifteen; it was weakest on dark ripe dates.

Look for a count against the criterion plus a specific weak spot, not a vague 'some were wrong'.

Describe the constraint you added and how you re-evaluated the model responsibly.

I added a 'flag possible mould' risk rule, re-scored the cards it affected, and still used only fruit photos - no personal data.

Accept any valid constraint with a re-evaluation and an intact responsible-use rule.

- Targets: the same integrated outcome, deepened - running the evaluation end-to-end, analysing reliability against the criterion, and adapting the model under a constraint (Analyse/Create | DOK 3 | SOLO: Relational).
- Check: review each pair's results log with accuracy before and after and mark the exit ticket.
- Success criteria: a complete log for twelve or more cards, a stated reliability against the criterion with a named weak spot, and one constraint added with the responsible-use rule kept.
- Evidence: the pair's full results log with before and after accuracy and the exit ticket showing both scores.
- Feedback / adjustment: if logs are incomplete, pause the run and re-model one clean scored row; if a change is claimed without evidence, prompt pairs to compare their before and after totals on a specific card.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Running end-to-end and scoring

Runs the full test set and records a scored result for every card with
Runs the set but the log has gaps or missing totals
Cannot yet complete the run and scoring without guidance step-by-step help
Analysing reliability States accuracy against the criterion and names a weak spot
Gives a score but no comparison with the criterion, or a weak spot but no score

Cannot yet judge reliability from the log

Adapting under a constraint Adds a constraint, re-evaluates the affected cards, and keeps the responsible-use rule

Adds a constraint but does not re-evaluate, or drops a responsible-use rule

Cannot yet adapt the model

Representation: provide a pre-formatted results-log table, one completed scored row, and a small menu of constraint options so pairs choose rather than invent.

Action and expression: accuracy can be reported as a count, a fraction, or a quick before-and-after bar; the constraint can be shown by re-sorting cards rather than writing.

People of Determination: allow a reduced test set of ten cards, pre-printed prediction cards, and a sentence frame for the reliability statement ('We got ... out of ...; the criterion was ...; the weakest was ...'). Give extra time for the re-evaluation.

Week 7 - Core | Signature Solution - Planning | Interactive lesson: 'Architect the Plan'

Real-World Connection (UAE): a Masdar City sustainability solution (e.g. monitoring energy or water use) is a good planning exemplar - define the task, the data, the checks and the responsible-use limits before building.

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Plan the signature solution by stating the goal, the ordered structure of steps, and the check steps that measure the output against clear performance criteria.

- Identify the inputs, teacher-operated tools and non-personal data the solution will use.
- Describe one risk in the plan and explain how a check step reduces it.

Student-facing success criteria:

- I can write a clear goal and set out my solution as ordered input-process-output steps with a check step at the end.
- I can name the inputs, the teacher-operated tool and the non-personal data my solution needs.
- I can state one performance criterion, name one risk, and show the check step that measures the output and catches the risk.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 7 you traced input-process-output workflows and placed a check step that tested whether an output was correct. This term your solution must also be measured and improved against clear performance criteria.

Activate: Show a Grade 7 workflow strip - Input, Process, Output, Check - on the board and ask the class what the check step was actually testing.

Teach: Explain the new idea for this year: a good plan does not just check that an output exists, it measures the output against a clear performance criterion, such as how often the label is correct, so the solution can be improved on purpose.

Apply: In pairs, students turn one vague success statement - it works well - into one measurable performance criterion.

Check: Cold-call three pairs; the rung is secure if their criterion is measurable, stating a number or a comparison, not just a feeling.

Critical & analytical thinking focus

Design & plan - architect a structured solution (Create (plan) | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A signature solution is a small multi-step AI workflow that completes one task and is judged against clear performance

criteria. Planning it means designing the architecture before building: a goal, an ordered input-process-output structure,

and check steps that measure the output, not merely confirm it exists. What marks Grade 8 apart is measurable criteria - for

example the share of labels that are correct - so the solution can be evaluated and improved deliberately rather than by

guesswork. The plan names the resources each step needs: inputs, a teacher-operated school-approved tool, and non-

personal data only. It anticipates at least one risk, such as a confident but wrong output, and positions a check step to

catch it. This is design thinking applied to AI: architect first, set the bar, build and measure second.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show two plans for the same task: one

ends at looks good to me, the other ends at a check step that measures the output against a stated performance criterion such as at least 8 of 10 labels correct. Think aloud about which plan you could actually improve.

Students vote with thumbs and give one reason for the plan they trust more.

- Which plan tells you, with a number, how good the output has to be?

Model the plan /

10 min

We-do: fill a planning frame on the board for a shared example task - goal, ordered steps, inputs, tool, non-personal data, one performance criterion
Students copy the headings onto their own blank planning frame and follow along.

- Where does the check step sit, and what number is it measuring against?

criterion, one risk, one check step - narrating each choice.

Guided plan / 14

min

Circulate as pairs draft the plan for their own signature solution, prompting them to keep the process step teacher-operated and to use only non-personal data.

In pairs, students write a goal, order their input-process-output steps, and add a check step tied to a performance criterion.

- What non-personal data does your process step actually need?

Criteria and risk /

10 min

Lead a round: each pair states one measurable performance criterion and one risk; the class suggests where a check step measures the output and catches the risk. Model a fallback as a stretch.

Students add a performance criterion, a named risk and the matching check step to their plan.

- If your check shows the output misses the criterion, what is your plan B?

Check and close

/ 5 min

Collect the planning frames and run the exit ticket; flag any plan whose check step has no measurable criterion for next lesson.

Students complete the exit ticket linking their goal, their criterion and their check step.

- Read me the sentence that proves your plan can measure good output from bad.

Anticipated misconceptions

Misconception How to address it

A plan is finished once the AI produces an answer - the answer is the output, so I am done.

Show a confident but wrong answer; make the point that the answer is one step inside the workflow and the plan must include a check step that measures it against a performance criterion.

Success criteria can just say it works well or it is good. Model turning it works well into a measurable criterion, such as at

least 8 of 10 labels correct; only a measurable criterion lets you evaluate and improve the solution.

More data always makes the plan better, so I should add student names or photos.

Reinforce the no personal data rule: the plan names only non-personal data, and adding personal data is never an improvement; it is a safeguarding failure.

Hands-on activity guide

Overview: Pairs architect the plan for their Term-1 signature solution, moving from a goal to an ordered input-process-output structure with named resources, a measurable performance criterion, one risk and a check step that measures the output.

Grouping / timing: Pairs for the planning build, whole class for the criteria-and-risk debrief.

Tool: A paper planning frame plus a teacher-operated, school-approved demo model shown on the board only; students

never log in, so the routine stays account-free and child-safe.

Facilitation steps:

1. Write your goal in one sentence: what task will your solution complete?
2. Draw three boxes - Input, Process, Output - and write the ordered steps inside them.
3. Under the boxes, list the inputs, the teacher-operated tool and the non-personal data each step needs.
4. Set one measurable performance criterion: how good must the output be, stated as a number or comparison?
5. Add a check step after the output that measures the result against your criterion.
6. Name one risk and draw an arrow from it to the check step that catches it.

Unplugged fallback: If no demo tool is available, run the process step as a simulated AI: a sealed envelope of pre-written response cards a student draws from, so the plan can still be built and measured against the criterion with no tool access.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write a one-sentence goal for a signature solution that sorts photos of school lunches into healthy or less healthy, then list its three ordered steps.

Goal: decide whether each lunch photo is healthy or less healthy. Steps: Input - collect the lunch photos; Process - the teacher-operated classifier labels each photo; Output

Award the mark if the goal names one task and the three steps are correctly ordered as input, process, output.

- a sorted list of the two groups.

State one measurable performance criterion for this solution and the check step that measures it.

Criterion: at least 8 of every 10 photos are labelled correctly. Check: compare a sample of the outputs against a trusted food guide and count how many match.

Full marks require a measurable criterion, a number or comparison, and a check step that counts or compares against it.

Describe one risk in this plan and explain how the check step reduces it.

Risk: the classifier confidently mislabels a healthy lunch as less healthy. Check: measuring a sample against the food guide reveals the mislabels before the result is trusted.

Award the mark if the check tests correctness of the output against the criterion, not merely that the tool ran.

- Targets: the Integration outcome for planning a signature solution - a stated goal, an ordered input-process-output structure, named resources, a measurable performance criterion and a check step that reduces a named risk.
- Check: a marked planning frame plus a one-sentence exit ticket linking the goal, the criterion and the check step.
- Success criteria: the plan states a clear goal, orders the three steps correctly, names non-personal inputs and data, sets a measurable criterion, and includes a check step tied to a named risk.
- Evidence: the collected planning frames and exit tickets, annotated for a present or missing measurable criterion.
- Feedback / adjustment: return plans with a single next-step prompt; any plan whose criterion is not measurable is re-opened at the start of the Lab period.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Goal and structure States a clear one-task goal and orders input, process and output steps correctly.

Goal present but broad, or one step out of order.

No clear goal or steps not in order.

Criterion and resources Sets one measurable performance criterion and names inputs, the teacher-operated tool and non-personal data.

Sets a vague criterion, or

omits the tool or the data type.

No criterion, or names personal data.

Risk and check Names one risk and a check step that measures the output against the criterion.

Names a risk or a check but does not link them.

No risk or check step identified.

Representation: provide the planning frame with the three boxes pre-drawn and labelled, a small word bank of step verbs, and a worked example criterion such as 8 of 10 correct.

Action & expression: let students plan by sketching, by sticky-note cards moved into order, or by dictating steps to a partner scribe; all routes produce the same frame.

People of Determination: offer a partially completed frame with the goal supplied so the learner focuses on ordering steps, setting the criterion and placing the check; allow extra time and a quiet space for the criteria-and-risk round.

Week 7 - Lab | Signature Solution - Planning (Lab) | Interactive lesson: 'Prototype and Pressure-Test'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Prototype the structure of your signature solution by running each planned step by hand and confirming the output reaches the check step.
 - Pressure-test the plan by feeding it a tricky input and measuring whether the check step catches an output that misses your performance criterion.
 - Strengthen the plan by adding a fallback path for when the output fails to meet the criterion.
- Student-facing success criteria:
- I can walk my whole workflow through once, step by step, and confirm the output reaches the check step.
 - I can test my plan with a hard input and use my check step to measure whether the output meets my criterion.
 - I can add a fallback path so my solution does something safer when the output misses the bar.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you planned your signature solution: a goal, an ordered input-process-output structure, named resources, a measurable performance criterion, one risk and a check step.

Activate: Ask each pair to hold up their Core planning frame and read their performance criterion aloud in ten seconds.

Teach: Explain that a plan on paper is a hypothesis; a prototype is a first walk-through that shows whether the chain holds together and whether the check step can actually measure the output against your criterion.

Apply: In pairs, students trace one input along their own chain with a finger and say each step out loud, ending at the check step.

Check: Listen for pairs who reach the output but skip the measuring at the check step; prompt them to finish the walk-through by comparing the output to the criterion.

Critical & analytical thinking focus

Design & plan - architect a structured solution (Create (plan) | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Prototyping is the bridge between a plan and a working solution: you run the planned steps by hand, with a real input, to

reveal gaps the paper plan hid. Pressure-testing goes further by deliberately feeding a hard or misleading input to see

whether the check step actually measures the output against the performance criterion and catches a result that falls

short, rather than waving it through. This is how engineers build confidence in a workflow before scaling it, and how they

decide what to improve. A key move is adding a fallback path - a defined action for when the output misses the criterion,

such as rerouting to a safer step, flagging for a human, or returning to the input - so the solution degrades gracefully instead

of trusting bad output. Throughout, the process step stays teacher-operated and account-free, and no personal data enters

the workflow; the prototype is tested on non-personal example inputs only.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Show a plan that looks perfect on paper

but breaks when a tricky input flows

through it - the check step waves

through an output that misses the

criterion. Think aloud: paper plans can

hide gaps.

Students predict where this

plan will break when a hard

input arrives.

- What input would push your

- output below its criterion?

Model the

prototype walk /

8 min

We-do: take the shared example plan and walk one input through every step on the board, pausing at the check step to measure the output against the stated criterion.

Students set up their own plan and a chosen test input, ready to walk it through.

- At the check step, what number are you comparing the output against?

Prototype run /

12 min

Circulate as pairs run their plan by hand with a normal input, using the teacher-

In pairs, students walk a normal input end to end and

- Did the output actually reach your check step, or did it get stuck? operated demo or the simulated-AI cards for the process step, and note any step that breaks.

mark every step that did not work as planned.

Pressure-test /

12 min

Hand each pair a tricky test input designed to produce a confident output that misses the criterion; support them in measuring it at the check step and logging the result.

Students feed the tricky input through, measure the output against their criterion, and tighten the check if it let a poor output pass.

- Did your check catch the output that missed the bar, and if not, how will you tighten it?

Fallback and

close / 8 min

Lead the fallback move: each pair adds a defined action for when the output misses the criterion, then run the exit

ticket collecting the revised plan.

Students write a fallback path onto their plan and complete the exit ticket describing one fix they made.

- Read me your fallback: what does your solution do the moment the output misses the bar?

Anticipated misconceptions

Misconception How to address it

If my plan looks complete on paper, the prototype will just work.

Show a tidy plan breaking on a real input; make the point that walking the steps by hand is what reveals the hidden gap a written plan cannot show.

Pressure-testing means picking an easy input so my output clears the bar.

Reframe testing as trying to break your own plan on purpose with a hard input; an output that only clears the bar on easy inputs has not really been tested.

A fallback is giving up, so a good plan does not need one. Model a fallback as engineering for failure - rerouting to a safer step or flagging for a human - so the solution stays safe when, not if, an output misses the criterion.

Hands-on activity guide

Overview: Pairs turn their Core plan into a working prototype: they walk a normal input through every step to the check step, then pressure-test with a tricky input, measure it against their criterion, and add a fallback path so the solution handles failure safely.

Grouping / timing: Pairs for the prototype and pressure-test, whole class for the fallback share-out.

Tool: The teacher-operated, school-approved demo model on the board for the process step, or the simulated-AI response cards; students stay account-free and enter no personal data, testing on non-personal example inputs only.

Facilitation steps:

1. Set up your Core plan and choose one normal, non-personal example input to test.
2. Walk that input through every step in order, saying each step aloud, until it reaches the output.
3. At the check step, measure the output against your performance criterion and record whether it passed.
4. Now take the tricky test input and send it through the same chain.

5. Measure the tricky output at the check step; if a poor output slipped through, tighten the check and note the fix.

6. Add a fallback path: write exactly what your solution does the moment the output misses your criterion.

Unplugged fallback: With no demo tool, run the whole prototype on paper: the process step is a simulated AI where a

partner draws a pre-written response card, including one deliberately poor card so the pressure-test and fallback still work with no tool access.

Lab tool integration (practical, hands-on)

Tool: Plan cards + Scratch storyboard (plan an evaluate-improve project) [WORKFLOW DESIGN]

Teacher setup: Provide plan cards (Goal, Inputs, Performance criteria, Evaluate method, Improve step, Risk) and a Scratch storyboard frame.

What students do:

1. Lay out the plan cards and fill each for your signature solution.
2. Write your performance criteria as numbers you can measure.
3. Decide exactly how you will test and score the solution.
4. Name one improvement you expect to make and how you will check it worked.
5. Sketch the solution as Scratch blocks on the storyboard.

Students produce: A measurable plan with a stated evaluate-and-improve method.

Data-safety: Planning only; no live tool or data.

Unplugged fallback: Cards alone are sufficient; the storyboard is optional.

Plan cards including performance criteria and an evaluate-then-improve method.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Walk one normal input through your workflow and record the output at each step. Did the output reach the check step and meet your criterion?

Input photo entered; process labelled it healthy; output listed it in the healthy group; check step compared a sample to the food guide - it reached the check step and met the criterion of 8 of 10 correct.

Award the mark if every step is recorded in order and the walk-through ends by measuring the output at the check step.

Feed the tricky input through and state whether your check step catches an output that misses your criterion.

The blurred photo was confidently labelled

healthy in error; measuring the sample against the food guide showed only 6 of 10 correct, below the criterion, so the check caught it.

Full marks require a clear judgement plus the measure used; a caught shortfall and a tightened check both earn credit.

Write the fallback path your solution follows when the output misses the criterion.

If the output falls below 8 of 10, do not use the labels: send the batch back to the input step for clearer photos and flag it for the teacher to review.

Award the mark if the fallback names a specific safer action rather than simply trusting or discarding the output.

- Targets: the Integration outcome applied practically - prototyping the planned structure and pressure-testing it so the check step reliably measures the output against the performance criterion, extended by a fallback path.

- Check: observed prototype walk-through plus the pressure-test log and the revised plan with a fallback path.

- Success criteria: the pair walks a normal input end to end to the check step, measures a tricky output against the criterion, tightens the check if needed, and adds a specific fallback action.

- Evidence: the annotated prototype log, the corrected check step, and the written fallback path on the revised plan.

- Feedback / adjustment: give one targeted prompt on the weakest step; pairs whose check step let a below-criterion output pass re-run the pressure-test before moving on.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Prototype walk-through Walks a normal input through every step in order and reaches the check step.

Walks most steps but skips one or does not reach the check.

Cannot trace an input through the steps.

Pressure-test Feeds a tricky input and measures

whether the output meets the criterion at the check step.

Runs the tricky input but does not measure it against the criterion.

Only tests an easy input or does not test.

Fallback path Adds a specific safer action for when the output misses the criterion.

Mentions a fallback but it is vague or unsafe.

No fallback path added.

Representation: provide a prototype log sheet with a row per step and a marked column for pass or fail against the criterion at the check step, plus a ready-made tricky test input for pairs who need one.

Action & expression: pairs may prototype by moving physical cards along a printed chain, by narrating each step to a scribe, or by annotating their frame; the recorded log is the same.

People of Determination: supply a pre-set tricky input and a fallback sentence stem - if the output misses the bar, my solution will

... - and allow extra time; a partner can operate the cards while the learner measures the output.

Week 8 - Core | Signature Solution - Building | Interactive lesson: 'The Build Loop'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Build the signature solution by following the agreed plan and structure.
- Test the solution and evaluate its outputs against the performance criteria.
- Make at least one improvement to the structure or inputs based on an evaluation.

Student-facing success criteria:

- I can build my solution step by step by following our plan and structure.
- I can test my solution and judge its outputs against our success criteria.
- I can make one clear improvement and explain what evidence led to it.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier Grade 8 lessons you planned a signature solution: you defined the task, chose the structure (the workflow and its AI step), listed the inputs, and wrote clear performance criteria for a good output.

Activate: Show the class plan from last lesson on the board - the workflow map with the AI step highlighted and the criteria list

beside it. Ask pairs to point to the one step where the AI does its work and the one criterion they think will be hardest to meet.

Teach: Remind students that building means turning the plan into a running routine we can test - not inventing anything new.

Model reading one plan step and stating exactly what the teacher will run at the board and what students prepare as inputs (with no personal data).

Apply: In pairs, students rewrite their next two plan steps as build instructions a partner could follow, and underline the criterion each step is meant to satisfy.

Check: Cold-call three pairs to read one build instruction and name its matching criterion. If a pair cannot link a step to a criterion, the plan rung is not yet secure - fix it before building.

Critical & analytical thinking focus

Create & evaluate - build, evaluate, iterate (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A signature solution here is a small, structured workflow with one AI step - for example, a classifier or a prompt-and-check

routine - that turns defined inputs into an output judged against fixed criteria. Building it is an engineering loop, not a single

act: assemble the steps, run them on test inputs, read the output, compare it to the criteria, then change the structure or

the inputs and run again. The AI step is only one component; most reliability comes from clear inputs and a sound structure

around it. Outputs are judged, never trusted on sight - a fluent, confident answer can still fail a criterion. Improvement is

driven by evidence from testing, so each change should be traceable to a specific gap the evaluation exposed, and every

result checked against a trusted reference.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Teacher shows two board-demo

outputs from the same solution - one

that meets the criteria and one that

does not - without saying which is

which.

Students vote and justify

using one criterion each.

- Which criterion separates a pass

from a fail here?

Build set-up / 8

min

Teacher we-do: assemble the first steps

of the class solution on the board,

stating the account-free tool and that

the AI step is teacher-run with no

personal data entered.

Students set up their build

sheet and prepare non-personal test inputs.

- What input must we never put into the tool, and why?

Run and evaluate / 14 min

Teacher circulates as students run their solution on test inputs (teacher operates any live AI step at the board on request).

Students record each output and score it against every criterion on the build sheet.

- Does a confident-looking output actually meet the criterion, or does it just sound right?

Improve / 12 min Teacher models one evidence-based change - link a failed criterion to a specific fix in structure or input - then re-run.

Students make one improvement, re-run, and log the before-and-after result.

- What exact evidence tells you this change helped?

Check and close / 5 min

Teacher collects build sheets and runs a 30-second share of one improvement per group.

Students write one sentence naming their change and the criterion it targeted.

- If your fix failed, what would you try next and how would you know it worked?

Anticipated misconceptions

Misconception How to address it

If the output sounds confident and detailed, the solution is working.

Show a fluent output that fails a criterion; require students to score against the criteria list, not their gut, and to verify one fact against a trusted source.

Building means adding new features or a cleverer AI until it

looks impressive.

Redirect to the plan: building is following the agreed structure and testing it. Improvements must fix a criterion that failed, not add unrequested extras.

An improvement is any change you make to the solution. Define improvement as an evidence-based change: it must be traceable to a specific evaluation gap and shown to move a criterion from fail toward pass on a re-run.

Hands-on activity guide

Overview: Groups run their planned signature solution on prepared non-personal test inputs, score each output against the shared criteria, then make and re-test one evidence-based improvement.

Grouping / timing: Pairs or trios for the build and evaluation; whole class for the debrief and improvement share.

Tool: The class solution runs as a teacher-mediated, account-free routine: the teacher operates any live school-approved

AI step at the board using a curated, pre-generated output set, while students assemble structure and inputs on a build sheet. This keeps all AI use teacher-run and free of student accounts.

Facilitation steps:

1. Open your build sheet and copy the workflow steps and the criteria list from the class plan.
2. Prepare three test inputs using only non-personal data - no names, faces, or voices.
3. Run the solution on each input (raise a hand for the teacher to operate the AI step at the board) and record the output.
4. Score every output against every criterion: pass or not-yet, with one word of evidence.
5. Pick the criterion that failed most and make one change to structure or input to target it; re-run and log the before-and-after.
6. Write one sentence stating your change and the criterion it was meant to improve.

Unplugged fallback: Simulated-AI fallback: the AI step becomes a paper rule card or a pre-printed output deck. Students

draw the matching card for each input, score it against the criteria exactly as above, and swap one rule card as their improvement - no device or account needed.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Given one solution output and the criteria list, mark each criterion pass or not-yet and cite one piece of evidence.

Criterion 1 pass (output names the required category); Criterion 2 not-yet (missing the location field); Criterion 3 pass (matches the trusted list).

Award the mark for correct

pass/not-yet judgements tied to

evidence, not for opinion.

Confident wording alone is not evidence.

Name one improvement you made and the exact evaluation evidence that led to it.

I added a 'location' field to the input because Criterion 2 failed on two of three test outputs; after the change both passed on re-run.

Look for a traceable link: failed criterion -> specific change -> re-run result. A change with no evidence scores not-yet.

State one thing that must never be entered into the solution and give the reason.

No personal data such as a classmate's name or photo, because the solution uses only non-personal inputs and student

Full marks for a correct rule plus a safeguarding reason (no personal data / teacher-mediated). accounts are never used.

- Targets: whether students can build to plan, evaluate outputs against fixed criteria, and make one evidence-based improvement.
- Check: teacher review of completed build sheets plus a 30-second per-group share of one before-and-after improvement.
- Success criteria: each group runs the solution, scores every output against every criterion, and logs one change traceable to a failed criterion.
- Evidence: the build sheet showing test inputs, per-criterion scores, and a before-and-after result for one improvement.
- Feedback / adjustment: for groups whose change is not evidence-linked, prompt 'which criterion failed and how do you know it improved?' and have them re-run before the close.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Builds to plan Follows the agreed structure to produce a running solution on test inputs.

Builds most steps but needs prompts to keep to the structure.

Cannot yet turn the plan into a running set of steps.

Evaluates against criteria Scores each output against every criterion with evidence, not gut feel.

Scores against some criteria or with weak evidence.

Judges outputs by how confident they sound.

Improves on evidence Makes one change traceable to a failed criterion and shows a re-run result.

Makes a change but the link to evidence or the re-run is unclear.

Changes things at random or adds unrequested features.

Representation: provide the criteria as a checklist with a worked pass/not-yet example, and show the workflow as a labelled map with the AI step highlighted so the build sequence is visible at a glance.

Action & expression: let students record scores by ticking a table, annotating a printed output, or explaining verbally to the teacher; sentence stems support the improvement write-up.

People of Determination: offer a reduced two-criteria version, pre-filled test inputs, a partner scribe, and extra time; for the stretch challenge, pair with a peer to talk through the edge case aloud.

Week 8 - Lab | Signature Solution - Building (Lab) | Interactive lesson: 'The Evaluation Log'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Complete a working version of the signature solution across the full workflow.
- Run a structured evaluation and keep a written log of outputs scored against every criterion.
- Iterate the solution through more than one improvement, each traced to logged evidence.

Student-facing success criteria:

- I can finish a working solution that runs end to end on test inputs.
- I can keep an evaluation log that scores each output against every criterion.
- I can make more than one evidence-based improvement and show the log that justifies each.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you built the first steps of your signature solution, scored some outputs against the criteria, and made one evidence-based improvement.

Activate: Show a group's Core build sheet on the board (personal data removed). Ask: which steps are still unbuilt, and which criterion has the weakest evidence so far?

Teach: Explain that the Lab is extended build time: the aim is a complete, working solution and a proper evaluation log - the same routine as Core, run more times and recorded more carefully.

Apply: In pairs, students list the remaining build steps and set a target of at least two improvements to make and log during the session.

Check: Each pair states its finish line for the Lab (working end to end) and names the two criteria it will focus its improvements on.

Redirect any pair aiming to add new features instead of completing and testing.

Critical & analytical thinking focus

Create & evaluate - build, evaluate, iterate (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

The Lab is dedicated iteration time. A solution is not finished when it first runs; it is finished when it runs end to end and has

been evaluated across a range of test inputs with the results logged. A disciplined evaluation log is the engine of

improvement: for each input it records the output, a pass or not-yet against every criterion, and the evidence. Patterns in

the log - a criterion that fails repeatedly - direct where to iterate next, so changes are prioritised by evidence rather than

guesswork. Handling edge cases matters: a robust solution should fail gracefully on a wrong or unexpected input and the fix

should be documented. Throughout, any AI step stays teacher-mediated and account-free, inputs carry no personal data,

and every output is verified against the criteria and a trusted source before it is trusted.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Re-entry / 5 min Teacher recaps the build loop and sets

the Lab finish line: a working solution

plus a completed evaluation log with at

least two logged improvements.

Students review their Core

build sheet and list the steps

still to build.

- What is the smallest thing left to make your solution run end to end?

Complete the

build / 14 min

Teacher circulates and operates any

live AI step at the board on request,

checking inputs stay non-personal.

Students finish the remaining workflow steps until the solution runs on every test input.

- Does every step from your plan now run, or is one still missing?

Structured
evaluation / 12
min

Teacher models one clean log row - input, output, per-criterion score, evidence - then monitors logging quality.

Students run a range of test inputs and complete the evaluation log fully for each.

- Which criterion is failing most across your logged rows?

Iterate twice / 9
min

Teacher prompts groups to prioritise the most-failed criterion and make Students make at least two improvements, logging a

- Can you point to the log row that justifies each change you made?
- evidence-based changes, re-running each time.

before-and-after for each.

Log review and
close / 5 min

Teacher spot-checks logs and asks two groups to read one edge-case they handled.

Students write a two-line summary: what works now, and the strongest evidence for one improvement.

- If a wrong output slipped through, how would your solution fail safely?

Anticipated misconceptions

Misconception How to address it

The Lab is for adding new, more impressive features to my

solution.

Reset the goal: the Lab completes and tests the planned solution.

New features are out of scope; the win is a working end-to-end solution with a full log.

Once I have made two changes, the log does not really matter.

Show that changes without log rows cannot be justified or reproduced. Every improvement must point to the logged evidence that prompted it and a re-run result.

An edge case is just a rare bug I can ignore for now. Demonstrate a wrong output on an odd input; require students to

make the solution fail gracefully and document the fix, since real solutions must handle bad inputs safely.

Hands-on activity guide

Overview: Extended build time: groups complete their signature solution so it runs end to end, keep a full evaluation log

across several test inputs, and iterate through at least two logged improvements including one edge-case fix.

Grouping / timing: Pairs or trios for the extended build and logging; short whole-class checkpoints at the evaluation and close.

Tool: Same teacher-mediated, account-free routine as Core: the teacher operates any school-approved live AI step at the board with curated outputs while students build structure, inputs, and the log. No student accounts and no personal data are used at any point.

Facilitation steps:

1. List the workflow steps still unbuilt from Core and build them until the solution runs on every test input.
2. Open a fresh evaluation log with a row per test input and a column per criterion.
3. Run at least five test inputs (raise a hand for the teacher-run AI step) and complete every log row: output, per-criterion pass or not-yet, evidence.
4. Find the criterion that fails most across your log and make an evidence-based improvement; re-run and log the result.
5. Make a second improvement, including one that handles a deliberately tricky or edge-case input gracefully; document the fix.
6. Write a two-line status: what now works end to end, and the strongest logged evidence for one improvement.

Unplugged fallback: Simulated-AI fallback: replace the live step with a pre-printed output deck or rule cards. Groups draw the matching card per input, complete the same evaluation log by hand, and swap rule cards as their two improvements - including one card that shows a graceful response to a bad input.

Lab tool integration (practical, hands-on)

Tool: Scratch + improvement-move cards (build and iterate) [WORKFLOW DESIGN]

Teacher setup: Open each group's Scratch project from L7; hand out the improvement-move cards.

What students do:

1. Build the solution following your plan.
2. Run it and score the outputs against your performance criteria.
3. Draw an improvement-move card (change the data, change a step, add a check, change the tool) and apply it.
4. Re-run and re-score to see if the move helped.
5. Log the move, the before score and the after score.

Students produce: A working solution and a log showing at least one evidence-based improvement.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Only the non-personal model is used.

Unplugged fallback: Build with block-cards; apply improvement-move cards by hand.

Pick an improvement move, apply it, and check the score changed.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Show one completed evaluation log

covering at least four test inputs, scored against every criterion.

A four-row log where each row has an input, an output summary, pass/not-yet for all three criteria, and a word of evidence per score.

Award marks for completeness

and consistency: every row

scored against every criterion with

evidence. Missing evidence or

unscored criteria score not-yet.

Point to the two log rows that justify your two improvements and state each change.

Row 2 shows Criterion 2 failing, so I added a location field; Row 5 shows an edge-case crash, so I added a 'no match found' safe response - both passed on re-run.

Look for two changes, each traced to a specific log row and a re-run result. One unjustified change scores approaching, not full.

Describe how your solution now handles a wrong or unexpected input, and where you documented the fix.

On an unknown input it returns a clear 'not recognised, please check the input' message instead of a false answer; the fix is noted in the log's improvement column.

Full marks for a graceful failure plus a documented fix. A solution that still gives a confident wrong answer scores not-yet.

- Targets: whether students can complete a working solution, keep a full evaluation log, and iterate through two evidence-based improvements including an edge-case fix.
- Check: teacher spot-review of the evaluation log plus a short read-aloud of one edge-case each from two groups.
- Success criteria: the solution runs end to end, every log row is scored against every criterion with evidence, and at least two improvements each trace to a log row.
- Evidence: the completed evaluation log showing multiple test inputs, per-criterion scores, and two before-and-after improvements with an edge-case fix documented.
- Feedback / adjustment: for logs with unjustified changes or unscored criteria, prompt 'which row is your evidence?' and have the group complete the missing rows and re-run before the close.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Completes the solution Finishes a solution that runs end to end on all test inputs. or two steps remain unbuilt. through the full workflow.

Keeps a full evaluation log Logs every test input scored against every criterion with evidence.

Logs some inputs or leaves criteria or evidence blank.

No usable log; outputs judged by feel.

Iterates on evidence Makes two or more changes, each traced to a log row, including a graceful edge-case fix.

Makes changes but not all are evidence-linked or the edge case is unhandled.

Changes are random or

add unrequested features.

Representation: provide a pre-formatted evaluation log template and a worked example row, plus the workflow map with the AI step highlighted so students see the full sequence they are completing.

Action & expression: allow logging by typing, ticking a printed table, or dictating rows to a scribe; sentence stems support the two-line status and the edge-case write-up.

People of Determination: offer a shorter log (three inputs, two criteria), pre-built earlier steps so effort goes into evaluation, a partner scribe, and extra time; keep the edge-case task as a guided talk-through with the teacher.

Week 9 - Core | Signature Solution - Refining & Finishing | Interactive lesson:
'Refining and Finishing'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E |
1 x 45 min

Objectives & success criteria

- Finalise the signature AI solution so it meets every agreed performance criterion.
- Evaluate the finished solution for accuracy, usefulness and risk against the shared performance criteria.
- Explain and justify one refinement that measurably improved the solution.

Student-facing success criteria:

- I can finish my AI solution so it meets our agreed performance criteria.
- I can evaluate my solution for accuracy, usefulness and risk against those criteria.
- I can explain one change I made that improved it, and show the evidence it worked.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In earlier Term-1 lessons you designed, built and iterated a signature AI solution and agreed clear performance criteria for it. In Grade 7 you learned to run an input-process-output workflow and to check an output against a trusted source.

Activate: Show two versions of the same solution's output on the board - an earlier iteration and a later one - and ask students to point to exactly which performance criterion improved and by how much.

Teach: Remind students that improving a model or solution means changing one variable at a time and re-measuring it against fixed performance criteria, not rebuilding the whole thing.

Apply: In pairs, students list their agreed performance criteria and tick which are already met, noting the evidence.

Check: Each pair names one criterion not yet met and one small, testable change that could meet it; the teacher listens for a specific, measurable change, not 'make it better'.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Improving an AI solution is a controlled iteration loop: change one variable - usually the prompt, the input data, or one

process step - re-run the solution, then measure the output against fixed performance criteria.

Sound evaluation separates

three qualities. Accuracy: is the output correct and verifiable against a trusted source, and how often. Usefulness: does it

actually complete the intended task for its user. Risk: could it mislead, exclude or expose

anyone. A confident, fluent output

can still be wrong, so verify-before-trust is the habit that matters most. A final risk check asks what happens if the output is

wrong and who could be affected. Because students never enter personal data and never hold accounts, refinement here is

about prompt, data and process quality, and every AI step stays teacher-operated on a school-approved tool.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 6 min Show the earlier vs improved output

pair; model naming one concrete

improvement against a specific performance criterion.

Compare the two outputs

and name which criterion

improved and how you can

tell.

- Which performance criterion

does this change move, and by

how much?

Set the criteria /

8 min

We-do: rebuild the shared

performance-criteria checklist on the

board and define accuracy, usefulness

and risk as measurable checks.

Copy the checklist and mark

which criteria their own

solution already meets, with

evidence.

- How would we know, with

evidence, that this criterion is met?

Refine / 14 min Circulate; coach each pair to change

one variable at a time and re-measure.

Operate any approved AI step at the board on request.

Make one targeted refinement, re-run their solution (or the unplugged version) and record the before and after against the criteria.

- What single variable are you changing, and what result do you expect?

Evaluate and risk check / 12 min

Lead the accuracy-usefulness-risk evaluation; model verify-before-trust against a trusted source.

Score their output against each performance criterion and complete a short risk check.

- If this output were wrong, who could be affected and how would you catch it?

Justify and close / 5 min

Cold-call pairs to justify one improvement citing a check; preview the Lab showcase rehearsal.

State one change that improved the solution and the evidence it worked.

- Which check proves your change actually helped?

Anticipated misconceptions

Misconception How to address it

Changing several things at once improves the model faster.

Show that changing several variables at once hides which change helped or hurt; require one change, then a re-measure against the criteria before the next.

If the AI output sounds confident and neat, it must be correct.

Demonstrate a fluent but wrong output and check it against a trusted source; reinforce verify-before-trust as the rule, not the

exception.

Low risk just means the tool did not crash. Reframe risk as impact on people - could the output mislead,

exclude or expose someone? A working solution can still produce a risky output.

Hands-on activity guide

Overview: Pairs run one final refinement cycle on their signature solution, then evaluate the finished output against the

shared performance criteria and complete a risk check, finishing with a one-line justification of what improved.

Grouping / timing: Pairs for the refine-and-evaluate cycle; whole class for the criteria set-up and the closing justifications.

Tool: A teacher-operated, school-approved AI step run at the board on request (or a pre-generated output set), plus a paper

performance-criteria checklist - account-free, so no student logs in and no personal data is entered.

Facilitation steps:

1. Read your agreed performance criteria aloud and tick the ones your solution already meets.
2. Choose ONE variable to change - the prompt wording, the input data, or one process step.
3. Re-run the solution (your teacher operates any AI step) or run the unplugged version, and keep both the before and after output.
4. Measure the new output for accuracy against a trusted source - do not trust it just because it sounds confident.
5. Complete the risk check: could this output mislead, exclude or expose anyone, and how would you catch it if it were wrong?
6. Write one sentence - the change I made, and the check that proves it improved the solution.

Unplugged fallback: Use the simulated-AI card routine - a pre-written set of earlier and improved output cards - so pairs

can run the full refine, evaluate and risk-check cycle on paper with no tool access.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Tick which of your performance criteria are met and give one piece of evidence for each met criterion.

Criterion 2 (output correct for 9 of 10 test cases) met - evidence: our tally sheet shows 9 correct. Criterion 3 not yet met.

Look for evidence tied to the criterion, not just a tick; a 'met' claim needs observable, measurable proof.

Name one change you made and the check that shows it helped.

I sharpened the prompt to one clear

instruction; the re-run output answered the actual task, which the first missed - checked against the task sheet.

Award if the change is specific

AND linked to a check or evidence, not just 'it looks better'.

Complete the risk check: if your output were wrong, who could be affected and how would you catch it?

If the summary were wrong, a reader could act on false information; I catch it by checking the key facts against the source before sharing.

Look for a named who plus a concrete catch method (verify against a trusted source).

- Targets: the Integration outcome - finalise against the performance criteria and evaluate for accuracy, usefulness and risk

(AIF·CU·G8·E, AIF·SD·G8·E, AIF·CE·G8·E, AIF·GE·G8·E).

- Check: pairs score their finished output against the shared performance criteria and justify one refinement in the closing cold-call.

- Success criteria: the student finishes the solution to meet the agreed criteria, evaluates it for accuracy, usefulness and risk, and explains one improving change with evidence.

- Evidence: the completed performance-criteria checklist, the before/after output pair, and the one-sentence justification.

- Feedback / adjustment: if a pair cannot cite a check, re-model verify-before-trust on one criterion and have them re-test that single change before the Lab.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Meets performance criteria Finishes the solution so all agreed criteria are met, with evidence.

Meets most criteria; evidence is partial.

Several criteria unmet or unchecked.

Evaluates accuracy, usefulness and risk

Judges the output on all three and completes the risk check.

Judges one or two of the three.

Judgement is 'looks fine'
with no check.

Explains an improving
change

Explains one change and cites the
check that proves it helped.

Names a change but the
justification is vague.

Cannot identify a change
or its effect.

Representation: provide the performance criteria as a picture checklist and keep the before/after
outputs side by side; pre-teach
accuracy, usefulness and risk with one icon each.

Action & expression: allow the justification to be spoken, recorded or written; offer a sentence
frame - 'I changed ... and I know it
helped because ...'.

People of Determination: give a reduced two-criterion checklist, extra time and a partner reader;
permit a verbal risk check scribed
by the teacher.

Week 9 - Lab | Signature Solution - Refining & Finishing (Lab) | Interactive lesson:
'Lock It and Show It'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E |
1 x 45 min

Objectives & success criteria

- Run a final iteration of the signature solution and version-lock the build to be showcased.
- Rehearse the showcase demonstration, narrating the workflow and the performance checks run.
- Justify why the finished solution is trustworthy and low-risk, citing specific checks.

Student-facing success criteria:

- I can run one final iteration and choose the version I will show.
- I can rehearse my demo and explain each step of my solution and its checks clearly.
- I can justify that my solution is trustworthy and low-risk using the checks I ran.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you finished your solution against the performance criteria and ran
an accuracy, usefulness
and risk evaluation. You have a before/after output pair and one recorded improvement.

Activate: Play a 30-second model demo (teacher-led or recorded) of a solution being shown
well, then ask students to name the
parts that made it clear and trustworthy.

Teach: Teach the demo spine - state the task, show the input, show the process, show the
output, name one performance check
you ran - all in under a minute.

Apply: In pairs, students draft their five demo sentences using that spine.

Check: Each pair does a 20-second dry run of just the opening two sentences; the teacher checks the task and input are stated clearly.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

A showcase is where a solution is judged in public, so the final iteration focuses on reproducibility and clear

communication rather than new features. Version-locking means choosing one output you can reproduce on demand and

not changing it again - late changes reintroduce untested risk and can quietly move a performance criterion. A good

demonstration narrates the workflow as input, process and output, then shows the evidence: which performance checks

were run and what they proved. Justifying trustworthiness means pointing to specific checks - accuracy verified against a

trusted source, and a risk check naming who could be affected - not simply asserting that the tool works. All AI steps stay

teacher-operated and account-free, and no personal data appears in the demo, so the showcase itself models safe

practice.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Model demo / 6

min

Play a short model showcase; name the demo spine - task, input, process, output, one check.

Watch and list the parts that made the demo clear and trustworthy.

- Which single check made the demo feel trustworthy?

Final iteration /

12 min

Coach pairs through ONE last iteration, then version-lock; operate any approved AI step at the board.

Run one final change, confirm it still meets the performance criteria, then lock the version they will show.

- Is this change worth the risk of re-testing, or should you lock now?

Build the demo /

12 min

We-do a sentence frame for the five-line demo; circulate to tighten wording.

Write and stage their five demo sentences with the before/after evidence to hand.

- Where in your demo do you show the check, not just the result?

Rehearse / 10

min

Run timed pair rehearsals; give one glow and one grow per pair.

Rehearse the full demo

twice, swapping the speaker

- Can a listener restate your workflow and your check after one and checker roles. viewing?

Justify and lock /

5 min

Cold-call the trustworthiness justification; confirm each pair has a locked version.

State why the solution is trustworthy and low-risk, citing two checks.

- Which two checks prove your solution is low-risk?

Anticipated misconceptions

Misconception How to address it

A better demo means adding more features right before we show it.

Show how a last-minute feature reintroduces untested risk and can move a performance criterion; require a version-lock and rehearsal instead of new building.

Rehearsing is just reading the output aloud. Model narrating the workflow and the checks; a demo explains

input-process-output and shows evidence, not only the final answer.

Saying 'it works' proves the solution is trustworthy. Require named checks - accuracy against a source and a risk

check - as the evidence; an assertion is not a justification.

Hands-on activity guide

Overview: Pairs run one final iteration and version-lock their solution, build a five-sentence demo that narrates the

workflow and its performance checks, then rehearse it twice and deliver a trustworthiness justification.

Grouping / timing: Pairs for iteration and rehearsal; brief whole-class rounds for the model demo and the closing justifications.

Tool: A teacher-operated, school-approved AI step at the board (or the reproducible pre-generated output for the locked version), plus paper demo cards - account-free, no student login, no personal data in the demo.

Facilitation steps:

1. Run ONE final iteration and confirm it still meets every performance criterion.
2. Version-lock: choose the output you can reproduce and agree not to change it again.
3. Write five demo sentences - the task, the input, the process, the output, and one check you ran.
4. Lay out your before/after evidence so you can point to what improved.
5. Rehearse the full demo twice, swapping who speaks and who runs the check.
6. Finish with your justification: why the solution is trustworthy and low-risk, citing two checks.

Unplugged fallback: Run the whole showcase on paper using the reproducible output cards and demo cards - pairs

rehearse the same five-sentence demo and justification with no live tool needed.

Lab tool integration (practical, hands-on)

Tool: Teachable Machine retrain + packs (measure, improve, re-measure) [MODEL TRAINING]

Teacher setup: Have each solution and a printed test-set of 10 inputs plus a comparison pack of before/after outputs.

What students do:

1. Run the solution across the 10-input test-set and count correct outputs.
2. Identify the two weakest cases and the limitation behind them.
3. Improve (retrain with cleaner examples, or add a check) targeting those cases.
4. Re-run the full test-set and count correct outputs again.
5. Record the before and after accuracy and confirm no personal data is used.

Students produce: A before-and-after accuracy measure with a completed risk check.

Data-safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand

gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson. Risk check is part of the task.

Unplugged fallback: Use printed inputs/outputs and tally accuracy by hand.

Accuracy measured before and after an improvement, side by side.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Write your five demo sentences (task,

input, process, output, one check).

Task: summarise a news paragraph. Input: the paragraph. Process: an AI summarise step run by our teacher. Output: a two-line summary. Check: we compared the key facts to the source for accuracy.

Award if all five parts are present and the check is a real, named performance check.

Version-lock statement: which output are you showing and why that one?

We are showing the second version because it met all our performance criteria and we can reproduce it; the later draft was untested.

Look for a reproducible choice plus a reason tied to the criteria, not just 'the newest'.

Give your trustworthiness justification citing two checks.

It is trustworthy because we verified the facts against the source (accuracy) and our risk check found only low impact with a way to catch errors before sharing.

Award if trustworthiness is argued from two specific checks, not asserted.

- Targets: the Integration outcome at showcase - finalise, evaluate for accuracy, usefulness and risk, and justify trustworthiness

(AIF·CU·G8·E, AIF·SD·G8·E, AIF·CE·G8·E, AIF·GE·G8·E).

- Check: a timed pair rehearsal plus a cold-called trustworthiness justification citing two checks.

- Success criteria: the student runs a final iteration, version-locks, demos the workflow clearly, and justifies it is trustworthy and low-risk from named checks.

- Evidence: the five-sentence demo script, the version-lock statement, and the recorded or observed justification.

- Feedback / adjustment: if a justification only asserts 'it works', prompt for the specific accuracy and risk checks and have the pair re-run the closing line before showcase day.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Final iteration and version-lock

Runs one final iteration, confirms the criteria, and locks a

reproducible version.

Locks a version but the
criteria check is partial.

Keeps changing; no
locked, reproducible
version.

Clear demonstration Narrates task, input, process,
output and one check in a tight
demo.

Covers most parts; the
check is missing or unclear.

Reads the output only;
the workflow is not
explained.

Justifies trustworthiness Justifies low-risk and trustworthy
citing two specific checks.

Cites one check or the link is
vague.

Asserts 'it works' with no
check.

Representation: provide the five-sentence demo spine as a printed frame with one icon per line;
show a captioned model demo
before pairs draft.

Action & expression: let pairs deliver the demo live, as a recording, or read from cards; offer a
justification frame - 'It is trustworthy
because ... and low-risk because ...'.

People of Determination: allow a three-sentence demo, a partner co-presenter, extra rehearsal
time and a teacher-scribed
justification.

Week 10 - Core | Showcase, Assessment & Reflection | Interactive lesson:
'Showcase and Reflect'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E |
1 x 45 min

Objectives & success criteria

- Present the finished AI solution and justify at least two design or evaluation choices made this term.
- Demonstrate the term's key outcomes across all four strands in a short assessment task.
- State one thing learned this term and set one goal for next term.

Student-facing success criteria:

- I can show my solution to the class and explain two decisions I made, saying why each one improved it.
- I can answer a short task that samples all four strands - concept, design, evaluation and ethics.
- I can name one thing I learned this term and write one clear goal for next term.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Across this term you built an AI solution, set clear performance criteria for it, evaluated it against those criteria, and improved it through at least one round of iteration.

Activate: Show a before-and-after pair of results from an anonymised example solution on the board and ask the class to call out what changed and, more importantly, how we would know it actually got better.

Teach: Remind students that a design choice is only worth showcasing when it is tied to evidence: a criterion, a measurement, and a comparison before and after the change.

Apply: In pairs, students point to one change they made to their own solution this term and say the criterion it was meant to improve.

Check: Cold-call three pairs; the rung is secure if each names a change and links it to a measurable criterion, not just to it feeling better.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

A showcase is not a sales pitch; it is the point where a builder makes their reasoning visible and defensible. By this lesson

students have designed a small AI solution, defined performance criteria such as accuracy on a sample or fairness across

groups, evaluated the solution against those criteria, and improved it by iterating. Showcasing means justifying design and

evaluation choices with evidence: what was measured, what changed, and how the result compared before and after. It

also means honest critical evaluation - naming a limitation and a risk that remain. This integrates all four strands:

conceptual understanding, solution design, critical evaluation, and ethics and governance.

Throughout, all AI use stayed

teacher-mediated and account-free, any model was trained only on non-personal data, and every output was verified

before it was trusted.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Show two thirty-second showcase clips

as scripts on the board: one says the

solution just works, the other names a

criterion, a measurement and a before-

and-after result. Think-aloud about

which builder you believe and why.

Students vote with thumbs

for the showcase they trust more and give one reason.

- Which showcase gives you evidence rather than just a claim?

Showcase presentations /

12 min

Run timed showcases: each pair presents its finished solution and states at least two design or evaluation choices, using the anonymised sample results on screen. Keep to a strict timer and model one sentence of feedback.

Pairs present their solution and justify two choices, each tied to a criterion and a before-and-after comparison; the audience notes one strength and one question.

- What did you measure to know that choice made the solution better?

Four-strand sampling task /

14 min

Hand out the short assessment sampling all four strands and circulate. Remind students it is individual and closed-book, and that any AI mention must respect the no personal data and verify-before-trust rules.

Individually, students answer one item per strand: explain a concept, describe a design improvement, evaluate a limitation, and name an ethics or data safeguard.

- Where in your answer do you show the output was checked, not just trusted?

Structured reflection / 9 min

Guide the reflection frame: one thing learned, one thing that was hard, one goal for next term. Model a specific goal versus a vague one on the board.

Students complete the reflection frame, turning a vague goal into a specific, measurable one for next term.

- Is your goal specific enough that you would know when you had reached it?

Check and close

/ 5 min

Collect the assessment and reflection frames; give a warm, whole-class summary of the term and flag what the Lab peer-review will build on.

Students hand in both sheets and read their next-term goal to a partner in one sentence.

- Read me the one design choice you are proudest of and the evidence behind it.

Anticipated misconceptions

Misconception How to address it

A good showcase just means my solution looks impressive or finished.

Reframe the showcase around evidence: a strong presentation names a criterion, a measurement and a before-and-after result, and it also admits one limitation that remains. Polish without evidence does not count.

Reflection is about saying the term went well, not about being critical of my own work.

Model an honest reflection that names something genuinely hard and one specific goal; make clear that spotting a weakness is a skill being assessed, not a failure.

Because I built it, my solution's outputs can be trusted without checking.

Show a confident but wrong output from an anonymised example; reinforce verify-before-trust - every output, including your own solution's, is checked against the task and a trusted source before it is relied on.

Hands-on activity guide

Overview: Each pair showcases its finished Term-1 solution and justifies two choices, then every student completes a short individual task sampling all four strands and a structured reflection that sets a next-term goal.

Grouping / timing: Pairs for the showcase presentations and audience feedback; individual work for the four-strand task and the reflection.

Tool: The showcase runs on the board using anonymised, pre-generated sample results only; any live model stays teacher-operated and school-approved, so students never log in and the routine remains account-free and child-safe.

Facilitation steps:

1. Set up your showcase: name your solution's goal and the performance criteria you evaluated it against.
2. Present two design or evaluation choices you made, each tied to what you measured before and after.
3. Admit one limitation that still remains, and one risk you would watch in real use.
4. Complete the individual four-strand task: one answer each for concept, design, evaluation and ethics.
5. In your evaluation answer, show how an output was checked against the task, not just accepted.
6. Finish the reflection frame: one thing learned, one thing that was hard, and one specific goal for next term.

Unplugged fallback: If no board demo is available, run the showcase entirely from printed before-and-after result cards

prepared earlier: pairs present from the cards and the four-strand task is answered on paper, so the whole lesson works with no tool access.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Concept strand: in one sentence, explain what a performance criterion measures

A performance criterion is a clear, measurable standard - such as accuracy on a

Award the mark if the answer names something measurable

and why a solution needs one. sample or fairness across groups - that lets you judge whether the solution is doing its job and whether a change made it better. and links it to judging or improving the solution.

Design strand: describe one

improvement you made to your solution

and the evidence it worked.

I balanced the example set so the classifier saw more of the rare category; accuracy on that category rose from a low score to a clearly higher one when I re-tested on the same sample.

Full marks require a specific change plus a before-and-after comparison against a criterion, not just it felt better.

Ethics strand: name one safeguard your solution follows and why it matters.

The model was trained only on non-personal example data - no names, faces or voices - because entering personal data would put real people at risk and is never allowed.

Award the mark if the safeguard is accurate - no personal data, teacher-operated tool, or verify-before-trust - and tied to a reason.

- Targets: the integration of all four Grade 8 strand outcomes - concept, design, critical evaluation and ethics - demonstrated through a justified showcase and a short sampling task.

- Check: timed showcase with two evidence-backed choices, plus a marked four-strand individual task and a structured reflection.

- Success criteria: the pair justifies two choices with before-and-after evidence and names one limitation, and each student answers all four strands correctly with safeguarding respected.

- Evidence: the collected four-strand task sheets, the reflection frames with a specific next-term goal, and audience feedback notes from the showcase.

- Feedback / adjustment: give each student one strand-specific next step; any reflection with a vague goal is reworked into a measurable one at the start of the Lab period.

Differentiation

Criterion Emerging (expected) Approaching Not yet
Justified showcase Presents the solution and justifies two choices, each tied to a criterion and a before-and-after result.

Presents and names two choices but ties only one to evidence.

Presents with no choices

justified by evidence.

Four-strand task Answers all four strands correctly, showing a checked output in the evaluation item.

Answers three strands correctly or omits the verify step.

Answers fewer than three strands correctly.

Reflection and goal States one thing learned and sets one specific, measurable next-term goal.

Reflects but the goal is vague or not measurable.

No clear learning point or goal set.

Representation: provide a showcase script frame with sentence stems - 'One choice I made was ... I measured ... before it was ...

after it was ...' - and a strand-labelled answer grid for the sampling task.

Action & expression: students may present live, from a recorded read-aloud, or by talking a partner through printed result cards; all routes deliver the same two justified choices.

People of Determination: offer a partially completed showcase script with the criteria supplied, allow extra time, and let a partner operate the board results while the learner speaks the justification.

Week 10 - Lab | Showcase, Assessment & Reflection (Lab) | Interactive lesson:

'Review, Measure, Aim'

Strand: Integration | Framework ref(s): AIF·CU·G8·E AIF·SD·G8·E AIF·CE·G8·E AIF·GE·G8·E | 1 x 45 min

Objectives & success criteria

- Give and receive structured peer feedback on a finished solution in a review carousel, judging choices against the performance criteria.

- Use the feedback to sharpen your justification of at least two design or evaluation choices.

- Consolidate your reflections into one specific, measurable Term-2 goal with a first step.

Student-facing success criteria:

- I can review a peer's solution against clear criteria and give one strength and one specific, kind next step.

- I can take the feedback I received and make my justification of my choices clearer and better evidenced.

- I can turn my reflections into a single measurable Term-2 goal and name the first step I will take.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core period you showcased your finished solution, justified two choices, completed the four-strand sampling task, and drafted a next-term goal.

Activate: Ask each pair to hold up their Core showcase sheet and read one justified choice aloud in ten seconds.

Teach: Explain that a carousel is a rotation where you review several solutions in turn, using the same criteria each time so feedback is fair and comparable rather than just an opinion.

Apply: In pairs, students practise the feedback frame on one anonymised example - one strength, one specific next step tied to a criterion.

Check: Listen for feedback that is vague or personal; prompt those pairs to anchor every comment to a named criterion.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

Peer review is how a community of builders raises the quality of everyone's work, and a carousel makes it systematic: each

solution is reviewed by several peers against the same performance criteria, so comparisons are fair and feedback is

evidence-based rather than opinion. Good feedback is specific, kind and criterion-anchored - it names a strength and one

actionable next step - and the reviewer practises the same critical evaluation they apply to their own work. Consolidation

then turns many scattered reflections into one focused, measurable goal with a concrete first step, which is far more useful

than a vague intention. Throughout the carousel, only anonymised, non-personal work is displayed, any live model stays

teacher-operated and account-free, and every claim on a poster is treated as verify-before-trust: a confident result on a

peer's sheet is still checked against its criterion.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook / 5 min Show two peer comments on the board:

one says nice work, the other names a criterion and one specific next step.

Think-aloud about which comment actually helps the builder improve.

Students decide which comment they would rather receive and say why in one sentence.

- Which comment could you

actually act on, and why?

Set up the

carousel / 6 min

Model the review frame and the criteria sheet, and set the rotation rules:

anonymised work only, one strength and one criterion-anchored next step per station, strict timer.

Students lay out their showcase sheet at a station and take a review frame to the first solution they will visit.

- What criterion will you judge this solution against first?

Peer-review

carousel / 18 min

Run three timed rotations, prompting reviewers to judge each solution against the shared criteria and to keep feedback specific, kind and evidence-

In rotation, students review three peer solutions, writing one strength and one specific next step for each,

- Is this a confident claim you should verify against its criterion before you trust it?

based; watch for and redirect any personal or vague comments. always tied to a criterion.

Consolidate into

a Term-2 goal /

11 min

Guide students to gather the feedback they received, spot the pattern across it, and consolidate their Core reflection into one measurable Term-2 goal with a first step. Model turning three comments into one focused goal.

Students read the feedback on their own sheet, sharpen their justification of two choices, and write one

specific, measurable Term-2 goal plus its first step.

- Does your goal name what you will improve and how you would measure it?

Close / 5 min Collect the review frames and consolidated goals; celebrate the term and read the criteria the class will carry into Term-2.

Students hand in their frames and read their Term-2 goal and first step to a partner.

- Read me your Term-2 goal and the very first step you will take toward it.

Anticipated misconceptions

Misconception How to address it

Peer feedback just means saying something nice so no one feels bad.

Model kind but useful feedback: one genuine strength and one specific next step tied to a criterion. Kindness and honesty go together; a comment that cannot be acted on does not help the builder.

I can judge a peer's solution on whether I personally like it. Reframe review around the shared criteria: every comment is

anchored to a named performance criterion, so feedback is fair and comparable rather than a matter of taste.

A confident-looking result on a peer's poster must be correct.

Reinforce verify-before-trust: even a peer's polished claim is checked against its stated criterion; a result that is not measured against the criterion is not yet evidence.

Hands-on activity guide

Overview: Students run a structured peer-review carousel, reviewing three anonymised peer solutions against shared

criteria, then consolidate the feedback they receive into a single measurable Term-2 goal with a first step.

Grouping / timing: Individuals rotating through stations for the carousel; brief pairs for the consolidation coaching, then individual goal-setting.

Tool: A paper review frame and a shared criteria sheet; all displayed work is anonymised and non-personal, and any live model stays teacher-operated and account-free, so students never log in.

Facilitation steps:

1. Place your anonymised showcase sheet at your station and take a review frame and the criteria sheet.
2. At each of three solutions, judge it against one shared criterion at a time.
3. Write one genuine strength and one specific next step, each tied to a named criterion.
4. Before trusting any confident claim on a sheet, check it against the criterion it names.
5. Return to your own sheet and read all the feedback you received; look for the pattern across it.
6. Consolidate that pattern into one measurable Term-2 goal and write the first step you will take.

Unplugged fallback: The carousel is already unplugged by design: it runs entirely on printed showcase sheets, review

frames and the criteria sheet, so it works fully with no tool or account access.

Lab tool integration (practical, hands-on)

Tool: Showcase the improved solution + peer-review cards

Teacher setup: Set up a showcase carousel; print peer-review cards (two stars and a wish, strand-tagged) with a space for the before/after score.

What students do:

1. Demonstrate your solution and show your before-and-after score.
2. Explain two evaluation or improvement choices you made.
3. Visit two other solutions and complete a peer-review card for each, noting their improvement.
4. Read the feedback you received.
5. Write one Term-2 goal based on the feedback.

Students produce: A demonstrated improved solution, peer-review cards, and a Term-2 goal.

Data-safety: Showcase uses only the non-personal solution; no data shared.

Unplugged fallback: Present the paper/block-card version and its hand-tallied scores.

A showcase carousel with peer-review cards noting each solution's improvement.

Formative assessment (aligned to the objectives)

Task / item Model response Marking note

Review this anonymised solution against the accuracy criterion and write one strength and one specific next step.

Strength: accuracy on the common category

is high and clearly measured. Next step:

report accuracy on the rare category too,

because the criterion asks for fairness across groups, not just an overall score.

Award the mark if both comments

are anchored to the named

criterion and the next step is

specific and actionable.

A peer's sheet claims their solution is

ninety-five per cent accurate. What

should you do before trusting that number?

Check what it was measured on and against which criterion: ninety-five per cent on a tiny or unbalanced sample may not be trustworthy, so I would verify the sample before accepting the claim.

Full marks require the verify-before-trust move - questioning what the number was measured against - not simply accepting or rejecting it.

Turn these three reflection notes into one measurable Term-2 goal with a first step.

Goal: improve my solution's accuracy on the hardest category to a set target by re-balancing the example data. First step: count how many examples of that category I currently have.

Award the mark if the goal is specific and measurable and the first step is concrete and doable.

- Targets: the integrated Grade 8 outcomes applied through structured critique - reviewing solutions against criteria, sharpening one's own justification, and consolidating reflection into a measurable goal.
- Check: completed review frames from three rotations plus the consolidated Term-2 goal sheet with a first step.
- Success criteria: each review gives a strength and a criterion-anchored next step, the student's own justification is improved, and the Term-2 goal is specific, measurable and paired with a first step.
- Evidence: the three annotated review frames, the sharpened justification, and the consolidated goal-and-first-step sheet.
- Feedback / adjustment: give one prompt on the weakest review; any Term-2 goal that is still vague is reworked with the teacher before the sheet is filed.

Differentiation

Criterion Emerging (expected) Approaching Not yet

Criterion-anchored

feedback

Gives a strength and a specific next step for each solution, each tied to a named criterion.

Gives feedback but some is vague or not linked to a

criterion.

Feedback is personal,
vague or missing.

Using feedback Uses the feedback received to make
a justification clearer and better
evidenced.

Notes the feedback but
changes little.

Ignores the feedback
received.

Term-2 goal Sets one specific, measurable goal
with a concrete first step.

Sets a goal but it is vague or
has no first step.

No goal or first step set.

Representation: provide a review frame with the criteria pre-listed and feedback stems - 'A
strength against ... is ...', 'One next step
against ... is ...' - and a goal template with a measurement blank.

Action & expression: students may write feedback, dictate it to a scribe, or place criterion cards
against each solution; the
recorded review is the same either way.

People of Determination: shorten the rotation to two stations, supply pre-listed criteria and a
partly drafted goal, allow extra time,
and let a partner read peers' sheets aloud.

Standards & references

This pack is designed against the following international standards for curriculum and
assessment quality.

Reference Used for

ADEK (2026). K-12 AI Fluency Framework (Draft V4). Source of all learning outcomes, strands,
tiers and codes.

AERA, APA & NCME (2014). Standards for Educational and
Psychological Testing.

Validity, reliability, and fairness of assessment.

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Formative assessment that supports learning.

OECD (2005). Formative Assessment. Evidence base for embedding formative assessment.

Anderson & Krathwohl (2001). Revised Bloom's Taxonomy. Cognitive-demand classification of
objectives.

Webb, N. (1997/2002). Depth of Knowledge. Depth of cognitive processing and grading of
demand.

Biggs & Collis (1982). SOLO Taxonomy. Structure of observed learning; anchors success
criteria.

CAST (2018). Universal Design for Learning Guidelines v2.2. Access, representation, and multiple means of engagement.

Reviewer-response addenda (external technical review, Term 1)

Key terms - consistent definitions (use these across all Grade 8 packs)

Model - the trained part of an AI system that has learned patterns from examples and turns an input into a prediction/output. Changing its training examples and training again produces a NEW, retrained (updated)

model - not the same model.

Machine learning (ML) - building a model by learning patterns from a dataset instead of writing fixed rules by hand.

Dataset - the whole collection of examples, split into training data and test data.

Training data / test data - training data are the examples the model learns from; test data are separate, new

examples held back to measure performance. Never test on training examples.

Training / inference / evaluation - training teaches the model from data; inference is one forward run that

produces an output from an input; evaluation measures output quality against criteria using the test data.

Process - a single step, or set of steps, that acts on an input (e.g. the model step, a translate step, a check step).

Workflow / pipeline - the planned, ordered sequence of steps (processes) that turns an input into an output. In

these materials 'pipeline' and 'workflow' mean the same thing.

Output - the direct result a single run produces (a label, number, sentence or image).

Outcome - the real-world effect of using the output (e.g. a parent receives a correct heat-safety alert). An output

can be good while the outcome is still poor if the workflow around it is weak.

Forward vs loop - one prediction runs forward (input -> model -> output); development runs in loops of testing,

evaluation and retraining, so AI building is not linear.

Prerequisite knowledge (maths & computing) - required before this unit

Students are assumed to be able to: (1) sequence steps and follow/trace a workflow

(computing); (2) calculate

simple percentages and read them as accuracy, e.g. correct / total x 100 (maths); (3) compare two results and

describe the difference (maths); (4) run a controlled/fair test by changing ONE factor while keeping all other test

conditions the same (science/computing). Use the Week 1 Prerequisite Primer to activate or reteach any gap;

the Bridging course rebuilds these before the main unit.

Label each activity by what students actually improve (Design & Creation)

MODEL TRAINING = changing the training examples and training again -> a new/retrained model (e.g. Teachable Machine train/retrain labs). WORKFLOW DESIGN = adding or reordering steps; the model is unchanged (e.g. Scratch multi-step solutions). PROMPT / INPUT REFINEMENT = changing the input given to the model; the model is unchanged (e.g. change-one-factor labs). INTERFACE DEVELOPMENT = changing how a user interacts with the solution. State the label in each lab header so students know whether they are changing the model or the surrounding workflow.

Fitting the Teachable Machine + Scratch labs into 45 minutes (Accessibility)
Combined train-build-test-reflect labs are too long for one period. Use ONE of these: (a) provide a PRE-TRAINED Teachable Machine model and a STARTER Scratch file so students go straight to testing and improving; (b) split setup and evaluation across two prepared stations (Station A trains/sets up, Station B evaluates); or (c) divide the activity across two periods. The ESSENTIAL task is: run new test examples -> measure accuracy -> make one improvement -> re-measure. Extra classes, extra test cases and interface polish are optional/extension.

Reduce over-scoped test sets (e.g. 6 test cases, 2 per class, instead of 10) when time is short.

Educational tools, not production platforms (Industry Relevance)
Teachable Machine and Scratch simulate key stages of professional AI development (classify, train, test, prototype) but are not full production platforms. Real industry systems use more advanced tools for large-scale training, deployment, monitoring and security. Present them as an honest introduction to the concepts.

Technical sources of bias (name the mechanisms in L5)
Go beyond 'one-sided data'. Name at least: (1) training-data/sampling bias; (2) selection bias; (3) label/measurement bias; (4) design/threshold bias. Each can produce unfair or unreliable outputs. Full procedure in the Fairness and Bias Technical Note.

Fairness evaluation step - add to the Signature Solution rubric (L7 planning sheet)
Require every student to test whether their workflow performs consistently for at least TWO input types or user groups. They record each group's accuracy or error count under the SAME test conditions, note any performance disparity, and propose a mitigation if a disparity is found. 'Cannot determine' is a valid result when

the evidence is too limited. Full method and rubric wording in the Fairness and Bias Technical Note.

Explainability note (use in L2/L3)

Structured, traceable workflows are more EXPLAINABLE than black-box models: each output can be traced to

the step that produced it, which makes the system auditable and trustworthy. This traceability is a design

advantage. See the Explainability Note.

Model record card (Future Readiness)

From L2 onward, have students complete the one-page Model Record Card (data source, purpose,

version/changes, known limitations, test results, and checks needed after the system is put into use). This

builds the habit of recording provenance and monitoring performance after deployment.

Cross-curricular labels for lesson headers (Interdisciplinary Connections)

Add these visible labels to lesson headers so subject coordinators can plan across subjects:

Science link - heat

safety (heat-safety notices, sports-day weather alerts); Language link - Arabic-English

translation and reading-

level (two-audience reply workflow); Maths link - accuracy percentages and before-and-after comparison;

Science link - fair test (change one factor). Spam filtering and litter-sorting classifiers connect to computing/science classification.

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder

Model Record Card

A one-page model-card template. Complete one per model/solution; keep with the project.

Fill in every row. Use non-personal data only. The teacher operates any live tool; students direct, observe and record.

Field Record

Solution / model name

Made by (student/group) and date

Purpose - what task is it for?

Data source - where did the training examples come from?

Dataset - number of classes and examples per class (balanced? duplicates removed?)

Test set - how many separate new examples? (never training examples)

Model version / changes made (original, or retrained - what changed?)

Test results - accuracy % and results BY class/group

Fairness check - two groups compared, any

disparity found?

Known limitations - where does it fail or become unreliable?

Responsible-use / privacy note - no names, faces, voices; cleared after lesson?

Checks needed after it is put into use (monitoring) - what to re-test, and when

Why 'monitoring after use' matters:

A model that works today can drift as conditions change (new items, new lighting, new users). Recording where the data and model came from, and scheduling a re-test, is how real teams keep a deployed system trustworthy over time.

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder

Privacy Mechanisms - Note

Emerging-level de-identification, data minimisation, retention, access control and encryption.

Aligned to UAE PDPL and ADEK policy.

Why privacy mechanisms matter

The Grade 8 rules already forbid names, faces and voices. This note adds the mechanisms that protect data even when some information must be kept, so students understand privacy as design, not just a 'do not' rule.

Key mechanisms

Data minimisation

Collect and keep only the data the task actually needs. If the litter classifier only needs the photo of the item, it does not need who took it or where. Less data collected means less that can leak.

De-identification / anonymisation

Remove or hide anything that identifies a person before data is stored or used - names, faces, voices, ID numbers. In these labs we go further and never collect them at all, using objects, gestures, drawings or neutral text.

Data retention

Decide in advance how long data is kept and delete it when it is no longer needed. In class: clear every trained model and project at the end of the lesson.

Access control

Only the people who need the data can reach it. In class: the teacher operates and owns any account or key; students never log in. Real systems use passwords, roles and permissions.

Encryption (preview of next-level content)

Encryption scrambles stored or sent data so that only someone with the key can read it. If a device is lost, encrypted data stays unreadable. Introduce it as the mechanism that protects data 'at rest' and 'in transit'.

UAE PDPL and ADEK policy

- UAE Personal Data Protection Law (PDPL, Federal Decree-Law No. 45 of 2021) sets out principles of lawful, minimal, purpose-limited processing, data-subject rights, and secure handling of personal data. These labs stay clearly inside it by collecting no personal data at all.

- ADEK policy: all school AI use is teacher-mediated and account-free for students; no personal data (names, faces, voices) enters any tool. Confirm every tool against the ADEK/school approved-tools list before use.

Classroom checklist

1. Minimise: is any personal data actually needed? If not, do not collect it.
2. De-identify: use objects/gestures/neutral text; never names, faces or voices.
3. Retain: clear models and projects at the end of the lesson.
4. Access: teacher owns accounts/keys; students direct only.
5. Protect: keep any teacher files on approved, secured (ideally encrypted) storage.

Grade 8 (Year 9) | Phase 3 / Cycle 2 | AI Solution Builder

Tooling Guide - Practical Kit for Term 1

Teachable Machine · Scratch · Pre-generated output packs · Card decks

TOOLING GUIDE

This guide equips the Grade 8 practical labs. The kit is deliberately small so it works on one teacher device and a

projector, keeps every student account-free, uses no personal data, and always has an unplugged fallback.

Confirm each tool against your school / ADEK approved-tools list and test it on the classroom network before the lesson.

How the kit maps to the unit

The Grade 8 unit is 'Evaluating & Improving AI Models'. Teachable Machine lets students train, test and MEASURE a

model; Scratch lets them build and iterate a solution; pre-generated output packs let them find limitations and score

quality safely; card decks make criteria, ethics and improvement moves concrete with no device at all.

Golden rules (every tool, every lesson)

- The teacher operates any live tool; students direct it but never log in.
- No personal data - never faces, voices or names. Use objects, gestures, drawings or neutral text.
- Measure, then improve, then re-measure: change one thing at a time and check the score.
- Verify before you trust: check every output against the task and a source.
- Every activity has an unplugged fallback; clear any trained model or project at the end of the lesson.

Google Teachable Machine

What it is: A free, browser-based tool to train and test a simple image / sound / pose classifier with no account.

Training and testing happen on the device and the samples do not leave it.

What it teaches: Grade 8's core move: train a model, TEST it on new examples, MEASURE how often it is right, then

improve the data and re-measure.

Access & accounts: teachablemachine.withgoogle.com - teacher-operated on the board. No login. Students direct it; they do not sign in.

Safety: Teachable Machine trains in the browser and the samples stay on the teacher's device. Use objects, hand gestures or drawings only - never faces, voices or names. Clear the project at the end of the lesson.

Used in: Main L2, L3, L6, L9; Bridging W1, W3, W5.

Unplugged fallback: A card-deck 'human classifier' scored by hand: agree a rule, test it on new cards, tally hits and misses.

Scratch (with optional Machine Learning for Kids)

What it is: A free block-based environment for building multi-step AI-powered solutions. An offline app means no

account is needed; ML for Kids can plug a trained model into a project (teacher-managed key).

What it teaches: Lets students BUILD a solution for a specific task and then add checks and improvements to it.

Access & accounts: scratch.mit.edu or the offline Scratch app. Account-free for the base tool.

ML for Kids

(machinelearningforkids.co.uk) needs a teacher-managed setup - check your approved-tools list first.

Safety: No personal data in projects. If connecting a model, reuse the non-personal Teachable Machine model from the same unit.

Used in: Main L3, L6, L7, L8; Bridging W1, W2, W5.

Unplugged fallback: Paper 'block cards' plus 'improvement-move cards': build and iterate the solution on the desk.

Pre-generated output packs

What it is: Sets of AI outputs the teacher prepares BEFORE the lesson - strong and weak versions of the same task,

or an output plus its known errors - printed or shown on the board.

What it teaches: The heart of Grade 8 evaluation: find limitations, measure quality against criteria, and decide what

to improve, with zero live-tool risk.

Access & accounts: Prepared in advance by the teacher (using the school-approved assistant on the teacher

device) and printed. Students never touch a tool.

Safety: Teacher curates every output in advance, so nothing unexpected or personal reaches students.

Used in: Main L4, L5, L6, L9; Bridging W3, W4, W6.

Unplugged fallback: This IS the offline route - it needs no live tool.

Card decks

What it is: Printed card sets: evaluation-criteria cards, limitation-spotting cards, ethics-scenario cards, bias-source

cards, plan cards and improvement-move cards.

What it teaches: Makes evaluation criteria, limitations, ethics and improvement moves concrete and fully

unplugged; the default fallback for every AI activity.

Access & accounts: School-printed. No devices, no accounts. Always available.

Safety: Fully offline; no data of any kind involved.

Used in: Main L1, L5, L7, L8; Bridging W4, W6 (and the fallback everywhere).

Unplugged fallback: This IS the unplugged tool.

Lab-by-lab tool map

Lab Primary tool Students produce

Bridging W1 - Lab Teachable Machine / Scratch (workflow designs affect outcomes)

A two-design comparison explaining how design affects outcome

Bridging W2 - Lab Scratch (refine a workflow's structure and sequencing)

A refined Scratch workflow and an explanation of the improvement

Bridging W3 - Lab Teachable Machine retrain + packs (improve through iteration)

A three-iteration score record explaining the impact of each

Bridging W4 - Lab Card decks (bias-source cards) + packs (reduce risks / responsible use)

A risk-reduction record: risk, cause, who is affected, and a

Bridging W5 - Integrated Teachable Machine + Scratch (integrated: design, build, first measure)

A built solution with a first measured score and a target likelihood

Bridging W6 - Integrated Pre-generated packs + improvement-move cards (integrated: improve, re-measure, justify)

A before-and-after score, a justified improvement, a risk check

Main L1 - Lab Card decks (evaluation-criteria cards) A ranked criteria layout and two measurable

checks the class

Main L2 - Lab Google Teachable Machine (train, test, measure)

A trained model and a first accuracy measure with a note on

Main L3 - Lab Scratch + Teachable Machine (develop a solution for a task)

A working AI-powered solution for a specific

task and a first

Main L4 - Lab Pre-generated output packs (find and measure limitations)

A scored pack, a limitations-by-type tally, and one targeted

Main L5 - Lab Card decks (ethics-scenario cards) + pre-generated packs

An ethics record: the concern, who is affected, its type, an

Main L6 - Lab Teachable Machine + Scratch + packs (build, evaluate, improve loop)

A before-and-after score showing an improvement, plus a response

Main L7 - Lab Plan cards + Scratch storyboard (plan an evaluate-improve project)

A measurable plan with a stated evaluate-and-improve method.

Main L8 - Lab Scratch + improvement-move cards (build and iterate)

A working solution and a log showing at least one evidence-based

Main L9 - Lab Teachable Machine retrain + packs (measure, improve, re-measure)

A before-and-after accuracy measure with a completed risk check

Main L10 - Lab Showcase the improved solution + peer-review cards

A demonstrated improved solution, peer-review cards, and a T

Setup checklist (before the term)

- Bookmark Teachable Machine and test the webcam on the teacher device; prepare non-personal example sets and held-back test sets.
- Install the offline Scratch app (or confirm scratch.mit.edu loads); save the starter projects.
- Pre-generate and print the output packs for the evaluation labs.
- Print the six card decks (see the Designer Image Brief, Section D, for full contents).
- Confirm all four tools against the approved-tools list; note any that need a teacher-managed key (e.g. ML for Kids).

Recommended AI assistant, versions and minimum requirements (Resource Adequacy)

Recommended school-approved AI assistant: use the ADEK/school-approved general assistant on the teacher

device only (for example a current, licensed version of a mainstream assistant such as ChatGPT, Microsoft

Copilot or Google Gemini - confirm the exact product and version against your school's approved-tools list

before the term). Students never log in; the teacher prepares outputs in advance.

Minimum requirements for teacher board demonstrations: a current version (updated within the last 12 months)

of Google Chrome, Microsoft Edge or Safari; a stable internet connection for Teachable Machine and

scratch.mit.edu (the offline Scratch app removes this need); a working webcam/microphone for Teachable

Machine image/sound demos; a device with at least 4 GB RAM and a projector or board.

Teachable Machine

version: current web build at teachablemachine.withgoogle.com. Scratch version: Scratch 3 (online or offline app).

Distribution: this Tooling Guide is distributed as part of the packaged Teacher Pack set for Grade 8, not as a stand-alone document.

Retrain vs reuse - keep tool instructions consistent (Technical Clarity)

The teacher operates every live tool; students direct, observe and record - they never click or log in. Where a worksheet step says 'Train', read it as 'ask the teacher to press Train'. Be explicit with students about what

changes: reusing an already-trained model on a new INPUT does NOT change the model (same model);

changing the TRAINING EXAMPLES and training again produces a NEW, retrained/updated model. When

Scratch reuses the unit's Teachable Machine model, that is reuse (same model), not retraining.

Machine Learning for Kids - setup and permissions checklist (optional)

Before using ML for Kids (machinelearningforkids.co.uk): 1) Confirm it is on your school/ADEK approved-tools list. 2) Teacher creates and manages the account and API key - students never see or enter keys. 3) Check the classroom network allows the site. 4) Pre-create the project and connect the non-personal model in advance. 5)

Use no personal data (no names, faces or voices). 6) Fallback: if access, permission or setup is not confirmed,

use standard Scratch (with the unit's Teachable Machine model) or the unplugged card-deck 'human classifier' -

the lesson never depends on ML for Kids.

Educational tools, not production platforms (Industry Relevance)

Teachable Machine and Scratch are EDUCATIONAL tools. They simulate key stages of professional AI

development - classifying, training, testing and prototyping - so students learn the concepts. They are not full

production AI platforms. Real industry systems use far more advanced tools for large-scale training,

deployment, monitoring, security and version control. Present these tools as an honest introduction to the

ideas, not the finished professional workflow.

Technical sources of bias (bias mechanisms note - use in L5 or here)

Bias is not only about one-sided training data. Name at least these mechanisms for students:

(1) Training-data /

sampling bias - the examples do not represent everyone the system will serve, so a whole group is under-

served. (2) Selection bias - how the data was chosen skews it (e.g. only daytime photos). (3)

Label /

measurement bias - human labels or the way something is measured are wrong or unfair. (4)

Design / threshold

bias - a design or cut-off choice (e.g. the confidence threshold) favours some cases over others.

Each

mechanism can produce unfair or unreliable outputs even when the code is correct. See the

Fairness and Bias

Technical Note for the full procedure.

Sample Core Lesson (without images attached)

From 'It Works' to 'How Well'

You already know an AI turns an input into an output. This unit asks the harder question: is that output any good, and how could we make the model better? | About 45 minutes

What you will learn

- 🕒 I can ask at least two good questions about how AI turns an input into an output and how we judge that output.
- 🕒 I can sort examples that relate to evaluating and improving AI models, and say why.
- 🕒 I can help write our unit success criteria and our safe, responsible-use rules for AI tools.

Lesson route

Work through each part in order.

- 1 Ask the harder question**
Not 'did it work' but 'how well'
- 2 Sort output, judge, improve**
Tell the three moves apart
- 3 Agree our criteria and rules**
Fair measures and safe use

Full lesson · Week 1 · Core

From 'It Works' to 'How Well'

Read the whole lesson, then try the practice below and complete your worksheet.

◀ Previously

Welcome to Term 1 - you know AI turns input into output; now we ask the bigger question: is it any good?

Next ▶

Next in the Lab: put two real outputs side by side and use evidence to name the winner.

 Concept link

Add a half-line reassuring pupils that 'model' and 'workflow' get fully unpacked in the very next lesson.

 I can...

I can ask at least two good questions about how AI turns an input into an output and how we judge that output.

I can sort examples that relate to evaluating and improving AI models, and say why.

I can help write our unit success criteria and our safe, responsible-use rules for AI tools.

Words to know

output - what the AI system gives back after it processes your input

criteria - the clear measures we agree on to judge whether an output is good, such as accuracy or usefulness

evaluate - to judge something fairly against the criteria, not just by how it looks

improve - to change the input or the process on purpose so more outputs meet the criteria

responsible use - using AI safely and fairly, with no personal data and always checking the result

Today we launch a new unit. You already know an AI turns an input into an output - now we ask the harder question: is that output any good, and how could we make the model better?

From 'it works' to 'how well does it work'

An AI system still follows the shape you know: input, process, output. This unit adds a new job on top. Instead of just asking 'did it give me something back', you ask 'is what it gave me good enough for the task'. To answer that fairly you need criteria - clear measures like accuracy, usefulness and fairness that you agree on before you judge. Judging against criteria is called evaluating. It turns a gut feeling into a decision you can explain and defend to someone else.

Judge it, then improve it

Once you can evaluate an output, you can improve the system behind it - but be precise about WHICH part you improve. You can improve the INPUT (a clearer prompt or cleaner example data - the trained model itself does not change); you can improve the WORKFLOW (add or reorder steps, such as a check step - the model still does not change); or you can improve the MODEL itself by changing its training examples and training again, which produces a new, RETRAINED (updated) model. So a clearer input improves the output, a better-designed workflow improves the outcome, and only retraining changes the model itself. Improving does not mean nagging the AI until it gives a nicer answer. It means changing something on purpose - a clearer input, a better example, a tuned step in the process - and then re-testing against the same criteria to see if the output really got better. Keep the charter in mind: no real names, faces or voices go into any tool, an adult runs it for us, and a confident-looking output still has to be checked before you trust it.

Lesson figure

The workflow still runs input to output - now we add a step that judges the output against clear criteria.

Why it matters

Why it matters - across the UAE, teams evaluate and improve AI every day: tuning an Arabic-English translator for government service desks, checking a smart-parking predictor in Abu Dhabi, or testing a recycling sorter until it meets a clear accuracy target; learning to judge and improve outputs helps you understand and question the tools around you.

Let's try it - hands-on

Watch the class stimulus and note one thing you noticed and one thing you wonder.
Turn your wondering into two clear questions about judging or improving AI outputs.
Sort each example card as an output, a way to judge, or a way to improve, with your partner.
Write one reason for each card you sorted.
Add one word to the vocabulary wall and suggest one safe-use rule.

 Think about it

Think about it - if two outputs both look neat, how would you decide fairly which one is actually better for the task?

 My reflection

My reflection: One question I still have about evaluating or improving AI models is ...

 Cross-curricular link · Social Studies / Moral Education

Co-writing the class success criteria and responsible-use charter mirrors drafting a shared code of conduct in Moral Education: everyone agrees clear, fair rules in advance so any decision can be judged against them and defended, not made on personal feeling.



Tool for this week

Unplugged - simulated-AI cards (teacher-led)

This week: a 'simulated AI' paper routine

This is an unplugged concept lesson: the teacher runs a simulated-AI card routine on the board (no live tool, no logins).

Students reason about pre-prepared example outputs on the cards - no personal data is used.

Use the worksheet to sort or judge the cards and explain the thinking with the key vocabulary.

The live tools (Teachable Machine and Scratch) are used in the matching Lab period.

Cards

input

process

output

Sort / match /

sequence

record on worksheet,

give the reason



Practice · interactive

Try it yourself

Three quick activities and case check-ins to lock in the lesson.

Part 1 of 3

How well does it work?

What turns a gut feeling into a decision?

1Figure 1. The workflow still runs input to output - now we add a step that judges the output against clear criteria.

Figure 1. The workflow still runs input to output - now we add a step that judges the output against clear criteria.

Judge against criteria

An AI system still follows the shape you know: input, process, output. This unit adds a new job on top. Instead of asking 'did it give me something back', you ask 'is what it gave me good enough for the task'.

To answer that fairly you need criteria - clear measures like accuracy, usefulness and fairness that you agree on before you judge. Judging against criteria is called evaluating. It turns a gut feeling into a decision you can explain and defend to someone else.



Which are good evaluation questions?

Objective 1 · ask sharp questions about outputs

Do it

Tap the THREE questions that judge how WELL an output does the task.



Does the output look neat and colourful on screen?



Do I simply like this one more than the other?



How accurate is this output when I check it against a trusted source?



Did it answer quickly, so it must be right?



Does it work as fairly for one group as for another?



Does this output actually do the task it was asked to do?



AI Evaluators · Case 1

The Two Translations

“

A UAE service desk has two AI translations of the same notice. Both read smoothly on screen. Evaluator, how should the team decide which one is better?”



same task



two outputs



one measure

How should the team decide fairly?

A

Pick whichever output simply looks the neatest on the screen.

B

Ask the team which one they personally happen to like the most.

C

Judge each output against a clear criterion, such as accuracy.

D

Choose the longer translation, assuming more text means effort.

Part 2 of 3

Output, judge, improve

Which part are you actually changing?

2Figure 2. Three different moves: making an output, judging an output, and improving the system behind it.

Figure 2. Three different moves: making an output, judging an output, and improving the system behind it.

Improve on purpose

Once you can evaluate an output, you can improve the system behind it - but be precise about WHICH part you improve. You can improve the input (a clearer prompt or cleaner example data - the model does not change); improve the workflow (add or reorder steps, such as a check step - the model still does not change); or improve the model itself by changing its training examples and training again, which produces a new, retrained model.

Improving does not mean nagging the AI until it gives a nicer answer. It means changing something on purpose, then re-testing against the same criteria to see if the output really got better.



Output, judge, or improve?

Objective 2 · sort the three moves and say why

Do it

Sort each card: is it an output, a way to judge an output, or a way to improve the model?



A translated notice the tool produced



Adding a check step to the workflow



Counting how many of 10 answers are correct



Comparing the output to a trusted source



The label 'recyclable' that the sorter returns



Rewriting the prompt to be clearer



Adding better training examples and retraining



An output



A way to judge



A way to improve



AI Evaluators · Case 2

The Clearer Prompt

“

A team reworded a messy prompt into a clear one and the output improved. They did not retrain anything. Evaluator, which part did they actually improve?”



new prompt



same model



better output

Which part did the team improve?

A

Change the input - a clearer prompt, with the model unchanged.

B

Change nothing, since none of these three stages really moved.

C

Change the model itself, assuming it was retrained on new data.

D

Change the output directly, by editing the answer text by hand.

Part 3 of 3

Our criteria and rules

What makes a criterion fair, and use safe?

Measure it, and stay safe

A fair success criterion is something you can actually measure against the task - like 'how many of 10 answers are correct' - not just 'it looks nicer'. Agreeing these before you judge keeps everyone honest.

Keep the charter in mind: no real names, faces or voices go into any tool, an adult runs it for us, and a confident-looking output still has to be checked before you trust it.



Fair measure, or gut feeling?

Objective 3 · agree criteria we can measure

Do it

Sort each idea: is it a criterion we can measure, or just a gut feeling?



Whether it works as well for different groups



Whether it sounds confident



Whether the output feels impressive



Whether the output does the exact task asked



How many of 10 answers match a trusted source



Whether the colours look nice



Fair measure



Gut feeling



AI Evaluators · Case 3

The Name in the Box

“

A student wants to type a classmate’s real name and photo into a tool to test it out. Evaluator, what should our responsible-use charter say?”



no personal data



adult runs it



always check

What should the charter say here?

A

No real names, faces or voices go into any tool you might use.

B

Enter any data you like, since the tool keeps it private anyway.

C

Trust confident outputs and skip checking them for any mistakes.

D

It is fine to enter a classmate's photo if the output convinces.



Away from the screen · Hands-on mission

AI Evaluators: launch the unit

Watch a curated stimulus, then sort example cards into outputs, ways to judge, and ways to improve - before building the class vocabulary wall and safe-use charter.

What you need:

A class stimulus (board demo or printed pair)

A curated card set (simulated-AI cards)

A vocabulary wall space

A pencil

Watch the stimulus and note one thing you noticed and one thing you wonder.

Turn your wondering into two clear questions about judging or improving AI outputs.

With your partner, sort each card as an output, a way to judge, or a way to improve.

Write one reason for each card - why does it belong there?

Add one word and its meaning to the class vocabulary wall.

Propose one responsible-use rule for our safe-use charter.

 No board demo? Describe one worked pair of outputs aloud, name the task, and have pairs sort the printed cards - no device needed.

Lesson summary

You launched the unit: from 'it works' to 'how well does it work'.

Key ideas to remember

Evaluating means judging an output against clear criteria, not by first impression.

Making an output, judging it, and improving the system are three different moves.

You can improve the input, the workflow, or the model - only retraining changes the model itself.

A fair criterion can be measured and repeated; responsible use keeps every test safe.

What to do next

Next in the Lab: put two real outputs side by side and use evidence to name the winner.



Print / Save worksheet as PDF

Type your answers in the boxes below, then use this button to print the completed worksheet or choose "Save as PDF".



Student worksheet

From 'It Works' to 'How Well'

Print this page or save it as a PDF to complete by hand.

Name: Class: Date:

Lesson figure

Sort each example card into the correct column and give a reason.

My two questions

Write at least two questions about how AI outputs are made, judged or improved. Aim at least one at how we decide an output is good.

Sort it: output, judge or improve

Write each example in the right column and add one reason.

Example	Output / Way to judge / Way to improve	Reason
---------	--	--------

A finished Arabic-English translation		
---------------------------------------	--	--

Checking the translation for accuracy against the task		
--	--	--

Giving the model a clearer input, then re-testing		
---	--	--

Our safe-use charter		
----------------------	--	--

Write one responsible-use rule for the class and why it keeps us safe.		
--	--	--

 Exit ticket

Exit ticket - Name one example, say whether it is an output, a way to judge, or a way to improve a model, and explain why. Then write one question you still have.

★ Challenge (stretch)

Challenge - Pick a real UAE situation (for example a smart-parking app or an Arabic-English translator) and predict what a strong AI solution would need to get right. Name two performance criteria you would judge it against and one change you might make to improve it. This stretches toward the Proficient idea of connecting the unit to a real context and predicting what a strong solution needs.

Sample Lab (without images)

Two Outputs, One Verdict

In the Core period you learned AI outputs can be judged against clear criteria. Today you become an investigator - comparing real outputs and using evidence to say which one is better. | About 45 minutes

What you will learn

- ✓ I can compare two AI outputs for the same task and write down how they differ.
- ✓ I can judge each output against a simple criterion like accuracy or usefulness.
- ✓ I can use what I observed to say what made one output better.

Lesson route

Work through each part in order.

- 1 Pick a criterion**
Choose one clear measure to judge by
- 2 Compare and log**
Find differences, judge against the measure
- 3 Prove your verdict**
Use evidence, then check it holds

Two Outputs, One Verdict

In the Core period you learned AI outputs can be judged against clear criteria. Today you become an investigator - comparing real outputs and using evidence to say which one is better. | About 45 minutes

M1L · Two Outputs, One Verdict

Lesson

Part 1

Part 2

Part 3

Summary

Worksheet

Cases cracked:



Full lesson · Week 1 · Lab

Two Outputs, One Verdict

Read the whole lesson, then try the practice below and complete your worksheet.

◀ Previously

Previously you learned outputs are judged against clear criteria - now test that as an investigator.

Next ▶

Next you open up the input-model-output pipeline so you can explain how any tool works.



Concept link

Add a one-line note that 'stage' is previewed here and fully unpacked next lesson, or soften to 'where you think it came from'.



I can...

I can compare two AI outputs for the same task and write down how they differ.

I can judge each output against a simple criterion like accuracy or usefulness.

I can use what I observed to say what made one output better.



Words to know

compare - look closely at two outputs to find how they are alike and different

criterion - one clear measure you judge an output against, such as accuracy or usefulness

difference - something that is not the same between the two outputs

evidence - what you actually observed, used to back up your judgement

conclusion - the idea you decide on after weighing the evidence against the criterion

In the Core period you learned that AI outputs can be judged against clear criteria. Today you become an investigator - comparing real outputs and using evidence to say which one is better.

Same task, different outputs

Two AI outputs can be made for the same task and still come out different. Maybe one input was clearer, or the process handled it differently. On its own, 'I like this one better' is not enough. So you pick a criterion - say accuracy or usefulness - and judge each output against it. When you compare them side by side and trace where the difference began, you can say which output wins and prove it. That is detective work: you are finding evidence for a judgement, not guessing.

Log it, judge it, then trust it

A good investigator writes things down. In your log you note each difference, the stage it came from - input, process or output - and how it affects the criterion. Then, before you trust a conclusion, you check it against the task: would it still be true for a new example? Remember the charter - the outputs are made by the teacher, with no personal data, and a confident-looking output still has to be checked against a trusted source before you rely on it.

Lesson figure

Same task, small change, different output - your job is to judge which better meets the criterion.



Why it matters

Why it matters - engineers in the UAE constantly compare AI outputs against criteria before they trust them, whether tuning a smart-traffic system in Abu Dhabi or checking an Arabic-English translation for accuracy; learning to judge and log differences is the first step of that job.



Let's try it - hands-on

Read the first output pair and the named criterion, and predict which output wins.

Judge each output against the criterion.

Log at least two differences and mark the stage each came from.

Rotate to a new pack with a new criterion and log again.

Write one conclusion about what made an output better against its criterion.



Today's tool - hands-on

Tool: Card decks (evaluation-criteria cards)

Deal the deck and read each criterion and its 'how to check' line aloud.

Rank the six criteria for a chosen everyday AI tool (for example, a translation app) from most to least important.

Agree, as a pair, what 'good' would look like for your top two criteria.

Write one measurable check for each of your top two (for example, 'count how many of 10 answers are correct').

Share your ranking with another pair and defend the order.

You will make: A ranked criteria layout and two measurable checks the class will reuse all term.

Stay safe: Fully unplugged; no devices or data.

Lesson figure

Evaluation-criteria cards ranked, each with a way to measure it.



Think about it

Think about it - if you changed only the input and kept everything else the same, could you make the weaker output meet the criterion? How would you test it fairly?



My reflection

My reflection: One difference I logged today, the criterion it changed, and where it came from, was ...

 Cross-curricular link · Mathematics

Turn the accuracy criterion into maths: score each output as a fraction like 7 out of 10, convert it to a percentage, and compare the two outputs numerically so you justify which one wins with data rather than opinion.



Tool for this week

Unplugged - card decks / output packs

This week: evaluation-criteria cards

Teacher lays out the pre-printed card deck / output pack for this task (account-free, no devices).

In pairs, students sort, match or sequence the cards as the worksheet directs.

For each card read its input, process and output and record it on the worksheet.

Compare piles, then agree the reason using the lesson's key words.

Cards

input

process

output

Sort / match /

sequence

record on worksheet,

give the reason



Practice · interactive

Try it yourself

Three quick activities and case check-ins to lock in the lesson.

Part 1 of 3

Pick a criterion

How do you turn a measure into a check?

1Figure 1. Each criterion is one clear measure, paired with a way to actually check it.

Figure 1. Each criterion is one clear measure, paired with a way to actually check it.

A measure and its check

Two AI outputs can be made for the same task and still come out different. On its own, 'I like this one better' is not enough. So you pick a criterion - one clear measure like accuracy or usefulness - and judge each output against it.

A criterion is only useful if you can check it. For every measure you name, agree a plain way to test it - for example, 'count how many of 10 answers are correct'. Then the verdict rests on evidence, not on a feeling.



Match the criterion to its check

Objective 2 · make each criterion measurable

Do it

Tap a criterion on the left, then the way you would check it on the right.

Criterion



Usefulness



Reliability



Accuracy



Clarity

How to check it



can a reader follow it without help



does it give the same answer when repeated



count how many of 10 answers are correct



does it do the exact task asked



AI Evaluators · Case 1

The Two Summaries

“

Two AI summaries of the same report sit side by side. The criterion is accuracy. Evaluator, which difference actually decides the verdict?”



same report



two summaries



accuracy

Which difference decides the verdict?

A

One summary opens with a noticeably longer first sentence here.

B

One summary states a wrong date the report does not support.

C

One summary uses a larger font and a more elegant heading style.

D

One summary simply feels a little friendlier and warmer to read.

Part 2 of 3

Judge each output

Meets it, partly, or fails?

Weigh it against the measure

When you compare two outputs side by side and judge each against the same criterion, you can say which one wins and prove it. That is detective work: you are finding evidence for a judgement, not guessing.

It helps to be more precise than 'good' or 'bad'. Against a named criterion, an output can meet it fully, meet it only partly, or fail it. Naming which one keeps your log honest and your verdict defensible.



Judge against the criterion

Objective 2 · rate each output against the measure

Do it

The criterion is accuracy. Sort each output: does it meet it, partly meet it, or fail it?



Several facts contradict the source



Correct overall, with one unclear claim



Most facts are right, but one date is wrong



The main claim is simply untrue



All 10 checked answers are correct



Every fact matches the trusted source



Meets it



Partly meets



Fails it



AI Evaluators · Case 2

The Useful Answer

“

A student asks a tool for bus times to the museum. The criterion is usefulness. One output lists correct times; the other lists correct facts about the museum’s history. Evaluator, which wins on usefulness?”



asked: bus times



usefulness



verdict

Which output wins on usefulness?

A

Whichever output happens to be the longer one on the screen here.

B

The bus-times output, because it does the exact task asked for.

C

The museum-history output, since it lists more interesting facts.

D

Neither of them, since both outputs do contain some correct facts.

Part 3 of 3

Log it, then trust it

Where did the difference begin?

2Figure 2. A good investigator records each difference, the stage it came from, and whether the verdict still holds.

Figure 2. A good investigator records each difference, the stage it came from, and whether the verdict still holds.

The investigator’s routine

A good investigator writes things down. In your log you note each difference, the stage it came from - input, process or output - and how it affects the criterion.

Then, before you trust a conclusion, you check it against the task: would it still be true for a new example? Remember the charter - outputs are made by the teacher, with no personal data, and a confident-looking output is still checked against a trusted source before you rely on it.



Run the compare-and-log routine

Objective 3 · draw an evidence-based conclusion

Do it

Put the investigator's steps in the order you would work through them.

Tap the steps in the right order, from first to last.

?



Check it against the task and a trusted source

?



Draw a conclusion about which output wins

?



Log each difference and the stage it came from

?



Name the criterion you will judge by

?



Judge each output against that criterion



AI Evaluators · Case 3

Would It Still Hold?

“

Your log says the clearer input gave the more accurate output. Before you trust that conclusion, you reach for one more move. Evaluator, what should you do?”



logged verdict



new example



verify

What should you do before trusting the conclusion?

A

Test it on a fresh new example to see whether it still holds up.

B

Accept it at once, since it is already written down in the log.

C

Delete the log entry so that nobody can question the result later.

D

Trust it because the output happened to sound quite confident.



Away from the screen · Hands-on mission

AI Evaluators: rank the criteria cards

Work through curated output-pair packs, judging each output against a named criterion and logging the differences - then rank the evaluation-criteria deck you will reuse all term.

What you need:

Output-pair packs (curated, paper)

An evaluation-criteria card deck (6 cards)

A log table (stage and criterion labels)

A pencil

Read the first output pair and the named criterion, and predict which output wins.

Judge each output against the criterion.

Log at least two differences and mark the stage each came from.

Rank the six criteria for a chosen everyday AI tool from most to least important.

Write one measurable check for each of your top two criteria.

Share your ranking with another pair and defend the order.

 Fully unplugged - no devices or data. Extend by adding a blank criterion card the pair defines themselves.

Lesson summary

You compared real outputs and used evidence to reach a verdict.

Key ideas to remember

A criterion is only useful when you pair it with a clear way to check it.

Against a named criterion an output can meet it, partly meet it, or fail it.

You log each difference and the stage - input, process or output - it came from.

Before trusting a conclusion, you test whether it still holds on a new example.

What to do next

Next you open up the input-model-output pipeline so you can explain how any tool works.



Print / Save worksheet as PDF

Type your answers in the boxes below, then use this button to print the completed worksheet or choose "Save as PDF".



Student worksheet

Two Outputs, One Verdict

Print this page or save it as a PDF to complete by hand.

Name: Class: Date:

Lesson figure

Judge each output against the criterion, then log the difference and where it began.

Comparison log

For each output pair, write one difference, name the criterion it affects, and mark the stage where it began.

Output pair (criterion) Difference I observed Stage (input / process / output)

Pair 1

Pair 2

Pair 3

My conclusion

Use your log as evidence. What one change most improved an output against its criterion?

Verify it

How could you check your conclusion is right for a new example on the same task?

Tool log - Card decks (evaluation-criteria cards)

Record as you work. Goal: A ranked criteria layout and two measurable checks the class will reuse all term.

Step	What I did	What I saw / result
1	Deal the deck and read each criterion and its 'how to check'	
2	Rank the six criteria for a chosen everyday AI tool (for example)	
3	Agree, as a pair, what 'good' would look like for your top two	
4	Write one measurable check for each of your top two (for example)	
5	Share your ranking with another pair and defend the order.	

Our evaluation criteria

Criterion	Why it matters here	How we will measure it
Safety check (tick): I used no personal data (no faces, voices or names).		

Exit ticket

Exit ticket - Name one difference you logged, the criterion it affected, and the workflow stage it came from, and say whether it made the output better or worse against that criterion.

★ Challenge (stretch)

Challenge - Design your own fair test: choose one thing to change in the input, name the criterion you will judge against, predict how the output should change, and describe how you would compare the before and after to prove it. Then explain what a strong UAE solution built on this idea would need to get right. This stretches toward the Proficient idea of predicting what a strong solution needs and evaluating outputs against criteria such as accuracy and usefulness.

ASSESSMENT

FINAL ASSESSMENT

Grade 8 (Year 9)

Unit: Evaluating and Improving AI Models - Compare, Refine, Evaluate, Govern

ASSESSMENT IDENTIFICATION

Assessment End-of-module final assessment

Module 10 of 14

Phase / Cycle Phase 3 - AI Solution Builder | Cycle 2

Term / Tier Term 1 | Emerging tier

Competency strands AI Conceptual Understanding; AI Solution Design & Development; Critical Evaluation & Informed Interaction; Ethics & Governance

Framework reference ADEK K-12 AI Fluency Framework - AIF - Grade 8 - Emerging

Format Performance-based (formative checkpoints + summative task)

Administration 60-70 minutes | individual written and oral work, with paired discussion

Version / Date Version 1.0 | 2026

Prepared by AI Curriculum Unit

A curriculum-aligned, performance-based assessment of the four AI Fluency competency strands, in which students compare two competing AI system designs, refine the stronger one into a structured workflow, evaluate it systematically against criteria, and produce a structured risk report with mitigation. It includes formative checkpoints, an authentic summative task, an analytic rubric, moderation guidance, annotated exemplars, and a full teacher guide.

For teacher use in ADEK schools. Aligned to the delivered Grade 8 Main Course (Term 1, ten weeks of Core and

Lab lessons). Not for distribution to students.

ADEK K-12 AI Fluency | Final Assessment | Grade 8 - Module 10: Evaluating and Improving AI Models

1. Assessment overview and design rationale

This final assessment measures how well Grade 8 students have met the four AI Fluency competency outcomes taught across the ten weeks of the Term 1 Main Course, 'Evaluating and Improving AI Models'. The unit runs a single produce-judge-improve cycle and deepens it each week, so the assessment is built to capture the move Grade 8 makes beyond earlier grades: students no longer ask only 'does this work?' but 'why does this design perform better than the alternative, and how do I prove it?'. The assessment is valid (it measures what was taught), reliable (two teachers reach the same judgement from the same evidence), and fair (every learner has an equitable route to show what they know). It is performance-based and comparative. Students reason across two competing AI system designs for the same goal, refine the stronger design into a structured workflow, evaluate it systematically against agreed criteria, and conduct a structured ethical risk analysis with mitigation, rather than recalling isolated facts. All AI use is teacher-mediated and account-free, and no personal data is entered at any point, exactly as in the delivered lessons.

What this assessment contains, at a glance

Component Purpose Evidence type Time

Formative checkpoints

(2)

Assessment FOR learning - check comparative and evaluative reasoning mid-unit and adjust teaching before the summative task.

Teacher observation and short written or oral responses

10-15 min

each, in-

lesson

Summative

performance task (4 parts)

Assessment OF learning - one authentic task covering all four strands through compare, refine, evaluate and govern.

Student worksheets, teacher observation, brief conversation

60-70 min

(may be split)

Analytic rubric Judge each strand at Emerging / Proficient / Advanced against observable descriptors.

Teacher judgement record

During or
right after
the task

Evidence and
moderation tools

Record evidence consistently and agree
judgements across teachers.

Observation record,
moderation protocol

As needed

Annotated exemplars Anchor judgements to worked examples at
each level.

Reference material Reference

Research-based design principles

The design draws on well-established assessment research so that the results can be
trusted and used to move learning forward:

Backward design (Wiggins and McTighe). The comparison worksheets, refinement
frame and rubric were built from the framework outcomes first, then the evidence,
then the activity, so every part earns its place against a Grade 8 outcome.

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Models

Constructive alignment (Biggs). Each task part is aligned to one primary strand, its
success criteria, and a cognitive-demand level, shown in the alignment map in Section
2.

Formative assessment (Black and Wiliam). Two checkpoints make comparative
and evaluative thinking visible early and carry specific feedback moves, so teaching
can adapt before the summative task.

Triangulation of evidence. Judgements draw on three sources - what students
produce, what they do, and what they say - because comparative reasoning shows
most clearly when a student defends a judgement aloud.

Cognitive demand (Webb's Depth of Knowledge and the SOLO taxonomy).

Parts progress from tracing and comparing pipelines (DOK 2) to systematic evaluation
and structured risk analysis with mitigation (DOK 3, SOLO Relational to Extended
abstract), matching the demand levels used in the lessons.

Universal Design for Learning (CAST). Multiple means of engagement,
representation, and expression are built in, so access arrangements are part of the
design rather than an afterthought.

Academic integrity. Parts 2, 3 and 4 (refine, evaluate, govern) are recorded individually so
each student's reasoning is their own. Students may discuss the two designs during Part 1
exploration, but recorded judgements, evaluation log entries and the risk report must be
independent. All AI steps are teacher-operated on the board with pre-generated, curated
outputs;

students never log in and enter no personal data. Teachers may scribe or accept alternative

response modes without changing the content of a student's answer.

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2. Curriculum alignment map

This section makes the link between the delivered curriculum and every part of the assessment explicit and auditable. A teacher or reviewer can trace each strand from the framework outcome, to where it was taught, to where it is assessed.

Module identification

Field Detail

Programme Grade 8 Main Course, Term 1 (Emerging tier)

Module 10 of 14

Phase / Cycle Phase 3 - AI Solution Builder | Cycle 2

Unit theme Evaluating and Improving AI Models - the produce-judge-improve cycle

Signature activity

assessed

A structured, multi-step AI classifier solution that is planned, built, evaluated against performance criteria and improved, then risk-audited (Weeks 7-10)

Lessons covered Week 1 to Week 10 (Core + Lab, twenty 45-minute periods)

Framework reference AIF - Grade 8 - Emerging (CU, SD, CE, GE)

Framework outcomes assessed (the four strands)

Competency strand Emerging outcome
(target)

Proficient outcome Advanced outcome
(stretch)

AI Conceptual

Understanding

Describes how AI
systems process inputs
and generate outputs.

Explains how different
factors (data, inputs,
design) affect
performance.

Compares approaches
and explains why some
solutions perform better.

AI Solution Design &
Development

Develops AI-powered
solutions for specific
tasks.

Designs structured
workflows combining

multiple steps or tools.

Designs and refines solutions considering how components interact.

Critical Evaluation & Informed Interaction
Identifies limitations in outputs.

Evaluates solutions using criteria such as accuracy, bias, and reliability.

Refines solutions systematically and justifies improvements using evidence.

Ethics & Governance Identifies ethical concerns in AI use.

Conducts structured evaluation of risks (bias, privacy, misuse).

Proposes mitigation strategies and explains their impact.

Alignment matrix: taught, then assessed

Each strand is mapped to the weeks that teach it and the assessment component that provides evidence, with the cognitive-demand level for the summative part.

Strand Where taught

Success criteria assessed Assessed by Demand

AI Conceptual W2 Core/Lab; Trace how each design turns Checkpoint 1; DOK 2-3 /

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ADEK K-12 AI Fluency | Final Assessment | Grade 8 - Module 10: Evaluating and Improving AI Models

Understanding W1 inputs into outputs; explain how data, inputs and design affect performance; compare two approaches and say why one performs better

Summative Part

1

SOLO

Relational

AI Solution Design & Development

W3 Core/Lab;

W7-W9

Design a structured multi-step workflow with a check step and fallback; explain how the components interact

Checkpoint 1;

Summative Part

2

DOK 3 /

SOLO

Relational

Critical Evaluation &

Informed Interaction

W4 Core/Lab;

W8-W9

Evaluate a solution against accuracy, reliability and bias using test-set evidence; refine systematically and justify improvements

Checkpoint 2;

Summative Part

3

DOK 3 /

SOLO

Relational-

Extended

Ethics & Governance W5 Core/Lab;

W6

Sort risks into bias, privacy and misuse; rate likelihood and impact; propose mitigation and explain its impact

Checkpoint 2;

Summative Part

4

DOK 3 /

SOLO

Relational-

Extended

Vertical coherence (progression)

From Grade 7: students designed multi-step AI workflows, evaluated outputs against criteria such as accuracy and usefulness, improved workflows through iteration, and identified responsible-use risks with governance rules. Grade 8 extends this into performance evaluation and optimization - students now compare different AI system designs and explain why one performs better, design structured workflows where multiple components interact, evaluate solutions systematically against criteria including accuracy, bias, and reliability, and conduct structured ethical risk analysis with mitigation strategies that explain their impact.

Towards Grade 9: students move into Phase 4 systems design, architecting larger AI systems, defending design trade-offs and governance at a system level. A strong Advanced performance here - comparing designs on evidence, refining components that interact, and prioritising mitigations by likelihood and impact - is early evidence of readiness for the Grade 9 outcomes.

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3. Learning outcomes and student-facing success criteria

Teachers share the student-facing 'I can' statements before and during the unit so learners know what success looks like. They are written in the same language used in the lessons.

Strand Student-facing success criteria ('I can...')

AI Conceptual

Understanding

I can trace how each design turns an input into an output through its pipeline.

I can explain how a factor such as the training data or an input changes a design's performance.

I can compare two designs for the same goal and explain, with a reason, why one performs better than the other.

AI Solution Design &

Development

I can take the stronger design and lay it out as an ordered, multi-step workflow.

I can add a check step and a fallback path, and explain how the steps connect so the parts work together.

I can use words like input, process, output, check, workflow and fallback correctly to describe my solution.

Critical Evaluation &

Informed Interaction

I can evaluate a solution against agreed criteria - accuracy, reliability and bias - using evidence from a test set.

I can name the limitation behind a wrong output and locate the step that caused it.

I can make a systematic, evidence-based refinement and show the before-

and-after result that proves it helped.

Ethics & Governance I can find the ethical concerns in an AI use case and sort them into bias, privacy and misuse.

I can rate each risk for likelihood and impact on a matrix and say which to mitigate first.

I can propose a specific mitigation for a risk and explain the impact it would have.

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4. Assessment blueprint

The blueprint shows how the summative task distributes emphasis and cognitive demand across the four strands. Each strand carries roughly equal weight so no single strand dominates the overall judgement, and every part demands comparative or systematic reasoning rather than recall.

Summative part Primary strand Cognitive demand

Evidence Weight

Part 1 - Compare two designs

AI Conceptual

Understanding

DOK 2-3 Two Designs

Comparison

worksheet + talk

25%

Part 2 - Refine the workflow

AI Solution Design & Development

DOK 3 Workflow

Refinement frame

(individual)

25%

Part 3 - Evaluate systematically

Critical Evaluation & Informed Interaction

DOK 3 Systematic

Evaluation log

(individual)

25%

Part 4 - Govern the risks Ethics & Governance DOK 3 Structured Risk Report + short

conversation

25%

Cognitive-demand spread. The task deliberately rises from tracing and comparing two pipelines (DOK 2-3) to systematic evaluation and structured risk analysis with prioritised mitigation (DOK 3, SOLO Relational to Extended abstract). Advanced probes in each part give every student a route to demonstrate higher-order comparative reasoning without penalising those still consolidating the Emerging outcome. Written and oral responses are expected throughout, as in the delivered lessons.

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5. Formative checkpoints (Assessment FOR learning)

Run these two short checkpoints during the unit, not on the summative day. Each names when to use it, what to look for at each level, and a specific feedback move that closes the gap before the summative task.

Checkpoint 1 - 'Which design performs better, and why?' (use after Week 3)

Strands: AI Conceptual Understanding and AI Solution Design & Development.

Activity. Show two design cards for the same goal - an AI that sorts photos of harvested dates into 'ready to pack' or 'leave to ripen'. Design A trains on photos of one date variety in bright light and outputs a label straight away. Design B trains on several varieties in mixed light and adds a check step that flags low-confidence photos for a person. Ask: 'Which design performs better, and

why? Name one factor Design A ignores, and say how it would change the output.' Allow two minutes of paired thinking, then take individual answers.

Look-for Emerging Proficient Advanced

Comparing the designs

Picks Design B but gives a general reason such as 'it looks better'

Picks Design B and names one factor (varied training data or the check step) that improves performance

Compares both designs and explains why B performs better, linking the factor to the output it changes

Reasoning about a factor

Names a factor only with prompting

Names a specific factor

Design A ignores and its likely effect

Explains how the missing factor would make Design

A biased or unreliable on a named case

Feedback move. If a student judges by appearance, re-teach the compare-and-log routine from W1 Lab: line up the two outputs against one named criterion and mark where the difference begins.

If they cannot name a factor, point to the pipeline diagram from W2 and ask which stage - the training data or the check - is different between A and B.

Checkpoint 2 - 'Evaluate and mitigate' (use after Week 5)

Strands: Critical Evaluation & Informed Interaction, and Ethics & Governance.

Activity. Give students a short test-set result card for the date sorter: on ten photos it scored 7 correct, but every miss was on a dark-skinned variety the model rarely saw in training. Ask students to (a) name the limitation and the criterion it fails on, and (b) name the ethical concern this raises, rate its likelihood and impact as low or high, and propose one mitigation and its impact.

Look-for Emerging Proficient Advanced

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ADEK K-12 AI Fluency | Final Assessment | Grade 8 - Module 10: Evaluating and Improving AI Models

Evaluating with criteria

Says the score is 'not good' without a criterion

Judges against accuracy and reliability and names the limitation using the test-set evidence

Weighs accuracy against bias and explains the model is unreliable for one variety, citing the misses

Risk and mitigation Says 'someone should check' with no reason

Names the concern (bias) and proposes a mitigation such as adding more examples of that variety

Rates likelihood and

impact, proposes a mitigation and explains the impact it would have on fairness

Feedback move. If a student stops at 'it is unfair', use the W5 sorting key and the likelihood-and-impact matrix: sort the concern into bias, then place it on the grid and ask what a mitigation would

change. If they cannot separate a slip from a limitation, show that all seven misses fall on one variety - a pattern, not a one-off - so the fix must target the training data, not a single photo.

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6. Summative performance task - 'The Date Harvest Sorter'

An authentic task built directly on the unit's signature activity: a structured, multi-step AI classifier that an Abu Dhabi date farm wants for its packing line. Two design teams have proposed competing solutions for the same goal. Students compare the two designs, refine the stronger one into a workflow, evaluate it systematically against criteria, and produce a structured risk report with mitigation. The four parts move through all four strands. Read the scene-setting script, then run the parts in order.

Scene-setting script (read aloud). 'A date farm near Al Ain wants an AI helper for its packing line. As each date passes a camera, the helper should sort it into ready to pack, leave to ripen, or remove (spoiled). Two teams have built competing designs for the same goal. Your job today is to

think like an AI solution builder: first work out which design performs better and why, then refine that design into a clear workflow, then evaluate it systematically using the test results, and finally write a short risk report that says how you would make it safe and fair to use. All the AI runs are already prepared for you on the board - you will reason about the results, not log in to anything, and we use only photos of fruit, never people. Explain your thinking in writing and out loud; a strong answer always says why, not just what.'

Materials to prepare (see Appendix)

Two Designs Comparison worksheet - a side-by-side card for Design A and Design B of the date sorter, with their pipelines, training data and a pre-generated test-set result for each (from the pre-generated output packs).

Workflow Refinement frame - a printed template with labelled boxes (Input, Process/model, Check, Output) plus a fallback path, based on the Week 7 planning frame and the Scratch storyboard.

Systematic Evaluation log - a pre-formatted table (input, output, accuracy, reliability, bias, responsible step) with one worked row, as used in Weeks 8-9.

Structured Risk Report worksheet - a risk table plus a blank likelihood-and-impact matrix and a mitigation column, from the Week 5 Ethics Audit.

Card decks for reference: evaluation-criteria cards, bias-source cards and a Model Record Card; coloured pens for annotating.

Teacher-operated board only: the pre-generated output packs for both designs (Google Teachable Machine images and Scratch runs curated in advance). No student logins; no personal data.

Part 1 - Compare two designs

Strand: AI Conceptual Understanding · DOK 2-3

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Give each student the Two Designs Comparison worksheet and read the shared goal aloud:

'Both designs sort date photos into ready to pack, leave to ripen, or remove.' Then ask:

'Trace each design: what is the input, what does the model do, what is the output?

Where do the two pipelines differ?'

'Using the test-set results on the card, which design performs better? Give the reason - name the factor (training data, inputs or design) that makes the difference.'

Advanced probe: 'Design A scored well on the sample it was trained on but poorly on new varieties. Explain why a design can look strong yet perform worse, and what that tells you about how to judge an AI solution.'

Part 2 - Refine the workflow

Strand: AI Solution Design & Development · DOK 3 · Completed individually

Give each student the Workflow Refinement frame. Say: 'Take the stronger design and refine it into a clear workflow. Lay out the ordered steps, add a check step that measures the output against a performance criterion, and add a fallback path for when a date cannot be sorted confidently. Then explain how your steps connect - how does the check step change what reaches the packer?' Prompt each student: 'Why did you place the check where you did, and what would break if you removed it?'

Refinement guidance (for the teacher). A complete refinement lays out Input (a date photo) -> Process (the trained classifier) -> Check (is the model's confidence above the agreed bar, and does the label match a quick human spot-check?) -> Output (the sorting decision), with a Fallback

path (send low-confidence dates to a person, do not auto-sort).

Strong work explains interaction: the check step catches confident-but-wrong labels before they reach the packer, and the fallback keeps an unusual date safe rather than mislabelling it. This targets the Advanced outcome - designs and refines solutions considering how components interact.

Part 3 - Evaluate systematically

Strand: Critical Evaluation & Informed Interaction · DOK 3 · Completed individually

Give each student the Systematic Evaluation log and the pre-generated test-set results for the refined design. Say: 'Evaluate the solution properly. For each test photo, record the output and score it against every criterion - accuracy, reliability and bias. Find the criterion that fails most, name the limitation and the step that caused it, then propose one systematic refinement and say what before-and-after evidence would prove it helped.'

Allow independent work, then take one judgement from each student with its evidence.

Planted evaluation evidence (for the teacher). The results show accuracy of 12 out of 15

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overall, but all three misses are on one under-represented variety (a bias pattern, not random slips), and two 'ready to pack' labels were confidently wrong (a reliability problem the check step should catch).

Strong work separates a one-off slip from a repeating limitation, links the bias miss to the training

data step and the confident-wrong miss to a missing or weak check, and proposes a targeted, testable refinement with a before-and-after measure - the Advanced outcome, refines systematically and justifies improvements using evidence.

Part 4 - Govern the risks

Strand: Ethics & Governance · DOK 3

Give each student the Structured Risk Report worksheet. Say: 'Now govern this solution before the farm uses it. Find the ethical and governance risks, sort each into bias, privacy or misuse, and plot it on the likelihood-and-impact matrix. For each risk, propose a specific mitigation and explain the impact it would have. Finish by naming the one mitigation you would build in first, and why.' Then ask:

'Which risk is highest priority on your matrix, and what makes it high likelihood, high impact?'

'Give one mitigation and explain its impact - what exactly does it reduce, and does it remove the risk or only lower it?'

'Where could privacy become a concern even though we only photograph fruit, and how would you prevent it?'

Advanced probe: 'A mitigation has a cost - it may slow the line or need a person. Justify why your first mitigation is still worth building in.'

How the task maps to the strands

Part Strand What strong evidence looks like Demand

1 AI Conceptual Understanding Traces both pipelines, compares them on the test-set evidence, and explains why the stronger design performs better by naming the factor

DOK 2-3

2 AI Solution Design & Development

Refines the design into an ordered workflow with a check step and fallback, and explains how the components interact

DOK 3

3 Critical Evaluation & Informed Interaction

Evaluates against accuracy, reliability and bias with test-set evidence, and proposes a systematic, evidence-based refinement

DOK 3

4 Ethics & Governance Sorts and rates risks on the matrix and proposes prioritised mitigations, explaining the impact of each

DOK 3

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7. Analytic rubric

Judge each strand independently against the descriptors below. Circle the observed level.

The rubric describes observable performance, so two teachers using it on the same evidence should reach the same level. Because Grade 8 is comparative, the Proficient and Advanced descriptors turn on whether a judgement is justified with evidence and a reason, not on whether an answer is present.

Competency

strand

Emerging Proficient Advanced

AI Conceptual

Understanding

(Part 1)

Identifies that Design B is the stronger design but gives a general or unsupported reason, and cannot yet trace where the two pipelines differ.

Traces both pipelines and explains how a factor such as the training data or the check step affects performance, selecting Design B for a specific reason.

Compares the two designs and explains why B performs better even though A looks strong, linking the factor to the exact output it changes and to how a solution should be judged.

AI Solution Design
& Development

(Part 2)

Lays out some workflow

steps, but the order is incomplete or the check step or fallback is missing.

Designs a complete, ordered multi-step workflow with a check step and a fallback path for the stronger design.

Designs and refines the workflow explaining how the components interact - how the check step and fallback change what reaches the output and keep an unusual case safe.

Critical Evaluation
& Informed
Interaction

(Part 3)

Completes part of the evaluation log or judges by appearance, and names a limitation only with support.

Evaluates outputs against accuracy, reliability and bias using the test-set evidence and names the limitation behind the failing criterion.

Refines systematically: separates a slip from a repeating limitation, traces it to the responsible step, and justifies a targeted improvement with a before-and-after measure.

Ethics &
Governance
(Part 4)

Identifies that one or more risks are problems but does not sort or rate them, and mitigations are vague.

Conducts a structured risk evaluation - sorts risks into bias, privacy and misuse and rates likelihood and impact on the matrix.

Proposes prioritised mitigation strategies and explains the impact of each, justifying which to build in first against its cost.

Overall level judgement

Overall level Decision rule

Emerging 0-2 strands at Proficient, the rest at Emerging. The student recognises the stronger design and that risks exist, but cannot yet justify a comparison with evidence, design an interacting workflow, evaluate systematically, or

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structure and prioritise risks.

Proficient

(expected)

At least 3 of 4 strands at Proficient or above. The student compares designs on a named factor, refines a complete workflow with a check and fallback, evaluates against accuracy, reliability and bias with evidence, and runs a structured risk evaluation.

Advanced At least 3 strands at Advanced, with no strand below Proficient. The student explains why one design performs better, refines for component interaction, justifies improvements with before-and-after evidence, and prioritises mitigations by likelihood, impact and cost.

Working towards Emerging. If a student cannot yet meet the Emerging descriptor in a strand even with support, record 'working towards' and use the linked bridging content (Section 11) before re-checking. This is diagnostic information, not a failure.

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8. Evidence recording and triangulation

Base each judgement on three sources of evidence, not one worksheet. This raises

reliability and gives students more than one way to show comparative reasoning, which often surfaces most clearly when a student defends a judgement aloud.

Evidence source What to collect Strands it best evidences

Products (what students make)

The Two Designs Comparison worksheet (Part 1), the Workflow Refinement frame (Part 2), the Systematic Evaluation log (Part 3) and the Structured Risk Report (Part 4)

All four strands

Observations (what students do)

How students annotate and compare the two design cards, place the check and fallback, and plot risks on the matrix

AI Solution Design & Development; Ethics & Governance

Conversations (what students say)

Answers to the Part 1 'why is it better' probe, the Part 3 evidence probe and the Part 4 priority-and-impact probe

AI Conceptual Understanding; Critical Evaluation & Informed Interaction

Teacher observation record (one per student)

Field Record

Student name / Class _____ Date: _____

AI Conceptual Understanding E P A Note: _____

AI Solution Design & Development

E P A Note: _____

Critical Evaluation & Informed Interaction

E P A Note: _____

Ethics & Governance E P A Note: _____

Overall judgement EMERGING PROFICIENT ADVANCED

Access arrangements used _____

Next step / bridging _____

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9. Standardisation and moderation guide

Moderation makes judgements consistent across teachers and classes. Run a short moderation meeting after the first set of students has been assessed.

1. Each teacher independently levels the same 2-3 anonymised student sets (all four worksheets) using the rubric in Section 7.
2. Compare levels. Where they match, note the rubric language that decided it. Where they differ, return to the observable evidence and the annotated exemplars in Section 10 to agree a level.
3. Agree 'borderline' decisions: a student who is Proficient on three strands but Emerging on one is Proficient overall; a student Advanced on two strands and Proficient on two is Proficient (Advanced needs three).
4. Record any task-specific decisions (for example, how much a comparison must justify to count as Proficient, or whether a matrix plotted without ratings counts) so the whole team applies them the same way.

Common pitfalls to avoid

Rewarding an answer over a justification. In a comparative task, picking Design B is not Proficient on its own; the reason and the named factor are what earn the level.

Rewarding fluency over reasoning. A confident talker who cannot cite test-set evidence in Part 3 is still Emerging on that strand.

Halo effect. Level each strand on its own evidence; a strong Part 1 comparison does not lift a weak Part 4 risk report.

Counting a plotted matrix as a structured evaluation. In Part 4, credit Proficient only when risks are sorted and rated with a mitigation, not merely placed on the grid.

Counting scaffolding as independence. If a check step or a refinement needed heavy teacher support, it is Emerging, not Proficient.

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10. Annotated exemplars

Use these worked responses to anchor judgements. Each shows a plausible Grade 8 performance across the four parts and the annotations that place it at a level.

Exemplar - EMERGING

Student response (observed) Annotation (why this level)

Part 1: 'Design B is better because it looks more

careful.' Cannot say where the pipelines differ. Part

2: lays out Input - model - Output but leaves out the

check step and fallback. Part 3: marks a few log

rows 'wrong' but judges by the picture, not the

criteria; names no limitation. Part 4: 'There could be

bias' but does not sort or rate the risks; mitigation is

'be careful'.

Recognises the stronger design and that risks

exist, but the comparison is unsupported, the workflow does not show interacting components, the evaluation is not systematic, and risks are neither structured nor mitigated. 0-2 strands at Proficient - Emerging overall. Next step: the W1 compare-and-log routine and the W5 sorting key and matrix.

Exemplar - PROFICIENT

Student response (observed) Annotation (why this level)

Part 1: 'Design B performs better because it trained on several varieties and adds a check step, so it does not mislabel new dates like Design A.'

Part 2: complete workflow - Input, model, Check (confidence bar plus spot-check), Output, and a fallback sending low-confidence dates to a person.

Part 3: scores the log against accuracy, reliability and bias and writes 'the misses are all on one variety - a bias limitation from the training data'.

Part 4: sorts three risks into bias, privacy and misuse, rates each on the matrix, and gives a mitigation for each.

Compares on a named factor, designs a complete workflow with check and fallback, evaluates against all three criteria with evidence, and runs a structured, rated risk evaluation. At least 3 strands at Proficient - Proficient overall, the expected Grade 8 standard.

Exemplar - ADVANCED

Student response (observed) Annotation (why this level)

As Proficient, and: Part 1 explains that Design A's high score on its own sample is misleading because it was never tested on new varieties, 'so a strong-looking score is not proof - you judge on a fair test set'. Part 2 explains the interaction: 'the check step catches confident-wrong labels before the packer, and the fallback keeps an unusual date safe instead of guessing.' Part 3 separates the two confident-wrong slips from the repeating bias miss, traces each to its step, and proposes retraining with more of the missing variety plus a tighter confidence bar, naming the before-and-after measure. Part 4 prioritises the bias risk as high likelihood, high impact, and justifies building it in first even though Explains why one design performs better, refines

for component interaction, justifies improvements with before-and-after evidence, and prioritises mitigations by likelihood, impact and cost. At least 3 strands at Advanced with none below Proficient - Advanced overall.

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retraining slows deployment.

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11. Reporting, feedback and next steps

Report each student against the four strands and the overall level, and turn the result into a next step. Feedback is most useful when it is specific and forward-looking.

If the student is... Feed-forward for the student Teacher next step / bridging link

Working towards /

Emerging

'You can spot the stronger design and that risks exist. Next, practise saying why - name the factor that makes one design better, and back it with evidence.'

Re-run the W1 compare-and-log routine and the W4 judge-with-criteria grid; use the bias-source cards and the likelihood-and-impact matrix from W5 before re-checking.

Proficient (expected) 'You compare on evidence, refine a working workflow and run a structured risk evaluation. Next, explain how your components interact and prioritise your mitigations by impact.'

Extend with the W8-W9 iterate-twice log and the fairness-evaluation step (test the workflow on two input types); introduce component-interaction thinking to prepare for Grade 9.

Advanced 'You justify why a design performs better and prioritise mitigations by cost. Next, defend design trade-offs across a whole system.'

Offer the W9 refine-and-justify stretch and the W10 showcase

justification; begin early Grade 9

Phase 4 systems-design tasks.

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12. Accessibility and Students of Determination

Access is designed in, following Universal Design for Learning and the school's Best Endeavours and Reasonable Adjustments (BERA). None of these changes the construct being assessed - the comparative and evaluative reasoning - they change how a student accesses the task or shows what they know.

Universal Design for Learning - built-in options

UDL principle Options offered in this assessment

Multiple means of engagement

A familiar UAE context (a date-farm packing line); choice of which test photos to reason about; a 'say why, not just what' framing that values reasoning over a single right answer.

Multiple means of representation

The two design cards read aloud; pipelines shown as labelled diagrams; the likelihood-and-impact matrix pre-drawn; key words (input, model, output, check, fallback, accuracy, reliability, bias, privacy, misuse, mitigation) on a word bank.

Multiple means of action and expression

Students may write, annotate the cards, arrange step-cards, plot the matrix, or explain aloud; a teacher may scribe; the workflow can be built with cut-and-place block cards instead of drawing.

Reasonable adjustments by need (BERA)

Need Adjustment

Visual impairment Large-print, high-contrast design cards and matrix; read all test-set results aloud; allow a magnifier or screen reader; use tactile step-cards for the workflow.

Hearing impairment Provide the scene-setting script and all prompts in writing; face the student when speaking; check understanding of the comparison visually.

Physical / fine-motor Allow pointing or verbal answers scribed by an adult; use cut-and-place step-cards and matrix tokens instead of writing and drawing.

Attention / cognitive load

Split into two sessions (Parts 1-2, then 3-4); one part at a time; extra processing time; a checklist of the four parts and a visible produce-judge-improve diagram.

Language / EAL Sentence stems ('Design B performs better because ____', 'This risk is high

impact because ____', 'My mitigation would ____'); bilingual key-word card; accept answers in the student's stronger language and scribe.

Sensory needs Quiet space; movement break between parts; reduce visual clutter on the printed cards and log.

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13. Administration guide

Field Guidance

Time 60-70 minutes total, about 15 minutes per part. May be split into two sessions (Parts 1-2, then 3-4).

Grouping Part 1 may be discussed in pairs; Parts 2, 3 and 4 are recorded individually, with a short conversation for Parts 1 and 4.

Conditions Calm, low-stakes. Reassure students that Part 1 discussion is expected; recorded judgements, the evaluation log and the risk report are their own.

When to run After Week 9, once the signature solution has been planned, built, evaluated and refined, so students have the workflow, evaluation and risk routines before the task.

Preparation Print the four worksheets and the reference cards (Appendix); pre-generate the output packs for both designs on the teacher device; display the word bank and the produce-judge-improve diagram.

Integrity Keep the two design cards and the planted test-set results consistent across the cohort; operate all AI steps yourself; enter no personal data; do not pre-mark the limitations or risks for students.

Before / during / after checklist

Before: share the success criteria; print materials; pre-generate both output packs; plan access arrangements; run both checkpoints earlier in the unit.

During: read the script; run parts in order; observe and note; ask the 'why is it better', 'how do the parts interact' and 'priority and impact' probes; scribe where needed.

After: complete the rubric and observation record; moderate a sample; record next steps and bridging links.

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Appendix. Printable stimulus materials

Print one copy of each per student. The test-set results are pre-generated in advance on the teacher device (Google Teachable Machine and Scratch) and printed; students reason about them and never log in. The comparison and evaluation results contain deliberate patterns (a bias miss on one variety and confident-wrong labels); do not mark them for students.

A. Two Designs Comparison worksheet (Part 1)

Shared goal: an AI helper sorts each date photo on the packing line into 'ready to pack', 'leave to ripen', or 'remove (spoiled)'.

Feature Design A Design B

Input One date photo from the line camera One date photo from the line camera

Training data Photos of one variety (Khalas) in bright light only

Photos of four varieties in mixed light

Process / model Trained classifier outputs a label straight away

Trained classifier plus a confidence score

Check step None - the label is used directly Flags low-confidence photos for a quick human spot-check

Output ready to pack / leave to ripen / remove ready to pack / leave to ripen / remove, or 'needs a person'

Test-set result (15 photos)

14 of 15 correct on Khalas; 6 of 15

correct on other varieties

12 of 15 correct across all four varieties;

flags the 3 it is unsure of

Questions: (1) Trace each design and mark where the pipelines differ. (2) Using the test-set results,

which design performs better, and why - name the factor. (3) Design A scored 14 of 15 on its own

variety. Explain why that high score does not prove it is the better solution.

B. Workflow Refinement frame (Part 2)

Refine the stronger design. Fill each box in order, then add the fallback path and answer the interaction question.

Step What happens here Why this step is needed

1. Input _____

2. Process / model _____

3. Check _____

4. Output _____

Fallback (if check fails)

Interaction question: how does the check step change what reaches the Output, and what would go

wrong if you removed it?

C. Systematic Evaluation log (Part 3)

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Score the refined design against every criterion. Row 1 is a worked example; complete the rest from the printed test-set results, then answer below.

Photo Output Accuracy

(right/wrong)

Reliability Bias check

(variety)

Responsible

step

1

(worked)

ready to pack right consistent balanced verified vs true

label

2

3

4

5

After the log: overall accuracy is 12 of 15, but the 3 misses fall on one under-represented variety and 2 'ready to pack' labels were confidently wrong. Name the limitation and the step that caused it,

then propose one systematic refinement and the before-and-after measure that would prove it worked.

D. Structured Risk Report worksheet (Part 4)

List each risk, sort it, rate likelihood and impact, then give a mitigation and its impact. Plot each risk

on the matrix by its number.

Risk in this AI use Type (bias /

privacy /

misuse)

Likelihood

(L/M/H)

Impact

(L/M/H)

Mitigation and

its impact

1 Mislabels an under-represented variety

2 A face is caught in the background of a line photo

3 Staff trust the label and stop

checking spoiled dates

Likelihood \ Impact Low impact High impact

High likelihood plot risk # here plot risk # here (build in first)

Low likelihood plot risk # here plot risk # here

Finish: name the one mitigation you would build in first and justify it against its cost (for example,

slowing the line or needing a person).

E. Model Record Card (reference)

Students may use the one-page Model Record Card from Week 2 onward to support Parts 3 and 4:

data source, purpose, version and changes, known limitations, test results, and the responsible-use

check (non-personal data only).

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